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***Equivalents of
the Axiom of Choice, II***

H. RUBIN and J.E. RUBIN

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EQUIVALENTS OF THE AXIOM OF CHOICE, II

Herman RUBIN

*Professor of Mathematics and Statistics
Purdue University
West Lafayette
Indiana
U.S.A.*

and

Jean E. RUBIN

*Professor of Mathematics
Purdue University
West Lafayette
Indiana
U.S.A.*



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In dedication to our children

Arthur Leonard Rubin

Leonore Anne Rubin Findsen

in hope of the future;

and to

Alfred Tarksi

in remembrance of the past.

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PREFACE (1963 Edition)

In 1955 at Stanford University Professor Patrick Suppes gave a course on axiomatic set theory, during which time Professor Herman Rubin was asked to give a series of lectures on the axiom of choice. During the course of these lectures it was noted that, while there was much in the literature on the axiom of choice, the material was available only in many diverse journals and books. It was suggested that we collect this material to make it more readily available. It seemed like a simple enough project to begin with; then it grew and grew some more and now has blossomed forth into a book.

The book consists of a selection of the forms of the axiom of choice which appeared in the literature together with additional forms which were obtained in the process of writing the book. It would have been a hopeless task to try to include all of the forms of the axiom of choice which appeared in the literature, so we chose the forms which in our opinion were either used often in practice, unusual, relatively unknown, or particularly weak or strong. We hope that we have included all of the interesting equivalents of the axiom of choice.

We assume a knowledge of logic and elementary set theory (von Neumann-Bernays-Gödel set theory), but we do include a list of definitions of set theoretical symbols and terms in the section entitled "Preliminary Definitions and Theorems".

In Part I we discuss propositions which are equivalent to the usual form of the axiom of choice. These equivalents will be referred to as set forms. In Part II we discuss stronger forms - essentially forms which are obtained from the set forms by changing the word "set" to "class". These latter forms are called class forms. The set forms of the axiom of choice are

the forms which are most often used in practice.

In preparing this monograph for publication we first prepared a draft and sent it to several people for their comments and corrections. We are very grateful to the people who did reply. We believe that the quality and usefulness of the book was greatly improved by their comments. In particular we should like to thank Professor Alfred Tarski for his many useful comments and corrections. Others whom we should like to thank are Professors E. W. Beth, A. Levy, D. Scott, and R. Vaught. Our typists, Ann Breen and Barbara Johnson, also deserve credit for bearing with us under strain, and we should like to thank the Mathematics and Statistics departments at Michigan State University for their cooperation.

HERMAN RUBIN

JEAN E. RUBIN

September 1961
East Lansing, Michigan

PREFACE (1985 Edition)

In the twenty years since "Equivalents" (Rubin & Rubin 1963) has been published, there have been so many new developments with respect to the axiom of choice that we thought it was time for a new book to describe these developments. Many new equivalents have been discovered since 1963. The section on algebraic equivalents has been greatly expanded to include statements about vector spaces, groups and other algebras. There are so many forms in logic, analysis and topology that a new section was added to include these forms. Another new section was added for those forms which are equivalent to the axiom of choice only under the axioms of extensionality and foundation. In addition, new forms were added to the other sections.

We have also rewritten all the material using modern notation in a style which, we hope, is easier to read.

We should especially like to thank the following people for their help: our son Arthur who proof read Part I and made significant contributions to Part II; Norbert Brunner who, while visiting Purdue during the 1983-84 academic year, made many suggestions for improving the manuscript; Paul Howard who proof read part of the manuscript and made important contributions; Wilfrid Hodges who sent us a list of some of the new algebraic equivalents of the axiom of choice; Andreas Blass who corrected some errors and sent us his unpublished proof that the "vector space theorem" implies the axiom of choice; and our typist, Judy Snider, who did an admirable job of typing and cutting and pasting.

June 1984
West Lafayette, Indiana

HERMAN RUBIN
JEAN E. RUBIN

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INTRODUCTION

Equivalents of the axiom of choice appear frequently in almost all branches of mathematics in a large variety of different forms. In this monograph we should like to describe some of these equivalents.

By the "Axiom of Choice" we mean a statement similar to the following statement given by Russell [1906] and Zermelo [1908].

AC: If x is a pair-wise disjoint family of non-empty sets, then there is a set C which consists of one and only one element from each set in x .

The reluctance that some mathematicians have about using the axiom of choice is due to its non-constructive nature. There is no rule for constructing the choice set, C . Since, in our opinion, it is more desirable to have a constructive proof rather than a non-constructive proof, the axiom of choice and other non-constructive principles should be avoided whenever possible. At times, of course, non-constructive principles are unavoidable and the least the mathematician can do at these times is to declare their use in his/her proof.

The set theory in which we prove most of the equivalences of the axiom of choice is called NBG^o (NBG with atoms but without the axiom of foundation). It is due to von Neumann, Bernays and Gödel and it is described in detail in the section entitled "Preliminary Definitions and Theorems".

The relative consistency and independence of the axiom of choice with NBG (and NBG^o) has now been completely determined.

(NBG is consistent if and only if NBG° is consistent). Kurt Gödel [1938,1939,1940] has shown that if NBG is consistent so is $\text{NBG} + \text{AC}$; Abraham Fraenkel [1922] proved that if NBG° is consistent so is $\text{NBG}^\circ + \neg \text{AC}$ (see also Fraenkel [1937], Lindenbaum & Mostowski [1938] and Mostowski [1939]); and Paul Cohen [1963,1964] proved that if NBG is consistent so is $\text{NBG} + \neg \text{AC}$ (see also Cohen [1966] and Shoenfield [1971]).

In Part II of this monograph we discuss "class" forms of the axiom of choice, CAC. Generally, these forms are obtained from the set forms in Part I by replacing the word "set" by "class". It is clear that CAC implies AC because each set is a class.

Using a Fraenkel-Mostowski model in which the class of atoms is a proper class it is easy to see that if NBG° is consistent, so is $\text{NBG}^\circ + \text{AC} + \neg \text{CAC}$. (This result is credited to E. Specker. See Bernays [1958] p.196.) Later Easton [1964] proved that if NBG is consistent, so is $\text{NBG} + \text{AC} + \neg \text{CAC}$. (See Felgner [1976a]). Thus AC does not imply CAC in NBG. However, it has been shown by a number of people that $\text{NBG} + \text{CAC}$ is a conservative extension of $\text{NBG} + \text{AC}$. That is, even though AC does not imply CAC, if a statement about sets, ϕ , is provable in $\text{NBG} + \text{CAC}$, ϕ is also provable in $\text{NBG} + \text{AC}$. (See Cohen [1966], p.77, Felgner [1971,1976a], Gaifman [1968], Grishin [1972], Jensen, Kripke and Solovay.)

The relative strength of the axiom of choice and other well-known propositions has now been demonstrated using Cohen's forcing techniques. For example, Cohen [1963] has shown that the axiom of choice does not imply the continuum hypothesis in NBG (assuming, of course, that NBG is consistent). It was known much earlier that the generalized continuum hypothesis does imply the axiom of choice in NBG° . (Lindenbaum & Tarski [1926], Sierpinski [1947]). Halpern & Levy [1971] proved that the Boolean prime ideal theorem, BPI, does not imply the axiom of choice in NBG. (Halpern [1961,1964] proved that BPI does not imply AC in NBG° .) Sageev [1975] proved that the $2^k = k$ principle, (that is, for all infinite cardinals k , $2^k = k$) does not imply AC in NBG. (Halpern & Howard [1974])

proved that the $2^k = k$ principle does not imply AC in NBG°..) Clearly, AC implies each of BPI and the $2^k = k$ principle.

In closing, we'll mention a few of the unsolved problems. It is well-known that the axiom of choice is used in the proof of each of the following propositions:

Every subgroup of a free group is free. (Nielson-Schreier Theorem.) (Lauchli [1962] proved that its proof is dependent on the axiom of choice and Howard [1983] proved that it implies the axiom of choice for finite sets.)

For all sets x and y , $\mathcal{P}(x) \prec \mathcal{P}(y)$ or $\mathcal{P}(y) \prec \mathcal{P}(x)$.

For all infinite cardinals k , $k^2 = k^3$.

For each set x , and partition P of x , $P \prec x$.

(Partition Principle).

However, it is not known if any of the above propositions imply AC in NBG° or NBG. Other unsolved problems are mentioned in the text.

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PRELIMINARY DEFINITIONS AND THEOREMS

We assume an elementary knowledge of logic and set theory.
We shall use the following logical symbols:

$\neg$	for	not
$\wedge$	for	and
$\vee$	for	or
$\rightarrow$	for	if ... then ...
$\leftrightarrow$	for	if and only if (iff)
$=$	for	identity
$(\exists X)$	for	there exists an X
$(\forall X)$	for	for all X .

The system of axioms for set theory which we adopt is NBG°
(von Neumann-Bernays-Gödel set theory with atoms and without
the axiom of foundation). (See von Neumann [1928], Bernays
[1937] and Gödel [1940].) Thus, our undefined terms are:

class, atom, and the element relation, $\in$.

The convention is made that capital letters, $X, Y, \dots$, denote
variables whose range is all classes and all atoms.

AXIOMS

A1: Characterization of Atoms

- (1) $\text{Atom}(X) \leftrightarrow \neg \text{Class}(X)$ and
- (2) $\text{Atom}(X) \rightarrow (\forall Y) (Y \notin X)$.

DEFINITION 0.1:

- (a) X is a set, $\text{set}(X)$ iff $\text{Class}(X) \wedge (\exists Y) (\text{Class}(Y) \wedge X \in Y)$.

(b) X is a proper class, $\text{Pr}(X)$ iff $\text{Class}(X) \wedge \neg \text{Set}(X)$.

Lower case letters, $x, y, z, \dots$, denote variables whose range is all sets and atoms.

A2: Axiom of Extensionality

$$\text{Class}(X) \wedge \text{Class}(Y) \rightarrow [(\forall u)(u \in X \leftrightarrow u \in Y) \rightarrow X = Y].$$

A3: Empty Set Axiom

$$(\exists X)[\text{Set}(X) \wedge (\forall Y)(Y \notin X)]$$

It follows from A2 that the X of A3 is unique and it is denoted by $\emptyset$.

A4: Class Construction Schema

If $\phi(u)$ is a formula in the language of NBG° which has no bound class variables and in which X is not a free variable then

$$(\exists X)[\text{Class}(X) \wedge (\forall u)(u \in X \leftrightarrow \phi(u))].$$

(Gödel [1940] has shown that in NBG° , A4 can be replaced by a finite number of axioms.)

It follows from A2 that the class X of A4 is unique and it is denoted by

$$X = \{u: \phi(u)\}.$$

DEFINITION 0.2: Let X and Y be classes.

(a) An element is a set or an atom.

(b) Subclass: $X \subseteq Y$ iff $(\forall u)(u \in X \rightarrow u \in Y)$.

(c) Proper Subclass: $X \subset Y$ iff $X \subseteq Y \wedge X \neq Y$.

(d) Power Class: $\mathcal{P}(X) = \{u: u \subseteq X\}$.

(e) Union:

$$(1) \cup X = \{u: (\exists v \in X)(u \in v)\}.$$

$$(2) X \cup Y = \{u: u \in X \vee u \in Y\}.$$

(f) Intersection:

$$(1) \cap X = \{u: (\forall v \in X)(u \in v)\}.$$

$$(2) \quad X \cap Y = \{u: u \in X \wedge u \in Y\}$$

(g) Class Difference: $X \sim Y = \{u: u \in X \wedge u \notin Y\}$.

(h) Complement: $\sim X = \{u: u \notin X\}$. (The complement of X will also be denoted by $V \sim X$ where V is the universe. See 0.4(a).)

(i) Disjoint: X and Y are disjoint if $X \cap Y = \emptyset$.

(j) Pair-wise Disjoint: X is pair-wise disjoint if each pair of distinct sets in X are disjoint.

A5: Power Set Axiom

$$\text{Set}(X) \rightarrow \text{Set}(\mathcal{P}(X)).$$

A6: Union Axiom

$$\text{Set}(X) \rightarrow \text{Set}(\cup X).$$

A7: Pairing Axiom

$$(\exists x)(\text{Set}(x) \wedge (\forall u)[u \in x \leftrightarrow (u = v \vee u = w)]).$$

It follows from A2 that the set x of A7 is unique and it is denoted by

$$x = \{v, w\}.$$

DEFINITION 0.3:

(a) Unordered Pair: $\{v, w\}$ is called the unordered pair of v and w .

(b) Unit Set: $\{u\} = \{u, u\}$ is called a unit set.

(c) Ordered Pair: $\langle u, v \rangle = \{\{u\}, \{u, v\}\}$ is called the ordered pair of u and v .

(d) Relation: A (binary) relation is a class of ordered pairs.

(e) Function:

$$\text{Func}(F) \leftrightarrow [F \text{ is a relation} \wedge (\forall u, v, w)(\langle u, v \rangle, \langle u, w \rangle \in F \rightarrow v = w)].$$

(f) Domain: $\mathcal{D}(X) = \{u: (\exists v)\langle u, v \rangle \in X\}$.

(g) Range: $\mathcal{R}(X) = \{v: (\exists u)\langle u, v \rangle \in X\}$.

If R is a relation we frequently write xRy instead of $\langle x, y \rangle \in R$; and if F is a function, we write $F(x) = y$ or

$F_x = y$ instead of xFy or $\langle x, y \rangle \in F$.

A8: Axiom of Replacement

$$(\text{Func}(F) \wedge \text{Set}(\mathcal{D}(F))) \rightarrow \text{Set}(\mathcal{R}(F)).$$

A9: Axiom of Infinity

$$(\exists x)[\text{Set}(x) \wedge \emptyset \in x \wedge (\forall u)(u \in x \wedge \text{Set}(u) \rightarrow u \cup \{u\} \in x)].$$

Axioms A1-A9 are the axioms of NBG°.

A10: Axiom of Foundation

$$(\text{Class}(X) \wedge X \neq \emptyset) \rightarrow (\exists u \in X)(\text{Set}(u) \wedge u \cap X = \emptyset).$$

Axioms A7 and A10 imply there do not exist any atoms.

A1-A10 are the axioms of NBG.

Another form of the axiom of foundation which is sometimes used is the following:

A'10: Axiom of Foundation, alternate form

$$[\text{Class}(X) \wedge X \neq \emptyset \wedge (\forall u \in X)\text{Set}(u)] \rightarrow (\exists u \in X)(u \cap X = \emptyset)$$

Of course, A10 is equivalent to A'10 if there do not exist any atoms.

As is well-known, the set form of A10, (that is, replace "Class(X)" by "Set(X)" in A10) is equivalent to the class form in NBG°. For suppose the set form of A10 holds and let X be a class. Suppose $u \in X$, u is a set and $X \cap u \neq \emptyset$. Let $z = X \cap u$. Then the transitive closure of z , $\text{TC}(z)$, is a set,

$$\text{TC}(z) = z \cup (Uz) \cup (U \cup z) \cup \dots$$

Let $w = \text{TC}(z) \cap X$. Then $w \neq \emptyset$ is a set so by the set form of the axiom of foundation, there is a set $v \in w$ such that $v \cap w = \emptyset$. Clearly $v \in X$. If $v \cap X \neq \emptyset$ then there exists an $a \in v \cap X$. By definition of $\text{TC}(z)$, $a \in v \in \text{TC}(z)$ implies $a \in \text{TC}(z)$. Thus $a \in v \cap w$, which is a contradiction.

All proofs, unless otherwise stated, are in NBG° and are independent of the existence of atoms.

DEFINITION 0.4:

- (a) Universe: $V = \{u: u = u\} [(\forall u)(u \in V)]$.
- (b) Unordered n-tuple: $\{x_1, x_2, \dots, x_{n-1}, x_n\} = \{x_1, x_2, \dots, x_{n-1}\} \cup \{x_n\}$.
- (c) Ordered n-tuple: $\langle x_1, x_2, \dots, x_n \rangle = \langle x_1, \langle x_2, \dots, x_n \rangle \rangle$.
- (d) If F is a function and $\phi(x_1, \dots, x_n)$ is a formula in the language of NBG^o,
- $$\{F(x_1, \dots, x_n): \phi(x_1, \dots, x_n)\} = \{u: (\exists x_1 \dots x_n)(u = F(x_1, \dots, x_n) \wedge \phi(x_1, \dots, x_n))\}.$$
- (e) Cartesian or Direct Product:
- $$\text{Class}(X) \wedge \text{Class}(Y) \rightarrow X \times Y = \{\langle u, v \rangle: u \in X \wedge v \in Y\}.$$
- (f) Image: $R''X = \{v: (\exists u \in X)(\langle u, v \rangle \in R)\}$.
- (g) Restriction: $R \upharpoonright X = \{\langle u, v \rangle: u \in X \wedge \langle u, v \rangle \in R\}$.
- (h) Inverse: $R^{-1} = \{\langle x, y \rangle: \langle y, x \rangle \in R\}$.
- (i) Indexed Classes: Suppose $(\forall i \in I)\text{Set}(X_i)$.

$$\bigcup_{i \in I} X_i = \cup \{X_i: i \in I\}.$$

$$\bigcap_{i \in I} X_i = \cap \{X_i: i \in I\}.$$

$$\bigcap_{i \in I} X_i = \{f: \text{Func}(f) \wedge \mathcal{D}(f) = I \wedge (\forall i \in I)(f(i) \in X_i)\}.$$

$$(\forall i \in I)(X_i = Y) \rightarrow \bigcap_{i \in I} X_i = Y^I.$$

We may also take the union, intersection, and Cartesian product of proper classes by using the following device:

Let R be a relation such that for each $i \in I$, X_i is a set. ($R''X_i$ could be a proper class.)

$$\bigcup_{i \in I} R''X_i = \{u: (\exists i \in I)(u \in R''X_i)\}.$$

$$\bigcap_{i \in I} R''X_i = \{u: (\forall i \in I)(u \in R''X_i)\}.$$

$$\bigcup_{i \in I} R''X_i = \{f: \text{Func}(f) \wedge \mathcal{J}(f) = I \wedge (\forall i \in I) (f(i) \in R''X_i)\}.$$

(j) Functional Notation:

$$F: X \rightarrow Y \text{ iff } \text{Func}(F) \wedge \mathcal{J}(F) = X \wedge \mathcal{R}(F) \subseteq Y.$$

$$F \text{ is 1-1 or an } \underline{\text{injection}} \text{ iff} \\ (\forall u, v, w) (\langle u, w \rangle, \langle v, w \rangle \in F \rightarrow u = v).$$

$$F: X \prec Y \text{ iff } \text{Func}(F) \wedge F \text{ is an injection.}$$

$$X \prec Y \text{ iff } (\exists F) (F: X \prec Y).$$

$$F: X \xrightarrow{\text{onto}} Y \text{ iff } F: X \rightarrow Y \wedge \mathcal{R}(F) = Y. (F \text{ is called a} \\ \underline{\text{surjection}}.)$$

$$F: X \prec_* Y \text{ iff } F: Y \xrightarrow{\text{onto}} X.$$

$$X \prec_* Y \text{ iff } X = \emptyset \vee (\exists F) (F: X \prec_* Y).$$

$$F: X \approx Y \text{ iff } F: X \prec Y \wedge \mathcal{R}(F) = Y. (F \text{ is called a } \underline{\text{bijection}} \\ \text{and } X \text{ and } Y \text{ are called } \underline{\text{equipollent}} \text{ or } \underline{\text{equinumerous}}.)$$

$$X \approx Y \text{ iff } (\exists F) (F: X \approx Y).$$

- (k) Class-Valued Functions: F is a class-valued function if whenever $X \in \mathcal{J}(F)$ and $\langle X, Z \rangle \in F$ then either $Z = \emptyset$ or Z is a unit set. Then $F[X] = \{y: \langle x, \{y\} \rangle \in F\}$.

($F[X]$ can either be $\emptyset$, a set or a proper class. Using this definition we can distinguish the two cases when $F[X] = \emptyset$, (1) if $X \notin \mathcal{J}(F)$, then $F''\{x\} = \emptyset$; (2) if $\langle x, \emptyset \rangle \in F$, then $F''\{x\} = \{\emptyset\}$.)

(l) Composite Product:

$$R \circ S = \{\langle u, w \rangle: (\exists v) (\langle u, v \rangle \in S \wedge \langle v, w \rangle \in R)\}.$$

- (m) An n-ary Relation is a class of ordered n-tuples.

We shall make the convention that whenever we write:

$$X \subseteq Y, X \cup Y, X \cap Y, X \sim Y, X \times Y, \cup X, \cap X, \bigcup_{i \in I} X_i, \bigcap_{i \in I} X_i, \\ \bigcup_{i \in I} X_i, \text{ etc., we shall assume that } X \text{ and } Y \text{ are classes and}$$

that for each $i \in I$, X_i is a class.

DEFINITION 0.5: Let R be a relation.

- (a) R is a reflexive on X if $(\forall u \in X)(u R u)$.
 R is irreflexive on X if $(\forall u \in X)(\neg u R u)$.
- (b) R is symmetric on X if $(\forall u, v \in X)(u R v \rightarrow v R u)$.
 R is anti-symmetric on X if $(\forall u, v \in X)(u R v \wedge v R u \rightarrow u = v)$.
- (c) R is transitive on X if $(\forall u, v, w \in X)(u R v \wedge v R w \rightarrow u R w)$.
- (d) R is a partial ordering on X if R is reflexive, anti-symmetric and transitive on X .
- (e) R is connected on X if $(\forall u, v \in X)(u R v \vee v R u \vee u = v)$.
- (f) R is a linear ordering on X if R is a connected partial ordering on X .
- (g) u is said to be an R -first or R -smallest element of X if $u \in X$ and $(\forall v \in X)(u \neq v \rightarrow u R v)$.
- (h) u is said to be an R -last or R -largest element of X if $u \in X$ and $(\forall v \in X)(u \neq v \rightarrow v R u)$.
- (i) u is said to be an R -minimal element of X if $u \in X$ and $(\forall v \in X)(v R u \rightarrow u R v)$.
- (j) u is said to be an R -maximal element of X if $u \in X$ and $(\forall v \in X)(u R v \rightarrow v R u)$.
- (k) If $Y \subseteq X$, u is said to be an R -lower [R -upper] bound of Y in X if $u \in X$ and $(\forall v \in Y)(u \neq v \rightarrow u R v)$ [$(\forall v \in Y)(u \neq v \rightarrow v R u)$].
- (l) If $Y \subseteq X$, u is said to be an R -greatest lower bound or infimum [R -least upper bound or supremum] of Y in X if u is an R -largest [R -smallest] lower [upper] bound of Y in X .

- (m) R is a well ordering on X if R is a partial ordering on X and every non-empty subset of X has an R -first element.
- (n) $\langle X, R \rangle$ is a partially (linearly, well) ordered class if X is a class and R is a partial (linear, well) ordering on X .
- (o) If $\langle X, R \rangle$ and $\langle Y, S \rangle$ are partially ordered classes, $\langle X, R \rangle$ is isomorphic to $\langle Y, S \rangle$, $\langle X, R \rangle \cong \langle Y, S \rangle$ iff there is a function F such that
- (1) $F: X \approx Y$, and
 - (2) $(\forall u, v \in X) (uRv \leftrightarrow F(u)SF(v))$.
- (p) $X \subseteq Y$ is an initial R -segment of Y if $(\forall u \in X) (\forall v \in Y) (vRu \rightarrow v \in X)$.
- (q) If $u \in Y$ then X is the initial R -segment of Y generated by u if $X = \{v \in Y: vRu\}$.
(If R is transitive, an initial R -segment of Y generated by u is an initial R -segment of Y .)

DEFINITION 0.6: Ordinal Numbers

- (a) A class X is transitive iff $(\forall u, v) (u \in v \wedge v \in X \rightarrow u \in X)$.
- (b) $X \in Y$ iff $X \in Y \vee X = Y$.
- (c) X is an Ordinal iff
- (1) $\langle X, \in \rangle$ is a well ordered class and
 - (2) $(\forall u \in X) (u$ is the initial $\in$ -segment of X generated by $u)$.
- (d) X is an Ordinal Number iff X is an Ordinal and X is a set.
- (e) X is called a successor class iff $\emptyset \in X \wedge (\forall u) (\text{Set}(u) \wedge u \in X \rightarrow u \cup \{u\} \in X)$.

- (f) $\omega = \cap \{x: x \text{ is a successor set}\}$ (ω is the set of natural numbers).
- (g) A set x is finite iff $(\exists n \in \omega)(x \approx n)$. Otherwise x is called infinite. A set x is called transfinite if $\omega \leq x$.
- (h) $0 = \emptyset$ and On is the class of all ordinal numbers.

Lower case greek letters, $\alpha, \beta, \gamma, \dots$, will be used to denote variables whose range is On .

- (i) $\alpha < \beta$ iff $\alpha \in \beta$.
 $\alpha \leq \beta$ iff $\alpha < \beta \vee \alpha = \beta$.
- (j) α is a successor ordinal iff $(\exists \beta)(\alpha = \beta \cup \{\beta\})$.
- (k) α is a limit ordinal iff $\alpha \neq \emptyset$ and α is not a successor ordinal.
- (l) $\alpha \geq \omega$ is an initial ordinal iff $(\forall \beta < \alpha) \beta < \alpha$.

(Initial ordinals are denoted by ω_α for $\alpha \in On$, where $\omega_0 = \omega$.)

We shall assume all the well-known properties of ordinal numbers, some of which are listed below. (All these properties are provable in NBG°.)

THEOREM 0.7:

- (a) Every ordinal is transitive.
- (b) On is an ordinal.
- (c) A set of ordinal numbers is an ordinal number iff it is transitive.
- (d) Recursion Theorem: If $g:V \rightarrow V$ then there is a function F with domain On such that for all $\alpha \in On$, $F(\alpha) = G(F''\alpha)$.

- (e) Transfinite Induction: If $X \subseteq \text{On}$ such that
 $(\forall \alpha \in \text{On})(\alpha \subseteq X \rightarrow \alpha \in X)$ then $X = \text{On}$.
- (f) $(\forall \alpha)(\alpha \notin \alpha)$.
- (g) $(\forall \alpha, \beta) \neg (\alpha \in \beta \wedge \beta \in \alpha)$.
- (h) $(\forall \alpha, \beta)(\alpha \in \beta \vee \alpha = \beta \vee \beta \in \alpha)$.
- (i) $(\forall \alpha \neq 0)(\alpha \text{ is a limit ordinal iff } \alpha = \cup \alpha)$.
- (j) If two well ordered sets are isomorphic there is a unique isomorphism mapping one onto the other.
- (k) For each well ordered set there is a unique ordinal number which is isomorphic to it.
- (l) $(\forall \alpha \geq \omega)(\alpha \times \alpha \approx \alpha)$.

THEOREM 0.8: Axiom of Foundation

- (a) If $V_0 = \emptyset$, $(\forall \alpha)(V_{\alpha+1} = \mathcal{P}(V_\alpha))$ and if α is a limit ordinal, $V_\alpha = \bigcup_{\beta < \alpha} V_\beta$ then

$$V = \bigcup_{\alpha \in \text{On}} V_\alpha \text{ iff } A10.$$
- (b) If $V_0 = A = \text{class of atoms}$, $(\forall \alpha)(V_{\alpha+1} = \mathcal{P}(V_\alpha))$ and if α is a limit ordinal then $V_\alpha = \bigcup_{\beta < \alpha} V_\beta$, then

$$V = \bigcup_{\alpha \in \text{On}} V_\alpha \text{ iff } A'10.$$

DEFINITION 0.9: Rank

- (a) If $u \in \bigcup_{\alpha \in \text{On}} V_\alpha$ and $\text{Set}(u)$ then the rank of u , $\rho(u)$, is the smallest ordinal α such that $u \subseteq V_\alpha$. (Each V_α is transitive so $u \in V_\alpha$ and $\text{Set}(u)$ implies $u \subseteq V_\alpha$.)

(b) If $u \in \bigcup_{\alpha \in \text{On}} V_\alpha$ and $\text{Atom}(u)$ then $\rho(u) = 0$.

THEOREM 0.10:

If $u \in \bigcup_{\alpha \in \text{On}} V_\alpha$, $\rho(u)$ is the smallest ordinal number α such that for all $v \in u$, $\rho(v) < \alpha$.

DEFINITION 0.11: Hartogs Function (Hartogs [1915]).

For each set x ,

$$\Gamma(x) = \{\alpha : \alpha \prec x\}.$$

THEOREM 0.12:

(a) For each set x , $\Gamma(x)$ is a set.

(b) For each infinite set x , $\Gamma(x)$ is an initial ordinal.

(c) For each set x , $\Gamma(x)$ is the smallest ordinal α such that it is not the case that $\alpha \prec x$.

It has been shown independently by Gauntt [1967] and Levy [1969] that the notion of cardinality is undefinable in NBG° . However, it is convenient for us to use cardinal notation even though all statements we make about cardinals can be replaced by statements about sets. Thus, we shall use the following notation.

If x is a set, $|x|$ is the cardinal number of x .

We assume cardinal numbers have the usual properties.

$$\text{Set}(x) \rightarrow \text{Set}(|x|)$$

If x and y are sets,

$$|x| = |y| \leftrightarrow x \approx y,$$

$$|x| \leq |y| \leftrightarrow x \prec y,$$

$$|x| < |y| \leftrightarrow x \prec y.$$

(It is well-known that cardinal numbers are definable in NBG. Using the axiom of choice, the cardinal number of a set x can be defined as the initial ordinal equipollent to x . (von Neumann [1928])). Alternatively, the cardinal number of a set x can be defined as the set of all sets of smallest rank which are equipollent to x (Scott [1955])).

We shall use lower case greek letters, $\kappa, \mu, \nu, \dots$, to denote cardinal number variables. We shall also make the convention that when we write $|x|$, we assume x is a set.

DEFINITION 0.13:

- (a) $|x| = \kappa \rightarrow |\Gamma(x)| = \kappa^*$.
- (b) $(\forall \alpha) (\aleph_\alpha = |\omega_\alpha|)$.
- (c) An aleph is the cardinal number of an infinite well ordered set.
- (d) A cardinal κ is transfinite, if $\aleph_0 \leq \kappa$.
- (e) If $|x| = \kappa$ and $|y| = \mu$ then
 - (1) Addition: If $x \cap y = \emptyset$, $\kappa + \mu = |x \cup y|$.
 - (2) Subtraction: If there is a unique γ such that $\kappa = \mu + \gamma$ then $\kappa - \mu = \gamma$.
 - (3) Multiplication: $\kappa \cdot \mu = |x \times y|$.
 - (4) Division: If there is a unique γ such that $\kappa = \mu \cdot \gamma$ then $\kappa \div \mu = \gamma$.
 - (5) Exponentiation: $\kappa^\mu = |x^y|$.
 $2^\kappa = |\mathcal{P}(x)|$.
- (f) If for each $i \in I$, $|x_i| = \kappa_i$, then
 - (1) $\sum_{i \in I} \kappa_i = |\bigcup_{i \in I} (x_i \times \{i\})|$.
 - (2) $\prod_{i \in I} \kappa_i = |\prod_{i \in I} x_i|$.

(The sum and product of an infinite number of cardinals is in general not definable without the axiom of choice.)

We shall use some of the following well-known properties of cardinal numbers.

THEOREM 0.14:

- (a) For each aleph, $\aleph$, there exists a unique ordinal α such that $\aleph = \aleph_\alpha$.
- (b) If $\kappa, \mu \geq 2$ then $\kappa + \mu \leq \kappa \cdot \mu$.
- (c) For each pair of alephs, $\aleph$ and $\aleph'$,
 - (1) $\aleph + \aleph' = \aleph \cdot \aleph' = \max(\aleph, \aleph')$.
 - (2) $\aleph^2 = \aleph$.
- (d) If $\kappa_1, \kappa_2, \mu_1$ and μ_2 are transfinite cardinals such that at least one is an aleph, then

$$\kappa_1 \cdot \kappa_2 \leq \mu_1 + \mu_2 \rightarrow (\kappa_1 \leq \mu_1 \vee \kappa_2 \leq \mu_2).$$
- (e) If κ and μ are transfinite cardinals and $\aleph$ is an aleph then

$$\aleph \leq \kappa + \mu \rightarrow \aleph \leq \kappa \vee \aleph \leq \mu \quad \text{and}$$

$$\aleph \leq \kappa \cdot \mu \rightarrow \aleph \leq \kappa \vee \aleph \leq \mu.$$
- (f) If κ is transfinite, $2^\kappa - \kappa = 2^\kappa$.
- (g) If κ is an infinite cardinal then κ^* is an aleph and it is the smallest aleph which is not $\leq \kappa$.
- (h) For all cardinals κ , $\kappa^* \leq 2^{2^{\kappa^2}}$.
- (i) For all cardinals κ , $\kappa^* \leq 2^{2^{2^\kappa}}$.

Even though we are using NBG^o as our axiom system, it follows from the work of Mostowski that NBG^o is a conservative extension of ZF^o, Zermelo-Fraenkel set theory without the axiom of foundation. That is, if ϕ is a formula in the

language of ZF° , then ϕ is provable in NBG° iff it is provable in ZF° . Thus nothing more about sets is provable in NBG° than in ZF° .

PART I, SET FORMS

§1. THE WELL-ORDERING THEOREM

Before 1904, when Zermelo [1904] published his proof that the axiom of choice implies the well-ordering theorem, the well-ordering theorem was considered as self-evident. Cantor and others used it without hesitation. WO 1-WO 3 are three classical forms of the well-ordering theorem. WO 4(m) and WO 5 are due to Azriel Levy [1962] and WO 6 is a generalization of WO 5.

WO 1: Every set can be well-ordered.

WO 2: Every set is equipollent to an ordinal number.

WO 3: Every set is equipollent to a subset of an ordinal number.

Let m be a natural number, $m \geq 1$.

WO 4(m): For every set x there exists an ordinal number α and a function f defined on α such that $f(\beta) \leq m$ for every $\beta < \alpha$ and $\bigcup_{\beta < \alpha} f(\beta) = x$.

WO 5: There exists a natural number $m \geq 1$ such that WO 4(m).

WO 6: For every set x there exists a natural number $m \geq 1$, an ordinal number α , and a function f defined on α such that $f(\beta) \leq m$ for every $\beta < \alpha$ and $\bigcup_{\beta < \alpha} f(\beta) = x$.

WO 4(m) could be restated in the following way:

Every set is the union of a well-ordered set of finite sets each of which has not more than m elements. Similarly for WO 5 and WO 6.

Later we shall see in section 5 that if we omit the bound m on the finite set, $WO\ 4(m)$ remains equivalent to the well-ordering theorem, but then the axiom of foundation is required in the proof.

It is clear that $WO\ 2 \rightarrow WO\ 3$ and $WO\ 3 \rightarrow WO\ 1$. To prove that $WO\ 1 \rightarrow WO\ 2$ see Gödel [1940]. (The proof is given in Gödel [1940] p. 27, theorem 7.7. This proof is quite similar to the proof of Theorem 2.8 which we shall give in detail.) The following equivalences are also clear: $WO\ 1 \leftrightarrow WO\ 4(1)$, $WO\ 4(m) \rightarrow WO\ 4(n)$ if $m \leq n$, and $WO\ 4(m) \rightarrow WO\ 5 \rightarrow WO\ 6$. Therefore, to prove that the above six forms are all equivalent it remains to prove the following theorem. (The proof is a modification of a proof due to Levy [1962].)

THEOREM 1.1. $WO\ 6 \rightarrow WO\ 1$.

PROOF. Let x be an arbitrary set. For each set y , define a set of natural numbers N_y as follows:

- (i) $m \in N_y$ if $(\exists f)(\exists \alpha) [f \text{ is a function, } \alpha \in On, \mathcal{D}(f) = \alpha, \bigcup_{\beta < \alpha} f(\beta) = y, \text{ and } (\forall \beta)[\beta < \alpha \rightarrow f(\beta) \leq m]]$. ($WO\ 6$ implies that $N_y \neq \emptyset$ for every set y .)

Next, we shall prove the main result:

- (ii) If $y \times y \subseteq y$ and m is a natural number such that $m > 1$, then $m \in N_y$ implies $m - 1 \in N_y$.

Suppose y and m satisfy the hypothesis of (ii) and $m \in N_y$. Then there exists a function f and an ordinal number α satisfying (i). Define

- (iii) $u_{\beta\gamma\delta} = (f(\beta) \times f(\gamma)) \cap f(\delta)$, where $\beta, \gamma, \delta < \alpha$.

Now, $u_{\beta\gamma\delta}$ is a set of ordered pairs, so that it is a relation and we may talk about its domain and range. Moreover, we have

$$\mathcal{D}(u_{\beta\gamma\delta}) \subseteq f(\beta) \leq m, \mathcal{R}(u_{\beta\gamma\delta}) \subseteq f(\gamma) \leq m, \text{ and } u_{\beta\gamma\delta} \subseteq f(\delta) \leq m.$$

Case 1. $(\forall \beta)[\beta < \alpha \text{ and } f(\beta) \neq \emptyset \rightarrow (\exists \gamma)(\exists \delta)[\gamma, \delta < \alpha, \mathcal{D}(u_{\beta\gamma\delta}) \neq \emptyset \text{ and } \mathcal{D}(u_{\beta\gamma\delta}) < m]]$.

For each $\beta < \alpha$ with $f(\beta) \neq \emptyset$, let λ_β and μ_β be the lexicographically $<$ - first ordinal numbers γ and δ such that $\mathcal{D}(u_{\beta\gamma\delta}) \neq \emptyset$ and $\mathcal{D}(u_{\beta\gamma\delta}) < m$. (That is, first find ordinal numbers γ and δ which satisfy the conditions. Then let λ_β be the $<$ - smallest such γ which satisfies the conditions. Then given λ_β , let μ_β be the $<$ - smallest δ which satisfies the conditions.) Now define:

$$v_\beta = \begin{cases} \mathcal{D}(u_{\beta, \lambda_\beta, \mu_\beta}) & \text{if } f(\beta) \neq \emptyset \\ \emptyset & \text{if } f(\beta) = \emptyset \end{cases},$$

and $w_\beta = f(\beta) \sim v_\beta$. Next we define a function g as follows:

$$\mathcal{D}(g) = \alpha + \alpha,$$

$$\text{if } \beta < \alpha \text{ then } g(\beta) = v_\beta,$$

$$\text{if } \alpha \leq \beta, \text{ and } \beta \sim \alpha \cong \gamma < \alpha \text{ then } g(\beta) = w_\gamma.$$

(The sum of two ordinal numbers is defined, for example in Bachmann [1955] p. 45.) We shall prove that $m - 1 \in N_y$ by proving that g is a function and $\alpha + \alpha$ an ordinal which satisfies (i). That is, we have to prove $\bigcup_{\beta < \alpha + \alpha} g(\beta) = y$ and if $\beta < \alpha + \alpha$ then $g(\beta) < m - 1$. The first equality follows from set theoretic considerations and from the fact that $\bigcup_{\beta < \alpha} v_\beta \subseteq \bigcup_{\beta < \alpha} f(\beta) = y$. If $\beta < \alpha$, we have $v_\beta < m$ and consequently $g(\beta) < m - 1$. If $\alpha \leq \beta < \alpha + \alpha$, then there is a $\gamma < \alpha$ such that $\beta \sim \alpha \cong \gamma$. If $f(\gamma) \neq \emptyset$ then $v_\gamma \neq \emptyset$, $f(\gamma) < m$ by (i), and $v_\gamma \subseteq f(\gamma)$. Therefore, $w_\gamma = f(\gamma) \sim v_\gamma < m$, so that $g(\beta) < m - 1$. If $f(\gamma) = \emptyset$ then $w_\gamma = \emptyset$, so that again $g(\beta) < m - 1$.

Case 2. $(\exists \beta)[\beta < \alpha, f(\beta) \neq \emptyset \text{ and } (\forall \gamma, \delta)[\gamma, \delta < \alpha \text{ and } \mathcal{D}(u_{\beta\gamma\delta}) \neq \emptyset \rightarrow \mathcal{D}(u_{\beta\gamma\delta}) \approx m]]$.

Let β be the $<$ - smallest ordinal number in $\{\lambda: \lambda < \alpha, f(\lambda) \neq \emptyset, \text{ and } (\forall \gamma, \delta)[\gamma, \delta < \alpha \text{ and } \mathcal{D}(u_{\lambda\gamma\delta}) \neq \emptyset \rightarrow \mathcal{D}(u_{\lambda\gamma\delta}) \approx m]\}$.

Let $s \in f(\beta)$.

If $\gamma, \delta < \alpha$ and $\mathcal{D}(u_{\beta\gamma\delta}) \neq \emptyset$ then $\mathcal{D}(u_{\beta\gamma\delta})$ has m elements. It follows from (i) and (iii) that $u_{\beta\gamma\delta}$ has at most m elements ($u_{\beta\gamma\delta} \subseteq f(\delta) \prec m$). Therefore $u_{\beta\gamma\delta} \approx m$ and $u_{\beta\gamma\delta}$ is a function.

Now, if in addition to $f(\beta) \neq \emptyset$ also $f(\gamma) \neq \emptyset$, then there exists a δ such that $u_{\beta\gamma\delta} \neq \emptyset$. (This follows from (iii) and the fact that $\gamma \times \gamma \subseteq \gamma$.) Let δ_γ be the $<$ -smallest ordinal number in $\{\delta: u_{\beta\gamma\delta} \neq \emptyset\}$.

Next, define

$$v_\gamma = \begin{cases} \{u_{\beta\gamma\delta_\gamma}(s)\} & \text{if } f(\gamma) \neq \emptyset \\ \emptyset & \text{if } f(\gamma) = \emptyset \end{cases} \quad \text{and } w_\gamma = f(\gamma) \sim v_\gamma.$$

(Note that since $f(\beta) \neq \emptyset$, if $\mathcal{D}(u_{\beta\gamma\delta}) \neq \emptyset$ then $s \in f(\beta) = \mathcal{D}(u_{\beta\gamma\delta})$ and $f(\gamma) \neq \emptyset$, therefore δ_γ exists.) Now define a function g on $\alpha + \alpha$ as follows:

If $\gamma < \alpha$ then $g(\gamma) = v_\gamma$.

If $\alpha \leq \gamma$ and $\gamma \sim \alpha \cong \lambda < \alpha$ then $g(\gamma) = w_\lambda$.

Again we shall show that g is a function and $\alpha + \alpha$ an ordinal number which satisfies (i) with m replaced by $m - 1$. As in case 1, we have $\bigcup_{\gamma < \alpha + \alpha} g(\gamma) = \gamma$. If $\gamma < \alpha$,

$g(\gamma) = v_\gamma \prec 1 \prec m - 1$. If $\alpha \leq \gamma$ and $\gamma \sim \alpha \cong \lambda$, $g(\gamma) = w_\lambda = f(\lambda) \sim v_\lambda$; if $f(\lambda) \neq \emptyset$, $v_\lambda \approx 1$ and $g(\gamma) \prec m - 1$, while if $f(\lambda) = \emptyset$, $g(\gamma) = \emptyset \prec m - 1$. Hence, if $\lambda < \alpha + \alpha$, then $g(\lambda) \prec m - 1$. This completes the proof of (ii).

It follows from (ii) and mathematical induction that if $\gamma \times \gamma \subseteq \gamma$ then γ can be well-ordered. Now, to complete the proof of the theorem we must show that our original set x can be well-ordered. Therefore, we prove:

- (iv) For every set x there exists a set γ such that $x \cup (\gamma \times \gamma) \subseteq \gamma$.

Construct γ as follows:

Let $z_0 = x$ and $z_{n+1} = z_n \cup (z_n \times z_n)$. Let $y = \bigcup_{n=0}^{\infty} z_n$.

If $m < n$ then $z_m \subseteq z_n$ and $z_m \times z_m \subseteq z_n$. Clearly, $x \subseteq y$. Also,

$$\begin{aligned} y \times y &= \bigcup_{n,m=0}^{\infty} z_n \times z_m \\ &\subseteq \bigcup_{n,m=0}^{\infty} z_{\max(m,n)} \times z_{\max(m,n)} \\ &\subseteq \bigcup_{n,m=0}^{\infty} z_{\max(m,n)+1} \\ &\subseteq y. \end{aligned}$$

Hence, $x \cup (y \times y) \subseteq y$.

Therefore, x is a subset of a set which can be well-ordered, so that x can be well-ordered also, q.e.d.

When Alfred Tarski was studying the notion of finite (see Tarski [1924a, 1938a]), he discovered several definitions of finite which were equivalent to the axiom of choice. The one we give here is equivalent to the well ordering theorem. (We give several others in §7).

It is easy to prove in NBG° that

(*) If x is finite and R well orders x then R^{-1} well orders x . The converse of (*) is equivalent to

(**) If x is infinite then there exists a relation R such that R well orders x but R^{-1} does not.

It is easy to see that (**) is equivalent to the well ordering theorem because each finite set can be well ordered and each infinite ordinal is well ordered by $\leq$ but not by $\geq$. Thus, we have the following equivalent to the well ordering theorem:

WO 7: For every set x , x is finite iff for each well ordering R of x , R^{-1} also well orders x .

The final form in this section, WO 8, was discovered through conversations with Norbert Brunner and he noticed a simple proof that WO 8 implies WO 1. We say f is a choice function on a set x of non-empty sets if $\mathcal{D}(f) = x$ and for all $u \in x$, $f(u) \in u$. It is well known that if there is a choice function on $\mathcal{P}(x) \sim \{\emptyset\}$ then x can be well ordered. (See Theorem 2.9). Therefore, it was interesting to observe that the following statement is equivalent to WO 1.

WO 8: Every set on which there is a choice function, can be well ordered.

Clearly $\text{WO 1} \rightarrow \text{WO 8}$. Conversely, $\text{WO 8} \rightarrow \text{WO 1}$ because if x is any non-empty set then there is a choice function on $\{\{u\}: u \in x\}$. Thus, WO 8 implies x can be well ordered.

§2. THE AXIOM OF CHOICE

Apparently, the first specific reference to the axiom of choice was given in a paper by G. Peano [1890]. In proving an existence theorem for ordinary differential equations, he ran across a situation in which such a statement is needed. Beppo Levi [1902], while discussing the statement that the union of a disjoint set t of non-empty sets has a cardinal number greater than or equal to the cardinal number of t , (The partition principle) remarked that its proof depended on the possibility of selecting a single member from each element of t . It is not known whether the partition principle implies the axiom of choice. Others, including Cantor, had used the principle earlier, but did not mention it specifically. We give here nineteen forms of the axiom of choice.

AC 1: If s is a set of non-empty sets, there is a function f such that for every $x \in s$, $f(x) \in x$.

AC 2: If t is a pairwise disjoint set of non-empty sets, there is a set c which consists of one and only one element from each set in t .

AC 3: For every function f there is a function g such that for every x , if $x \in \mathcal{D}(f)$ and $f(x) \neq \emptyset$, then $g(x) \in f(x)$.

AC 4: For every relation r there is a function f such that $\mathcal{D}(f) = \mathcal{D}(r)$ and $f \subseteq r$. (See AC 20 in section 5.)

AC 5: For every function f there is a function g such that $\mathcal{D}(g) = \mathcal{R}(f)$ and for every $x \in \mathcal{D}(g)$, $f(g(x)) = x$. (If f is 1-1 then g is its inverse.)

AC 6: The Cartesian product of a set of non-empty sets is non-empty.

AC 7: The Cartesian product of a set of non-empty sets of the same cardinality is non-empty.

AC 8: If s is a set of pairs of equipollent sets then there is a function which associates with each pair a bijection mapping one onto the other.

AC 9: If s is a set of sets of the same cardinality then there is a function which associates with each pair a bijection mapping one onto the other.

Let $n \geq 2$ be a natural number.

AC 10(n): If s is a set of infinite sets, there is a function f such that for each $x \in s$, $f(x)$ is a decomposition of x into a set of disjoint sets of size between 2 and n .

AC 11: There exists a natural number $n \geq 2$ such that AC 10(n).

AC 12: If s is a set of infinite sets there is a natural number $n \geq 2$ and a function f such that for each $x \in s$, $f(x)$ is a decomposition of x into a set of disjoint sets of size between 2 and n .

Each of AC 10(n), AC 11 and AC 12 remains equivalent to the axiom of choice if we require that the size of each of the sets in the partition be exactly n .

Let $m \geq 1$ be a natural number.

AC 13(m): If s is a set of non-empty sets there is a function f such that for each $x \in s$, $f(x)$ is a non-empty subset of x with $f(x) \prec m$.

AC 14: There is a natural number $m \geq 1$ such that AC 13(m).

AC 15: If s is a set of non-empty sets then there is a natural number $m \geq 1$ and a function f such that for each $x \in s$, $f(x)$ is a non-empty subset of x with $f(x) \prec m$.

Let n and k be natural numbers with $n > 2$ and $1 < k < n$.

AC 16(n, k): If s is an infinite set then there is a set t_n of n -element subsets of s such that each k -element subset of s is a subset of exactly one element of t_n .

AC 17: If s is a set, $t = \mathcal{P}(s) \sim \{\emptyset\}$, F is the set of all functions from t to s and g is a function from F to t then there is an $f \in F$ such that $f(g(f)) \in g(f)$.

AC 18: For any non-empty set A , and for every function B whose domain is A and B_a is a non-empty set for each $a \in A$,

$$\bigcap_{a \in A} \bigcup_{b \in B_a} X_{a,b} = \bigcup_{f \in \prod_{a \in A} B_a} \bigcap_{a \in A} X_{a,f(a)} .$$

AC 18 is equivalent to the axiom of choice if $X_{a,b}$ is a set or a proper class or $X_{a,b} = b$. As we shall see later in section 5, if we allow B to be a relation so that for each $a \in A$, B_a is a class, the resulting form is equivalent to the axiom of choice in NBG. (See AC 21).

AC 19: For any non-empty set A , each of whose elements is non-empty,

$$\bigcap_{a \in A} \bigcup_{b \in a} b = \bigcup_{f \in C(A)} \bigcap_{a \in A} f(a) ,$$

where $C(A)$ is the set of all choice functions on A .

Dual forms of AC 18 and AC 19 can be obtained by interchanging union and intersection. The dual forms are also equivalent to the axiom of choice.

In 1904, Zermelo [1904] stated a principle of choice similar to AC 1 and proved that it implied the well-ordering theorem. (Zermelo took s to be the set of all non-empty subsets of a given set and he stated in his paper that the proof of the theorem came out of conversations with Erhardt Schmidt.) AC 3 and AC 6 are essentially restatements of AC 1.

Bertrand Russell [1906] gave a principle analogous to AC 2. He announced this principle as a possible substitute

for Zermelo's but he believed that it was weaker. (Russell actually stated his principle in the form of AC 6 with disjoint sets. However, by "product" Russell meant the following: $x \in \prod_{a \in A} S_a$ if $x \subseteq \bigcup_{a \in A} S_a$ and $x \cap S_a$ is a unit set for each $a \in A$. This definition of product is due to Whitehead [1902]. Hence Russell's form is the same as AC 2. See 0.4 (i) in the section entitled Preliminary Definitions and Theorems for our definition of product.) Zermelo, in [1908a] stated, and in [1908b] proved, that Russell's and his formulations of the axiom of choice are equivalent. The name "axiom of choice" is due to Zermelo [1904], and the name "multiplicative axiom" or "multiplicative principle" is due to Russell [1906]. AC 4 and AC 5 were given by Bernays [1941]. (Bernays actually gives CAC 4 and CAC 5 which are stronger than AC 4 and AC 5. CAC 4 is given as an axiom and CAC 5 is proved equivalent to it.) AC 7 is an interesting variation of AC 6 which is based on some topological results due to Ward [1962]. (See TOP 2 in Section 8.) AC 8 and 9 follow from work of Pincus [1974]. AC 10(2) is due to Sierpinski [1965] and AC 10(n) for $n > 2$, AC 11 and 12 are generalizations which follow from results of A. Levy [1962]. AC 13(m) and AC 14 are due to Levy [1962] and AC 15 is a generalization of AC 14. (If the bound m on the finite sets is omitted in AC 10(m) and AC 13(m), the resulting statements remain equivalent to the axiom of choice, but the axiom of foundation is required in the proof. See MC 1 and MC 3 in section 5.) Sierpinski [1940] proved that the axiom of choice implies AC 16(3,2) and Sobocinski [1964] proved the converse. Frascella [1965,1966] and H. Rubin and J. Rubin [1967] independently proved that AC 16(n,k) for $n > 2$ and $1 < k < n$ is equivalent to the axiom of choice. Tarski [1951] proved that AC 17 is equivalent to the axiom of choice. AC 18 is due independently to Collins [1954] and Farah [1955]; and AC 19 is a generalization of AC 18.

We shall first show that AC 1 - AC 7 are all equivalent. AC 1 obviously implies AC 2, and it is clear

from the definition of product that AC 1 and AC 6 are equivalent.

THEOREM 2.1: $AC\ 2 \rightarrow AC\ 1$.

PROOF. Let s be a set of non-empty sets. Define $t = \{\{x\} \times x : x \in s\}$. Then the choice set of AC 2 is the required choice function on s .

THEOREM 2.2: $AC\ 1 \rightarrow AC\ 3$.

PROOF: Let f be an arbitrary function. Let $s = \mathcal{R}(f)$ and let F be the choice function on s . Define a function g so that for each $x \in \mathcal{D}(f)$, $g(x) = F(f(x))$. Then g is the required function.

THEOREM 2.3: $AC\ 3 \rightarrow AC\ 1$.

PROOF: Let s be a set of non-empty sets. Let f be a 1-1 function such that $\mathcal{R}(f) = s$. Define a function F such that for each $x \in s$, $F(x) = g(f^{-1}(x))$, where g is defined by AC 3. Then F is the required choice function.

THEOREM 2.4: $AC\ 4 \rightarrow AC\ 5$.

PROOF: Let f be an arbitrary function and let $r = \{\langle x, y \rangle : \langle y, x \rangle \in f\}$. AC 4 implies that there is a function g such that $\mathcal{D}(g) = \mathcal{D}(r)$ and $g \subseteq r$. Clearly, for every $x \in \mathcal{R}(f) = \mathcal{D}(g)$, $f(g(x)) = x$.

THEOREM 2.5: $AC\ 5 \rightarrow AC\ 4$.

PROOF: Let r be an arbitrary relation and define a function h as follows:

$$h = \{\langle \langle x, y \rangle, x \rangle : \langle x, y \rangle \in r \}.$$

AC 5 implies that there is a function g such that $\mathcal{D}(g) = \mathcal{R}(h)$ and for every $x \in \mathcal{D}(g)$, $h(g(x)) = x$. Now, $g(x)$ is an ordered pair, so we define $f(x)$ to be the second coordinate of $g(x)$, for each $x \in \mathcal{D}(g) = \mathcal{D}(r)$. Clearly, $\mathcal{D}(f) = \mathcal{D}(r)$, f is a function, and $f \subseteq r$.

THEOREM 2.6: $AC\ 4 \rightarrow AC\ 3$.

PROOF: Let f be an arbitrary function. Define a relation r as follows:

$$r = \{\langle x, y \rangle : y \in f(x)\}.$$

AC 4 implies that there is a function g such that $\mathcal{J}(g) = \mathcal{J}(r)$ and $g \subseteq r$. g is the required function.

THEOREM 2.7: AC 3 $\rightarrow$ AC 4.

PROOF: Let r be an arbitrary relation. Define a function h as follows:

For each $x \in \mathcal{J}(r)$, $h(x) = \{y : \langle x, y \rangle \in r\}$. AC 3 implies that there is a function f such that if $x \in \mathcal{J}(h)$ and $h(x) \neq \emptyset$, then $f(x) \in h(x)$. f is the required function.

It is clear that AC 6 $\rightarrow$ AC 7.

THEOREM 2.8: AC 7 $\rightarrow$ AC 6.

PROOF: Suppose s is a set and for each $i \in s$, x_i is a non-empty set. Let $y = (\bigcup_{i \in s} x_i)^\omega$. Define $F: y \times x_i \rightarrow y$ for $i \in s$ such that for $f \in y$ and $u \in x_i$, $F(f, u) = g$, where $g(0) = u$ and $g(n+1) = f(n)$. Then F is a 1-1 function mapping $y \times x_i$ into y . Since $y \prec y \times x_i$, it follows that for each $i \in s$, $y \times x_i \approx y$. Thus, AC 7 implies that $\bigcup_{i \in s} (y \times x_i) \neq \emptyset$. But if $\bigcup_{i \in s} (y \times x_i) \neq \emptyset$ then $\bigcup_{i \in s} x_i \neq \emptyset$, so AC 6 holds.

Next, we shall show that each of AC 1 - AC 7 is equivalent to the well-ordering theorem. WO 1 implies AC 1 because WO 1 implies U_s can be well-ordered. Then define the f of AC 1 to be the function which assigns to each $x \in s$ its first element.

THEOREM 2.9: AC 1 $\rightarrow$ WO 2.

(The intuitive idea of the proof is as follows: Let x be a non-empty set and f a choice function on the set of all non-empty subsets of x . We well-order x as follows: Let $f(x) = a$ be the first element of x ; $f(x \sim \{a\}) = b$ be the second element of x ; $f(x \sim \{a, b\}) = c$ be the third element of x ; etc. The proof given here is a modification of Zermelo's [1904] proof using ordinal numbers instead of

well-ordered sets.)

PROOF: Let x be a non-empty set. Let f be a choice function on the set of all non-empty subsets of x and define $f(\emptyset) = u$ where $u \notin x$. We define a function G as follows: For all ordinal numbers α , $G(\alpha) = f(x \sim G''\alpha)$. (It follows by transfinite induction that G is defined for all $\alpha \in \text{On}$.)

(1) G^{-1} is 1-1 on $\mathcal{R}(G) \cap x$. If $\alpha < \beta$, and $G(\alpha), G(\beta) \in x$, then $G(\alpha) \neq G(\beta)$ since $G(\beta) = f(x \sim G''\beta) \in x \sim G''\beta$ and $G(\alpha) \in G''\beta$.

(2) There is an ordinal number α such that $G''\alpha = x$. For there must be some ordinal number β such that $G(\beta) \notin x$. If this were not so, by (1) G would be a 1-1 mapping of On into x , which contradicts x being a set. Let α be the smallest ordinal number such that $\alpha \in \{\beta: G(\beta) \notin x\}$. Then by the definition of α , $G''\alpha \subseteq x$, and by the definition of G , $x \sim G''\alpha = \emptyset$.

From (1) we see that an ordinal number satisfying (2) yields the desired result, q.e.d.

Next, we shall show AC 8 is equivalent to the axiom of choice. It is clear that $\text{AC } 1 \rightarrow \text{AC } 8$.

THEOREM 2.10: $\text{AC } 8 \rightarrow \text{AC } 1$.

PROOF: Let s be any set of non-empty sets. Then for each $x \in s$, $x \times \omega \approx (x \times \omega) \cup \{\emptyset\}$. Thus, AC 8 implies that there is a function f such that for each $x \in s$, f_x is a bijection from $(x \times \omega) \cup \{\emptyset\}$ onto $x \times \omega$. Define a function g on s such that for each $x \in s$, $g(x)$ is the first coordinate of $f_x(\emptyset)$. The function g is the required choice function.

To prove that $\text{AC } 9 \rightarrow \text{AC } 1$, we use techniques given in Theorems 2.8 and 2.10.

THEOREM 2.11: $\text{AC } 9 \rightarrow \text{AC } 1$.

PROOF: Let s be a set of non-empty sets. Let $y = (Us)^\omega$. Then we prove as in Theorem 2.8 that for each

$x \in s$, $x \times y \approx y$. Also, it is easy to see that for each $x \in s$, $x \times y \approx (x \times y) \cup \{\emptyset\}$. Let

$$t = \{x \times y : x \in s\} \cup \{(x \times y) \cup \{\emptyset\} : x \in s\}.$$

The set t satisfies the hypothesis of AC 9. Let f be a function such that for each $x \in s$, f_x is a bijection from $(x \times y) \cup \{\emptyset\}$ onto $x \times y$. Define a choice function g on s such that for each $x \in s$, $g(x)$ is the first coordinate of $f_x(\emptyset)$.

Thus, $AC\ 1 \leftrightarrow AC\ 9$, since it is clear that $AC\ 1 \rightarrow AC\ 9$.

The following implications are also clear:

$WO\ 1 \rightarrow AC\ 10(n) \rightarrow AC\ 11 \rightarrow AC\ 12$, $AC\ 1 \leftrightarrow AC\ 13(1)$,
 $AC\ 13(m) \rightarrow AC\ 13(n)$ if $m \leq n$, and $AC\ 13(n) \rightarrow AC\ 14 \rightarrow AC\ 15$.
 We shall show that $AC\ 10(n) \rightarrow AC\ 13(n - 1)$, $AC\ 11 \rightarrow AC\ 14$
 and $AC\ 12 \rightarrow AC\ 15 \rightarrow WO\ 6$, which is sufficient to show that
 each of $AC\ 10(n) - 15$ is equivalent to the axiom of choice.

THEOREM 2.12: If n is a natural number, $n \geq 2$ then
 $AC\ 10(n) \rightarrow AC\ 13(n - 1)$.

PROOF: Suppose s is a set of non-empty sets. Let
 $t = \{(x \times \omega) \cup \{a\} : x \in s\}$ where $a \notin x \times \omega$ for any $x \in s$.
 The set t satisfies the hypothesis of $AC\ 10(n)$ so there
 is a function f such that for each $x \in s$, $f(x)$ is a
 decomposition of $(x \times \omega) \cup \{a\}$ into disjoint sets of
 cardinality between 2 and n . There is exactly one such
 set u , which has a as an element. Then the set of first
 coordinates of $u \sim \{a\}$ is a non-empty subset of x which
 has at most $n - 1$ elements, q.e.d.

The same proof gives us that $AC\ 11 \rightarrow AC\ 14$ and
 $AC\ 12 \rightarrow AC\ 15$.

THEOREM 2.13: $AC\ 15 \rightarrow WO\ 6$.

PROOF: (The proof is similar to 2.9.) Let x be a set
 and let s be the set of all nonempty subsets of x . By
 $AC\ 15$, there is a natural number m and a function y such
 that for each $y \in s$, $g(y) \neq \emptyset$, $g(y) \subseteq y$ and $g(y) \prec m$.

Define $g(\emptyset) = u$ where $u \notin x$. Define a function G as follows: For all ordinal numbers α , $G(\alpha) = g(x \sim \bigcup_{\beta < \alpha} G(\beta))$.

Now, we proceed as in the proof of 2.9 and prove:

(1) G^{-1} is 1-1 on $\mathcal{K}(G) \cap \mathcal{P}(x)$.

(2) There is an ordinal α such that $\bigcup G''\alpha = x$. Then define $f = G|_{\alpha}$, and we have m is the natural number, α is the ordinal number and f is the function which satisfies WO 6.

Next, we shall show that if $n > 2$ and $1 < k < n$ then $WO\ 2 \rightarrow AC\ 16(n, k) \rightarrow WO\ 4(n - k)$.

THEOREM 2.14: If n and k are natural numbers such that $n > 2$ and $1 < k < n$ then $WO\ 2 \rightarrow AC\ 16(n, k)$.

PROOF: Suppose s is an infinite set. $WO\ 2$ implies that there is an initial ordinal ω_α such that $s \approx \omega_\alpha$. Let K be the set of all k -element subsets of s and N , the set of all n -element subsets of s . Then $K \approx N \approx \omega_\alpha$. Suppose $K = \{k_\beta: \beta < \omega_\alpha\}$ and $N = \{n_\beta: \beta < \omega_\alpha\}$.

Now, we shall construct a subset $t \subseteq N$ which has the desired property. Define a sequence of sets, T_γ , by transfinite induction as follows: $T_0 = \emptyset$. Suppose T_γ has the property that for all $\beta < \gamma < \omega_\alpha$, k_β is a subset of exactly one element of T_γ and for all β such that $\gamma \leq \beta < \omega_\alpha$, k_β is a subset of at most one element of T_γ . If k_γ is a subset of exactly one element of T_γ define $T_{\gamma+1} = T_\gamma$. If k_γ is not a subset of any element of T_γ , let n'_γ be the smallest element $x \in N$ such that $k_\gamma \subseteq x$, but for all $\beta < \gamma$, $k_\beta \not\subseteq x$ and for all $\beta > \gamma$ if there is a $y \in N$ such that $k_\beta \subseteq y$ then $k_\beta \not\subseteq x$. (An example of such an x is $x = k_\gamma \cup \{u_1, u_2, \dots, u_{n-k}\}$ where the u_i 's are distinct elements of $s \sim (UT_\gamma \cup k_\gamma)$.) Now, define $T_{\gamma+1} = T_\gamma \cup \{n'_\gamma\}$. If γ is a limit ordinal define $T_\gamma = \bigcup_{\beta < \gamma} T_\beta$. Then, $t = \bigcup_{\gamma < \omega_\alpha} T_\gamma$ is the required set.

THEOREM 2.15: If n and k are natural numbers such that $n > 2$ and $1 < k < n$ then $AC\ 16(n,k) \rightarrow WO\ 4(n-k)$.

PROOF: Suppose x is an infinite set, and n and k satisfy the hypothesis. (There is no loss of generality in proving $WO\ 4(n-k)$ for an infinite set x .) Let y be a well-ordered set such that $y \cap x = \emptyset$ and $y \not\prec w$ where

$$w = \{u \subseteq x: u \approx n - k\}.$$

(For example, let $y \approx \Gamma(\mathcal{P}(x))$, where Γ is Hartogs' function; see 0.11.) Apply $AC\ 16(n,k)$ to $x \cup y$. Let t_n be a set of n -tuples of elements of $x \cup y$ such that each k -tuple of elements of $x \cup y$ is a subset of exactly one element of t_n . For each $u \in x$, let

$$s_u = \{v \in t_n: u \in v \wedge k - 1 \leq v \cap y\}.$$

Then there is a $v \in s_u$ such that $k \leq v \cap y$. For suppose not. Let a be any $(k-1)$ -element subset of y . Then $a \cup \{u\}$ is a subset of exactly one element in s_u . Moreover, each element of s_u contains exactly one subset $a \cup \{u\}$ where a is a $(k-1)$ -element subset of y . Consequently,

$$s_u \approx \{a \subseteq y: a \approx k - 1\} = K.$$

Since y can be well-ordered, $K \approx y$. We claim $K \leq w$. The function F such that for each $a \in K$, $F(a) = v_a \sim (a \cup \{u\})$, where v_a is the unique element of s_u with $a \cup \{u\} \subseteq v_a$, is a 1-1 function. This contradicts the definition of y , for $y \not\prec w$. Therefore, there is a $v \in s_u$ such that $k \leq v \cap y$.

Now let

$$M = \{v \in t_n: k \leq v \cap y\}$$

and

$$L = \{v \cap y: v \in M\}.$$

We have shown that for each $u \in x$, $M \cap s_u \neq \emptyset$. Furthermore, since each element of L has at least k elements, each

element of L is contained in exactly one element of M . Since y can be well-ordered and L is a set of finite subsets of y , L can be well-ordered. For each $a \in L$, let $G(a) = u_a \sim a$, where u_a is the unique element of M such that $a \subseteq u_a$. Then G yields a well-ordering of a collection of subsets of x , each of which has at most $n - k$ elements and whose union is x . Thus $WO\ 4(n - k)$ holds.

We note that $AC\ 16(n, k)$ can be generalized similarly to the generalizations of $AC\ 10(n)$ ($AC\ 11$ and $AC\ 12$) and $AC\ 13(m)$ ($AC\ 14$ and $AC\ 15$), but we shall not state the generalization here.

We shall show next that $AC\ 1 \leftrightarrow AC\ 17$.

THEOREM 2.16: $AC\ 1 \rightarrow AC\ 17$.

PROOF: Let $s, t = \mathcal{P}(s) \sim \{\emptyset\}$, F and g satisfy the hypothesis of $AC\ 17$. Then if f is a choice function on t , then $f \in F$, $g(f) \in t$ and $f(g(f)) \in g(f)$.

THEOREM 2.17: $AC\ 17 \rightarrow AC\ 1$.

PROOF: Suppose $AC\ 1$ is false. Then there is a set s such that if $t = \mathcal{P}(s) \sim \{\emptyset\}$, and F is the set of all functions mapping t into s then for all $f \in F$ there is a $u \in t$ such that $f(u) \notin u$. Suppose $a \notin s$. For each ordinal α and for each $f \in F$, define

$$h_f(\alpha) = \begin{cases} f(s \sim h_f''\alpha), & \text{if } f(s \sim h_f''\alpha) \in s \sim h_f''\alpha \\ a & \text{otherwise} \end{cases}.$$

Just as in the proof of Theorem 2.9 (for G) we can prove that h_f^{-1} is 1-1 on $\mathcal{R}(h_f) \cap s$ and therefore, there is an α such that $h_f(\alpha) = a$. Let α_0 be the smallest such α . If $h_f''\alpha_0 = s$ then s can be well-ordered, which implies t has a choice function, contradicting our assumption. Thus, $h_f''\alpha_0 \subset s$. Define $g(f) = s \sim h_f''\alpha_0$. Then g is a function with domain F , range contained in t and $f(g(f)) \notin g(f)$ for all $f \in F$. This contradicts $AC\ 17$, q.e.d.

To complete the proofs of equivalences in this section we shall prove $AC\ 1 \rightarrow AC\ 18 \rightarrow AC\ 19 \rightarrow AC\ 1$.

THEOREM 2.18: $AC\ 1 \rightarrow AC\ 18$.

PROOF: Suppose

$$(i) \quad u \in \bigcap_{a \in A} \bigcup_{b \in B_a} X_{a,b}.$$

Then, for each $a \in A$ there is a $b \in B_a$ such that $u \in X_{a,b}$. Let

$$C_a = \{b \in B_a : u \in X_{a,b}\}.$$

Let f be a choice function on $\{C_a : a \in A\}$. Then $f \in \prod_{a \in A} B_a$ and $u \in X_{a,f(a)}$ for all $a \in A$, so

$$(ii) \quad u \in \bigcap_{a \in A} X_{a,f(a)} \quad \text{for some } f \in \prod_{a \in A} B_a.$$

Conversely, suppose (ii) holds. Then there is an $f \in \prod_{a \in A} B_a$ such that for all $a \in A$, $u \in X_{a,f(a)}$. Therefore, for each $a \in A$, there is a $b \in B_a$, namely, $b = f(a)$, such that $u \in X_{a,b}$. Consequently, (i) is true, q.e.d.

The axiom of choice was not used in the second part of the proof of 2.18.

Since $AC\ 19$ is a special case of $AC\ 18$, it remains to be shown that $AC\ 19 \rightarrow AC\ 1$.

THEOREM 2.19: $AC\ 19 \rightarrow AC\ 1$.

PROOF: Let A be a set of non-empty sets. For each $a \in A$, let $u_a = \{c \cup \{u\} : c \in a\}$, and let $B = \{u_a : a \in A\}$. Then $u \in \bigcup_{b \in u_a} b$ for each $u_a \in B$. Therefore,

$$\bigcap_{u_a \in B} \bigcup_{b \in u_a} b \neq \emptyset; \quad AC\ 19 \text{ implies } \bigcup_{f \in C(B)} \bigcap_{u_a \in B} f(u_a) \neq \emptyset, \text{ where}$$

$C(B)$ is the set of choice functions on B . Consequently, $C(B) \neq \emptyset$ and a choice function on B determines a choice function on A .

In closing, in this section, we should like to say a few words about the principle of dependent choice.

DC(α): If R is a relation between subsets and elements of a set x such that $(\forall y \subseteq x)(y < \alpha \rightarrow (\exists u \in x)\langle y, u \rangle \in R)$ then there is a function f mapping α into x such that $(\forall \beta < \alpha)\langle f''\beta, f(\beta) \rangle \in R$.

The first mention of the principle of dependent choices was in Bernays [1942] and rediscovered independently by Tarski [1948]. The statement in these papers has the following form:

DC: If $R \neq \emptyset$ is a relation such that $\mathcal{R}(R) \subseteq \mathcal{D}(R)$ then there is a function f with domain ω such that for all $n \in \omega$ $\langle f(n), f(n+1) \rangle \in R$.

It is easy to see that $DC \leftrightarrow DC(\omega)$. The forms $DC(\alpha)$ were given by Levy [1964] and Theorems 2.20 and 2.21 are also due to Levy [1964]. (See also Felgner [1971] pp. 146-166.)

THEOREM 2.20: $WO 1 \rightarrow (\forall \alpha)DC(\alpha)$.

PROOF: Let α be an ordinal and let x and R satisfy the hypothesis of $DC(\alpha)$. $WO 1$ implies x can be well-ordered. Define a function f on α so that for all $\beta < \alpha$, $f(\beta)$ is the first element $a \in x$ such that $\langle f''\beta, a \rangle \in R$. It follows from the hypothesis that f is defined on α and f is clearly the desired function.

THEOREM 2.21: $(\forall \alpha)DC(\alpha) \rightarrow WO 3$.

PROOF: Let x be any set and let Γ be Hartogs' function. Then $\Gamma(x) = \{\alpha: \alpha < x\}$ is the ordinal number, λ , such that for all $\alpha < \lambda$, $\alpha < x$ but $\lambda \nless x$. Define a relation $R \subseteq \mathcal{P}(x) \times x$ such that

$$R = \{\langle y, u \rangle: y \subset x \wedge y < \lambda \wedge u \in x \sim y\}.$$

If $x < \lambda$ then $WO 3$ holds, so suppose it is not the case that $x < \lambda$. Then for all $y \subseteq x$ such that $y < \lambda$ there is a $u \in x \sim y$ with $\langle y, u \rangle \in R$.

Thus x and R satisfy the hypotheses of $DC(\lambda)$. Therefore, there is a function f such that for all $\beta < \lambda$, $\langle f''\beta, f(\beta) \rangle \in R$. The function f is 1-1 because by the definition of R , $f(\beta) \notin f''\beta$. Thus if $f(\alpha) = f(\beta)$ with $\alpha < \beta$ we would have $f(\beta) = f(\alpha) \in f''\beta$. This implies $\lambda \leq x$, which contradicts the definition of λ . Consequently, $x < \lambda$, q.e.d.

§3. THE LAW OF THE TRICHOTOMY

The equivalence of the axiom of choice and the trichotomy was given by Hartogs [1915]. As in the case of the well-ordering theorem, the trichotomy was considered self-evident and was used without hesitation before 1915.

T1: For all sets x and y , $x < y$, $x \approx y$ or $y < x$.

It is clear from the properties of ordinal numbers that WO 2 implies T1. To prove the implication the other way we use Hartog's function Γ , where $\Gamma(x) = \{\alpha: \alpha < x\}$. (See 0.11 and 0.12).

THEOREM 3.1: $T1 \rightarrow WO\ 3$.

PROOF: By T1, either $x \leq \Gamma(x)$ or $\Gamma(x) < x$. The first inequality implies WO 3 since $\Gamma(x)$ is an ordinal number. The second is impossible because it implies $\Gamma(x) \in \Gamma(x)$. (No ordinal number can be an element of itself.)

We shall next consider a proposition which is similar to T1. It was stated by Lindenbaum in Lindenbaum and Tarski [1926] and later proved by Sierpinski [1948].

T2: For every two non-empty sets, there is a mapping of one onto the other.

Clearly T1 implies T2. Before proving that T2 implies the axiom of choice we prove the following lemma.

LEMMA 3.2: For any two sets x and y , if there is a mapping of x onto y then there is a 1-1 function which maps $\mathcal{P}(y)$ into $\mathcal{P}(x)$.

PROOF: Let f map x onto y . For $t \in \mathcal{P}(y)$, define $g(t) = f^{-1}t$. Then g is a function from $\mathcal{P}(y)$ into $\mathcal{P}(x)$.

To show that g is 1-1, we note that if $s, t \in \mathcal{P}(Y)$, $g(s) \sim g(t) = f^{-1}''s \sim f^{-1}''t = f^{-1}''(s \sim t)$. Therefore, if $g(s) = g(t)$, $f^{-1}''(s \sim t) = \emptyset$ and $f^{-1}''(t \sim s) = \emptyset$; which implies $t \subseteq s$ and $s \subseteq t$; so that $s = t$.

THEOREM 3.3: $T_2 \rightarrow WO\ 3$.

PROOF: Consider the non-empty sets x and $\Gamma(\mathcal{P}(x))$. By T_2 , either there is a mapping of x onto $\Gamma(\mathcal{P}(x))$ or there is a mapping of $\Gamma(\mathcal{P}(x))$ onto x . If the first mapping exists, then 3.2 implies there is a 1-1 function which maps $\mathcal{P}(\Gamma(\mathcal{P}(x)))$ into $\mathcal{P}(x)$. But since $\Gamma(\mathcal{P}(x)) < \mathcal{P}(\Gamma(\mathcal{P}(x)))$, this implies that $\Gamma(\mathcal{P}(x)) < \mathcal{P}(x)$, which is impossible. Hence, there must be a mapping f of $\Gamma(\mathcal{P}(x))$ onto x . For each $t \in x$, define $g(t)$ to be the smallest element of $f^{-1}''\{t\}$. We see that g is a 1-1 function from x into $\Gamma(\mathcal{P}(x))$. Hence $x < \Gamma(\mathcal{P}(x))$, which proves $WO\ 3$.

There are two additional forms of the trichotomy due to Tarski [1964], $T_3(n)$, and Kruse [1962], T_4 . For convenience, we shall state these forms in terms of cardinal numbers rather than sets. (All statements about cardinal numbers which we use may be replaced by statements about sets.) In what follows, κ, μ, ν and these letters with subscripts, represent cardinal numbers. (See Definition 0.13 and the discussion which precedes it.) For example, T_1 stated in terms of cardinal numbers would be:

For all cardinals κ and μ , $\kappa < \mu$, $\kappa = \mu$, or $\mu < \kappa$.

If $n \geq 2$ is a natural number,

$T_3(n)$: If x is a set of cardinal numbers such that $|x| = n$ then there exists $\kappa, \mu \in x$ such that $\kappa < \mu$ or $\mu < \kappa$.

$T_3(2)$ is T_1 and it is clear that $T_1 \rightarrow T_3(n)$ for all natural numbers $n \geq 2$. We shall show that $T_3(n) \rightarrow WO\ 3$ for each natural number $n \geq 2$,

THEOREM 3.4: For each natural number $n \geq 2$, $T_3(n) \rightarrow WO\ 3$.

PROOF: Suppose x is an infinite set such that $|x| = \kappa$ and $\kappa \geq \aleph_0$. (There is no loss of generality in assuming $\kappa \geq \aleph_0$, for, if not, replace x by $x \cup \omega$. If $x \cup \omega$ can be well-ordered then so can x .) Let $n \geq 2$ be a natural number such that $T3(n)$ holds. For each $m \in \omega$, define $P_m(\kappa)$ so that $P_0(\kappa) = 0$, $P_1(\kappa) = \kappa$, and for all $m \geq 1$, $P_{m+1}(\kappa) = 2^{P_m(\kappa)}$. Let $\aleph_\alpha$ be any aleph such that $\aleph_\alpha \nless P_n(\kappa)$. (For example, let $\aleph_\alpha = |\Gamma(P_n(\kappa))|$.) Let

$$y = \{\aleph_{\alpha+j} + P_{n-j}(\kappa) : 0 \leq j < n\}.$$

Then y is a set of cardinal numbers such that $|y| \leq n$. Suppose there is a j and k , with $0 \leq j < k < n$ such that

$$\aleph_{\alpha+j} + P_{n-j}(\kappa) = \aleph_{\alpha+k} + P_{n-k}(\kappa).$$

Then $\aleph_{\alpha+k} \leq \aleph_{\alpha+j} + P_{n-j}(\kappa)$, so by, 0.14(e),

$$(1) \quad \aleph_{\alpha+k} \leq \aleph_{\alpha+j} \quad \text{or} \quad \aleph_{\alpha+k} \leq P_{n-j}(\kappa).$$

The first inequality is impossible because $j < k$ and the second is also impossible because it implies $\aleph_\alpha \leq P_n(\kappa)$ which contradicts the definition of $\aleph_\alpha$. Thus, $|y| = n$.

Applying $T3(n)$ to y , there exists natural numbers $j, k, j \neq k$, and $0 \leq j, k < n$ such that

$$(2) \quad \aleph_{\alpha+k} + P_{n-k}(\kappa) < \aleph_{\alpha+j} + P_{n-j}(\kappa).$$

If $j < k$ then we obtain the inequalities (1) which have been shown to imply a contradiction. Therefore, we must have $k < j$. Let $\mu = P_{n-j}(\kappa)$, then from (2) we obtain

$$\begin{aligned} \aleph_{\alpha+j} + \mu &> \aleph_{\alpha+k} + P_{n-k}(\kappa) \\ &\geq P_{n-k}(\kappa) \\ &\geq P_{n-(j-1)}(\kappa) = 2^\mu. \end{aligned}$$

Thus it follows from 0.14(d) that $2^\mu \leq \aleph_{\alpha+j}$, and this implies $\kappa \leq \aleph_{\alpha+j}$, q.e.d.

The form of the trichotomy due to Kruse [1962] is the following:

T4: For all cardinals μ , if $\mu \leq \kappa_1$, $\kappa_2 \leq 2^\mu$ then either $\kappa_1 \leq \kappa_2$ or $\kappa_2 \leq \kappa_1$.

It is clear that $T1 \rightarrow T4$. Moreover, once we prove that T4 implies the axiom of choice, we will have another proof that the generalized continuum hypothesis (GCH) implies the axiom of choice, because it is clear that $GCH \rightarrow T4$.

Before proving that T4 implies AC we acquire some preliminary results.

DEFINITION 3.5: For any set x , we define $K(x)$ to be the set of all 1-1 functions mapping an ordinal into x .

$$K(x) = \{f: (\exists \alpha \in \text{On}) f: \alpha \rightarrow x\}.$$

If $|x| = \kappa$, let $|K(x)| = \kappa^+$. We shall call K Kruse's function. It is easy to prove that K has the following properties

LEMMA 3.6: For each set x ,

- (a) $K(x) \approx \{R \subseteq x \times x: (\exists y \subseteq x) (\langle y, R \rangle \text{ is a well-ordered set})\}$.
- (b) $K(x) \leq \mathcal{P}(x \times x)$.
- (c) $K(x) \leq \mathcal{P}(\mathcal{P}(x))$.
- (d) $x \leq K(x)$.
- (e) If $|x|$ is an aleph then $K(x) \approx \mathcal{P}(x)$.

LEMMA 3.7: For all cardinals κ and μ ,

- (a) $\kappa^+ \cdot \mu^+ \leq (\kappa + \mu)^+$
- (b) $\kappa \geq \aleph_0 \rightarrow \kappa^+ + \kappa^+ = \kappa^+$
- (c) $\kappa + \kappa = \kappa \rightarrow (\kappa^+)^2 = \kappa^+$

PROOF: (a) Show that if $x \cap y = \emptyset$ then $K(x) \times K(y) \leq K(x \cup y)$.

(b) If $\kappa \geq \aleph_0$ then $\kappa + 1 = \kappa$. Moreover,
 $\kappa^+ + \kappa^+ = \kappa^+ \cdot 2 = \kappa^+ \cdot 1^+.$ Therefore, using part (a) we see that

$$\kappa^+ + \kappa^+ = \kappa^+ \cdot 1^+ \leq (\kappa + 1)^+ = \kappa^+.$$

(c) Suppose $\kappa + \kappa = \kappa$. Then using 3.7 (a) we obtain

$$(\kappa^+)^2 = \kappa^+ \cdot \kappa^+ \leq (\kappa + \kappa)^+ = \kappa^+.$$

LEMMA 3.8: If $\aleph_0 \leq \kappa$, $\aleph$ is an aleph and $\kappa^+ \leq \kappa + \aleph$ then $\kappa < \kappa^+ = 2^\kappa \leq \aleph$.

PROOF: Suppose κ and $\aleph$ satisfy the hypothesis. Then by 3.6 (d) $\kappa < \kappa^+$, thus using 3.7 (b) and hypotheses, we obtain,

$$\kappa + \kappa \leq \kappa^+ + \kappa^+ = \kappa^+ \leq \kappa + \aleph.$$

Therefore, there exists $\mu \leq \aleph$ such that $\kappa + \kappa = \kappa + \mu$. This implies $\mu \leq \kappa$, (0.14(e)) so $\kappa + \mu = \kappa$. Consequently, $\kappa + \kappa = \kappa$. It follows from 3.7 (c) that $(\kappa^+)^2 = \kappa^+$. Since $\kappa^+ \leq \kappa + \aleph$, there exist cardinals $\kappa_1 \leq \kappa$ and $\mu \leq \aleph$ such that $\kappa^+ = \kappa_1 + \mu$. Thus,

$$\kappa^+ \cdot \mu \leq (\kappa^+)^2 = \kappa^+ \leq \kappa + \aleph,$$

from which it follows that either $\kappa^+ \leq \aleph$ or $\mu \leq \kappa$. (See 0.14(d)). Suppose $\mu \leq \kappa$, then $\kappa^+ = \kappa_1 + \mu \leq \kappa + \kappa = \kappa$ which contradicts 3.6 (d). Thus, $\kappa^+ \leq \aleph$ which implies κ^+ is an aleph. Therefore, by 3.6 (e) $\kappa^+ = 2^\kappa \leq \aleph$, q.e.d.

To prove T4 implies the well-ordering theorem we shall prove two things:

(1) For every cardinal κ , there is an aleph, $\aleph$, such that $2^\kappa \leq 2^\aleph$.

(2) If $\aleph$ is an aleph, so is $2^\aleph$.

Then since $\kappa < 2^\kappa$, it follows from (1) that $\kappa < 2^\aleph$, and from (2) that κ is the cardinal number of a well-ordered set.

THEOREM 3.9: T4 $\rightarrow$ For each cardinal κ there is an aleph $\aleph$ such that $2^\kappa \leq 2^\aleph$.

PROOF: We may assume, without loss of generality that $\kappa \geq \aleph_0$. Let $\aleph = (2^\kappa)^*$, where if $|x| = \kappa$, $\aleph = |\Gamma(\mathcal{P}(x))|$. We have,

$$\kappa + \aleph \leq 2^\kappa + \aleph, \quad \kappa + 2^\aleph \leq 2^{\kappa+\aleph}.$$

T4 implies that

$$\kappa + 2^\aleph \leq 2^\kappa + \aleph \quad \text{or} \quad 2^\kappa + \aleph \leq \kappa + 2^\aleph.$$

If the first inequality holds then $2^\aleph \leq 2^\kappa + \aleph$, which by 0.14(d), implies $2^\aleph \leq 2^\kappa$, contradicting the definition of $\aleph$. Similarly, the second inequality implies $2^\kappa \leq 2^\aleph$, which is what we wanted to prove.

THEOREM 3.10: T4 $\rightarrow$ For every aleph, $\aleph$, $2^\aleph$ is an aleph.

PROOF: First we shall show that T4 implies

$$(*) \quad (\forall \alpha \in \text{On}) \quad \aleph_{\alpha+1} \leq 2^{\aleph_\alpha}.$$

Let $\kappa = 2^{\aleph_\alpha}$. If $\aleph_{\alpha+1} \not\leq \kappa$ then κ is not an aleph and $\kappa^* = \aleph_{\alpha+1}$. If we could find a cardinal μ such that $\mu < \kappa^* < 2^\mu$, then since $\mu \leq \aleph_\alpha$, we would obtain the contradiction $\kappa^* \leq 2^{\aleph_\alpha}$. Let us now construct this cardinal μ .

First,

$$\kappa^2 = (2^{\aleph_\alpha})^2 = 2^{\aleph_\alpha \cdot 2} = 2^{\aleph_\alpha} = \kappa.$$

Therefore, it follows from 3.6 (b) and (d) that

$$(1) \quad \kappa < \kappa^+ \leq 2^{\kappa^2} = 2^\kappa.$$

Moreover, we have by 0.14 (g) and (h),

$$(2) \quad \aleph_{\alpha+1} = \kappa^* = \aleph_\alpha^* \leq 2^{2^{\aleph_\alpha}} = 2^{2^{\aleph_\alpha}} = 2^\kappa.$$

Therefore,

$$(3) \quad \kappa < \kappa + \kappa^* \leq 2^\kappa + 2^\kappa = 2^\kappa.$$

By (1) and (3), both κ^+ and $\kappa + \kappa^*$ are between κ and 2^κ so T4 implies

$$\kappa^+ \leq \kappa + \kappa^* \quad \text{or} \quad \kappa + \kappa^* \leq \kappa^+.$$

If the first inequality holds it follows from 3.8 that $\kappa < \kappa^+ = 2^\kappa \leq \kappa^*$ which implies that κ is an aleph contradicting the hypothesis. If the second inequality holds then $\kappa^* \leq \kappa^+$. Let x be a set so that $|x| = \kappa$. Then there is a set $w \subseteq K(x)$ such that $\Gamma(x) \approx w$. Each element of w is a function from an ordinal number onto a subset of x . Let $y = \bigcup_{f \in w} \mathcal{R}(f) \subseteq x$.

Since w can be well-ordered, y can also be well-ordered. Let $\mu = |y| \leq \kappa$. Then, $w \subseteq K(y)$ and $\mu < \mu^* \leq \kappa^*$. Moreover, since $\kappa^* = |w|$, $\kappa^* \leq \mu^+$ and since μ is an aleph it follows from 3.6 (e) that $\mu^+ = 2^\mu$. Thus, μ is the required cardinal because $\mu < \kappa^* \leq 2^\mu$.

Now we shall show that for each ordinal α , $2^{\aleph_\alpha}$ is an aleph. Let $\aleph_\beta = (2^{\aleph_\alpha})^*$. Then, $\aleph_\alpha < \aleph_\beta$ and $\aleph_\beta \nless 2^{\aleph_\alpha}$. By (*), we have

$$\aleph_\beta < \aleph_{\beta+1} \leq 2^{\aleph_\beta}.$$

Also, $\aleph_\beta < 2^{\aleph_\alpha} + \aleph_\beta \leq 2^{\aleph_\beta} + 2^{\aleph_\beta} = 2^{\aleph_\beta}$. Therefore, both $\aleph_{\beta+1}$ and $2^{\aleph_\alpha} + \aleph_\beta$ are between $\aleph_\beta$ and $2^{\aleph_\beta}$. T4 implies

$$\aleph_{\beta+1} \leq 2^{\aleph_\alpha} + \aleph_\beta \text{ or } 2^{\aleph_\alpha} + \aleph_\beta \leq \aleph_{\beta+1}.$$

The first inequality is impossible because of the definition of $\aleph_\beta$ and the second inequality implies $2^{\aleph_\alpha} \leq \aleph_{\beta+1}$, q.e.d.

THEOREM 3.11: T4 $\rightarrow$ WO 3.

PROOF: 3.9 and 3.10.

Next, in this section on the trichotomy, we shall add three forms due to Sobocinski. T5 and T6 are in Sobocinski [1961a], and T7 in [1961b].

T5: For all cardinals κ and μ , if $\mu < 2^\kappa$ then $\kappa \leq \mu$ or $\mu \leq \kappa$.

T6: For all cardinals κ , and alephs $\aleph$, if $\aleph < 2^\kappa$ then $\kappa \leq \aleph$ or $\aleph \leq \kappa$.

T7: For all cardinals κ_1, κ_2 , and μ , if $\aleph_0 \leq \kappa_1$, $\kappa_2 < \mu$ and if for all ν_1 and ν_2 , $\kappa_1 + \nu_1 = \kappa_2 + \nu_2 = \mu$ implies $\nu_1 = \nu_2$, then $\kappa_1 \leq \kappa_2$ or $\kappa_2 \leq \kappa_1$.

Clearly, T1 implies each of T5, T6, and T7. It is clear that T5 implies T6. We shall show T6 implies WO 3 and that T7 implies T1.

THEOREM 3.12: $T6 \rightarrow WO\ 3$.

PROOF: Let $\kappa \geq \aleph_0$ be a cardinal. It follows from properties of cardinals (0.14(i)) that

$$\kappa^* \leq 2^{2^\kappa}.$$

If equality holds, then $\kappa < \kappa^*$ which implies κ is an aleph. Otherwise, T6 implies

$$\kappa^* \leq 2^{2^\kappa} \quad \text{or} \quad 2^{2^\kappa} \leq \kappa^*.$$

The second alternative implies κ is an aleph. If the first alternative holds, repeat the process. If $\kappa^* < 2^{2^\kappa}$ apply T6 to obtain

$$\kappa^* \leq 2^\kappa \quad \text{or} \quad 2^\kappa \leq \kappa^*.$$

The second alternative implies κ is an aleph. If the first alternative holds repeat the process, obtaining

$$\kappa^* \leq \kappa \quad \text{or} \quad \kappa \leq \kappa^*.$$

The first alternative is impossible. The second implies κ is an aleph, q.e.d.

THEOREM 3.13: $T7 \rightarrow T1$.

PROOF: Let $\kappa_1, \kappa_2 \geq \aleph_0$ be arbitrary cardinals. Let $\mu = 2^{\kappa_1 + \kappa_2}$. Then it follows from 0.14 (f) that

$$\kappa_1 + \nu_1 = 2^{\kappa_1 + \kappa_2} \rightarrow \nu_1 = 2^{\kappa_1 + \kappa_2}$$

and

$$\kappa_2 + \nu_2 = 2^{\kappa_1 + \kappa_2} \rightarrow \nu_2 = 2^{\kappa_1 + \kappa_2}.$$

Thus, the hypothesis of T7 is satisfied, so κ_1 and κ_2 are comparable. (The proof given here is simpler than the one given in Sobocinski [1961b].)

The final forms of the trichotomy which we shall consider here are due to Alas [1968].

Let $\mathcal{P}_{<\omega}(x)$ be the set of all finite subsets of x .

T8: For all sets x and y , $\mathcal{P}_{<\omega}(x) \prec \mathcal{P}_{<\omega}(y)$ or $\mathcal{P}_{<\omega}(y) \prec \mathcal{P}_{<\omega}(x)$.

T9: For all sets x and y , $x \prec \mathcal{P}_{<\omega}(y)$ or $\mathcal{P}_{<\omega}(y) \prec x$.

T10: For all sets x and y , $x \prec \mathcal{P}_{<\omega}(y)$ or $y \prec \mathcal{P}_{<\omega}(x)$.

T1 implies each of T8, T9 and T10, and each of T8 and T9 implies T10. We shall show $T10 \rightarrow WO\ 3$.

THEOREM 3.14: $T10 \rightarrow WO\ 3$.

PROOF: Let x be an infinite set and let $y = \Gamma(\mathcal{P}_{<\omega}(x))$. Then since y can be well-ordered, $\mathcal{P}_{<\omega}(y) \approx y$. T10 implies $x \prec y$ or $y \prec \mathcal{P}_{<\omega}(x)$. The second alternative is impossible and the first implies $WO\ 3$.

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§4. MAXIMAL PRINCIPLES

As mathematics developed further there also developed a need for another non-constructive proposition; a principle, which Kuratowski, Hausdorff, Zorn, and others, used to replace transfinite induction and the well-ordering theorem. It appears, at first glance, unrelated to the axiom of choice, but actually is equivalent to it.

This principle and principles similar to it are often referred to as forms of Zorn's Lemma. (According to Zorn (see P. Campbell [1978]), Artin and Chevalley first referred to the principle as Zorn's Lemma in 1933.) We prefer to call these forms maximal principles since it is a more descriptive title. Also, we discuss forms which are far afield from Zorn's original form, but are still maximal principles.

The history of maximal principles is quite tangled. (See P. Campbell [1978] for a more detailed description.) The earliest reference to a maximal principle in the literature is in Hausdorff [1907] in which a special case of M 6 is derived. Later in Hausdorff [1909], M 6 itself is proved. In his book, Hausdorff [1914], M 5 is derived from the well-ordering theorem. In 1927, in the second edition of his book, Hausdorff derives a form similar to M 4 from the axiom of choice. (In 1910, Janiszewski [1910], Mazurkiewicz [1910] and Zoratti [1910, 1912] published a special case of M 4 in the form of a theorem in topology. Brouwer [1911] derived a more general topological theorem from the well-ordering theorem. Reference was made in Hausdorff [1927] to Janiszewski [1910, 1911, 1912] and Brouwer [1910], but not to Brouwer [1911] which was the second half of Brouwer [1910].)

Kuratowski [1922] derived minimal principles equivalent to M 3 and M 4 from the well-ordering theorem. (In our forms M 1 - M 4 there is an obvious duality between minimal and maximal principles. We shall not discuss the minimal principles here.) Kuratowski used a minimal principle to prove a theorem in analysis. (Some of the preliminary work in Kuratowski's paper is due to Hessenberg [1909] and Kuratowski referred to the work of Brouwer [1911].) R. L. Moore [1932] derived a minimal principle equivalent to M 3 from the well-ordering theorem. (Moore lists Kuratowski [1922] in his bibliography.)

It wasn't until 1935 that Zorn published his paper. He was the first one to use a maximal principle in algebra. He stated without proof that M 3 is equivalent to the axiom of choice (to be proven in another paper which was never published). According to Zorn himself (as stated in Campbell [1978] p. 83) he saw a copy of Moore's book [1932] but did not see the maximal principle. He also knew about Kuratowski [1922] but had "formed an erroneous impression about the thrust of the article". Also, Zorn credits Artin for noticing that M 3 implies the axiom of choice.

O. Teichmüller [1939] and J. W. Tukey [1940] independently gave M 7 and proved it equivalent to the axiom of choice. Tukey also gave M 1, a generalization of M 3. Teichmüller was unaware of Zorn's work while Tukey refers to Zorn. N. Bourbaki [1939] gave a proposition similar to M 1 and states, without proof, that it is equivalent to the well-ordering theorem. (He stated a lemma similar to Kuratowski's lemma (see 4.17) to indicate how one would prove that M 1 implies the well-ordering theorem. M 7 is also stated by Bourbaki, but no proofs are given.) M 2 was given by Kneser [1950] and Szele [1950].

There are two different forms of the maximal principles. M 1 - M 4 postulate the existence of a maximal element, and the remaining forms postulate the existence of a maximal set.

M 1: If R is a partial ordering relation on a non-empty set x and if every subset of x which is linearly ordered by R has an R -upper bound, then x has an R -maximal element.

M 2: If R is a partial ordering relation on a non-empty set x and if every subset of x which is well-ordered by R has an R -upper bound, then x has an R -maximal element.

DEFINITION 4.1: A nest is a class which is linearly ordered by inclusion, $\subseteq$.

M 3: If every non-empty nest which is a subset of a non-empty set x has its union an element of x , then x has a $\subseteq$ -maximal element.

M 4: If every non-empty well-ordered nest which is a subset of a non-empty set x has its union an element of x , then x has a $\subseteq$ -maximal element.

Clearly $M 2 \rightarrow M 1 \rightarrow M 3$ and $M 2 \rightarrow M 4 \rightarrow M 3$. There are many other maximal principles we could have chosen just by changing the ordering relation or the conditions on the upper bound. We chose $M 1$ since it is a common form and often used in practice and $M 4$ for historical reasons. We shall discuss other variations of $M 1 - M 4$ and also of $M 5$ and $M 6$ later.

M 5: If R is a partial ordering relation on x , then there exists a $\subseteq$ -maximal subset of x which is linearly ordered by R .

M 6: For every set x , there exists a $\subseteq$ -maximal subset of x which is a nest.

Next, we define a property of finite character. By a property we mean a formula in the language of NBG which has at least one free variable.

DEFINITION 4.2: $P(X)$ is said to be a property of finite character if $P(\emptyset)$ and for each class X ,

$$P(X) \leftrightarrow \text{For each finite } x \subseteq X, P(x).$$

Equivalently, if we let

$$Q = \{x: x \text{ is finite} \wedge P(x)\}$$

Then $P(X)$ is a property of finite character iff

$$P(X) \leftrightarrow (\forall x)(x \subseteq X \wedge x \text{ is finite} \rightarrow x \in Q) .$$

For example, if $P(X)$ is any one of the following then it is a property of finite character:

1. X is partially ordered by R .
2. X is linearly ordered by R .
3. $u \notin X$.
4. The elements of X are pairwise disjoint.

On the other hand, the following are not properties of finite character:

5. X can be linearly ordered. (Note the difference between 5 and 2).
6. X is well-ordered by R .
7. $u \in X$.
8. X is infinite.
9. X is directed by R , where R is a directed relation on X iff R is a partial ordering on X and for all $u, v \in X$ there is a $w \in X$ such that $u R w$ and $v R w$.

(Note that if each finite subset of X is directed by R then X is directed by R , (in fact X is linearly ordered by R) but if X is directed it does not follow that each finite subset of X is directed. On the other hand, if X is well-ordered by R then each finite subset of X is well-ordered by R , but the converse does not necessarily hold.)

M 7: For every set x and every property P of finite character, there exists a $\subseteq$ -maximal subset of x which has the property P .

Clearly $M 7 \rightarrow M 5 \rightarrow M 6$. In order to prove these seven forms are equivalent, it remains to be shown that $M 3 \rightarrow M 7$ and $M 6 \rightarrow M 2$. It turns out that it is easier to prove the latter implication in two steps, $M 6 \rightarrow M 5$ and $M 5 \rightarrow M 2$.

THEOREM 4.3: $M 3 \rightarrow M 7$.

PROOF: Let x be a set and P be a property of finite character. Let $y = \{t: t \subseteq x \text{ and } P(t)\}$. We shall show that if $n \subseteq y$, $n \neq \emptyset$, and n is a nest then $\bigcup n \in y$. Clearly $\bigcup n \subseteq x$. Let u be a finite subset of $\bigcup n$. Then since n is a nest, u is a finite subset of some element of n . Since every element of n has the property P , it follows that $P(u)$. Therefore, we obtain $P(\bigcup n)$ by the definition of finite character. Furthermore, the empty set has property P . Thus y satisfies the hypothesis of M 3, so y has a maximal element.

THEOREM 4.4: $M 6 \rightarrow M 5$.

PROOF: Let R be a partial ordering relation on x . Let y be the set of all subsets of x which are linearly ordered by R . By M 6, there is a maximal subset n of y which is a nest. Let $m = \bigcup n$. We wish to show that m is a maximal element of y . Clearly $m \subseteq x$. Let $u, v \in m$, $u \neq v$; then $(\exists w)(u, v \in w \text{ and } w \in n)$. R linearly orders w , and hence either $u R v$ or $v R u$. Consequently R linearly orders m . Suppose m is not maximal; then for some $z \in y$, we have $m \subset z$. We observe that $n \cup \{z\}$ is a nest since every element of n is a subset of z . Since n is maximal, we must have $\{z\} \subseteq n$. That is, $z \in n$, whence $z \subseteq m$, which contradicts our assumption about z . Consequently, m is maximal, and therefore M 5 holds.

THEOREM 4.5: $M 5 \rightarrow M 2$.

PROOF: Let R be partial ordering relation on x . Let y be the set of all subsets of x which are well-ordered by R . Define a relation S on y as follows:

$$S = \{\langle t, u \rangle: t \in y \wedge u \in y \wedge t \text{ is an initial } R\text{-segment of } u\}.$$

S is clearly a partial ordering relation on y ; therefore, by M 5 there is a maximal subset n of y which is linearly ordered by S . Let $m = \bigcup n$. We shall show that m has an R -upper bound, and that such an element is an R -maximal element of x .

First let us show that $m \in y$. Clearly $m \subseteq x$. Let z be any non-empty subset of m . Then for some $t \in n$, $z \cap t \neq \emptyset$. Let u be the R -first element of $z \cap t$. It remains to be shown that u is R -first in z . Now let $v \in z$, $v \neq u$. If $v \in t$, we have $u R v$ and not $v R u$ by the well-ordering property. If $v \notin t$, $v \in w$ for some $w \in n$. By the definition of S and the construction of n , t is an initial R -segment of w . Hence $u R v$ and not $v R u$. Therefore, u is the unique R -first element of z so that R is a well-ordering on z and $m \in y$.

By an argument similar to that of the preceding theorem, m is maximal in y . Consequently, m cannot have a strict R -upper bound. For if b were a strict R -upper bound for m , we would have $m' = m \cup \{b\} \in y$ which contradicts the maximality of m . Since $m \in y$, it follows from the hypothesis of M_2 that m has an R -upper bound. Let b be such a bound. Suppose $z \in x$ and $b R z$. If not $z R b$, z would be a strict R -upper bound for m , which is impossible. Therefore, b is an R -maximal element of x , q.e.d.

There are some additional maximal principles which we shall discuss briefly. They are quite similar to the forms $M_1 - M_7$ and first appeared in the literature in Kuratowski [1922]. We shall call them $M'_1 - M'_7$. The hypotheses for these forms are the same as that of the corresponding unprimed forms, but the conclusions are stronger. The conclusions for $M'_1 - M'_4$ state that there exists a maximal element larger than any given element; the conclusions for $M'_5 - M'_7$ state that there exists a maximal subset containing any given subset with the required property. We shall state only two of these additional forms in a formal way; it will be clear what the other forms are.

M'_1 : If R is a partial ordering relation on x and if every subset of x which is linearly ordered by R has an R -upper bound and if $y \in x$, then there is an R -maximal element z in x such that $y R z$.

M'7: For every set x and every property P of finite character and for every subset y of x such that $P(y)$, there is a $\subseteq$ -maximal subset z of x such that $y \subseteq z$ and $P(z)$.

It is clear that $M'n \rightarrow M_n$ for $n = 1, \dots, 7$. Next we prove the implications the other way.

THEOREM 4.6: $M_n \rightarrow M'n$ for $n = 1, 2, 3, 4$.

PROOF: Let $y \in x$ and let w be the set of all upper bounds for $\{y\}$. Then apply M_n to the set w and thereby obtain $M'n$.

THEOREM 4.7: $M_7 \rightarrow M'n$ for $n = 5, 6, 7$.

PROOF: We give the proof for $M'5$. The others are analogous. Let y be a subset of x which is linearly ordered by R . "Linearly ordered" is a property of finite character. Call it P . We define a new property Q as follows. For any subset w of x , $Q(w)$ if and only $P(w \cup y)$. Q is clearly a property of finite character. By M_7 there is a maximal subset z of x which has the property Q . Therefore $z \cup y$ is a maximal subset of x which contains y and is linearly ordered by R .

We shall next look at some of the ways the ordering relation R in M_1 and M_5 can be changed and still obtain a maximal principle which is equivalent to the axiom of choice. The results are due to Harper and Rubin [1976]. Consider the following statements:

$M_1(Q, U)$: Every non-empty set which is Q -ordered by R in which each U -ordered subset has an R -upper bound, has an R -maximal element.

As possibilities for Q and U we shall take the following:

A: arbitrary (any binary relation)

TR: transitive

AS: antisymmetric

C: connected

P: partial ordering (reflexive, antisymmetric and transitive)

L: linear ordering (connected partial ordering)

W: well-ordering

D: directed (upwards), where X is directed by R if R is a partial ordering on X and each finite subset of X has an R -upper bound. (It is sufficient that each two element subset has an R -upper bound.)

RA: ramified, where X is ramified by R if R is a partial ordering on X and each initial R -segment of X is linearly ordered by R .

F: forest, where X is a forest under R if R is a partial ordering on X and every initial R -segment of X is well-ordered by R .

T: tree, where X is a tree under R if X is a forest under R and X has an R -first element.

Each forest is a union of pairwise disjoint trees. In what follows in this section, the maximal principles which hold for forests also hold for trees and conversely. We shall just state the results for forests.

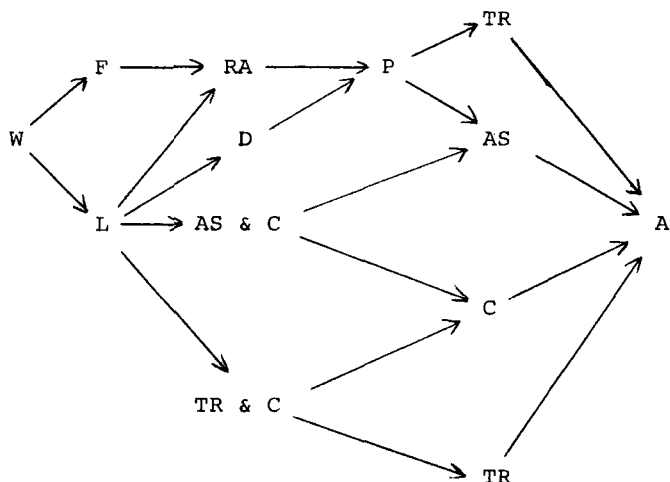
Thus, for example, $M1(P,L)$ is $M1$ and $M1(P,W)$ is $M2$.

If Q_1 and Q_2 are properties of relations we write $Q_1 \rightarrow Q_2$ if every relation with the property Q_1 also has the property Q_2 . Thus, for example, $L \rightarrow P$ because every linear ordering is a partial ordering.

The following facts hold:

- (1) If $U_1 \rightarrow U_2$ then $M1(Q,U_1) \rightarrow M1(Q,U_2)$ for all Q .
- (2) If $Q_1 \rightarrow Q_2$ then $M1(Q_2,U) \rightarrow M1(Q_1,U)$ for all U .

(3)



We shall show first that $M1(Q,U)$ is equivalent to the axiom of choice if

- (*) $Q = TR, P, RA$, or F and $U = C, AS \& C, TR \& C, L, D$, or W .
- (4) $M1(TR, W) \rightarrow M1(Q, U)$, for all Q and U in (*) above.
- (5) $M1(Q, U) \rightarrow M1(F, U)$, for all Q and U in (*) above.
- (6) The statements $M1(F, U)$, for all U in (*) are equivalent.

Thus, to prove that each pair of statements $M1(Q, U)$ with Q and U in (*) are equivalent, it is sufficient to prove $M1(F, C) \rightarrow M1(TR, W)$. (The proof is similar to one given by Feigner [1967].)

THEOREM 4.8: $M1(F, C) \rightarrow M1(TR, W)$.

PROOF: Let R be a transitive relation on x and suppose each R -well-ordered subset of x has an R -upper bound. Let

$$y = \{u : u \subseteq x \text{ and } R \text{ well-orders } u\}.$$

Define a relation S on y such that for $u, v \in y$, $u S v$ iff u is an R -initial segment of v . Then $\langle y, S \rangle$ is a

forest. Let z be a connected subset of y . Then Uz is an R -upper bound of z in y . Thus, $M1(F,C)$ implies that y has an R -maximal element, w . Since w is well-ordered by R , w has an R -upper bound. This R -upper bound is an R -maximal element of x .

Now we have proved the following result:

THEOREM 4.9: Each pair of statements in the following set are equivalent:

$$\{M1(Q,U): Q = TR, P, RA, \text{ or } F \text{ and } U = C, AS\&C, TR\&C, L, D \text{ or } W\}.$$

The statement $M1(D,W)$ is also equivalent to $M1(TR,W)$. Clearly, $M1(TR,W) \rightarrow M1(D,W)$. To prove the converse we shall show that $M1(D,W) \rightarrow M1(F,W)$. Then it follows from 4.9 that $M1(D,W) \leftrightarrow M1(TR,W)$.

THEOREM 4.10: $M1(D,W) \rightarrow M1(F,W)$.

PROOF: Let x be a forest under a relation R in which each well-ordered subset has an R -upper bound.

Case 1. Suppose x can be well-ordered by a relation S .

In this case we use S to construct a subset $y \subseteq x$ which is well-ordered by R such that an R -upper bound for y is an R -maximal element of x . The construction is by transfinite recursion as follows: For each ordinal α , define

$$f(\alpha) = \begin{cases} \text{The } S\text{-first element } v \in x \sim f''\alpha \text{ such} \\ \text{that } (\forall u \in f''\alpha) u R v, \\ a, \text{ if no such element exists, where } a \notin x. \end{cases}$$

Let α_0 be the smallest α such that $f(\alpha) = a$. Let $y = f''\alpha_0$. Just as in 2.9 we show $f|_{\alpha_0}$ is a bijection. Moreover, it is clear from the construction that $\langle \alpha_0, \varepsilon \rangle \cong \langle y, R \rangle$, so y is well-ordered by R . Also, by construction, an R -upper bound of y is an R -maximal element of x .

Case 2. x cannot be well-ordered.

Define

$$y = \{f: (\exists \alpha \in \text{On}) (\exists u \subseteq x) f: \alpha \approx u\}.$$

We define a relation S on y so that for $f, g \in y$, $f S g$ iff $\mathcal{R}(f) \subset \mathcal{R}(g) \vee f = g$. Since x cannot be well-ordered, S is directed on y . Moreover, if $z \subseteq y$ is well-ordered by S , then z has an S -upper bound constructed as follows: Let $u = \bigcup_{f \in z} \mathcal{R}(f)$. Well-order u by T defined by: if $a, b \in u$, $a \neq b$, and f_a and f_b are the S -first elements of z for which $a \in f_a$ and $b \in f_b$, $a T b$ if $f_a S f_b$, and either $f_a \neq f_b$ or $f_b^{-1}(a) \in f_b^{-1}(b)$. T well-orders u so there is a function g and an ordinal α so that $g: \alpha \approx u$. The function g is then an S -upper bound of z .

Thus, $\langle y, S \rangle$ satisfies the hypothesis of $M1(D, W)$ so y has an S -maximal element, f . It is easy to see that $\mathcal{R}(f) = x$, contradicting the fact that x cannot be well-ordered, q.e.d.

Thus, $M1(D, W)$ is equivalent to each of the statements given in 4.9. We shall show later, that each of these statements is equivalent to some form of the axiom of choice. (All of these statements also have equivalent primed forms. See p 36.)

Next, we consider variations of $M5$. Wallace [1944] and Gottschalk [1952] had forms similar to $M5(A, AS)$ and $M5(A, C)$. Hőft & Howard [1973] have a form similar to $M5(C, RA)$. The remaining forms are due to Harper & Rubin [1976].

$M5(Q, U)$: Every set which is Q -ordered by R contains a $\subseteq$ -maximal subset which is U -ordered by R .

For example, $M5(P, L)$ is $M5$.

THEOREM 4.11: Every pair of the following statements are equivalent:

$M5(A, U)$, where $U = AS, C, AS \& C, TR \& C, P, L, D$, or RA .

$M5(AS, U)$, where $U = C, AS \& C, TR \& C, L, D$, or RA .

$M5(TR, U)$, where $U = AS, C, AS \& C, TR \& C, P, L, D$, or RA .

$M \ 5(C,U)$, where $U = AS, AS\&C, P, L, D$, or RA .

$M \ 5(P,U)$, where $U = C, AS\&C, TR\&C, L, D$, or RA .

$M \ 5(D,U)$, where $U = C, AS\&C, TR\&C, L$, or RA .

$M \ 5(RA,U)$, where $U = C, AS\&C, TR\&C, L$, or D .

$M \ 5(F,U)$, where $U = C, AS\&C, TR\&C, L, D$, or W .

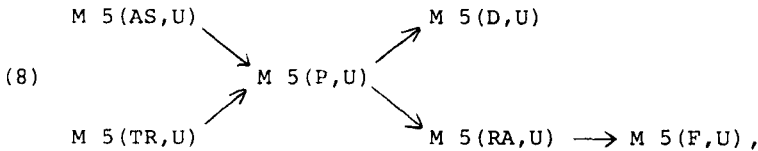
PROOF: By using a proof similar to the proof of 4.3, it is easy to show that $M \ 3$ implies each of the above 50 statements.

The following facts are easily verified:

- (1) $M \ 5(AS,C) \leftrightarrow M \ 5(AS,AS\&C), M \ 5(AS,TR) \leftrightarrow M \ 5(AS,P),$
 $M \ 5(AS,TR\&C) \leftrightarrow M \ 5(AS,L).$
- (2) $M \ 5(TR,C) \leftrightarrow M \ 5(TR,TR\&C), M \ 5(TR,AS) \leftrightarrow M \ 5(TR,P),$
 $M \ 5(TR,AS\&C) \leftrightarrow M \ 5(TR,L).$
- (3) $M \ 5(C,AS) \leftrightarrow M \ 5(C,AS\&C), M \ 5(C,TR) \leftrightarrow M \ 5(C,TR\&C),$
 $M \ 5(C,P) \leftrightarrow M \ 5(C,D) \leftrightarrow M \ 5(C,L) \leftrightarrow M \ 5(C,RA).$
- (4) $M \ 5(Q,C) \leftrightarrow M \ 5(Q,AS\&C) \leftrightarrow M \ 5(Q,TR\&C) \leftrightarrow M \ 5(Q,L),$
 if $Q = P, D$ or RA .
- (5) $M \ 5(RA,D) \leftrightarrow M \ 5(RA,L), M \ 5(D,RA) \leftrightarrow M \ 5(D,L).$
- (6) $M \ 5(F,C) \leftrightarrow M \ 5(F,AS\&C) \leftrightarrow M \ 5(F,TR\&C) \leftrightarrow M \ 5(F,L) \leftrightarrow$
 $M \ 5(F,D) \leftrightarrow M \ 5(F,W).$

Also, using the fact that if $Q_1 \rightarrow Q_2$ then $M \ 5(Q_2,U) \rightarrow M \ 5(Q_1,U)$ we obtain the following implications:

- (7) $M \ 5(A,U) \rightarrow M \ 5(Q,U)$, for all Q and U .



for all U .

Also a relation R is connected on x iff $\bar{R} = (x \times x) \sim R$ is antisymmetric on x , and $R \cup I \upharpoonright x$ is a linear ordering on x iff $(x \times x) \sim (R \sim I)$ is a linear ordering on x . Using these facts it is easy to prove (9) and (10) below.

$$(9) \quad M5(AS, C) \leftrightarrow M5(C, AS), \quad M5(AS, AS \& C) \leftrightarrow M5(C, AS \& C), \\ M5(AS, L) \leftrightarrow M5(C, L).$$

$$(10) \quad M5(A, AS) \leftrightarrow M5(A, C).$$

It follows from the preceding results that to complete the proof of the theorem it is sufficient to prove that

- (a) $M5(D, L) \rightarrow M5$, (b) $M5(F, L) \rightarrow M2$, and
 (c) $M5(TR, AS) \rightarrow AC2$. (We shall show later that the axiom of choice implies a maximal principle.)

$$(a) \quad M5(D, L) \rightarrow M5.$$

PROOF: Suppose x is a set which is partially ordered by R . Define

$$y = \{u \subseteq x: u \text{ is linearly ordered by } R\}.$$

Define a relation S on y so that if $u, v \in y$, $u S v$ iff $v \subseteq u$. Then S is a directed relation on y . ($\emptyset$ is an S -upper bound for every subset of y .) Let z be a $\subseteq$ -maximal S -linearly ordered subset of y . Then it is easy to see that $\cup z$ is a $\subseteq$ -maximal R -linearly ordered subset of x .

$$(b) \quad M5(F, L) \rightarrow M2.$$

PROOF: Suppose x is partially ordered by R and each R -well-ordered subset has an R -upper bound. Let

$$y = \{u \subseteq x: u \text{ is well-ordered by } R\}.$$

Define S on y so that for $u, v \in y$, $u S v$ iff u is an R -initial segment of v . Then y is a forest under S and the union of a $\subseteq$ -maximal S -linearly ordered subset of y is an R -well-ordered subset of x with an R -maximal element.

$$(c) \quad M5(TR, AS) \rightarrow AC2.$$

PROOF: Let x be a set of non-empty pairwise disjoint sets. Define a relation R on $\cup x$ as follows: For all $a, b \in \cup x$, where $a \in u \in x$ and $b \in v \in x$, $a R b$ iff $u = v$. Then R is transitive and a $\subseteq$ -maximal R -antisymmetric subset of $\cup x$ is a choice set for x , q.e.d.

All of these maximal principles have primed forms:

$M'5(Q,U)$: In every Q -ordered set, every U -ordered subset can be extended to a $\subseteq$ -maximal U -ordered set.

For all of the above choices for U , except $U = W$, $M'6 \rightarrow M'5(Q,U)$ for all Q . Also, $M'5(Q,U) \rightarrow M5(Q,U)$. Thus, Theorem 4.11 holds for the primed forms. In Theorem 4.12, below, Paul Howard sent us the proof of (a). In part (b), we prove that a primed form of $M5(Q,U)$ implies the axiom of choice, while it is unknown whether the corresponding unprimed form does.

THEOREM 4.12:

(a) $M5(A,TR) \rightarrow AC2$.

(b) $M'5(A,TR + \text{Reflexive}) \rightarrow AC2$.

PROOF: (a) Let x be a set of pairwise disjoint non-empty sets. Define R on $\cup x$ as follows:

$$a, b \in \cup x \rightarrow (a R b \leftrightarrow (\exists u \in x)(a, b \in u \wedge a \neq b)).$$

$M5(A,TR)$ implies that x has a $\subseteq$ -maximal subset y such that R is transitive on y . For each $u \in x$, $u \cap y$ is a singleton because if $a, b \in u \cap y$, we would have $a R b$ and $b R a$ but not $a R a$. It follows from the maximality of y that $u \cap y \neq \emptyset$.

(b) Let x be a set of pairwise disjoint non-empty sets. Suppose x has at least two elements. Define a relation R on $\cup x$ such that if $a, b \in \cup x$, where $a \in u \in x$, $b \in v \in x$ then

$$a R b \text{ iff } u \neq v \vee a = b.$$

Let $u, v \in x$, $u \neq v$, and $a \in u$, $b \in v$. Then $\{a, b\}$ is an R -transitive subset of $\cup x$. Any $\subseteq$ -maximal R -transitive

subset of Ux which contains $\{a,b\}$ is a choice set for x .

A result due to Gottschalk [1952] can also be written in the form $M5(Q,U)$ if we add to our definition of types of relations, UN, universal: a relation R is universal on x if $x \times x \subseteq R$. Gottschalk's form was $M5(A,UN)$. It is clear that $M7 \rightarrow M5(A,UN)$ because the property of being universal is a property of finite character.

THEOREM 4.13: $M5(A,UN) \rightarrow M5(A,C)$.

PROOF: If R is any relation then $R \cup R^{-1} \cup I$ is connected. An $R \cup R^{-1} \cup I$ -maximal universal set is an R -maximal connected set.

The form $M5(A,UN)$ can be generalized to n -ary relations.

$M5(A,UN(n))$: For any set x and n -ary relation R there is a $\subseteq$ -maximal subset $y \subseteq x$ such that $y^n \subseteq R$.

$M5(A,UN(n))$ is equivalent to the axiom of choice if $n \geq 2$ because $M7 \rightarrow M5(A,UN(n)) \rightarrow M5(A,UN(n-1))$ and since $M5(A,UN(2))$ is $M5(A,UN)$, by 4.13, $M5(A,UN) \rightarrow M5(A,C)$.

It is also clear that the following two variations of $M5(A,UN(n))$ are equivalent to the axiom of choice.

- (1) There is a natural number $n \geq 2$ such that $M5(A,UN(n))$.
- (2) For every set x there is a natural number $n \geq 2$ such that for each n -ary relation R there is a $\subseteq$ -maximal subset $y \subseteq x$ such that $y^n \subseteq R$.

$M5(A,UN(n)) \rightarrow (1) \rightarrow (2) \rightarrow M5(A,UN)$.

(We shall say more about the relation UN and its complement in section 5.)

We shall next show that these maximal principles are equivalent to the other forms of the axiom of choice we have given. Since there is so much in the literature about the equivalence of a maximal principle and the axiom of choice or the well-ordering theorem we shall give several of these proofs. The first two are well known.

THEOREM 4.14: $M_3 \rightarrow AC_1$.

PROOF: Let S be a collection of non-empty sets, and let C be the collection of all choice functions on subsets of S . It is clear that every nest of functions in C has its union in C , and therefore, by M_3 , C has a maximal element f . Clearly, f is the required choice function.

A similar proof could be given that $M_3 \rightarrow WO_1$. Instead we shall prove

THEOREM 4.15: $M_1 \rightarrow WO_1$.

PROOF: Let x be an arbitrary set, and let P be the set of all ordered pairs $\langle y, w \rangle$ such that $y \subseteq x$, $w \subseteq y \times y$ and w is a well-ordering on y . Let us define a relation R as follows: If $\langle y, w \rangle$ and $\langle y', w' \rangle$ are elements of P , then $\langle y, w \rangle R \langle y', w' \rangle$ if $y \subseteq y'$, $w = w' \cap (y \times y)$, and $y \times (y' \sim y) \subseteq w'$. That is, w' is an extension of w , $w = w' \upharpoonright y$, and y is an initial w' -segment of y' . If s is an R -linearly ordered subset of P , then an R -upper bound for s is $\langle \bigcup \mathcal{D}(s), \bigcup \mathcal{R}(s) \rangle$. (We leave the proof that $\langle \bigcup \mathcal{D}(s), \bigcup \mathcal{R}(s) \rangle \in P$ and is an R -upper bound for s , as an exercise.) Therefore, by M_1 , P has an R -maximal element $\langle z, v \rangle$. Suppose $m \in x \sim z$. We define $z' = z \cup \{m\}$ and $v' = v \cup (z' \times \{m\})$, and we observe that $\langle z, v \rangle R \langle z', v' \rangle$ and not $\langle z', v' \rangle R \langle z, v \rangle$. This contradicts the maximality of $\langle z, v \rangle$, so $z = x$ and hence v well-orders x , q.e.d.

We now proceed to the slightly more difficult task of proving that the axiom of choice implies a maximal principle. We shall first prove that the well-ordering theorem implies a maximal principle and then give several proofs that the axiom of choice implies a maximal principle. The proof of the next theorem was given in Hausdorff [1914].

THEOREM 4.16: $WO_2 \rightarrow M_5$.

The intuitive idea of the proof is as follows: Let x be a set and let R be a partial ordering relation on x . By WO_2 , there is an ordinal number α such that α is equipollent to x . We construct a maximal R -linearly ordered

subset of x by the following method: Let the first element of x belong. If the second element of x stands in the relation R to the first element, the second element is included, otherwise not. We continue by including those elements which are R -connected to the elements already included.

Formally, let F be a 1-1 function from α onto x . Define the function G as follows:

$$G(\beta) = F(\beta) \text{ if } \beta < \alpha \text{ and } (\forall \gamma)[\gamma < \beta \rightarrow (F(\beta) R G(\gamma) \text{ or } G(\gamma) R F(\beta))], \quad G(\beta) = F(0) \text{ otherwise.}$$

It is clear that the range of G is a maximal R -linearly ordered subset of x .

The proof of Theorem 4.17 is due to Kuratowski [1922]. Similar proofs were given by Hausdorff [1927], Bourbaki [1950], Kneser [1950] and Szele [1950].

THEOREM 4.17: $AC_3 \rightarrow M^4$.

Let us first prove the following lemma, which is independent of the axiom of choice:

LEMMA: Let $a \in x$ and suppose that every non-empty well-ordered nest of elements of x has its union in x . Furthermore, let f be a function on x to x , such that for all $y \in x$, $y \subseteq f(y)$. Then there is a $z \in x$ such that $a \subseteq z$ and $z = f(z)$.

To prove this let f be a function on x to x such that for all $y \in x$, $y \subseteq f(y)$. Define a function G by transfinite induction as follows:

$$G(0) = a,$$

$$G(\alpha + 1) = f(G(\alpha)),$$

$$G(\alpha) = \bigcup_{\beta < \alpha} G(\beta), \text{ if } \alpha \text{ is a limit ordinal.}$$

We observe that if $\alpha < \beta$, $G(\alpha) \subseteq G(\beta)$. Since every non-empty well-ordered nest of elements of x has its union in x , G is a function from On into x . Since x is a set, the function G cannot be 1-1. Hence for some ordinals

$\alpha < \beta$, $G(\alpha) = G(\beta)$. Therefore, $G(\alpha) = G(\alpha + 1)$, and thus $G(\alpha) = f(G(\alpha))$. Since $G(0) = a$, $a \subseteq G(\alpha)$ and this completes the proof of the lemma.

Now let us use this lemma to prove the theorem. Let x be a non-empty set with the property that every well-ordered non-empty nest of elements of x has its union in x , and let $a \in x$. Define a function h on x as follows: For all $y \in x$, $h(y) = \{z: z \in x \text{ and } y \subset z\}$. By AC 3, there exists a function f such that $\mathcal{J}(f) = \mathcal{J}(h)$ and for all $y \in x$, if $h(y) \neq \emptyset$ then $f(y) \in h(y)$. If $h(y) = \emptyset$ define $f(y) = y$. Now, f satisfies the hypothesis of the lemma - f is a function from x to x such that for all $y \in x$, $y \subseteq f(y)$. Therefore, there is a $z \in x$ such that $a \subseteq z$ and $f(z) = z$. Such an element z is a maximal element of x .

The proof of Theorem 4.18 is due to Frink [1952] and is similar to the proof given by Zermelo [1908].

THEOREM 4.18: AC 3 $\rightarrow$ M'5.

PROOF: Let x be an arbitrary set, R a partial ordering relation on x , and y an R -linearly ordered subset of x . Let $L = \{z: z \subseteq x \text{ and } R \text{ is a linear ordering on } z \text{ and } y \subseteq z\}$. Suppose M'5 does not hold. Then there is no maximal element of L . By AC 3, there is a function f on L such that for each $z \in L$, $f(z) \in x \sim z$ and $z \cup \{f(z)\} \in L$. (Formally we obtain this function f as follows: For each $z \in L$ define $h_1(z) = \{w: w \in L \text{ and } z \subset w\}$. Because we are assuming that L has no maximal element, $h_1(z) \neq \emptyset$. AC 3 implies there exists a function h_2 such that $\mathcal{J}(h_1) = \mathcal{J}(h_2)$ and $h_2(z) \in h_1(z)$ for all $z \in L$. Now define $h_3(z) = h_2(z) \sim z$. Since $h_3(z) \neq \emptyset$ for all $z \in L$, again by AC 3, there exists a function f such that $\mathcal{J}(f) = \mathcal{J}(h_3) = L$ and for all $z \in L$, $f(z) \in h_3(z)$.) Let us denote $z \cup \{f(z)\}$ by z' , which we shall call the successor of z .

Let $K \subseteq L$. We shall call K complete if

- (a) $y \in K$,
- (b) $z' \in K$ whenever $z \in K$,

(c) if N is a non-empty nest and $N \subseteq K$, then $UN \in K$.

Let $\mathcal{K}$ be the set of all complete sets. Since $L \in \mathcal{K}$, $\mathcal{K}$ is non-empty. Since the intersection of any non-empty set of complete sets is complete, $J = \bigcap \mathcal{K}$ is the smallest complete set. Let us now demonstrate that J is a nest.

If $z \in J$, z is normal if for every $u \in J$, $z \subseteq u$ or $u \subseteq z$. We shall prove that every element of J is normal, thereby proving that J is a nest.

Let z be a normal element of J . (J has a normal element since $y \in J$ is normal.) Define $K(z)$ to be the set of all $u \in J$ such that $u \subseteq z$ or $z' \subseteq u$. Clearly $y \in K(z)$ since $y \subseteq z$. Suppose $u \in K(z)$. If $z' \subseteq u$, then $z' \subseteq u'$ and $u' \in K(z)$. If $u \subseteq z$ and $u' \not\subseteq z$ then $z \subseteq u'$ since z is normal. But we have $u \subseteq z \subseteq u'$, which implies $u = z$ or $u' = z$. The latter equality is impossible since we have assumed $u' \not\subseteq z$. Consequently $u = z$, so $z' \subseteq u'$ and $u' \in K(z)$. It is also clear that if $N \subseteq K(z)$ is a nest then $UN \in K(z)$, because if there is an $n \in N$ such that $z' \subseteq n$ then $z' \subseteq UN$, otherwise $UN \subseteq z$. Therefore, $K(z)$ is a complete subset of J . Since J is the smallest complete set it follows that $K(z) = J$.

Now we are able to show that the set of normal elements form a complete set. First, y is normal. Next, suppose z is normal, then $J = K(z)$. Therefore, for all $u \in J$ either $u \subseteq z$, which implies $u \subseteq z'$, or $z' \subseteq u$. Consequently z' is normal. Finally, suppose N is a nest of normal elements, (every subset of normal elements is a nest). Suppose $u \in J$. If every element of N is a subset of u then $UN \subseteq u$. If there is an $n \in N$ such that $u \subseteq n$ then $u \subseteq UN$. Hence, UN is normal. Therefore, the set of normal elements of J forms a complete set which implies all elements of J are normal and thus J is a nest.

Since J is a nest and a complete class, by (c) $UJ \in J$, so that by (b) $(UJ)' \in J$, which implies $(UJ)' \subseteq UJ$, which is a contradiction, q.e.d.

The proof of the following theorem is due to Weston [1957].

THEOREM 4.19: $AC\ 1 \rightarrow M\ 1$.

PROOF: Let x be an arbitrary non-empty set and R a partial ordering relation on x . For each R -linearly ordered subset y of x , let $\hat{y}$ be the set of all R -upper bounds of y in $x \sim y$. Let f be the choice function on the set of all non-empty sets of the type $\hat{y}$. If y is an R -linearly ordered subset of x , y is said to be an f -chain if $y \neq \emptyset$ and if for all $z \subset y$ with $\hat{z} \cap y \neq \emptyset$, $f(\hat{z})$ is an R -minimal element of $\hat{z} \cap y$. (At least one f -chain exists since $\{f(\hat{\emptyset})\}$ is an f -chain.)

The following properties hold for f -chains:

(a) If y is an f -chain, $z \subseteq y$, and $\hat{z} \cap y = \emptyset$ then $\hat{z} = \hat{y}$. (In fact all linearly ordered sets satisfy this condition.)

(b) If y is an f -chain and $\hat{y} \neq \emptyset$ then $y' = y \cup \{f(\hat{y})\}$ is also an f -chain.

To see this observe that if $z \subset y'$ and $\hat{z} \cap y' \neq \emptyset$ then either

- (i) $z \subset y$ and $\hat{z} \cap y \neq \emptyset$
- or (ii) $z \subseteq y$ and $\hat{z} \cap y = \emptyset$
- or (iii) $z \not\subseteq y$.

Since y is an f -chain (i) implies that $f(\hat{z})$ is an R -minimal element of $\hat{z} \cap y'$. If (ii) holds then $\hat{z} \cap y' = \{f(\hat{y})\}$, but (a) implies $\hat{y} = \hat{z}$. Therefore, $f(\hat{z})$ is an R -minimal element of $\hat{z} \cap y'$. And (iii) is impossible because if $z \not\subseteq y$ and $z \subset y'$, then $f(\hat{y}) \in z$ which implies $\hat{z} \cap y' = \emptyset$. Since (i), (ii), and (iii) are the only possibilities it follows that y' is an f -chain.

(c) For any two f -chains, one is an initial R -segment of the other.

Let y and z be f -chains and suppose $t \in z \sim y$. Let $u = \{s : s \in z \cap y \text{ and } s R t\}$.

We shall show that $u = y$. Clearly $u \subseteq y$. Let us assume

$$(i) \quad u \subset y.$$

It is also clear that $u \subseteq z$. Since $t \in z$ and $t \notin u$, we have

$$(ii) \quad u \subset z.$$

Also, $t \in \hat{u} \cap z$, and therefore

$$(iii) \quad \hat{u} \cap z \neq \emptyset.$$

Since z is an f -chain, it follows from (ii) and (iii) that $f(\hat{u})$ is an R -minimal element of $\hat{u} \cap z$. In particular, since $t \in \hat{u} \cap z$,

$$(iv) \quad f(\hat{u}) R t.$$

Now, if $f(\hat{u}) \in y$ then since $f(\hat{u}) \in z$, by (iv) we would have $f(\hat{u}) \in u$, which is impossible. Therefore,

$$(v) \quad f(\hat{u}) \notin y.$$

If $\hat{u} \cap y \neq \emptyset$, since y is an f -chain, (i) implies $f(\hat{u}) \in \hat{u} \cap y \subseteq y$, which contradicts (v). Therefore,

$$(vi) \quad \hat{u} \cap y = \emptyset.$$

By (i) and (vi) it follows from (a) that

$$(vii) \quad \hat{u} = \hat{y}.$$

Now, suppose $r \in y \sim u$, then it follows from (vii) that there is an $s \in u$ such that $r R s$ (if not $r \in \hat{u}$ but $r \notin \hat{y}$, contradicting (vii)). Since $s \in u$, $s R t$, so that the transitivity of R implies $r R t$. Now, we have $r \in y$ and $r R t$, so that since $r \notin u$ we must have $r \in y \sim z$. Now, if we define a set

$$v = \{s: s \in y \cap z \text{ and } s R r\},$$

then by an argument analogous to the argument used to derive (vi), we obtain

$$(viii) \quad \hat{v} \cap z = \emptyset.$$

But since $r R t$ and $t \in z$, $t \in \hat{v} \cap z$ which contradicts

(viii). Therefore, $u = y$ and y is an initial R-segment of z .

Since the set of all f-chains is a nest, its union is linearly ordered by R . Let us next show that the union is an f-chain. Let $c = \cup\{y: y \text{ is an f-chain}\}$. Suppose $y \subset c$ and $\hat{y} \cap c \neq \emptyset$. Then there is f-chain z such that $\hat{y} \cap z \neq \emptyset$ and an f-chain w such that $y \subseteq w$. We wish to show first that $y \subseteq z$. Since z and w are both f-chains it follows from (c) that either $w \subseteq z$ or z is an initial R-segment of w . In the former case, $y \subseteq z$. In the latter case, if $y \not\subseteq z$ then there exists an element $s \in y$ such that $s \in w \sim z$ and $s \in \hat{z}$. This implies that for every $t \in \hat{y}$, $s R t$, which implies $\hat{y} \cap z = \emptyset$, which contradicts the choice of z . Hence, $y \subseteq z$. ($y \neq z$ since $\hat{y} \cap z \neq \emptyset$.)

Now, we see that there exists an f-chain z such that $y \subset z$ and $\hat{y} \cap z \neq \emptyset$, therefore $f(\hat{y})$ is an R-minimal element of $\hat{y} \cap z \subseteq \hat{y} \cap c$. It remains to be shown that $f(\hat{y})$ is an R-minimal element of $\hat{y} \cap c$. Suppose not, then there is a $t \in \hat{y} \cap c$ such that $t \notin z$ and $t R f(\hat{y})$. Also, there is an f-chain w such that $t \in w$. Since z and w are both f-chains and since $t \in w \sim z$, it follows from (c) that z is an initial R-segment of w , so that for every $s \in z$, $s R t$. This implies, in particular, $f(\hat{y}) R t$. Consequently, since R is anti-symmetric, $t = f(\hat{y})$, which is a contradiction. Hence, $f(\hat{y})$ is an R-minimal element of $\hat{y} \cap c$ and c is an f-chain. Moreover, $\hat{c} = \emptyset$ because, if $\hat{c} \neq \emptyset$, then we could construct $c' = c \cup \{f(\hat{c})\}$ which is also an f-chain by (b), and $c \subset c'$ which contradicts the choice of c .

Since c is linearly ordered by R the hypothesis of M 1 implies that c has an R-maximal element m . Clearly m is an R-maximal element of x , because if there exists an $n \neq m$ such that $m R n$ then $n \in \hat{c}$, which is a contradiction, q.e.d

Of the forms we have already given of the maximal principle, only two do not involve a relation or a property and state that there is a maximal element or subset with

respect to a natural relation among sets (M 3 and M 6). We shall concern ourselves only with the maximal subset form M 6. The question can then be raised as to whether the corresponding forms with other natural relations among sets are of equal strength. Let x and y be two arbitrary sets. These sets are disjoint unions of the appropriate two of $x \cap y$, $x \sim y$, $y \sim x$. ($x = (x \sim y) \cup (x \cap y)$ and $y = (y \sim x) \cup (x \cap y)$). The immediate symmetric set-theoretical relations derivable from these are whether $x \cap y$ is the empty set, and how many of $x \sim y$ and $y \sim x$ are the empty set. If x and y are distinct and non-empty, it can readily be seen that exactly one of the relations $x \cap y = \emptyset$, one of $x \sim y$ and $y \sim x$ is empty ($x \subseteq y$ or $y \subseteq x$), and none of the three sets are empty ($x \not\subseteq y$ and $y \not\subseteq x$ and $x \cap y \neq \emptyset$), must hold. Accordingly we introduce the following notation:

DEFINITION 4.20: Let X and Y be any two classes.

Then

- (i) $X D Y$ if $X \cap Y = \emptyset$
- (ii) $X \bar{D} Y$ if $X \cap Y \neq \emptyset$
- (iii) $X K Y$ if $X \subseteq Y \vee Y \subseteq X$
- (iv) $X \bar{K} Y$ if $X \not\subseteq Y \wedge Y \not\subseteq X$
- (v) $X J Y$ if $X \not\subseteq Y \wedge Y \not\subseteq X \wedge X \cap Y \neq \emptyset \wedge X \cup Y \neq V$
- (vi) $X \bar{J} Y$ if $X \subseteq Y \vee Y \subseteq X \vee X \cap Y = \emptyset \vee X \cup Y = V$.

Note that the properties $\bar{D}$, $\bar{K}$, and $\bar{J}$ are the complements of the properties D , K , and J respectively, and as we have previously observed, exactly one of D , K , and J must hold if the classes are non-empty. Therefore, $\bar{D} = K$ or J , $\bar{J} = D$ or K , and $\bar{K} = D$ or J .

We shall also use the following consistent ambiguity: We shall say that a class has a given one of the above properties if every distinct pair of its elements has the property. If a class X has the property D , we shall write $D[X]$ and similarly for the others.

Let us now introduce the following notation:

$M_8(A)$: Every set has a $\subseteq$ -maximal subset with the property A .

G. Kurepa [1952] has shown that $M_8(A)$ is equivalent to the axiom of choice if A is $\bar{D}$, J , or $\bar{J}$. R. Vaught [1952] proved that $M_8(D)$ is equivalent to the axiom of choice. $M_8(K)$ is M_6 . J. D. Halpern [1961] in his Ph.D. thesis constructed a Fraenkel-Mostowski model in which $M_8(\bar{K})$ is true but the axiom of choice is false. (In section 5 we shall show $NBG \vdash M_8(\bar{K}) \leftrightarrow AC$. The result is due to Felgner and Jech [1973].) On the other hand Kurepa [1952] has shown that the following statement is equivalent to the axiom of choice:

$M_9(\bar{K})$: Every set can be linearly ordered, and every set has a $\subseteq$ -maximal subset with the property $\bar{K}$.

Clearly M_7 implies each of the forms $M_8(A)$, $A = D, \bar{D}, K, J, \bar{J}$, and $M_9(\bar{K})$, since the properties considered are all properties of finite character. (While "can be linearly ordered" is not a property of finite character, M_7 implies every set can be well-ordered which, in turn, implies every set can be linearly ordered.)

THEOREM 4.21: $M_8(\bar{D}) \rightarrow M_8(D)$.

PROOF: Let x be an arbitrary set. For each element $s \in x$ we define a set N_s as follows:

$$N_s = \{u: u = \{s\} \text{ or } (\exists t)[t \in x, u = \{s, t\}, \text{ and } s \bar{D} t]\}.$$

Now if $s \neq t$, then $N_s \bar{D} N_t$ if and only if $s \bar{D} t$. $M_8(\bar{D})$ implies that $\{N_s: s \in x\}$ has a maximal subset w which has the property $\bar{D}$. We also have that $N_s = N_t$ if and only if $s = t$; therefore, $\{s: N_s \in w\}$ is well-defined and is clearly a maximal subset of x which has the property D . (The proof shows that $\bar{D}$ is a canonical symmetric relation. The same is true for D .)

THEOREM 4.22: $M_8(J) \rightarrow M_8(\bar{D})$.

PROOF: Let x be an arbitrary set. For each $s \in x$, define $s_u = s \cup \{\langle s, u \rangle\}$, where u is a set such that $(\forall t)(\forall s)[(s \in x \text{ and } t \in x) \rightarrow \langle s, u \rangle \notin t]$. Therefore, if $s \neq t$

then $s_u \bar{K} t_u$, which implies $s_u J t_u \leftrightarrow s_u \bar{D} t_u$. If $s \neq t$, $s_u \cap t_u = s \cap t$, so that $s_u \bar{D} t_u \leftrightarrow s \bar{D} t$. $M 8(J)$ implies that $\{s_u: s \in x\}$ has a maximal subset which has the property $\bar{D}$.

THEOREM 4.23: $M 8(D) \rightarrow AC 2$.

PROOF: Let s be a collection of non-empty disjoint sets. Let $x = \{\langle\{0,v\}, \{1,w\}\rangle: w \in x \text{ and } v \in w\}$. The elements $\langle\{0,v\}, \{1,w\}\rangle$ and $\langle\{0,v'\}, \{1,w'\}\rangle$ of x are disjoint if and only if $w \neq w'$. By $M 8(D)$, there exists a maximal subset y of x which has the property D . Then the choice set for the collection s can be taken to be $\{v: (\exists w) \langle\{0,v\}, \{1,w\}\rangle \in y\}$, q.e.d.

THEOREM 4.24: $M 8(\bar{J}) \rightarrow M 8(K)$.

PROOF: Let x be an arbitrary set and let $u \notin Ux$. For each $s \in x$, define $s_u = s \cup \{u\}$. Then for all $s, t \in x$, $s_u \cap t_u \neq \emptyset$, so that $s_u \bar{J} t_u \leftrightarrow s_u K t_u$. Also $s_u K t_u \leftrightarrow s K t$ since $u \notin Ux$. Therefore, if $\{s_u: s \in x\}$ has a maximal subset which has the property $\bar{J}$ then x has a maximal subset which has the property K , q.e.d.

THEOREM 4.25: $M 9(\bar{K}) \rightarrow AC 2$.

PROOF: Let s be a family of non-empty disjoint sets. Let R be a linear ordering of Us . If $x \in u \in s$, define

$$x_R = \{z: z \in u \text{ and } z R x\}.$$

Since R is antisymmetric, if $x \neq y$ then $x_R \neq y_R$ for all $x, y \in Us$. Also

(a) $x_R \bar{K} y_R$ if and only if x and y belong to different elements of s , because if x and y belong to different elements of s then $x_R \cap y_R = \emptyset$ which implies $x_R \bar{K} y_R$. Conversely, suppose x and y belong to the same element of s . Since R is connected on Us , either $x R y$ or $y R x$. But $x R y$ implies $x \in y_R$ which implies $x_R \subseteq y_R$, and $y R x$ implies $y \in x_R$ which implies $y_R \subseteq x_R$. Therefore, if x and y belong to the same element of s then $x_R K y_R$.

Hence, it follows from (a) that a maximal subset of $\{x_R: x \in U_s\}$ which has the property $\bar{K}$ yields a choice set for s , q.e.d.

We have now shown that the following implications hold:

$$M_7 \rightarrow M_8(J) \rightarrow M_8(\bar{D}) \rightarrow M_8(D) \rightarrow AC_2,$$

$$M_7 \rightarrow M_8(\bar{J}) \rightarrow M_8(K) = M_6, M_7 \rightarrow M_9(\bar{K}) \rightarrow AC_2.$$

We have shown previously that $M_6 \rightarrow M_7$ and $AC_2 \rightarrow M_7$.

It has been shown by C. C. Chang [1960] and Azriel Levy [1962] that $M_8(D)$ can be generalized by extending the property D to D_m , and, in an analogous way, we were able to generalize $M_8(\bar{D})$. (Chang actually obtained a stronger form, $M'_8(D_m)$ but Levy by using $AC_{13}(m)$, was able to weaken it by eliminating the prime. We give here the weaker form, $M_8(D_m)$.)

DEFINITION 4.26: If m is a natural number, $m \geq 2$, a class X is said to have the property D_m ($\bar{D}_m$) if every subset of X which has m distinct elements has an empty (non-empty) intersection. ($D_2 = D$ and $\bar{D}_2 = \bar{D}$.)

Now we claim that $M_8(D_m)$ and $M_8(\bar{D}_m)$ are equivalent to the axiom of choice. Clearly $M_7 \rightarrow M_8(\bar{D}_m)$ and the proof that $M_8(\bar{D}_m) \rightarrow M_8(D_m)$ is similar to the proof that $M_8(\bar{D}) \rightarrow M_8(D)$, 4.21.

THEOREM 4.27: $M_8(D_m) \rightarrow AC_{13}(m-1)$.

PROOF: Let x be a set of non-empty sets. We can assume without loss of generality that the elements of x are disjoint, since otherwise we can replace each element y of x by $\{\langle y, t \rangle: t \in y\}$.

Let

$$u = \{\{\langle 0, t \rangle, \langle 1, y \rangle\}: t \in y \text{ and } y \in x\}.$$

Let v be a maximal subset of u which has the property D_m . For each $y \in x$, define

$$f(y) = \{t: \{\langle 0, t \rangle, \langle 1, y \rangle\} \in v\}.$$

Then $f(y) \subseteq y$ since $v \subseteq u$; $f(y) \leq m-1$ since v has the

property D_m ; and $f(y) \neq \emptyset$ since $y \neq \emptyset$ and v is maximal, q.e.d. (See 4.23).

Just as in the case of Levy's forms of the well-ordering theorem and the axiom of choice (WO 4(m) and AC 7(m)) we are able to obtain weaker forms of $M 8(D_m)$.

$M 10$: There exists a natural number $m \geq 2$ such that $M 8(D_m)$.

$M 11$: For every set x there exists a natural number $m \geq 2$ such that x has a $\subseteq$ -maximal subset which has the property D_m .

Clearly, we have $M 8(D_m) \rightarrow M 10 \rightarrow M 11$. Also, $M 11 \rightarrow AC 15$; the proof is analogous to the proof of 4.27.

There is some difficulty involved in trying to obtain similar results for $\bar{D}_m$. Because of the definition of $\bar{D}_m$, (definition 4.26) the statement:

S : For every set x , there exists a natural number $m \geq 2$, such that x has a $\subseteq$ -maximal subset which has the property $\bar{D}_m$, holds independently of the axiom of choice. To see this let x be an arbitrary set.

Case 1. Every finite subset of x has a non-empty intersection. Then x itself is the required maximal set for every $m \geq 2$.

Case 2. There exists a finite subset y of x with n distinct elements which has an empty intersection. Then y is a maximal subset of x which has the property $\bar{D}_{n+1}$, so that $m = n + 1$.

To eliminate this rather trivial case we modify the definition of $\bar{D}_m$ slightly as follows:

DEFINITION 4.28: If m is a natural number, $m \geq 2$, a class X is said to have the property $\bar{D}_m^!$ if every subset of X which does not have more than m distinct elements has a non-empty intersection.

Therefore, if a class has the property $\bar{D}!_m$ every subset of m elements has a non-empty intersection, every subset of $m - 1$ elements has a non-empty intersection, etc., so that if $\bar{D}_m$ is replaced by $\bar{D}!_m$ in S , the proof of case 2 is not valid. In fact, we are able to prove that the following propositions are equivalent to the axiom of choice.

M 12: There exists a natural number $m \geq 2$ such that $M 8(\bar{D}!_m)$.

(It is rather easy to show that M 12 with $\bar{D}!_m$ replaced by $\bar{D}_m$ is also equivalent to the axiom of choice. The proof that it implies M 10 is analogous to the proof of 4.20.)

M 13: For every set x , there exists a natural number $m \geq 2$ such that x has a $\subseteq$ -maximal subset which has the property $\bar{D}!_m$.

It is clear that $M 7 \rightarrow M 12 \rightarrow M 13$. We shall prove $M 13 \rightarrow M 11$. The proof is a generalization of the proof of 4.21.

THEOREM 4.29: $M 13 \rightarrow M 11$.

PROOF: Let us first observe that if every element of x is disjoint from every element of y , then a subset w of $x \cup y$ has the property D_m if and only if $w \cap x$ and $w \cap y$ both have the property D_m . Consequently, to prove that x has a maximal subset with the property D_m it is sufficient to prove the fact for

$$u = (\{\omega\} \times x) \cup \{\langle \alpha, \alpha \rangle : \alpha \in \omega\},$$

where ω is the set of natural numbers.

In what follows the range of the variables m, n , etc., is the set of natural numbers greater than 1.

The set u satisfies the following property:

(i) There is a 1-1 function ϕ from ω into u such that for each $s \in u$, there is an $\alpha \in \omega$ such that $s \cap \phi(\alpha) = \emptyset$.

Note that (i) enables us to constructively enlarge any subset v of u with less than n elements to an n -element subset $\psi(n, v)$ with $\cap \psi(n, v) = \emptyset$.

Define

$$(ii) \quad z_n = \{t: t \subseteq u, t \approx n, \text{ and } \cap t = \emptyset\},$$

and for each $s \in u$, and n ,

$$(iii) \quad f_n(s) = \{h: h \in \bigcap_{m=2}^{\infty} z_m \text{ and } s \in h(n)\}.$$

Let

$$(iv) \quad y = \bigcup_n f_n''u.$$

We observe that

- (v) If $v \subseteq y$ then $\cap v \neq \emptyset$ if and only if for each n ,
 (a) $v \cap f_n''u < n$ or (b) $v \cap f_n''u \approx n$ and $\cap f_n^{-1}v = \emptyset$.

(v) is clearly true if $v = \emptyset$. Otherwise, if $h \in \bigcap_{m=2}^{\infty} z_m$

$$\begin{aligned} h \in \cap v &\leftrightarrow (\forall w) [w \in v \rightarrow h \in w] \\ &\leftrightarrow (\forall n) (\forall s) [s \in f_n^{-1}v \rightarrow h \in f_n(s)] \\ &\leftrightarrow (\forall n) (\forall s) [s \in f_n^{-1}v \rightarrow s \in h(n)] \\ &\leftrightarrow (\forall n) [f_n^{-1}v \subseteq h(n)]. \end{aligned}$$

Now $f_n^{-1}v \approx v \cap f_n''u$ since f_n is 1-1, and if $n \neq m$ then $f_n''u \cap f_m''u = \emptyset$. By (ii) and (iii) $h(n) \approx n$ so that $f_n^{-1}v < n$. If $f_n^{-1}v \approx n$ then $f_n^{-1}v = h(n)$ and consequently has an empty intersection. Conversely, if for each n (a) or (b) is satisfied, we can define h by

$$(vi) \quad h(n) = \begin{cases} \psi(n, f_n^{-1}v) & \text{if } f_n^{-1}v < n \\ f_n^{-1}v & \text{if } f_n^{-1}v \approx n. \end{cases}$$

Then $h \in \cap v$.

Now, let v be a maximal subset of y with the property $\bar{D}_m$. This is equivalent to $v \cap f_n^"u$ being a maximal subset of $f_n^"u$ with the property $\bar{D}_m$ for every n . Also, it follows from (i), that $v \cap f_m^"u$ has at least m elements. But if $w \subseteq f_m^"u$ has at least m elements, w has the property $\bar{D}_m$ if and only if $f_m^{-1}"w$ has the property D_m . Thus, $f_m^{-1}"v$ is a maximal subset of u with the property D_m . The theorem follows easily.

The concept $\bar{D}_m$ can be extended further.

DEFINITION 4.30: A class X has the finite intersection property (f.i.p.) if each finite intersection of its elements is non-empty.

M 14: Every set contains a $\subseteq$ -maximal subset with the f.i.p.

We shall show M 14 is equivalent to the axiom of choice. (Klimovsky [1956] also has a proof that M 14 implies the axiom of choice). Since the f.i.p. property is a property of finite character, $M 7 \rightarrow M 14$.

THEOREM 4.31: $M 14 \rightarrow M 4$.

PROOF: Suppose x is a set in which each $\subseteq$ -well-ordered nest has a $\subseteq$ -upper bound. Let

$$y = \{u \subseteq x: u \text{ is a } \subseteq\text{-well-ordered nest}\}$$

and for each $u \in y$, let

$$A_u = \{v \in y: v \text{ is an initial } \subseteq\text{-segment of } u\}.$$

Then $A_u \cap A_v \neq \emptyset$ iff either u is an initial $\subseteq$ -segment of v or v is an initial $\subseteq$ -segment of u . Let $w = \{A_u: u \in y\}$. M 14 implies that w has a $\subseteq$ -maximal subset t with the f.i.p. Then $\cup t$ is a $\subseteq$ -well-ordered nest in x . A $\subseteq$ -upper bound of $\cup t$ is a $\subseteq$ -maximal element of x , q.e.d.

In most cases for statements equivalent to the axiom of choice in NBG° , if the word "set" is changed to "class", the resulting statement is stronger. M 14 is a notable exception.

The following statement, which we shall call M 14S, is also equivalent to the axiom of choice in NBG°.

M 14S: Every class contains a $\subseteq$ -maximal subclass with the f.i.p.

Clearly M 14S $\rightarrow$ M 14.

THEOREM 4.32: M 14 $\rightarrow$ M 14S.

PROOF: Let X be a class of non-empty sets. (There is no loss of generality in assuming $\emptyset \notin X$.) Let $x \in X$ and let

$$Y = \{x \cap y : y \in X\}.$$

Since $Y \subseteq \mathcal{P}(x)$, Y is a set. M 14 implies that Y has a $\subseteq$ -maximal subset W with the f.i.p. property. Then let

$$Z = \{y \in X : x \cap y \in W\}.$$

We claim that Z is a $\subseteq$ -maximal subclass of X with the f.i.p. It is clear that Z has the f.i.p. because if u is a finite subset of Z then for all $a \in u$, $a \cap x \in W$. Since W has the f.i.p.,

$$\emptyset \neq \bigcap_{a \in u} (a \cap x) = x \cap \bigcap_{a \in u} a \rightarrow \bigcap u = \bigcap_{a \in u} a \neq \emptyset.$$

Suppose Z is not $\subseteq$ -maximal. Then there is a $z \in X$ such that $Z \cup \{z\}$ has the f.i.p. Since every element of W is a non-empty subset of x and W is a $\subseteq$ -maximal subset of Y with the f.i.p., $x \in W$. Consequently, $x \in Z$. Therefore, if u is any finite subset of Z then $x \cap z \cap (\bigcap u) \neq \emptyset$. This implies that $x \cap z \in W$, which in turn implies that $z \in Z$, q.e.d.

M 14 also has a primed form which is equivalent to the axiom of choice.

M'14: If $y \subseteq x$ and y has the f.i.p. then there is a $\subseteq$ -maximal set z with the f.i.p. such that $y \subseteq z \subseteq x$.

Next, we turn to some more generalizations of the form M 8(A) and thereby answer some of the questions asked in

Kurepa [1952]. In order to simplify the notation we define a property K^* as follows:

DEFINITION 4.33: A class X is said to have the property K^* if X can be linearly ordered and X has the property $\bar{K}$.

We shall show that $M_8(D \vee \bar{D} \vee J \vee \bar{J} \vee K \vee K^*)$ is equivalent to the axiom of choice. (The proof is due to A. Rubin and H. Rubin.) Thus it follows that if A is a disjunction of a subset of the six properties $D, \bar{D}, J, \bar{J}, K$, or K^* then $M_8(A)$ is equivalent to the axiom of choice because

$$M_7 \rightarrow M_8(A) \rightarrow M_8(D \vee \bar{D} \vee J \vee \bar{J} \vee K \vee K^*).$$

(While K^* is not a property of finite character, K^* is the conjunction of $\bar{K}$ and X can be linearly ordered. $\bar{K}$ is a property of finite character and $M_7 \rightarrow WO_1 \rightarrow X$ can be linearly ordered.)

Let B stand for the disjunction $D \vee \bar{D} \vee J \vee \bar{J} \vee K \vee K^*$. We shall prove:

THEOREM 4.34: $M_8(B) \rightarrow AC_2$.

PROOF: Let S be an infinite set of pairwise disjoint non-empty sets, no set of which is an element of another or of itself. (There is no loss of generality because, for example, we could replace each element $x \in S$ by $x \times \{\alpha\}$ where α is an infinite ordinal such that $\alpha \notin U_S$.) We define the following sets:

C is the set of all choice functions on subsets of S .

$T = \{\{u, x\}: u \in x \in S\}$

P is the set of all linear orderings on U_S .

$M = \{\{p\} \times x_u: p \in P \wedge u \in x \in S \wedge x_u = \{v \in x: v p u\}\} \cup \{\{q\} \times S\},$

where q is a fixed element such that $q \notin P$.

If $r \in T$ and $f \in C$,

$$N(r, f) = \{w: w = \{\langle r, f \rangle\} \vee (\exists r' \in T)(\exists f' \in C)(r' \cap r = \emptyset \\ \wedge w = \{\langle r, f \rangle, \langle r', f' \rangle\}) \vee (\exists f' \in C)(f \bar{K} f' \wedge w = \{\langle r, f \rangle, \langle r, f' \rangle\})\}.$$

If $f \in C$ and $m \in M$,

$$H(f, m) = \{\langle f', m' \rangle: f' \subset f \vee (f' = f \wedge m' \subseteq m)\}$$

$$Y = \{N(r, f) \times (a \cup H(g, m)): r \in T, f, g \in C, m \in M\},$$

where $a \neq \emptyset$ is a fixed element such that $a \notin \cup H(g, m)$ for any $g \in C$ and $m \in M$. An element of Y will be designated by y_{rfgm} .

The idea of the proof is to show that a $\subseteq$ -maximal subset of Y with one of the six properties $D, \bar{D}, J, \bar{J}, K$ or K^* induces either (a) a $\subseteq$ -maximal nest in C (The union of such a nest is a choice function on S .) or (b) a $\subseteq$ -maximal subset of T whose elements are pairwise disjoint (This set also induces a choice function on S .) or (c) a $\subseteq$ -maximal subset of $M \sim \{\{q\} \times S\}$ with the property $\bar{K}$. (If $\{p\} \times x_u, \{p'\} \times x'_u \in M \sim \{\{q\} \times S\}$, $\{p\} \times x_u \subseteq \{p'\} \times x'_u$, iff $p = p'$, $x = x'$ and $u p u'$. Thus a $\subseteq$ -maximal subset $N \subseteq M$ satisfying $\bar{K}$ has the property that if $\{p\} \times x_u, \{p\} \times y_v \in N$ then $x \neq y$. Therefore, $\{u: (\exists x \in S)\{p\} \times x_u \in N\}$ is a choice set for S .)

We make the following observations:

(1) If Y can be linearly ordered, this induces a linear ordering on $\cup S$.

(2) $y_{r_1 f_1 g_1 m_1}, y_{r_2 f_2 g_2 m_2} \in Y$ are disjoint iff

$$N(r_1, f_1) \cap N(r_2, f_2) = \emptyset.$$

This is true because

$$\begin{aligned} y_{r_1 f_1 g_1 m_1} \cap y_{r_2 f_2 g_2 m_2} &= [N(r_1, f_1) \times (a \cup H(g_1, m_1))] \\ &\quad \cap [N(r_2, f_2) \times (a \cup H(g_2, m_2))] \\ &= [N(r_1, f_1) \cap N(r_2, f_2)] \\ &\quad \times [a \cup (H(g_1, m_1) \cap H(g_2, m_2))]. \end{aligned}$$

A direct product $A \times B = \emptyset$ iff $A = \emptyset \vee B = \emptyset$. In our case $B \neq \emptyset$ because $a \neq \emptyset$.

(3) $y_{r_1 f_1 g_1 m_1} K y_{r_2 f_2 g_2 m_2}$ iff $r_1 = r_2$, $f_1 = f_2$ and $H(g_1, m_1) K H(g_2, m_2)$.

$H(g_1, m_1) K H(g_2, m_2)$ iff $(g_1 K g_2 \wedge g_1 \neq g_2)$

$\vee (g_1 = g_2 \wedge m_1 K m_2)$.

Case 1: Y has a $\subseteq$ -maximal subset Z with the property K or with the property $\bar{J}$.

It follows from (3) that if $y_{rfgm} \in Z$, $\{g \in C: y_{rfgm} \in Z\}$ is a $\subseteq$ -maximal nest contained in C .

Case 2: Y has a $\subseteq$ -maximal subset Z with the property J or with the property $\bar{D}$.

If $r_1 \neq r_2$, $N(r_1, f_1) \cap N(r_2, f_2) \neq \emptyset$ iff $r_1 \cap r_2 = \emptyset$. Thus, it follows from (2) and (3) that if $f, g \in C$ and $m \in M$, $\{r \in T: y_{rfgm} \in Z\}$ is a $\subseteq$ -maximal subset of T whose elements are pairwise disjoint.

Case 3. Y has a $\subseteq$ -maximal subset Z with the property D .

If $r_1 = r_2$, $N(r_1, f_1) \cap N(r_2, f_2) = \emptyset$ iff $f_1 K f_2$ and $f_1 \neq f_2$. Let $r \in T$ so that there are $f, g \in C$ and $m \in M$ with $y_{rfgm} \in Z$. (It follows from (2) that such an $r \in T$ exists.) Then, $\{f \in C: y_{rfgm} \in Z\}$ is a $\subseteq$ -maximal nest in C .

Case 4: Y has a $\subseteq$ -maximal subset Z with the property K^* .

By (1), $P \neq \emptyset$. Therefore, there exists a $p \in P$ and $u \in x \in S$ such that $\{p\} \times x_u \in M$, and for all $p \in P$, $u \in x \in S$, $(\{p\} \times x_u) \bar{K} (\{q\} \times S)$. Let $g \in C$ be such that there exist $r \in T$, $f \in C$ and $m \in M$ with $y_{rfgm} \in Z$. (It follows from (3) that such a $g \in C$ exists.) Then $\{m \in M \sim \{\{q\} \times S\}: y_{rfgm} \in Z\}$ is a $\subseteq$ -maximal subset of $M \sim \{\{q\} \times S\}$ with the property $\bar{K}$, q.e.d.

All the $M 8(A)$ forms have primed forms which are also equivalent to the axiom of choice.

The following propositions are put in here since they deal with the properties $D, \bar{D}, J, \bar{J}, K$ and $\bar{K}$, and they were suggested by some problems proposed by Kurepa [1952]. They are not the same type of maximal principles which were considered previously. Let us introduce the following schemata.

$M 15(A:B)$: Any set which contains a $\subseteq$ -maximal subset with the property A contains a $\subseteq$ -maximal subset with the property B .

$M 16(A:B)$: Every set can be linearly ordered and any set which contains a $\subseteq$ -maximal subset with the property A contains a $\subseteq$ -maximal subset with the property B .

Under the following conditions $M 15(A:B)$ is equivalent to the axiom of choice:

$A \neq B, A = D, \bar{D}, J, \bar{J}, K; \text{ or } \bar{K}, \text{ and } B = D, \bar{D}, J, \bar{J}, \text{ or } K.$

It follows from J. D. Halpern's Ph.D. thesis [1961] that $M 15(A, \bar{K})$ is not equivalent to the axiom of choice in NBG° if $A = D, \bar{D}, J, \bar{J}, \text{ or } K$ (See remarks following $M 8(A)$).

However, $M 16(A:B)$ is equivalent to the axiom of choice for all A and $B, A \neq B$, such that $A = D, \bar{D}, J, \bar{J}, K, \text{ or } \bar{K}$ and $B = D, \bar{D}, J, \bar{J}, K, \text{ or } \bar{K}$.

Clearly, $M 16(A:B) \rightarrow M 15(A:B); M 8(B) \rightarrow M 15(A:B); \text{ and } M 9(\bar{K}) \rightarrow M 16(A:\bar{K}).$

THEOREM 4.35: $M 15(A:B) \rightarrow M 8(B)$, for $A = K, J, \text{ or } \bar{D}$ and $B = \bar{K}, \bar{J}, \text{ or } D.$

PROOF: Let x be an arbitrary set and let a be a non-empty set such that for all $u \in x, a \cap u = \emptyset$. Let $y = x \cup \{a\}$. Then $\{a\}$ is a $\subseteq$ -maximal subset of y with the property K, J , and $\bar{D}$. If M is a $\subseteq$ -maximal subset of y with the property $\bar{K}, \bar{J}$ or D then $M \sim \{a\}$ is a $\subseteq$ -maximal subset of x with the same property.

COROLLARY 4.36: $M 15(\bar{J}:\bar{K}) \rightarrow M 8(\bar{K}).$

PROOF: Let x , a , and y be defined as in the proof of 4.35. Let b be an element such that $b \notin U y$. Let $z = \{u \cup \{b\} : u \in y\}$. Then $\{a \cup \{b\}\}$ is a $\subseteq$ -maximal subset of z with the property $\bar{J}$. A $\subseteq$ -maximal $\bar{K}$ subset of z induces a $\subseteq$ -maximal $\bar{K}$ subset of x .

COROLLARY 4.37: $M 16(A:K) \rightarrow M 9(K)$, for $A = K, J, \bar{J}$, or $\bar{D}$.

THEOREM 4.38:

(a) $M 15(A:B) \rightarrow M 8(B)$, for $A = D$ or $\bar{D}$, and $B = K$ or $\bar{K}$.

(b) $M 15(D:\bar{J}) \rightarrow M 8(K)$.

(c) $M 15(\bar{D}:J) \rightarrow M 8(\bar{K})$.

PROOF: Let x be a set and a be an element such that $a \notin U x$. Let $y = \{u \cup \{a\} : u \in x\}$. Then, $\subseteq$ -maximal subsets of y with the property D are one element sets. If M is a $\subseteq$ -maximal subset of y with the property K or $\bar{K}$ then $\{u : u \cup \{a\} \in M\}$ is a $\subseteq$ -maximal subset of x with the same property.

The proof is similar if " D " is replaced by " $\bar{D}$ " because a $\subseteq$ -maximal subset of y with the property $\bar{D}$ is y itself. This proves (a). Part (b) follows because $\bar{J}$ and K are identical on y . To prove (c) we note that a $\subseteq$ -maximal subset of y with the property J induces a $\subseteq$ -maximal subset of x with the property $\bar{K}$.

COROLLARY 4.39:

(a) $M 16(A:\bar{K}) \rightarrow M 9(\bar{K})$, for $A = D$ or $\bar{D}$.

(b) $M 16(\bar{D}:J) \rightarrow M 9(\bar{K})$.

THEOREM 4.40: $M 15(A:K) \rightarrow M 8(K)$, for $A = \bar{K}$ or J .

PROOF: Let x , a , and y be defined as in 4.38. Let $z = y \cup \{a\}$. Then $\{a\}$ is a $\subseteq$ -maximal subset of z with the properties $\bar{K}$ and J . A $\subseteq$ -maximal chain in z induces a $\subseteq$ -maximal chain in x .

THEOREM 4.41:

(a) $M 15(A:B) \rightarrow M 8(\bar{D})$, for $A = D, \bar{J}, K$, or $\bar{K}$, and $B = J$ or $\bar{D}$.

(b) $M 15(\bar{K}:B) \rightarrow M 8(D)$, for $B = D$ or $\bar{J}$.

PROOF: Let x be an arbitrary set and choose an element a such that for all $u, v \in x$, if $u \neq v$ then $\langle u, a \rangle \notin v$. Let $u_a = u \cup \{\langle u, a \rangle\}$. Then, if $u, v \in x$ and $u \neq v$, $u_a \cap v_a = u \cap v$. Let $y = x \times \{a\}$; then $u_a \cap y = \{\langle u, a \rangle\}$. Define $z = \{u_a : u \in x\}$ and $w = z \cup \{y\}$. Then, for all $u, v \in x$, if $u \neq v$, $u_a \bar{K} v_a$. Therefore,

$$u_a \bar{D} y \text{ and } u_a J v_a \leftrightarrow u_a \bar{D} v_a \leftrightarrow u \bar{D} v,$$

$$u_a J y \text{ and } u_a \bar{J} v_a \leftrightarrow u_a D v_a \leftrightarrow u D v.$$

The set $\{y\}$ is a $\subseteq$ -maximal subset of w with the property D and $\bar{J}$. If M is a $\subseteq$ -maximal subset of w with the property $\bar{D}$ then $\{u \in x : u_a \in M \sim \{y\}\}$ is a $\subseteq$ -maximal subset of x with the property $\bar{D}$. Moreover, J and $\bar{D}$ are identical on w . Thus, we obtain $M 15(A:B) \rightarrow M 8(\bar{D})$, for $A = D$ or $\bar{J}$ and $B = \bar{D}$ or J .

To obtain the result when $A = K$ or $\bar{K}$, we note that $\subseteq$ -maximal subsets of z with the property K are one element sets and a $\subseteq$ -maximal subset of z with the property $\bar{K}$ is z itself. If M is a $\subseteq$ -maximal subset of z with the property $\bar{D}$ or J then $\{u \in x : u_a \in M\}$ is a $\subseteq$ -maximal subset of x with the property $\bar{D}$. This completes the proof of (a) and the proof of (b) is similar.

THEOREM 4.42: $M 15(A:B) \rightarrow WO 1$, for $A = J$ or $\bar{J}$, and $B = \bar{D}$ or K .

PROOF: Let x be an arbitrary set. Let $\mathcal{R}$ be the set of all well-orderings on subsets of x . For each $R \in \mathcal{R}$ define

$$T_R = \{S \in \mathcal{R} : S = R \vee S \text{ is an extension of } R\}$$

and let $\mathcal{S} = \{T_R : R \in \mathcal{R}\}$. The set $\mathcal{S}$ has the property that

each pair of its elements either has the relation K or D . Therefore, $\subseteq$ -maximal sets in $\mathcal{S}$ with the property J are one element sets, and $\mathcal{S}$ itself is a $\subseteq$ -maximal set with the property $\bar{J}$. $M 15(A:\bar{D})$, for $A = J$ or $\bar{J}$, implies that there is a $\subseteq$ -maximal subset M of $\mathcal{S}$ which has the property $\bar{D}$. Since $\bar{D}$ and K are identical on $\mathcal{S}$, this implies that M is a $\subseteq$ -maximal nest in $\mathcal{S}$.

Let

$$S = \bigcup_{R \in M} R.$$

Since S is a union of comparable well-orderings of subsets of x , it is a well-ordering of a subset of x . Suppose $u \in x \sim \mathcal{J}(S)$. Let

$$S^* = S \cup (\mathcal{J}(S) \times \{u\}) \cup \{\langle u, u \rangle\},$$

and let $M^* = M \cup \{S^*\}$. Since S is an extension of R for each $R \in M$ and S^* is an extension of S , M^* is a nest and $M \subset M^*$. This contradicts the maximality of M . Thus, $\mathcal{J}(S) = x$ and S well-orders x .

THEOREM 4.43:

$$(a) \quad M 15(\bar{J}:D) \rightarrow M 8(D).$$

$$(b) \quad M 15(\bar{D}:J) \rightarrow M 8(J).$$

PROOF: Let x be a set, let In be the set of integers and define y as follows:

$$y = \{u \times \{i\} \times \{1,2\} : u \in x \wedge i \in In\} \cup$$

$$\{(Ux) \times \{j \in In : j \leq i\} \times \{1\} : i \in In\}.$$

For notational convenience, let $u_i = u \times \{i\} \times \{1,2\}$ and let $v_i = (Ux) \times \{j \in In : j \leq i\} \times \{1\}$. Each of the following is easy to verify for all $i, j \in In$:

$$(1) \quad u_i A u'_i \leftrightarrow u A u', \quad \text{for } A = D \text{ or } J,$$

$$(2) \quad v_i \subseteq v_j, \quad \text{for } i \leq j,$$

$$(3) \quad u_i J v_j, \quad \text{for } i \leq j,$$

$$(4) \quad u_i D v_j, \quad \text{for } i > j,$$

(5) $u_i D u_j^!$, for $i \neq j$.

The set $\{v_i : i \in \text{In}\}$ is both a $\subseteq$ -maximal subset of y with the property $\bar{J}$ (by (2) and (4)) and also a $\subseteq$ -maximal subset of y with the property $\bar{D}$ (by (2) and (3)). Therefore, the hypotheses of both $M 15(\bar{J}:D)$ and $M 15(\bar{D}:J)$ are satisfied.

$M 15(\bar{J}:D)$ implies that y has a $\subseteq$ -maximal subset z_1 with the property D . It follows from (2) that at most one v_i can be an element of z_1 . If no v_i is an element of z_1 then $\{u : u_i \in z_1\}$ is a $\subseteq$ -maximal subset of x with the property D for each $i \in \text{In}$ (by (1)). If $v_i \in z_1$ then for each $j > i$, $\{u : u_j \in z_1\}$ is a $\subseteq$ -maximal subset of x with the property D (by (1) and (3)). This proves part (a).

To prove part (b), we use the fact that $M 15(\bar{D}:J)$ implies that y has a $\subseteq$ -maximal subset z_2 with the property J . It follows from (2), (3) and (5) that there is exactly one $v_i \in z_2$, and it also follows from (5) that if $u_i, u_j^! \in z_2$ then $i = j$. Thus, there is a $j \in \text{In}$, $j \leq i$ such that if $u_k \in z_2$ then $k = j$. Then by (1) $\{u : u_j \in z_2\}$ is a $\subseteq$ -maximal subset of x with the property J , q.e.d.

This completes the discussion of $M 15(A:B)$ and $M 16(A:B)$ where $A \neq B$ and $A, B = D, \bar{D}, J, \bar{J}, K$, or $\bar{K}$. We shall show in section 5 that in NBG $M 8(\bar{K}) \rightarrow AC$.

The next form to be considered in this section is an obvious extension of the idea of constructing maximal subsets with respect to "natural" relations. The relation to be considered is the ζ -relation.

M 17: Every set x contains a $\subseteq$ -maximal subset y such that for every $s, t \in y$, $s \neq t$, either $s \in t$ or $t \in s$.

It is clear that $M 7 \rightarrow M 17$ since $M 17$ deals with a property of finite character. We shall prove that $M 17 \rightarrow WO 1$.

THEOREM 4.44: $M 17 \rightarrow WO 1$.

PROOF: Let x be an arbitrary set and let $\mathcal{W} = \{W : (\exists y)[y \subseteq x \text{ and } W \text{ is a well-ordering of } y]\}$. We define a relation R on $\mathcal{W}$ as follows:

If $W, W' \in \mathcal{W}$ then

$$W R W' \leftrightarrow W \subset W' \quad \text{and} \quad \mathcal{D}(W) \times (\mathcal{D}(W') \sim \mathcal{D}(W)) \subseteq W'.$$

W' is an extension of W . For all $W \in \mathcal{W}$ we define a function f on $\mathcal{W}$ as follows:

$$f(W) = \{\{W\}\} \cup \{f(W') : W' R W \text{ and } W' \in \mathcal{W}\}.$$

We claim: (i) $f(W) = f(U) \leftrightarrow W = U$ and

(ii) $f(W) \in f(U) \leftrightarrow W R U$.

It is clear that if $W = U$ then $f(W) = f(U)$. To prove the converse it is sufficient to prove that

$$(iii) \quad \{\{W\}\} \cap \{f(W') : W' R W \text{ and } W' \in \mathcal{W}\} = \emptyset$$

for all $W' \in \mathcal{W}$. We leave the proof of (iii) as an exercise.

To prove (ii) we note that if $W R U$ then clearly $f(W) \in f(U)$. Suppose $f(W) \in f(U)$; then either $f(W) = \{U\}$ or $W R U$, but the first alternative is impossible.

M 17 implies that $\mathcal{K}(f)$ contains a maximal subset Q such that for every $S, T \in Q$, if $S \neq T$ then either $S \in T$ or $T \in S$. Let $\mathcal{U} = \{U : f(U) \in Q\}$. Since f is a 1-1 mapping of $\mathcal{W}$ onto $\mathcal{K}(f)$ preserving order (by (i) and (ii)) it follows that $\mathcal{U}$ is a maximal subset of $\mathcal{W}$ such that for all $U, W \in \mathcal{U}$, either $U R W$ or $W R U$.

We claim that $\cup \mathcal{U}$ is a well-ordering on x . Since $\cup \mathcal{U}$ is a union of R -comparable well-orderings of subsets of x , $\cup \mathcal{U}$ is clearly a well-ordering on a subset of x . Suppose there exists an $s \in x$ such that $s \notin \mathcal{D}(\cup \mathcal{U})$. Then for all $U \in \mathcal{U}$, $s \notin \mathcal{I}(U)$. Let

$$V = \cup \mathcal{U} \cup (\mathcal{D}(\cup \mathcal{U}) \times \{s\}) \cup \{\langle s, s \rangle\},$$

and let $\mathcal{U}^* = \mathcal{U} \cup \{V\}$. Since for each $U \in \mathcal{U}^*$, $U R V$, and $\mathcal{U} \subset \mathcal{U}^*$, the maximality of $\mathcal{U}$ is contradicted. Hence, $\cup \mathcal{U}$ is a well-ordering on x .

M 18, the next maximal principle in this section, is put in primarily as a curiosity. Blair and Tomber [1960] discuss

two forms of the axiom of choice for finite sets. It was pointed out to them by H. Rubin that if a slight change was made in the wording, one would obtain forms of the general axiom of choice. M 18 is a combination of these two forms.

M 18: Suppose x is partially ordered by R . For each $t \in x$, let

$$\vec{t} = \{s \in x : t R s \wedge t \neq s\}.$$

Suppose every R -linearly ordered subset of x has an R -upper bound. Then there is a $t \in x$ such that $\vec{t}$ has no R -first element.

We shall show M 18 is equivalent to M 1.

THEOREM 4.45: $M 1 \rightarrow M 18$.

PROOF: If x satisfies the hypothesis of M 18. Then it also satisfies the hypothesis of M 1. Consequently, x has an R -maximal element, t . Since $\vec{t} = \emptyset$, $\vec{t}$ has no R -minimal element.

THEOREM 4.46: $M 18 \rightarrow M 1$.

PROOF: Suppose M 1 is false. Then there is a set x and a partial ordering relation R such that every R -linearly ordered subset of x has an R -upper bound, but x has no R -maximal element. Define a relation S on $x \times \omega$ as follows: If $\langle u, m \rangle, \langle v, n \rangle \in x \times \omega$ then

$$\langle u, m \rangle S \langle v, n \rangle \text{ iff } (u \neq v \wedge u R v) \vee (u = v \wedge m \leq n).$$

Then $\langle x \times \omega, S \rangle$ satisfies the hypothesis of M 18. For suppose u is an S -linearly ordered subset of $x \times \omega$. Then $\mathcal{J}(u)$ is an R -linearly ordered subset of x . Every R -linearly ordered subset of x has a strict R -upper bound. If not, then an R -upper bound would be an R -maximal element of x . Let $b \in x$ be a strict R -upper bound of $\mathcal{J}(u)$. Then $\langle b, 0 \rangle$ is an S -upper bound of u .

Suppose $\langle u, m \rangle \in x \times \omega$. Then we claim $\langle u, m+1 \rangle$ is an R -first element of $\langle \vec{u}, m \rangle$, because $\langle u, m \rangle S \langle u, m+1 \rangle$ and if $\langle u, m \rangle S \langle v, n \rangle$ and $\langle u, m \rangle \neq \langle v, n \rangle$, then $\langle u, m+1 \rangle S \langle v, n \rangle$. This contradicts M 18, q.e.d.

We end this section with fix-point theorems which are equivalent to maximal principles. Abian [1983] gave $M_{19}(P,W)$ in an abstract. The other forms and all the proofs are our own.

$M_{19}(Q,U)$: Every set x such that x is Q -ordered by R and every U -ordered subset has an R -upper bound, has the property that every function f from x to x such that for all $u \in x$, $u R f(u)$, has a fixed point.

We shall show $M_{19}(Q,U) \leftrightarrow M_1(Q,U)$. It is clear that $M_1(Q,U) \rightarrow M_{19}(Q,U)$ because an R -maximal element must be a fixed point.

THEOREM 4.47: $M_{19}(Q,U) \rightarrow M_1(Q,U)$.

PROOF: Suppose x is a set and R a relation which satisfy the hypotheses of $M_1(Q,U)$. Suppose x has no R -maximal element. Let $y = x \times \omega$ and define a relation S on y so that

$$\langle u, m \rangle S \langle v, n \rangle \text{ iff } (u \neq v \wedge u R v) \vee (u = v \wedge m \leq n).$$

It is clear that when one considers the possible values for Q and U given on pp. 37-8, that if R is a Q -ordering on x then S is a Q -ordering on y . Also, since x has no R -maximal element, if each U -ordered subset of x has an R -upper bound then each U -ordered subset of y has an S -upper bound.

Let f be a function defined on y such that for all $\langle u, m \rangle \in y$, $f(u, m) = \langle u, m+1 \rangle$. Then for all $\langle u, m \rangle \in y$, $\langle u, m \rangle S f(u, m)$, but f has no fixed point. This contradicts $M_{19}(Q,U)$. Thus, x must have an R -maximal element, q.e.d.

COROLLARY 4.48: $M_{19}(Q,U)$ is equivalent to the axiom of choice if

$Q = TR, P, RA$ or F and $U = C, AS \ \& \ C, TR \ \& \ C, L, D$, or W and also if $Q = D$ and $U = W$.

PROOF: Theorem 4.47 and preceding remarks, and Theorems 4.9 and 4.10.

§5. FORMS EQUIVALENT TO THE AXIOM OF CHOICE UNDER THE
AXIOMS OF EXTENSIONALITY AND FOUNDATION

As we state in the Introduction, all proofs in this book, unless otherwise specified, are in NBG° . Thus, the proofs are valid whether or not there are atoms and whether or not the axiom of foundation holds. H. Rubin [1960] discovered two forms of the Axiom choice, PW and LW below, in which he assumed there were no atoms (or the class of atoms was a set and could be well-ordered) and the axiom of foundation to prove they implied the axiom of choice. Also, since both PW and LW are true in the Fraenkel-Mostowski model given for example in Halpern [1962] and the axiom of choice is false in most Fraenkel-Mostowski models, it was clear the proof of equivalence did not hold in NBG° . Later Felgner & Jech [1973], Howard & Rubin [1977] and others discovered other statements in the same category. This section is devoted to such statements.

Our proofs below which require the stronger assumptions will be prefaced by " $\text{NBG} \vdash$ ". Thus, we shall assume in these proofs that there are no atoms and that the axiom of foundation holds. (In most instances, the conditions on atoms can be weakened.)

PW: The power set (the set of all subsets) of a well-ordered set can be well-ordered.

LW: Every linearly ordered set can be well-ordered.

INJ: If x is a non-empty set and A is a proper class then $x < A$.

PROJ: If x is a non-empty set and A is a proper class then A can be mapped onto x .

MC 1: If s is a set of non-empty sets then there is a function f such that for each $x \in s$, $f(x)$ is a non-empty finite subset of x .

MC 2: Every set is the union of a well-ordered set of finite sets.

MC 3: If s is a set of infinite sets, then there is a function f such that for each $x \in s$, $f(x)$ is a partition of x into pairwise disjoint finite sets, each of which has at least 2 elements.

DEFINITION 5.1: If R is a partial ordering relation on X then $Y \subseteq X$ is called an R -antichain iff for all $u, v \in Y$, if $u \neq v$ then $\neg uRv \wedge \neg vRu$.

For example, a set with the property $\bar{K}$ (see 4.20) is a $\subseteq$ -antichain.

A: Every partially ordered set contains a $\subseteq$ -maximal antichain.

MC 1 is usually called the multiple choice axiom and A. Levy [1962] proved that it does not imply the axiom of choice in NBG°. MC 2 and MC 3 are variations of MC 1. (See WO 4(m) and AC 10(m).) INJ and PROJ are forms similar to a form given by von Neumann [1928]. (They are called the injection and projection principles in Felgner & Jech [1973].) The statement A appears in Kurepa [1952]. If R is taken to be $\subseteq$, then A is $M 8(\bar{K})$. (As is well known, there is no loss of generality in assuming that a partial order is $\subseteq$, because if R is any partial ordering relation on a set x and for each $u \in x$, $\hat{u} = \{v \in x: vRu\}$, then if $y = \{\hat{u}: u \in x\}$, $\langle x, R \rangle \cong \langle y, \subseteq \rangle$.)

Felgner & Jech [1973] prove the following implications:

$$\text{INJ} \rightarrow \text{PROJ} \rightarrow \text{AC} \rightarrow \text{MC 1} \rightarrow \text{A} \rightarrow \text{LW} \rightarrow \text{PW}.$$

(They also proved that none of the arrows are reversible in NBG°). H. Rubin [1960] proved PW implies AC in NBG and

Rubin & Rubin [1963] prove that $AC \rightarrow INJ$ in NBG.

The following implications are clear: $INJ \rightarrow PROJ$, $AC \rightarrow MC \rightarrow MC \rightarrow MC$. The proofs that $MC \rightarrow MC$ and $MC \rightarrow MC$ are similar to the proof of 2.9. The proof that $MC \rightarrow MC$ is similar to the proof of 2.12.

THEOREM 5.2: $PROJ \rightarrow WO$.

PROOF: Let x be a non-empty set. $PROJ$ implies that there is a function F mapping On , the class of ordinal numbers, onto x . For each $s \in x$, $F^{-1}\{s\}$ is a non-empty class of ordinal numbers. Let α_s be the first element of $F^{-1}\{s\}$. Then $x \approx \{\alpha_s : s \in x\}$.

THEOREM 5.3: $MC \rightarrow A$.

PROOF: Suppose x is partially ordered by R . MC implies there is a function f such that for each non-empty subset $u \subseteq x$, $f(u)$ is a non-empty finite subset of u . Define a function g such that for all u such that $\emptyset \neq u \subseteq x$,

$$g(u) = \{a : a \text{ is an } R\text{-minimal element of } f(u)\}.$$

Then $g(u)$ is finite, non-empty and an R -antichain. Now, using transfinite induction, we construct a $\subseteq$ -maximal antichain, $A \subseteq x$. $A = \bigcup_{\alpha \in On} A_\alpha$, where

$$A_\alpha = g(\{a \in x : a \text{ is } R\text{-incomparable with all } b \in \bigcup_{\beta < \alpha} A_\beta\}).$$

THEOREM 5.4: $A \rightarrow LW$.

PROOF: Suppose x is linearly ordered by R . Let

$$y = \{\langle u, a \rangle : \emptyset \neq u \subseteq x \wedge a \in u\}.$$

Define S on y so that

$$\langle u, a \rangle S \langle v, b \rangle \text{ iff } u = v \wedge a R b.$$

A implies that y has a $\subseteq$ -maximal S -antichain, B . If $\langle u, a \rangle, \langle v, b \rangle \in B$ then $a = b$, so B is a choice function on the non-empty subsets of x . Then it follows that x can be well-ordered. (See 2.9.)

THEOREM 5.5: $LW \rightarrow PW$.

PROOF: Suppose x can be well-ordered by R . We shall show $\mathcal{P}(x)$ can be linearly ordered. Define a relation S on $\mathcal{P}(x)$ such that if $u, v \in \mathcal{P}(x)$,

$$u S v \text{ iff } u = v \vee (u \neq v \wedge \text{the } R\text{-first element of } (u \cup v) \sim (u \cap v) \text{ is an element of } u).$$

Then $\langle \mathcal{P}(x), S \rangle$ is a linearly ordered set so LW implies $\mathcal{P}(x)$ can be well-ordered, q.e.d.

THEOREM 5.6: $NBG \vdash AC\ 3 \wedge WO\ 2 \rightarrow INJ$.

PROOF: Let x be a non-empty set and A a proper class. For each $s \in x$ let $\rho(s)$ be the rank of s , (See 0.9) and let $B = \{\rho(s) : s \in x\}$. A is a proper class and the class of all sets with a given rank is a set so B is a proper class of ordinal numbers. Every proper class of ordinal numbers is equipollent to On . Let F be a bijection mapping On onto B .

By $WO\ 2$ there is an ordinal number α and a function f such that $f: x \approx \alpha$. By $AC\ 3$ there is a function g such that for each $\beta \in \alpha$, $g(\beta) \in \rho^{-1}\{F(\beta)\}$. Define a function h such that for each $u \in x$, $h(u) = g(f(u))$. Then, h is a 1-1 function which maps x into A .

(Actually, 5.6 is provable in $NBG^\circ + \text{axiom of foundation} + \text{the class of atoms is a set.}$)

THEOREM 5.7: $NBG \vdash PW \rightarrow WO\ 1$.

PROOF: Let s be a set. The axiom of foundation implies that each set has a rank so suppose $\rho(s) = \alpha$. We shall prove by transfinite induction that there exist a function F such that for each $\beta < \alpha$, F_β well-orders $z(\beta) = \{x : \rho(x) = \beta\}$. For all $u \in s$, $\rho(u) < \alpha$. Thus, the above function F can be used to well-order s .

For every β , let $V_\beta = \{x : \rho(x) < \beta\}$ and let

$$\lambda = \Gamma(V_\alpha) = \{\gamma : \gamma \prec V_\alpha\}$$

(Γ is Hartogs' function.)

λ is an ordinal number which doesn't depend upon β . Moreover, λ can be well-ordered so PW implies that there is a relation R which well-orders $\mathcal{P}(\lambda)$.

The induction hypotheses is that for all $\gamma < \beta$, where $\beta < \alpha$, F_γ is defined so that F_γ well-orders $z(\gamma)$.

Now, we well-order V_β in the natural way. That is, define S_β on V_β so that if $x \in z(\gamma_1)$, $y \in z(\gamma_2)$ then $x S_\beta y$ iff $\gamma_1 < \gamma_2 \vee (\gamma_1 = \gamma_2 \wedge x F_{\gamma_1} y)$. There exists a unique ordinal $\delta < \lambda$ and a unique function G_β such that $G_\beta: \langle V_\beta, S_\beta \rangle \cong \langle \delta, \subseteq \rangle$. Suppose $x, y \in z(\beta)$. Then $x, y \in \mathcal{P}(V_\beta)$. Define F_β on $z(\beta)$ so that if $x, y \in z(\beta)$,

$$x F_\beta y \leftrightarrow G_\beta''x R G_\beta''y.$$

Since $G_\beta''x, G_\beta''y \subseteq \delta \subseteq \lambda$, R is defined, so F_β is defined, and since R is a well-ordering relation, so is F_β . Moreover, the definition of R does not depend upon β , therefore, F is a function, q.e.d.

(Note: it is easy to give an incorrect proof of this theorem if one is not careful. For example, we might try to well-order V_β for each β using transfinite induction as follows: $V_0 = \emptyset$, so clearly V_0 can be well-ordered. Suppose V_β can be well-ordered. Then PW implies that $V_{\beta+1} = \mathcal{P}(V_\beta)$ can be well-ordered. Suppose β is a limit ordinal and for all $\gamma < \beta$, V_γ can be well-ordered. Then $V_\beta = \bigcup_{\gamma < \beta} V_\gamma$ "can be well-ordered", but the problem is that it requires some form of the axiom of choice to "construct" that well-ordering of V_β when β is a limit ordinal. In the correct proof of the theorem we construct the well-ordering using the relation R .)

It follows from 4.35, 4.36, and 4.38 that each of $M 15(A; \bar{K})$, for $A = D, \bar{D}, J, \bar{J}$ or K , imply $M 8(\bar{K})$, so each is equivalent to AC in NBG.

We continue with some forms of the axiom of choice which are due to P. Howard & J. E. Rubin [1977]. Let AC_Q^P and MC_Q^P be defined as follows:

AC_Q^P : For each set x , if x is a P -ordered set of non-empty sets such that each $u \in x$ can be Q -ordered, then there is a function f such that for each $u \in x$, $f(u) \in u$.

MC_Q^P : For each set x , if x is a P -ordered set of non-empty sets such that each $u \in x$ can be Q -ordered, then there is a function f such that for each $u \in x$, $f(u)$ is a non-empty, finite subset of u .

For P and Q we take the following:

LO: linear ordering

DLO: defined linear ordering.

For example, AC^{LO} is the axiom of choice for a linearly ordered family of non-empty sets and MC_{DLO} is the multiple choice axiom for a family of non-empty sets, each of which has a defined linear ordering.

ORD and W-ORD are due to J. Truss [1972].

ORD: If x is a set and f is a function such that for each $u \in x$, $f(u)$ is a non-empty set of linear orderings of u , then $\{f(u): u \in x\}$ has a choice function.

W-ORD: If x is a set and f is a function such that for each $u \in x$, $f(u)$ is a non-empty set of well-orderings of u , then $\{f(u): u \in x\}$ has a choice function.

TCL is $M 5(TR\&C, L)$ of section 4 and appeared in P. E. Howard & J. E. Rubin [1977].

TCL: Every transitive and connected set contains a $\subseteq$ -maximal linear ordered set.

Thus, $TCL \leftrightarrow M 5(TR\&C, AS) \leftrightarrow M 5(TR\&C, AS\&C) \leftrightarrow M 5(TR\&C, P) \leftrightarrow M 5(TR\&C, L) \leftrightarrow M 5(TR\&C, D) \leftrightarrow M 5(TR\&C, RA)$. (Here, D means directed.)

We shall prove the following implications in NBG° :

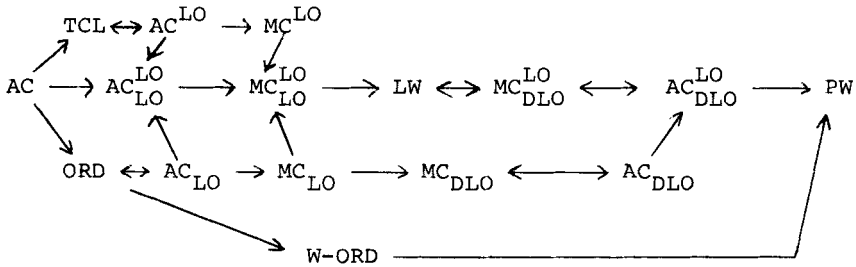
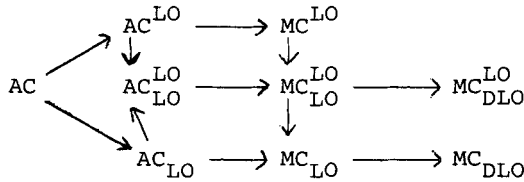


FIGURE 5.1

It can be shown using results in Howard & Rubin [1977] and a model of R. J. Gauntt that none of the arrows in the above diagram are reversible in NBG° . However we have shown that $\text{PW} \rightarrow \text{AC}$ in NBG , so all the statements are equivalent in NBG .

The following implications are easily verified:



THEOREM 5.8: $\text{TCL} \leftrightarrow \text{AC}^{\text{LO}}$.

PROOF: Suppose $\langle x, R \rangle$ is a linearly ordered set of pairwise disjoint non-empty sets. Define a relation S on $\cup x$ such that if $a, b \in \cup x$, where $a \in u \in x$ and $b \in v \in x$, $a S b$ iff $u R v$. Then S is transitive and connected on $\cup x$. A $\subseteq$ -maximal S -linearly ordered subset of $\cup x$ is a choice set for x .

Conversely, suppose $\langle x, R \rangle$ is a transitive and connected set. Define a relation S on x so that

$$u S v \text{ iff } u = v \vee (u R v \wedge v R u)$$

S is an equivalence relation on x . For each $u \in x$ let $[u] = \{v \in x : u S v\}$. Let $y = \{[u] : u \in x\}$. Since R is transitive, we can define a relation T on y so that

$[u]T[v]$ iff $u R v$. (If $u' \in [u]$ and $v' \in [v]$, then $u' R v'$ iff $u R v$.) The relation T linearly orders y . A choice set for y is a $\subseteq$ -maximal linearly ordered subset of x , q.e.d.

THEOREM 5.9: $AC_{DLO} \leftrightarrow MC_{DLO}$.

PROOF: Clearly, $AC_{DLO} \rightarrow MC_{DLO}$. To prove the converse, let x be a set of non-empty pairwise disjoint sets, each of which is linearly ordered by R . Let f be a multiple choice function on x . Then $\{a: (\exists u \in x) (a \text{ is the } R\text{-first element of } f(u))\}$ is a choice set for x .

The proof that $AC_{DLO}^{LO} \leftrightarrow MC_{DLO}^{LO}$ is similar to the proof of 5.9.

THEOREM 5.10: $LW \leftrightarrow AC_{DLO}^{LO}$.

PROOF: Suppose $\langle x, R \rangle$ is a linearly ordered set of non-empty pairwise disjoint sets such that each set in x is linearly ordered by a relation S . Then $\cup x$ can be linearly ordered, because if $a, b \in \cup x$ where $a \in u \in x$ and $b \in v \in x$, define

$$a T b \text{ iff } \begin{cases} u R v & \text{if } u \neq v, \\ a S b & \text{if } u = v. \end{cases}$$

The relation T linearly orders $\cup x$. Then LW implies $\cup x$ can be well-ordered. Using this well-ordering a choice set for x can be defined.

Conversely, suppose $\langle x, R \rangle$ is a linearly ordered set. Define

$$y = \{u \times \{s\}: u \subseteq x \wedge s \text{ well-orders } u\}.$$

If $\langle u, s \rangle$ is a well-ordered set, let $\text{ord}\langle u, s \rangle$ be the ordinal number of $\langle u, s \rangle$. If $\langle u_1, s_1 \rangle$ and $\langle u_2, s_2 \rangle$ are well-ordered sets with the same ordinal number then there is a unique isomorphism f mapping u_1 onto u_2 , $f: \langle u_1, s_1 \rangle \cong \langle u_2, s_2 \rangle$.

We define a linear ordering, T , on y as follows, if $u_1 \times \{s_1\} \neq u_2 \times \{s_2\}$, then

$$(u_1 \times \{s_1\}) T (u_2 \times \{s_2\}) \text{ iff } \text{ord}\langle u_1, s_1 \rangle < \text{ord}\langle u_2, s_2 \rangle$$

or $(f: \langle u_1, s_1 \rangle \cong \langle u_2, s_2 \rangle$ and if a is the s_1 -first element of $\{b \in u_1: b \neq f(b)\}$ then $a R f(a)$).

The set

$$z = \{(x \sim u) \times \{s\}: u \subset x \wedge s \text{ well-orders } u\}$$

is a linearly ordered set (T induces a linear ordering on z) of non-empty sets, each of which is linearly ordered by R' , where $\langle v_1, s \rangle R' \langle v_2, s \rangle$ iff $v_1 R v_2$. Thus, AC_{DLO}^{LO} implies there is a choice function f on z . Then, well-order x so that the α^{th} element of x , a_α = the first coordinate of $f((x \sim \bigcup_{\beta < \alpha} \{a_\beta\}) \times \{\leq\})$, where $\leq$ is the well-ordering of $\bigcup_{\beta < \alpha} \{a_\beta\}$ induced by α .

THEOREM 5.11: $ORD \leftrightarrow AC_{LO}$.

PROOF: Suppose x is a set of non-empty sets, each of which can be linearly ordered. For each $u \in x$, let $f(u)$ be the set of all linear orderings of u which have a first element. (Clearly, if u can be linearly ordered then it can be linearly ordered so it has a first element.) If g is a choice function on $\{f(u): u \in x\}$, then we can define a choice function, h , on x such that for each $u \in x$, $h(u)$ is the first element in the ordering $g(f(u))$.

Conversely, suppose x is a set and for each $u \in x$, $f(u)$ is a non-empty set of linear orderings on u . We shall show that each $f(u)$ can be linearly ordered. Then, it follows that $AC_{LO} \rightarrow ORD$.

Clearly, $AC_{LO} \rightarrow AC_{LO}^{LO} \rightarrow AC_{DLO}^{LO}$. By 5.10, $AC_{DLO}^{LO} \rightarrow LW$, and by 5.5, $LW \rightarrow PW$. Therefore, $AC_{LO} \rightarrow LW \rightarrow PW$. Suppose $u \in x$. Then, by hypothesis, u can be linearly ordered. Thus, LW implies u can be well-ordered. Therefore, $u \times u$ can also

be well-ordered. PW implies $\mathcal{P}(u \times u)$ can be well-ordered. Since $f(u) \subseteq \mathcal{P}(u \times u)$, $f(u)$ can be well-ordered, and therefore linearly ordered, q.e.d.

Clearly $\text{ORD} \rightarrow \text{W-ORD}$.

The proof of the following theorem is due to J. Truss [1972].

THEOREM 5.12: $\text{W-ORD} \rightarrow \text{PW}$.

PROOF: Suppose α is any ordinal. We shall show that $\mathcal{P}(\alpha)$ can be well-ordered. (This is sufficient since every set which can be well-ordered is equipollent to an ordinal.) Let $y = \alpha \cup \{x\}$ where $x \notin \alpha$. Let

$$z = \{y \times \{t\} : \emptyset \neq t \subseteq \mathcal{P}(\alpha)\}.$$

For $y \times \{t\} \in z$, let $f(y \times \{t\})$ be the set of all well-orderings, R , of $y \times \{t\}$ such that there is an $s \in t$ ($s \subseteq \alpha$) such that

- (i) $s \times \{t\}$ comes first,
- (ii) $\langle x, t \rangle$ is next,
- (iii) $(\alpha \sim s) \times \{t\}$ is last,

where the ordering in each of (i) and (iii) is the natural ordering induced by $<$ on α .

The set z and function f satisfy the hypotheses of W-ORD. A choice function g , on $\{f(y \times \{t\}) : \emptyset \neq t \subseteq \mathcal{P}(\alpha)\}$ enables us to choose for each $t \in \mathcal{P}(\alpha) \sim \{\emptyset\}$ an $s \in t$. (Namely, the s from the $s \times \{t\}$ which precedes the element $\langle x, t \rangle$ in the well-ordering $g(f(y \times \{t\}))$.) Thus, we have a choice function on $\mathcal{P}(\alpha) \sim \{\emptyset\}$. This choice function allows us to well-order $\mathcal{P}(\alpha)$. (See 2.9).

We have proved earlier, 5.5, that $\text{LW} \rightarrow \text{PW}$ so this completes the proof of the implications given in Figure 5.1.

The following weaker version of ORD:

If x is a set such that for each $u \in x$, u can be linearly ordered, then there is a function f such that for each $u \in x$, $f(u)$ linearly orders u

does not imply the axiom of choice. (This was proved by Halpern & Levy [1967].)

There is also a weaker version of MC 1 which was shown by Halpern & Levy [1967] not to imply the axiom of choice in NBG, namely the Kinna-Wagner principle (see Kuratowski [1922]):

KW 1: If x is a set of sets, each of which has at least 2 elements, then there exists a function f such that for each $u \in x$, $\emptyset \neq f(u) \subset u$.

Kinna & Wagner [1955] have shown that $KW\ 1 \leftrightarrow KW\ 2$, where
 KW 2: For each set x , there is an ordinal α such that $x \prec \mathcal{P}(\alpha)$.

However, in each Fraenkel-Mostowski model PW is true. Therefore, in each Fraenkel-Mostowski model, $KW\ 2$ is equivalent to the axiom of choice and is therefore false. But trivially we have $KW\ 2 + PW \leftrightarrow AC$.

M. N. Bleicher [1963,1965] discusses some variations of the multiple choice axiom, MC 1, which are useful in the study of vector spaces. For $m, n \in \omega$ we define

$$(m,n) = \begin{cases} m & \text{if } n = 0 \\ n & \text{if } m = 0 \\ \text{greatest common divisor of } m \text{ and } n, & \text{otherwise} \end{cases}$$

MC 4(n): If s is a set of non-empty sets then there is a function f such that for all $x \in s$, $f(x)$ is a non-empty finite subset of x and $(|f(x)|, n) = 1$.

Thus, according to the definition of (m,n) , MC 4(0) is AC 1 and MC 4(1) is MC 1. It is clear that for all $n \in \omega$, $AC\ 1 \rightarrow MC\ 4(n) \rightarrow MC\ 1$. We have shown that $NBG \vdash MC\ 1 \leftrightarrow AC$. Therefore, $NBG \vdash MC\ 4(n) \leftrightarrow AC$ for each $n \in \omega$.

Since every finite set can be well ordered the well-ordering theorem is equivalent to the statement:

(*) If x is infinite then x can be well ordered.

The word "infinite" means "not equipollent to a natural number". Mathematicians have studied various other definitions of "infinite" or equivalently "not finite". (See, for example, Tarski [1924], Mostowski [1938], Levy [1964] and Truss [1974]). All alternate definitions considered are equivalent if the axiom of choice is assumed. Since the well-ordering of a set implies the well-ordering of any subset, for any definition of infinite which has the properties that a superset of an infinite set is infinite and that there is an infinite set, (*) is equivalent to the well-ordering principle. Brunner [1982] has obtained some results along this line which we shall describe below:

DEFINITION 5.13:

i) A set x is said to be Dedekind-infinite (D-infinite) if it contains a countably infinite subset.

ii) A set is said to be structured if it is the union of two disjoint infinite sets.

Consider the following statements:

INF 1: For every set x , if every infinite subset of x is D-infinite, then x can be well-ordered.

INF 2: For every set x , if every infinite subset of x is structured, then x can be well-ordered.

Clearly, $WO\ 1 \rightarrow INF\ 2 \rightarrow INF\ 1$. Brunner [1982] proves that INF 1 does not imply INF 2 in NBG and that INF 2 does not imply $WO\ 1$ in NBG° . It follows from results of Levy [1958] that $INF\ 2 \rightarrow LW$ in NBG° . Thus, in NBG, $INF\ 2 \leftrightarrow AC$.

THEOREM 5.14: $INF\ 2 \rightarrow LW$.

PROOF: It is sufficient to prove that every infinite linearly ordered set is structured. Suppose x is an infinite linearly ordered set. If x has an infinite number of finite initial segments or an infinite number of finite terminal segments then x has a countably infinite subset, so x is structured. Otherwise, there is some $u \in x$ such that both the initial segment and terminal segment generated by u are

infinite so, again, x is structured. (A similar proof gives us that $\text{INF } 2 \rightarrow \text{AC}_{\text{LO}}^{\text{LO}}$.)

U. Felgner [1974] proved that the statement:

The union of a well-ordered family of sets, each of which can be well-ordered, can be well-ordered,

does not imply the axiom of choice in NBG. However, it is easy to see (J. E. Rubin [1982]) that if either of the "well-ordered" in the above statement is replaced by "linearly ordered", or anything weaker than linearly ordered, the resulting statement is equivalent to the axiom of choice in NBG. Thus, we make the following definition:

U_Q^P : The union of a P-ordered family of sets, each of which can be Q-ordered, can be well-ordered.

For P and Q we take:

LO: linear ordering,

DLO: defined linear ordering,

WO: well-ordering,

DWO: defined well-ordering.

It is easy to see that the following implications are all provable in NBG°:

$$\text{U}_Q^P \rightarrow \text{AC}_Q^P, \text{ for all } P \text{ and } Q.$$

$$\text{AC} \leftrightarrow \text{U}_{\text{LO}}^{\text{LO}} \leftrightarrow \text{U}_{\text{WO}}^{\text{WO}},$$

(because $x = \text{U}\{x\}$).

$$\text{AC} \leftrightarrow \text{U}_{\text{LO}}^{\text{LO}} \leftrightarrow \text{U}_{\text{WO}}^{\text{WO}} \leftrightarrow \text{U}_{\text{DLO}}^{\text{DLO}} \leftrightarrow \text{U}_{\text{DWO}}^{\text{DWO}},$$

(because $x = \text{U}\{u : u \in x\}$).

$$\text{LW} \leftrightarrow \text{U}_{\text{DLO}}^{\text{LO}} \leftrightarrow \text{U}_{\text{DLO}}^{\text{WO}} \leftrightarrow \text{U}_{\text{DWO}}^{\text{LO}}.$$

$$\begin{array}{ccccc} \text{AC} & \rightarrow & \text{U}_{\text{LO}}^{\text{LO}} & \begin{array}{c} \nearrow \text{U}_{\text{LO}}^{\text{WO}} \\ \searrow \text{U}_{\text{WO}}^{\text{LO}} \end{array} & \rightarrow & \text{LW}. \end{array}$$

Since $LW \rightarrow AC$ in NBG, all the above forms are equivalent to AC in NBG.

In section 4, we defined a universal, UN, relation as follows:

$$R \text{ is UN on } X \text{ iff } X \times X \subseteq R.$$

We proved that $M5(A, UN) \leftrightarrow AC$, where

$M5(A, UN)$: Given any relation R , every set contains a $\subseteq$ -maximal subset on which R is UN.

We say that a relation R is vacuous, VC, iff $\bar{R}$ is UN. As defined previously in section 4, TR means transitive; AS, anti-symmetric; and C, connected. Let ID mean irreflexively directed and IRA, irreflexively ramified. We shall show that each of $M5(Q, VC)$, for $Q = TR, AS, AS \& TR, ID$, or IRA is equivalent to AC in NBG. Clearly, $M7 \rightarrow M5(Q, VC)$.

THEOREM 5.15: $M5(AS \& TR, VC) \leftrightarrow A$.

PROOF: If x is irreflexively partially ordered by R , $y \subseteq x$ is an R -anti-chain iff $y \times y \subseteq \bar{R}$.

COROLLARY 5.16: $M5(AS, VC) \rightarrow A$.

COROLLARY 5.17: $M5(TR, VC) \rightarrow A$.

COROLLARY 5.18: $M5(C, UN) \rightarrow A$.

PROOF: $M5(C, UN) \leftrightarrow M5(AS, VC)$ because a relation R is connected iff $\bar{R}$ is anti-symmetric, and R is UN iff $\bar{R}$ is VC.

THEOREM 5.19: $M5(IRA, VC) \rightarrow AC_{DLO}$.

PROOF: Let x be a family of pairwise disjoint sets such that for each $u \in x$, R_u linearly orders u . Define a relation R on $\cup x$ such that

$$a R b \text{ iff } (\exists u \in x)(a, b \in u \wedge a R_u b \wedge a \neq b).$$

The relation R is IRA on $\cup x$. A $\subseteq$ -maximal VC subset of $\cup x$ is a choice set for x .

THEOREM 5.20: $M \ 5 (ID, VC) \rightarrow AC_{DLO}$.

PROOF: Let x be a family of pairwise disjoint sets such that for each $u \in x$, R_u is an irreflexive linear ordering of u . Let

$$y = \{ \langle a, n \rangle : n \in \omega \wedge (\exists u \in x) (a \in u) \}.$$

Define a relation R on y such that

$$\langle a, n \rangle R \langle b, m \rangle \text{ iff } n < m \vee [n = m \wedge (\exists u \in x) (a, b \in u \wedge a R_u b)].$$

Clearly, R is ID on y . Thus, if z is a $\subseteq$ -maximal VC subset of y , the set of all first coordinates of elements of z is a choice set for x .

Our next statement, M 19, is obtained from M 8($\bar{D}$) by changing the word "set" to "class". (By 4.20, $x \bar{D} y$ iff $X \cap Y \neq \emptyset$.) With the aid of the axiom of foundation, it can be shown that M 19 is equivalent to the set form of the axiom of choice in NBG.

M 19: Every class has a $\subseteq$ -maximal subclass with the property $\bar{D}$.

Clearly, $M \ 19 \rightarrow M \ 8(\bar{D})$.

THEOREM 5.21: $NBG \vdash WO \ 2 \rightarrow M \ 19$.

PROOF: Suppose $X \neq \emptyset$ is a class and let $x \in X$. WO 2 implies there is an ordinal α such that $\alpha \approx x$. Let f be a bijection mapping α onto x . Also, $On \approx \alpha \times On$. Let F be a bijection of On onto $\alpha \times On$. For each $\beta \in On$, define $a(\beta)$ and $b(\beta)$ by $F(\beta) = \langle a(\beta), b(\beta) \rangle$.

For each $\beta \in On$, let $S(\beta)$ be the set of all elements of X which contains the $a(\beta)^{th}$ element of x , have rank $b(\beta)$, and have a non-empty intersection with each element of $S(\gamma)$ for $\gamma < \beta$. In symbols,

$$S(\beta) = \{ y \in X : (\forall \gamma \in \beta) (\forall z \in S(\gamma)) (y \cap z \neq \emptyset \wedge \rho(y) = b(\beta) \wedge f(a(\beta)) \in y) \}.$$

In NBG, $S(\beta)$ is a set for each $\beta \in On$. Let $Y = \bigcup_{\beta \in On} S(\beta)$,

then Y is a $\subseteq$ -maximal subclass of X with the property $\bar{D}$. Clearly, $Y \subseteq X$ and by construction, Y has the property $\bar{D}$. Suppose Y is not $\subseteq$ -maximal. Then there is a $z \in X \sim Y$ such that for all $y \in Y$, $z \cap y \neq \emptyset$. Since $x \in Y$, $z \cap x \neq \emptyset$. Since $f: \alpha \approx x$, there is a $\beta < \alpha$ such that $f(\beta) \in z$ and since $F: \text{On} \approx \alpha \times \text{On}$, there is an ordinal γ such that $F(\gamma) = \langle \beta, \rho(z) \rangle$. Consequently, $z \in S(\gamma) \subseteq Y$. This is a contradiction, q.e.d.

Moreover, if the word "set" is changed to "class" in each of $M 8(\bar{D}_m)$, $M 12$ and $M 13$, a similar proof shows that each is equivalent to the set form of the axiom of choice in NBG. However, as we saw earlier the class form of $M 14$ is equivalent to the set form of the axiom of choice in NBG°.

Our next statement in this section is a class form of AC 4.

AC 20: For every relation R whose domain is a set (its range may be a proper class) there is a function f such that $\mathcal{D}(f) = \mathcal{D}(R)$ and $f \subseteq R$.

Clearly, $\text{AC } 20 \rightarrow \text{AC } 4$. (It was shown in Felgner & Flannagan [1978] that this arrow is not reversible in NBG°.)

THEOREM 5.22: $\text{NBG} \vdash \text{AC } 4 \rightarrow \text{AC } 20$.

PROOF: Let R be an arbitrary relation whose domain is a set. Define a relation R' as follows:

$$R' = \{ \langle x, y \rangle \in R : (\forall z) (\langle x, z \rangle \in R \rightarrow \rho(y) \leq \rho(z)) \}.$$

In NBG, the class of all elements with a given rank is a set, so R' is a set. Applying AC 4 to R' gives us the desired function.

AC 21 is a class form of AC 18.

AC 21: For any non-empty set A , and for each relation B whose domain is A ,

$$\bigcap_{a \in A} \bigcup_{b \in B_a} x_{a,b} = \bigcup_{f \in \prod_{a \in A} B_a} x_{a,f(a)},$$

where B_a is defined to be $B''\{a\}$ for each $a \in A$.

We shall prove that $AC\ 20 \leftrightarrow AC\ 21$. The proof that $AC\ 20 \rightarrow AC\ 21$ is similar to the proof of 2.18, except we use $AC\ 20$ to construct f rather than $AC\ 1$.

THEOREM 5.23: $AC\ 21 \rightarrow AC\ 20$.

PROOF: Let R be a relation whose domain is the set A . (The range of R may be a proper class.) For each $a \in A$, define

$$B_u[a] = \{\langle u, t \rangle : \langle a, t \rangle \in R\}.$$

Then

$$u \in \bigcap_{a \in A} \bigcup_{b \in B_u[a]} b.$$

$AC\ 21$ implies that there exists an $f \in \prod_{a \in A} B_u[a]$ such that

for all $a \in A$, $u \in f(a)$. (We are using $AC\ 21$ with $X_{a,b} = b$.) Now, from the definition of $B_u[a]$, $f(a) = \{u, t\}$, where $\langle a, t \rangle \in R$. Define a function g with domain A such that for all $a \in A$, $g(a) = t$ iff $f(a) = \{u, t\}$. Then $\mathcal{J}(g) = \mathcal{J}(R)$ and $g \subseteq R$, q.e.d.

Next, we give a class form of $DC(\alpha)$ (see section 2, p. 19).

$DCC(\alpha)$: If R is a relation between subsets and elements of a class X such that $(\forall y \subseteq X) (y < \alpha \rightarrow (\exists u \in X) \langle y, u \rangle \in R)$ then there is a function f mapping α into x such that $(\forall \beta < \alpha) \langle f''\beta, f(\beta) \rangle \in R$.

Clearly, $DCC(\alpha) \rightarrow DC(\alpha)$ for all $\alpha \in On$. Felgner [1976a] proves that it is not the case that $DC(\alpha) \rightarrow DCC(\alpha)$ in NBG° but that $DC(\alpha) \rightarrow DCC(\alpha)$ in NBG . Thus, it follows from 2.20 and 2.21 that in NBG , $(\forall \alpha) DCC(\alpha)$ is equivalent to the set form of the axiom of choice. We give Felgner's proof of $DC(\alpha) \rightarrow DCC(\alpha)$ here.

THEOREM 5.24: $NBG \vdash (\forall \alpha) (DC(\alpha) \rightarrow DCC(\alpha))$.

PROOF: Suppose X and R satisfy the hypotheses of $DCC(\alpha)$. For each $\beta < \alpha$, let

$$z(\beta) = \{y \subseteq X: y \approx \beta \wedge (\forall z \subseteq X)(z \approx \beta \rightarrow \rho(y) \leq \rho(z))\}.$$

It follows from the axiom of foundation that if there is a $y \subseteq X$ such that $y \approx \beta$ then $z(\beta) \neq \emptyset$ and that $z(\beta)$ is a set for all $\beta < \alpha$. Therefore, $v = \bigcup_{\beta < \alpha} z(\beta)$ is a set. Next for each $y \in z(\beta)$, let

$$w(y, \beta) = \{a \in X: \langle y, a \rangle \in R \wedge (\forall b \in X)(\langle y, b \rangle \in R \rightarrow \rho(a) \leq \rho(b))\}.$$

Then $w(y, \beta)$ is a set and $s = \bigcup_{\beta < \alpha} \bigcup_{y \in z(\beta)} w(y, \beta)$ is a subset of X . Let $R' = R \cap (v \times s)$. Then apply $DC(\alpha)$ to s and R' to get the desired result.

The last forms we shall discuss in this section are forms of the trichotomy due to Kruse [1963], each of which is equivalent to PW.

For each cardinal κ , let $T(\kappa)$ stand for the following statement:

$T(\kappa)$: For all cardinals μ_1 and μ_2 , if $\kappa \leq \mu_1$, $\mu_2 \leq 2^\kappa$ then either $\mu_1 \leq \mu_2$ or $\mu_2 \leq \mu_1$.

Thus, $T(\kappa)$ is the trichotomy for cardinals between κ and 2^κ . We show that each of the following statements are equivalent to PW.

T 11: $\{\alpha \in \text{On}: T(\aleph_\alpha) \wedge \aleph_{\alpha+1} \leq 2^{\aleph_\alpha}\}$ is unbounded.

T 12: $\{\alpha \in \text{On}: T(\aleph_\alpha) \wedge T(2^{\aleph_\alpha})\}$ is unbounded.

T 13: For all cardinals κ and μ , if there is an ordinal α such that $\kappa, \mu \leq 2^{|\alpha|}$ then $\kappa \leq \mu$ or $\mu \leq \kappa$.

Clearly PW implies each of T 11, T 12 and T 13.

THEOREM 5.25: T 11 $\rightarrow$ PW.

PROOF: Let $\aleph$ be an aleph. Let α be an ordinal number such that $\aleph < \aleph_\alpha$, $\aleph_\alpha \not\leq 2^\aleph$, $T(\aleph_\alpha)$ and $\aleph_{\alpha+1} \leq 2^{\aleph_\alpha}$.

Thus, we have that both $\aleph_{\alpha+1}$ and $2^\aleph + \aleph_\alpha$ are between $\aleph_\alpha$ and $2^{\aleph_\alpha}$. Therefore, $T(\aleph_\alpha)$ implies that $\aleph_{\alpha+1}$ and $2^\aleph + \aleph_\alpha$ are comparable; that is,

$$\aleph_{\alpha+1} \leq 2^{\aleph} + \aleph_{\alpha} \quad \text{or} \quad 2^{\aleph} + \aleph_{\alpha} \leq \aleph_{\alpha+1}.$$

The first inequality is impossible by the definition of $\aleph_{\alpha}$ and the second inequality implies $2^{\aleph} \leq \aleph_{\alpha+1}$ which implies PW.

THEOREM 5.26: $T \ 12 \rightarrow PW$.

PROOF: Let $\aleph$ be any aleph and let α be an ordinal such that $\aleph < \aleph_{\alpha}$, $\aleph_{\alpha} \not\leq 2^{\aleph}$, $T(\aleph_{\alpha})$ and $T(2^{\aleph_{\alpha}})$. Using the proof of (*) in Theorem 3.10, we see that $T(2^{\aleph_{\alpha}})$ implies that $\aleph_{\alpha+1} \leq 2^{\aleph_{\alpha}}$. Now use 5.25 to obtain PW.

THEOREM 5.27: $T \ 13 \rightarrow PW$.

PROOF: Let $\beta \in \text{On}$, let $\kappa = 2^{\aleph_{\alpha}}$ and let $\mu = (2^{\aleph_{\beta}})^*$. Let $\alpha = \max(\omega_{\beta}, \mu)$. Then $\kappa, \mu \leq 2^{|\alpha|}$. Therefore, $T \ 13$ implies that $\kappa \leq \mu$ or $\mu \leq \kappa$. By the definition of μ , $\mu \not\leq \kappa$, and $\kappa \leq \mu$ implies PW.

The cofinality index of a limit ordinal λ , $\text{cof}(\lambda)$, is the smallest ordinal α such that there exists a 1-1, order preserving function f mapping α into λ , with $\mathcal{K}(f)$ cofinal in λ , that is, for each $\gamma \in \lambda$ there is a $\beta \in \alpha$ such that $\gamma \leq f(\beta)$.

T 14: $\{\alpha: T(\aleph_{\alpha}) \wedge \text{cof}(\omega_{\alpha+1}) \geq \beta\}$ is unbounded for each $\beta \in \text{On}$.

It is easy to show that PW implies T 14. For the proof that T 14 implies PW, the reader is referred to Kruse [1963, Theorem 5.3, p.217].

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§6. ALGEBRAIC FORMS

This section is devoted to algebraic statements which are equivalent to the axiom of choice. Many of these statements were used in algebra long before it was known they were equivalent to the axiom of choice. It wasn't until 1953 that J. Schmidt [1953] and D. Scott [1953] independently proved that several of these statements imply the axiom of choice. Subsequently, a wide variety of statements about many types of algebras were proved equivalent to the axiom of choice.

DEFINITION 6.1: Algebra

(1) $\mathcal{A} = \langle A, \{O_k : k \in K\} \rangle$ is called an (abstract) algebra or an algebraic system if A is a set and for each $k \in K$, O_k is a finitary operation defined on A . (A 0-ary operation is a constant and for $n > 0$, an n -ary operation is a function mapping the direct product of A with itself n -times into A .)

(2) $\mathcal{B} = \langle B, \{O_k \upharpoonright B : k \in K\} \rangle$ is called a sub-algebra of $\mathcal{A}$ if $B \subseteq A$ and, if O_k is an n -ary operation then for all $b_1, b_2, \dots, b_n \in B$, $O_k(b_1, b_2, \dots, b_n) \in B$. (That is, B is closed with respect to the operations O_k for $k \in K$.) $\mathcal{B}$ is called a proper subalgebra if $\mathcal{B} \neq \mathcal{A}$.

When it will not lead to any confusion, we shall call the set " A " of $\mathcal{A}$ the algebra and the set " B " of $\mathcal{B}$ the sub-algebra.

(The following definition of a lattice is given in Birkhoff [1948] p. 19.)

DEFINITION 6.2: Lattice

(1) $\mathcal{A} = \langle L, \{\vee, \wedge\} \rangle$ is called a lattice if the following

conditions are satisfied for every a, b , and c in L :

(a) L is a set and $\vee$ and $\wedge$ are binary operations,

(b) $a \vee b \in L$ and $a \wedge b \in L$,

(c) $a \vee a = a$ and $a \wedge a = a$,

(d) $a \vee b = b \vee a$ and $a \wedge b = b \wedge a$,

(e) $a \vee (b \vee c) = (a \vee b) \vee c$ and $a \wedge (b \wedge c) = (a \wedge b) \wedge c$,

(f) $a \vee (a \wedge b) = a$ and $a \wedge (a \vee b) = a$;

$\vee$ is called the join and $\wedge$ the meet. (It should be clear from the context when these symbols are lattice operations and when they are sentential connectives.)

(2) $1 \in L$ is called an identity element if for all $a \in L$, $a \wedge 1 = a$.

(3) $0 \in L$ is called a zero element if for all $a \in L$, $a \vee 0 = a$.

(4) $E \subseteq L$ is called an ideal if whenever $a, b \in E$, $a \vee b \in E$ and whenever $a \in L$ and $b \in E$, $a \wedge b \in E$. E is called a proper ideal if $E \neq L$. (If E is a proper ideal then $1 \notin E$.)

(5) $F \subseteq L$ is called a filter if whenever $a, b \in F$, $a \wedge b \in F$ and whenever $a \in L$ and $b \in F$, $a \vee b \in F$. F is called a proper filter if $F \neq L$. F is called an ultrafilter if F is a $\subseteq$ -maximal proper filter.

(6) If $a, b \in L$, $a \leq b$ if $a \vee b = b$ (It follows from 6.2 (1) (f) that $a \vee b = b$ if and only if $a \wedge b = a$).

(7) $x = \bigvee_{a \in A} a$ if $a \leq x$ for all $a \in A$ and if $a \leq y$ for all $a \in A$ then $x \leq y$. (x is called the least upper bound of A . $\bigvee_{a \in A} a$ may be written as $\bigvee A$.)

$x = \bigwedge_{a \in A} a$ if $x \leq a$ for all $a \in A$ and if $y \leq a$ for all $a \in A$ then $y \leq x$. (x is called the greatest lower bound of A . $\bigwedge_{a \in A} a = \bigwedge A$.)

(8) The lattice $\mathcal{L}$ is called complete if for each non-empty subset A of L , $\bigvee_{a \in A} a \in L$ and $\bigwedge_{a \in A} a \in L$.

(9) $c \in L$ is called compact if whenever $c \leq \bigvee_{a \in A} a$ where $A \subseteq L$, then there exists a finite subset B of A such that $c \leq \bigvee_{a \in B} a$.

(10) $E \subseteq L$ is called a prime ideal if E is an ideal and for all $a, b \in L$, if $a \wedge b \in E$ then either $a \in E$ or $b \in E$.

(Note: A lattice is an algebraic system, but as defined a subalgebra is not necessarily an ideal. However, if the lattice were defined as follows a subalgebra would be an ideal.

$\mathcal{L} = \langle L, \{\vee\} \cup \{O_a : a \in L\} \rangle$ where O_a is a unary operation such that $O_a(b) = a \wedge b$, for all $a \in L$.)

AL 1: Every lattice with an identity element and at least one other element contains a $\subseteq$ -maximal proper ideal.

AL'1: Every proper ideal of a lattice with an identity element can be extended to a $\subseteq$ -maximal proper ideal.

(If in AL 1 and AL'1 we replace "identity" by "zero" and " $\subseteq$ -maximal proper ideal" by "ultrafilter", the resulting statements, DAL 1 and DAL'1, are equivalent.)

AL'2: If A is an algebraic system, B a subalgebra, and $a \in A$, but $a \notin B$, then there exists a $\subseteq$ -maximal subalgebra which contains B as a subset, but does not contain a as an element.

DEFINITION 6.3: C is called a finitary closure operator if C is a monotone unary operation such that for every class X :

$$C(X) = \bigcup \{C(y) : y \subseteq X \text{ and } y \text{ is finite}\}.$$

If $C(X) \subseteq X$ then X is called C-closed.

AL'3: If x is an arbitrary set, C a finitary closure operator, P a property of finite character, and y a

C-closed subset of x which has the property P , then there is a $\subseteq$ -maximal C-closed subset of x which contains y as a subset and has the property P .

AL'1 was given by D. Scott [1953] and AL'2 was given by Robert Blair in 1957 in an unpublished paper. AL'3 is a generalization of AL'1 and AL'2. AL'2 and AL'3 are false in the corresponding forms without primes as there may exist no subsets with the desired property. Dzik [1981] also published some equivalents similar to AL'3.

We shall prove the following implications:

$$M\ 3 \rightarrow AL'3 \rightarrow AL'2 \rightarrow AL'1 \rightarrow AL\ 1 \rightarrow M\ 7.$$

In section 4 it was shown that $M\ 3$ and $M\ 7$ are equivalent. The proof that $M\ 3 \rightarrow AL'3$ is similar to the proof of 4.3. AL'1 is a special case of AL'2 and it is clear that $AL'1 \rightarrow AL\ 1$.

THEOREM 6.4: $AL'3 \rightarrow AL'2$.

PROOF: Let $\mathcal{A} = \langle A, \{O_k : k \in K\} \rangle$ be an algebraic system. If $x \subseteq A$, define $C(x)$ as follows: $a \in C(x)$ if and only if there exist elements $a_1, a_2, \dots, a_n \in x$ and an operation $O_k, k \in K$ such that $a = O_k(a_1, a_2, \dots, a_n)$. Since O_k is a finitary operation for all $k \in K$, C is a finitary closure operator, and B is a subalgebra of $\mathcal{A}$ iff B is C -closed. Moreover, the property of excluding an element from a set is a property of finite character. Hence, AL'2 is a special case of AL'3.

THEOREM 6.5: $AL\ 1 \rightarrow M\ 7$.

PROOF: Let x be a set and P a property of finite character. Let us assume that x does not have the property P , for if it does then it is the maximal subset which has the property P . Let us also assume that at least one non-empty subset of x has the property, otherwise the empty set is the required maximal set. Let $y = \{t : t \subseteq x \text{ and } P(t)\} \cup \{x\}$. If $s, t \in y$, define $s \wedge t = s \cap t$ and

$$s \vee t = \begin{cases} s \cup t & \text{if } P(s \cup t) \\ x & \text{otherwise} \end{cases},$$

then $\langle y, \{\vee, \wedge\} \rangle$ is a lattice. The identity element of the lattice is x . By AL 1, $\langle y, \{\vee, \wedge\} \rangle$ has a maximal proper ideal z . We shall show that $\cup z$ is a maximal subset of x with the property P . Clearly, $\cup z \subseteq x$. Next, $x \notin z$ since z is a proper ideal. Let t be a finite subset of $\cup z$. Then there is a finite number of elements $s_1, s_2, \dots, s_n$ of z such that $t \subseteq \bigcup_{i=1}^n s_i$. Since z is an ideal and $x \notin z$, $\bigcup_{i=1}^n s_i \in z$, which implies $P(\bigcup_{i=1}^n s_i)$. Consequently, $P(t)$, which implies $P(\cup z)$. $\cup z$ is maximal since z is maximal, q.e.d.

In the proof of 6.5 we just required AL 1 to hold for complete lattices.

AL'4 was given in 1959 by E. W. Beth. It is a set theoretical interpretation of the syntactical form that Beth actually states. It is one of the most general forms of the axiom of choice that we have come across. Before stating it, we give the following definition:

DEFINITION 6.6:

(a) Let m and n be natural numbers and let R be an n -ary relation. Then for $m < n$, $\langle x_1, x_2, \dots, x_m \rangle \in \mathcal{P}_n^m(R)$ if there exist elements $x_{m+1}, x_{m+2}, \dots, x_n$ such that $\langle x_1, x_2, \dots, x_n \rangle \in R$.

(b) Let R be a quinary relation, then $R:X$ if for all natural numbers m and n , for all $u \in X^m$ and for all w such that $\langle w, m, n \rangle \in \mathcal{P}_5^3(R)$ there exists a $v \in X^n$ such that $\langle w, m, n, u, v \rangle \in R$. (Let $X^0 = \{\emptyset\}$ and $X^1 = \{0\} \times X$.)

AL'4: If x is an arbitrary set, $y \subseteq x$, and $R:y$ then there is a $\subseteq$ -maximal subset z of x such that $y \subseteq z$ and $R:z$.

The proof that $M_3 \rightarrow AL'4$ is similar to the proof of 4.3. We shall show that $AL'3$ is a special case of $AL'4$.

THEOREM 6.7: $AL'4 \rightarrow AL'3$.

PROOF: Let x be an arbitrary set, C a finitary closure operator, P a property of finite character, and y a C -closed subset of x which has the property P . Define a quinary relation R as follows:

$\langle w, m, n, u, v \rangle \in R$ iff m and n are natural numbers;
 $u \in x^m$; $v \in x^n$; and either $n = 0$, $w = \emptyset$, and $P(\mathcal{C}(u))$,
 or $n = 1$, $w \in C(\mathcal{C}(u))$, and $v = \langle 0, w \rangle$.

Then, we claim, $R:z$ if and only if $P(z)$ and z is C -closed. (We leave the details of the proof to the reader.) Now, it is clear that $AL'4 \rightarrow AL'3$.

The following maximal principle, $AL\ 5$, is due to Mrowka [1955, 1958]. (Mrowka [1955] showed that the axiom of choice implies $AL\ 5$, but his proof of the converse was incorrect. This was corrected in Mrowka [1958].) $AL\ 5$ deals with a special type of lattice, a Boolean algebra (see 6.2).

DEFINITION 6.8: Boolean Algebra

$\mathcal{B} = \langle B, \{\vee, \wedge\} \rangle$ is said to be a Boolean algebra if the following conditions are satisfied:

- (a) $\mathcal{B}$ is a lattice.
- (b) $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$ and $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$ for all $a, b, c \in B$.
- (c) 0 and 1 are in B .
- (d) For every $a \in B$ there exists an element $b \in B$ such that $a \vee b = 1$ and $a \wedge b = 0$. (It can be shown that b is unique. b is called the complement of a and is denoted by $\sim a$.)

$AL\ 5$: If B is a Boolean algebra and $S \subseteq B$ such that $0 \notin S$, then there exists a $\subseteq$ -maximal proper ideal disjoint from S .

(See 6.2(4) for the definition of a proper ideal. Every non-empty ideal E in a Boolean algebra B contains 0 , for if $a \in E$ then $\sim a \wedge a = 0 \in E$.)

The proof that $AL'3 \rightarrow AL\ 5$ is analogous to the proof of 6.4 since a Boolean algebra can be defined as an algebraic system in such a way that an ideal is a subalgebra (see the remarks following 6.2).

THEOREM 6.9: $AL\ 5 \rightarrow M\ 7$.

PROOF: Let x be an arbitrary set and P a property of finite character. Let $\mathcal{B} = \langle B, \{\vee, \wedge\} \rangle$ be defined as follows: $B = \mathcal{P}(x)$, $\vee = \cup$, and $\wedge = \cap$, ($0 = \emptyset$ and $1 = x$), then $\mathcal{B}$ is the Boolean algebra of the set of all subsets of x . Let $S = \{y: y \subseteq x \text{ and } \neg P(y)\}$. Then $S \subseteq B$ and $\emptyset \notin S$ since $P(\emptyset)$ for every property of finite character P . Therefore, $AL\ 5$ implies that there exists a maximal proper ideal E disjoint from S . It is clear that $\cup E$ is the desired maximal subset of x .

The Boolean Prime Ideal Theorem (BPI) is the statement:

BPI: Every Boolean algebra contains a proper prime ideal.

In a Boolean algebra, a proper ideal is prime iff it is $\subseteq$ -maximal. We obtain $AL\ 1$ from BPI by changing "Boolean algebra" to "lattice with a unit element and at least one other element", and we obtain BPI from $AL\ 5$ by substituting, " $\emptyset$ " or " $\{1\}$ " for " S ". Thus, $AL\ 1$ and $AL\ 5$ appear to be minor variations of BPI. However, in 1967, Halpern & Levy [1971] proved that BPI does not imply the axiom of choice in NBG. (Earlier, Halpern [1961, 1964] proved that BPI does not imply the axiom of choice in NBG o .) The changes made in BPI to obtain $AL\ 1$ and $AL\ 5$ are not as minor as one may think at first glance.

P. Howard [1977] discussed variations of $AL\ 5$ obtained by placing conditions on the set S .

$AL\ 5(\vee)$: If B is a Boolean algebra and $S \subseteq B$ such that $0 \notin S$ and S is closed with respect to $\vee$ (join), then

there exists a $\subseteq$ -maximal ideal disjoint from S .

AL 5($\sim$): If B is a Boolean algebra and $S \subseteq B$ such that $0 \notin S$ and S is closed with respect to $\sim$ (complementation), then there exists a $\subseteq$ -maximal ideal disjoint from S .

It is clear that $AL\ 5 \rightarrow AL\ 5(V)$ and $AL\ 5 \rightarrow AL\ 5(\sim)$.

THEOREM 6.10: $AL\ 5(V) \rightarrow AL\ 5$.

PROOF: Suppose B is a Boolean algebra, $S \subseteq B$ and $0 \notin S$. Let $\bar{S}$ be the closure of S with respect to $\vee$. Then for any ideal $I \subseteq B$, $I \cap S = \emptyset$ iff $I \cap \bar{S} = \emptyset$. Thus, $AL\ 5(V)$ implies $AL\ 5$.

THEOREM 6.11: $AL\ 5(\sim) \rightarrow AC\ 2$.

PROOF: Suppose x is a set of pairwise disjoint sets each of which has at least four elements. (There is no loss of generality because each $u \in x$ can be replaced by $u \times 4$.) Let $B = \mathcal{P}(U\ x)$, where $\vee$ is $\cup$, $\wedge$ is $\cap$, 0 is $\emptyset$ and 1 is Ux . Let

$$S' = \{w \in B: (\forall u \in x) (|w \cap u| \neq 1 \text{ and } |\sim w \cap u| \neq 1)\}$$

and let $S = S' \sim \{0,1\}$. Then S is closed with respect to complementation. Let I be a $\subseteq$ -maximal ideal in B disjoint from S . Then for each $u \in x$ there is exactly one $a \in u$ such that $\{a\} \in I$. There is at least one because of the maximality of I and the definition of S . Suppose there is a $u \in x$ and $a, b \in u$, $a \neq b$, such that $\{a\}, \{b\} \in I$. Then $\{a,b\} \in I$. But $\{a,b\} \in S$, contradicting the fact that $S \cap I = \emptyset$. A choice set for x is

$$\{a \in Ux: (\exists u \in x) (a \in u \text{ and } \{a\} \in I)\}.$$

In $AL\ 5$, if we require that S be closed with respect to $\wedge$ (meet) the resulting statement is equivalent to BPI . That is the statement:

SM: If B is a Boolean algebra and $S \subseteq B$ such that $0 \notin S$ and S is closed with respect to $\wedge$, then there is a

$\subseteq$ -maximal ideal disjoint from S .

To show that $SM \rightarrow BPI$, take $S = \{1\}$. To prove that $BPI \rightarrow SM$, first note that $BPI \leftrightarrow BPI'$, where

BPI' : If I is a proper ideal in a Boolean algebra B , then I can be extended to a proper $\subseteq$ -maximal ideal.

(Clearly, $BPI' \rightarrow BPI$. To prove the converse, show that if I is a proper ideal in B , then applying BPI to the factor algebra, B/I , induces a $\subseteq$ -maximal ideal in B which contains I .)

Let $S \subseteq B$ be such that $0 \notin S$ and S is closed with respect to $\wedge$. Let $J = \{a \in B : (\exists b \in S)(a \leq \sim b)\}$. The set J is a proper ideal in B . Use BPI' to extend J to a $\subseteq$ -maximal proper ideal, I . Then I is a $\subseteq$ -maximal proper ideal disjoint from S .

On the other hand, suppose we restate SM as follows:

S : If B is a Boolean algebra and $S \subseteq B$ such that S is closed with respect to $\wedge$ then there is a $\subseteq$ -maximal proper ideal I such that $I \cap S \subseteq \{0\}$.

It is clear that S implies BPI , but it is not known whether BPI implies S . A corresponding change in the wording of AL 5, AL 5($\vee$), or AL 5($\sim$) does not change the strength of these statements. They remain equivalent to the axiom of choice.

Andreas Blass raised the question whether the following weakened form of AL 1 implies the axiom of choice.

AL 1D: Every distributive lattice with an identity element and at least one other element contains a $\subseteq$ -maximal proper ideal.

The dual to AL 1D, DAL 1D, is clearly equivalent.

DAL 1D: Every distributive lattice with a zero element and at least one other element contains an ultrafilter.

Clearly, $AL\ 1 \rightarrow AL\ 1D \leftrightarrow DAL\ 1D$. We shall prove $DAL\ 1D \rightarrow M\ 14$. (The proof is due to Klimovsky [1958].)

THEOREM 6.12: $DAL\ 1D \rightarrow M\ 14.$

PROOF: Let x be a non-empty set and let $B = \mathcal{P}(Ux)$. Then $\mathcal{B} = \langle B, \{U, \cap, \sim\} \rangle$ is a Boolean algebra, where U is set union, $\cap$ is set intersection, $\sim$ is the relative complement ($\sim u = Ux \setminus u$), $0 = \emptyset$ and $1 = Ux$. Also $x \subseteq B$. Let $\bar{x}$ be the closure of x with respect to U and $\cap$ in $\mathcal{B}$. If x has the f.i.p. (every finite intersection is non-empty) then x itself is the $\subseteq$ -maximal subset. If not then $0 \in \bar{x}$ so $\langle \bar{x}, \{U, \cap\} \rangle$ is a distributive lattice which satisfies the hypothesis of DAL 1D. Thus, there is an ultrafilter $u \subseteq \bar{x}$. Then, it is easy to see that $u \cap x$ is a $\subseteq$ -maximal subset of x with the f.i.p., q.e.d.

It follows from the work of Rubin & Scott [1954] that if the words " $\subseteq$ -maximal proper" in AL 1D are replaced by "prime", the resulting statement is equivalent to BPI.

We shall consider next seven more algebraic equivalents of the axiom of choice.

AL 6: If $\mathcal{L} = \langle L, \{V, \wedge\} \rangle$ is a complete lattice, $a, b, c \in L$, $a \leq b$, c is compact, $c \leq b$ and $c \not\leq a$ then there exists an element $m \in L$ such that $a \leq m \leq b$, $c \not\leq m$ and if there is a $d \in L$ such that $m < d \leq b$ then $c \leq d$ (In other words, m is a maximal element in the interval between a and b which is not greater than or equal to c).

Before stating the remaining forms in this section we require some further definitions.

DEFINITION 6.13: Let $\mathcal{A} = \langle A, \{O_k : k \in K\} \rangle$ be an algebraic system (see 6.1).

(1) A subalgebra B is said to be meet-irreducible if whenever $B = \cap \mathcal{S}$ where $\mathcal{S}$ is a set of subalgebras then $B \in \mathcal{S}$.

(2) $B \subseteq A$ is said to generate A if for each $a \in A$ either $a \in B$ or there exists a natural number $n \geq 1$, $k \in K$ and $b_1, b_2, \dots, b_n \in A$ such that B generates $\{b_1, b_2, \dots, b_n\}$ and $a = O_k(b_1, b_2, \dots, b_n)$.

(3) $B \subseteq A$ is said to be algebraically dependent if there exists a natural number $n \geq 1$, $k \in K$, $b \in B$ and $a_1, a_2, \dots, a_n \in A$ such that $B \sim \{b\}$ generates $\{a_1, a_2, \dots, a_n\}$ and $b = 0_k(a_1, a_2, \dots, a_n)$. Otherwise, B is said to be algebraically independent.

(4) $B \subseteq A$ is said to be a basis for A if B is algebraically independent and generates A . (A basis is a $\subseteq$ -maximal algebraically independent subset of A .)

(5) An element $x \in A$ is said to be absolutely dispensible if for all sets $S \subseteq A$, whenever A is the smallest subalgebra containing S as a subset then A is the smallest subalgebra containing $S \sim \{x\}$ as a subset.

AL 7 - AL 12 all refer to an algebraic system $\mathcal{A}$ and subalgebras in that system.

AL 7: Every subalgebra is the intersection of all meet-irreducible subalgebras containing it as a subset.

AL 8: Every proper subalgebra has a proper meet-irreducible subalgebra containing it as a subset.

AL 9: The set of absolutely dispensible elements is equal to the intersection of the set of all $\subseteq$ -maximal proper subalgebras. (We make the convention that $\cap \emptyset$ is the whole algebra.)

AL 10: For every element in a basis there exists a $\subseteq$ -maximal subalgebra which does not contain it as an element.

AL 11: If A has a finite basis and $B \subset A$ is a subalgebra then there exists a $\subseteq$ -maximal proper subalgebra which contains B as a subset.

AL 12: Every algebra has a basis.

Diener [1956] proved AL 6 and AL 8 equivalent to a form of the axiom of choice. AL 7 was derived from the axiom of choice by Birkhoff & Frink [1948]. However, there is an error in their proof (the proof of Lemma 4, pp. 301-2, is incorrect)

which was later noted and corrected by Diener [1956]. McCoy [1938] proved a special case of AL 7 and Fuchs [1949] gave a correct proof of AL 7. (Actually, Fuchs proved a special case, but his proof extends to the general case virtually unchanged.) The proof given here is similar to Diener's. AL 9 is due to Frattini [1885] and Neumann [1937]. Frattini in 1885 only proved the theorem for finite groups and did not require the axiom of choice. The proof given here is essentially the same as the proof given by Neumann in 1937. Krull [1929] derived a proposition which is similar to AL 11 from the well-ordering theorem. Schmidt [1953] proved that AL 7, AL 9, AL 10, and AL 11 imply a form of the axiom of choice. (See also Schmidt [1962].) AL 12 is related to a form given in Schmidt [1962].

We shall prove

$$M\ 3 \rightarrow AL\ 6 \rightarrow AL\ 7 \rightarrow AL\ 8 \rightarrow AC\ 3,$$

$$AL'3 \rightarrow AL\ 9 \rightarrow AL\ 10 \rightarrow M\ 7,$$

$$AL'3 \rightarrow AL\ 11 \rightarrow M\ 7,$$

$$M\ 7 \rightarrow AL\ 12 \rightarrow AC\ 2.$$

It is clear that $AL\ 7 \rightarrow AL\ 8$ and since an element in the basis is not absolutely dispensable, $AL\ 9 \rightarrow AL\ 10$. Being algebraically independent is a property of finite character, so $M\ 7 \rightarrow AL\ 12$.

THEOREM 6.14: $M\ 3 \rightarrow AL\ 6$.

PROOF: Let $\mathcal{L} = \langle L, \{\vee, \wedge\} \rangle$ be a complete lattice and for $a, b \in L$, $a \leq b$, let $[a, b] = \{d : d \in L \text{ and } a \leq d \leq b\}$. Suppose $c \in L$ is compact, $c \leq b$ and $c \not\leq a$. Let $\mathcal{D} = \{D : D \subseteq [a, b] \text{ and } D \neq \emptyset \text{ and } c \not\leq \bigvee D\}$. $\mathcal{D} \neq \emptyset$ since $\{a\} \in \mathcal{D}$. Let $\mathcal{N}$ be a non-empty nest contained as a subset in $\mathcal{D}$. We shall show that $\bigcup \mathcal{N} \in \mathcal{D}$. Clearly, if $\mathcal{N} \neq \emptyset$, then $\bigcup \mathcal{N} \neq \emptyset$ and $\bigcup \mathcal{N} \subseteq [a, b]$. Suppose $c \leq \bigvee \bigcup \mathcal{N}$, then since c is compact there exist $d_1, d_2, \dots, d_k, n_1, n_2, \dots, n_k$, $d_i \in n_i \in \mathcal{N}$ such that $c \leq \bigvee \{d_1, d_2, \dots, d_k\}$. Since $\mathcal{N}$ is

a nest, for some $j \in \{1, 2, \dots, k\}$, $N_i \subseteq N_j$ for all $i = 1, 2, \dots, k$, therefore $c \subseteq \bigvee N_j$. This is a contradiction since $N_j \in \mathcal{D}$. Hence $\bigcup \mathcal{D} \in \mathcal{D}$ so that by M 3, $\mathcal{D}$ has a maximal element M .

Let $m = \bigvee M$; then $a \leq m \leq b$ and $c \not\leq m$. Suppose there is a $d \in [a, b]$ so that $m < d \leq b$. This implies $M \subset M \cup \{d\}$ so that $M \cup \{d\} \notin \mathcal{D}$. Therefore, $c \leq \bigvee (M \cup \{d\}) = d$, q.e.d.

Before proceeding with the proof that AL 6 $\rightarrow$ AL 7 we prove a lemma due to Buchi [1952].

$\mathcal{L} = \langle L, \{\vee, \wedge\} \rangle$ is said to be an algebraic lattice if L is the set of all subalgebras of an algebraic system. The meet of two elements of L is ordinary set intersection, but the join of two elements is the smallest subalgebra containing them both. (This is usually not ordinary set union.)

It is easily verified that an algebraic lattice is a complete lattice.

LEMMA 6.15: Every element in an algebraic lattice is the join of compact elements.

PROOF: Let $\mathcal{L} = \langle L, \{\vee, \wedge\} \rangle$ be an algebraic lattice and let $\mu \in q \in L$. We wish to show that the subalgebra generated by μ , (the smallest subalgebra containing μ), denoted by (μ) , is a compact element of L . Since (μ) is a subalgebra and L is the set of all subalgebras of an algebra, $(\mu) \in L$. Suppose $M \subseteq L$ and $(\mu) \leq \bigvee_{a \in M} a$, then $\mu \in \bigvee_{a \in M} a$. This implies that there exists a finite subset N of M such that $\mu \in \bigvee_{a \in N} a$. Since $\bigvee_{a \in N} a$ is a subalgebra it follows that $(\mu) \leq \bigvee_{a \in N} a$. Hence (μ) is compact.

Let $q \in L$; then clearly $q = \bigvee_{\mu \in q} (\mu)$, q.e.d.

THEOREM 6.16: AL 6 $\rightarrow$ AL 7.

PROOF: Let A be an algebraic system and $\langle L, \{\vee, \wedge\} \rangle$ the algebraic lattice of the set of all subalgebras of A . Let $B \in L$ and let D be the intersection of all meet-irreducible

subalgebras containing B . Suppose $B < D$. Then there exists a compact element $C \in L$ such that $C \not\leq B$ and $C \leq D$. For suppose $\mu \in D \sim B$, then by 6.15, (μ) is compact and clearly $(\mu) \not\leq B$ but $(\mu) \leq D$. Let $C = (\mu)$.

By AL 6, there exists an $M \in L$ such that $B \leq M \leq A$, $C \not\leq M$ and M is maximal. Suppose M is not meet-irreducible. Then there exists a subset $K \subseteq L$ such that $M = \bigwedge K$ and $M \notin K$. Let $N \in K$; then $M < N$. Since M is maximal $C \leq N$ for all $N \in K$. Therefore $C \leq \bigwedge K = M$. This is a contradiction so that M is meet-irreducible.

$B \leq M$ and D is the intersection of all meet-irreducible ideals which contain B , therefore $D \leq M$. Since $C \leq D$ it follows that $C \leq M$ and this again is a contradiction. Hence $B = D$, q.e.d.

THEOREM 6.17: AL 8 $\rightarrow$ AC 3.

PROOF: Let f be an arbitrary function whose domain is s and assume for all $x \in s$, $f(x)$ has at least two elements. We define an algebra $\mathcal{A} = \langle A, \{O_u : u \in A\} \rangle$ as follows:

$$A = \{ \langle x, y \rangle : x \in s \text{ and } y \in f(x) \}$$

$$O_u(v, w) = \begin{cases} u & \text{if } v = \langle x, y \rangle, w = \langle x, z \rangle, y \neq z, \\ v & \text{otherwise.} \end{cases}$$

A subset of A is a proper subalgebra if it is a function. It is clear that a subalgebra of $\mathcal{A}$ is meet-irreducible if and only if it is maximal. AL 8 implies, therefore, that there exists a maximal subalgebra $B \subset A$. If we let $g = B$, then g is a function, $\mathcal{I}(g) = \mathcal{I}(f) = s$, and for all $x \in s$, $g(x) \in f(x)$, q.e.d.

THEOREM 6.18: AL'3 $\rightarrow$ AL 9.

PROOF: Let A be an algebraic system. Let $D \subseteq A$ be the set of all absolutely dispensible elements and let $\mathcal{M}$ be the set of all maximal proper subalgebras. We wish to prove $D = \bigcap \mathcal{M}$.

Suppose $x \notin \cap \mathcal{M}$. Then there is a $M \in \mathcal{M}$ such that $x \notin M$. Therefore the smallest subalgebra containing $M \cup \{x\}$ is A , so that $x \notin D$. This implies $D \subseteq \cap \mathcal{M}$.

Now, suppose $x \notin D$. For any set $S \subseteq A$ let $C(S)$ be the smallest subalgebra containing S ; then C is a finitary closure operator. If $x \notin D$, then there exists a set $T \subset A$ such that $C(T) \subset A$ and $C(T \cup \{x\}) = A$. We define a property of finite character P as follows: for any $S \subseteq A$, $P(S) \leftrightarrow x \notin S$. Clearly, $P(C(T))$. Therefore AL'3 implies there is a maximal subalgebra M which contains $C(T)$ and $x \notin M$. Hence $x \notin \cap \mathcal{M}$. This implies $\cap \mathcal{M} \subseteq D$, q.e.d.

THEOREM 6.19: AL 10 $\rightarrow$ M 7.

PROOF: Let A be an arbitrary set and P a property of finite character. Assume A does not have the property P . For each $x \in A$ define n -ary operators on

$K = \{\langle y_1, y_2, \dots, y_n \rangle : y_1, y_2, \dots, y_n \in A\}$ as follows:

If $\langle y_1, y_2, \dots, y_n \rangle \in K$,

$$O_x^n(y_1, y_2, \dots, y_n) = \begin{cases} x & \text{if } \neg P(\{y_1, y_2, \dots, y_n\}), \\ y_1 & \text{otherwise.} \end{cases}$$

$S \subset A$ is a subalgebra if and only if $P(S)$. There exists a finite set which does not have the property P , otherwise A would have the property P . Therefore, there is a smallest finite set B which does not have the property P . B is a basis for A since no proper subalgebra contains B , and every proper subset of B is a subalgebra.

Suppose $b \in B$. AL 10 implies that there exists a maximal subalgebra M such that $b \notin M$.

If $M \subset N \subseteq A$ and $P(N)$, we have $P(N \sim \{b\})$. Since $M \subseteq N \sim \{b\}$, $N \sim \{b\} = M$ from the maximality of M . Hence $N = M \cup \{b\}$. Thus either M or $M \cup \{b\}$ is a maximal subset of A with the property P .

THEOREM 6.20: AL'3 $\rightarrow$ AL 11.

PROOF: Suppose A has a finite basis $B = \{b_1, b_2, \dots, b_n\}$ and $B_0 \subset A$ is a proper subalgebra. For any set $S \subseteq A$ we

define $C(S)$ as the smallest subalgebra containing S . Then C is a finitary closure operator. Define the property P_j , $j = 1, 2, \dots, n$ as follows: for any set $S \subseteq A$, $P_j(S) \leftrightarrow b_j \notin S$. P_j is a property of finite character for each $j = 1, 2, \dots, n$.

AL'3 implies that we may construct the following finite sequence of subalgebras:

If $b_j \in B_{j-1}$ then $B_j = B_{j-1}$, $j = 1, 2, \dots, n$.

If $b_j \notin B_{j-1}$ then B_j is a maximal subalgebra

containing B_{j-1} as a subset and $b_j \notin B_j$, $j = 1, 2, \dots, n$.

By construction, each B_j is a proper subalgebra and $B_{j-1} \subseteq B_j$ for each $j = 1, 2, \dots, n$. Hence each B_j contains B_0 as a subset. We claim B_n is a maximal proper subalgebra containing B_0 . For suppose B_n is not maximal. Then there exists a proper subalgebra D such that $B_n \subset D$. Since $D \neq A$, for some $b_k \in B$, $b_k \notin D$. But then $B_k \subset D$ which contradicts the definition of B_k .

THEOREM 6.21: AL 11 $\rightarrow$ M 7.

PROOF: Similar to 6.19. Take the subalgebra of AL 11 to be $\emptyset$.

THEOREM 6.22: AL 12 $\rightarrow$ AC 2.

PROOF: Let x be a set of non-empty pairwise disjoint sets. Let $A = \bigcup x$. For each $a \in A$ we define a unary operation O_a on A such that for each $b \in A$

$$O_a(b) = \begin{cases} a & \text{if } (\exists u \in x) a, b \in u, \\ b & \text{otherwise.} \end{cases}$$

A subset $B \subseteq A$ is algebraically independent iff all the elements of B belong to different elements in x . Thus, a basis for A is a choice set for x .

It is also clear that the following variations of AL 12 are equivalent to the axiom of choice.

AL'12: Any algebraically independent subset of an algebra can be extended to a basis.

AL 13: Any subset of an algebra which generates the algebra contains a basis.

We shall next consider some statements about groups and related structures which are equivalent to the axiom of choice. For convenience we shall define the following algebras:

DEFINITION 6.23:

(a) A groupoid is an algebra $\langle A, \{o\} \rangle$: where o is a binary operation.

(b) A cancellative groupoid, $\langle A, \{o\} \rangle$ is a groupoid which satisfies the cancellation laws:

$$(\forall a, b, c \in A) [aob = aoc \rightarrow b = c) \wedge (aoc = boc \rightarrow a = b)].$$

(c) An abelian (commutative) groupoid, $\langle A, \{o\} \rangle$ is a groupoid which satisfies the commutative law:

$$(\forall a, b \in A) (aob = boa)$$

(d) A semigroup, $\langle A, \{o\} \rangle$ is a groupoid which satisfies the associative law: $(\forall a, b, c \in A) (ao(boc) = (aob)oc)$.

(e) A quasigroup, $\langle A, \{o\} \rangle$ is a cancellative groupoid such that

$$(\forall a, b \in A) (\exists x, y \in A) (aox = b \wedge yoa = b) .$$

(f) A loop is an algebra $\langle A, \{o, e\} \rangle$ such that $\langle A, \{o\} \rangle$ is a quasigroup and e is the identity element:

$(\forall a \in A) (aoe = eoa = a)$. (The cancellation law implies that the identity element is unique.)

(g) A group is an algebra $\langle A, \{o, e\} \rangle$ where $\langle A, \{o\} \rangle$ is a quasigroup, o is associative, and e is the identity element. (In a group, each element a has a unique inverse, a^{-1} , such that $aoa^{-1} = a^{-1}oa = e$, and the cancellation laws hold. Frequently it is necessary to consider a group as the algebra $\langle A, \{o, e, {}^{-1}\} \rangle$.)

(h) A ring is an algebra $\langle A, \{+, \cdot, 0\} \rangle$ such that $\langle A, \{+, 0\} \rangle$ is an abelian group, $\cdot$ is associative and the distributive laws hold:

$$(\forall a, b, c \in A) [(a \cdot (b+c) = a \cdot b + a \cdot c) \wedge ((a+b) \cdot c = a \cdot c + b \cdot c)].$$

(We call a ring commutative if $\cdot$ is commutative.)

(i) An integral domain, $\langle A, \{+, \cdot, 0\} \rangle$, is a commutative ring such that $(\forall a, b \in A) (a \neq 0 \wedge b \neq 0 \rightarrow a \cdot b \neq 0)$.

(j) A field is an algebra $\langle A, \{+, \cdot, 0, 1\} \rangle$ such that $\langle A, \{+, \cdot, 0\} \rangle$ is a commutative ring and $\langle A \setminus \{0\}, \{\cdot, 1\} \rangle$ is an abelian group.

(k) If $\langle A, \{+, \cdot, 0, 1\} \rangle$ is a field and there is an $n \in \omega \setminus \{0\}$ such that $\underbrace{a + \dots + a}_{n \text{ times}} = na = 0$, then the

smallest positive integer n with this property is called the characteristic of the field. (Since a field is an integral domain, no divisors of zero, the characteristic is a prime.) If there is no such positive integer n such that $na = 0$, we say the field has characteristic zero.)

The first result is due to Hajnal and Kertesz [1972] who prove the following statement is equivalent to the axiom of choice.

AL 14: On every non-empty set there exists a cancellative groupoid.

THEOREM 6.24: WO 2 $\rightarrow$ AL 14.

PROOF: Suppose x is a non-empty set. If x is finite, say x has n elements, then we can take as our groupoid, the natural numbers modulo n . (In fact, this is a commutative ring with multiplicative identity). If x is infinite, it follows from the well ordering theorem that there is a set y such that x is equipollent to $z = \mathcal{P}_{<\omega}(y)$ = the set of all finite subsets of y . (By WO 2, there is an infinite ordinal α such that $x \approx \alpha$. It is easy to show $\mathcal{P}_{<\omega}(\alpha) \approx \alpha$.)

Define the operation $\circ$ on z to be symmetric difference. That is, if $u, v \in z$, $u \circ v = (u \cup v) \setminus (u \cap v)$.

It is easy to show that $\langle z, \{0, \emptyset\} \rangle$ is an abelian group. Thus, since $x \approx z$, we can define operations on x to make it an abelian group.

THEOREM 6.25: AL 14 $\rightarrow$ WO 1.

PROOF: Let x be a non-empty set. Let y be a set such that $y \approx \Gamma(x)$ (Hartogs' function) and $x \cap y = \emptyset$. Let $z = x \cup y$. AL 14 implies that there is a binary operation, $+$, defined on z such that $\langle z, \{+\} \rangle$ is a cancellative groupoid.

For each $u \in x$ there is a $v \in y$ such that $u + v \in y$. If not, the function f such that for each $v \in y$, $f(v) = u + v$ is a 1-1 function mapping y into x . (The cancellation law implies f is 1-1.) Thus, $y \prec x$ which contradicts the definition of y .

Let $w = y \times y$; then since y can be well-ordered, so can w . Define a function $g: x \rightarrow w$ so that for each $u \in x$, $g(u)$ is the smallest element in

$$\{\langle v_1, v_2 \rangle: v_1, v_2 \in y \wedge u + v_1 = v_2\}.$$

It follows from the cancellation law that g is 1-1. Thus, x can be well-ordered. (Actually, only a one-sided cancellation law is required.)

It is clear from the proofs of 6.24 and 6.25 that if in AL 14 we replace "cancellative groupoid" by any one of the following, the statement remains equivalent to the axiom of choice: (i) cancellative semigroup; (ii) cancellative abelian semigroup; (iii) quasigroup; (iv) loop; (v) group; (vi) abelian group; and (vii) ring. Also, for infinite sets, a transcendental extension of the field of rational numbers will, with the well-ordering theorem, give a field of any infinite cardinality. (The number of elements in a finite field is always the power of a prime.) So for infinite sets we can add the following to the above list: (viii) integral domain and (ix) field. Clearly, any other algebraic structure between a cancellative groupoid and a field will work.

Krull [1929] noted that the axiom of choice implies that every commutative ring with a multiplicative identity element

has a $\subseteq$ -maximal proper ideal and Scott [1954] asked whether the converse holds. Hodges [1979] proved that the answer is yes. In fact he proved a stronger statement.

DEFINITION 6.26: Suppose $\langle A, \{+, \cdot, 0, 1\} \rangle$ is an integral domain (see 6.23(i)) with a multiplicative identity element 1, $0 \neq 1$.

(a) $a \in A$ is called a unit element if there is a $b \in A$ such that $a \cdot b = 1$. (Since multiplication is commutative, $b \cdot a = 1$ also. Thus, a unit element is an element which has a multiplicative inverse.)

(b) $a \in A$ is called irreducible if a is not a unit element and whenever $a = b \cdot c$, either b or c is a unit element.

(c) $\langle A, \{+, \cdot, 0, 1\} \rangle$ is a unique factorization domain if it is an integral domain with multiplicative identity element 1, $0 \neq 1$, satisfying the following properties:

i) If $a \in A \setminus \{0\}$ and a is not a unit element, then a can be written as a product of irreducible elements.

ii) If $a = b_1 \cdot b_2 \cdots b_j = c_1 \cdot c_2 \cdots c_k$ where the b 's and c 's are irreducible elements, then $j = k$ and the b 's and c 's can be reordered so that for each i there is a unit element e_i such that $b_i = e_i c_i$, $i = 1, 2, \dots, j$.

(d) $I \subseteq A$ is called an ideal if I is a subalgebra such that

$$(\forall a \in A) (\forall b \in I) (a \cdot b \in I)$$

$I \subseteq A$ is called a proper ideal if I is an ideal and $I \neq A$. (Thus $1 \notin I$ if I is proper.)

(e) $I \subseteq A$ is called a prime ideal if I is an ideal and for all $a, b \in A$ if $a \cdot b \in I$ then $a \in I$ or $b \in I$.

Hodges' [1979] form is the following:

AL 15: Every unique factorization domain contains a $\subseteq$ -maximal proper ideal.

Using a proof similar to the proof of 4.3, it is easy to prove that $M_3 \rightarrow AL_{15}$. (In 4.3 take

$$y = \{I \subseteq A: 1 \notin I \wedge I \text{ is an ideal}\}.$$

Then, the same proof shows that y has a $\subseteq$ -maximal element.)

THEOREM 6.27: $AL_{15} \rightarrow M_5(RA, L)$.

PROOF: Suppose $\langle x, \leq \rangle$ is a ramified set. (That is, $\langle x, \leq \rangle$ is a partially ordered set such that initial segments are linearly ordered by $\leq$.) We shall prove that x contains a $\subseteq$ -maximal linearly ordered subset.

Let F be the field of rational numbers and let $F[x]$ be the abelian polynomial ring with elements $u \in x$ as indeterminates. (That is, $F \subseteq F[x]$ and each element of $F[x] \sim F$ is a sum of elements of the form $au_1u_2\dots u_m$ where $a \in F$, $u_i \in x$ and the u_i are not necessarily distinct.) $F[x]$ is a unique factorization domain and for $y \subseteq x$ if $yF[x]$ is the set of all finite sums from $\{uv: u \in y \wedge v \in F[x]\}$ then $yF[x]$ is a prime ideal in $F[x]$.

Let L be the set of all $\leq$ -linearly ordered subsets of x and let

$$S = F[x] \sim \bigcup_{y \in L} yF[x].$$

S is a subalgebra of $F[x]$. Let

$$R = S^{-1}F[x] = \{u/s: u \in F[x] \wedge s \in S\}.$$

If addition and multiplication are defined in the natural way,

R is a unique factorization domain. $u/s \in R$ is a unit (invertible) iff $u \in S$. If, for each $w \in x$, $\bar{w} = \{v \in x: v \leq w\}$ then $u/s \in R$ is a unit iff $(\forall v \in x)u \notin \bar{v}F[x]$.

AL_{15} implies that R contains a $\subseteq$ -maximal proper ideal M . Let $c = u/s \in M$, so that u and s have no common factors. Then

$$u = a_1 t_1 + a_2 t_2 + \dots + a_j t_j,$$

where $a_i \in F$ and t_i is a monomial in x . Moreover, c is not a unit, for if it were we would have $cc^{-1} = 1 \in M$. Thus, if $u \neq 0$ then $u \notin S$, so $u \in \bigcup_{y \in L} yF[x]$. This implies the

monomials, t_i , have factors in some linearly ordered subset of x . Therefore, if $c \neq 0$, there must be at least one finite non-empty $\leq$ -linearly ordered set $A \subseteq x$ such that (i) each monomial, t_i , has a factor in A , and (ii) each element of A occurs as a factor of some t_i . The set A is not unique, but there are at most a finite number of them, $A_1, A_2, \dots, A_k$. Let

$$E(c) = \{v: (\exists i \in \omega) (1 \leq i \leq k \wedge v \text{ is the } \leq\text{-largest element of } A_i)\}.$$

If $v \in E(c)$ then $c \in \overleftarrow{v}R$. If $c = 0$ then $E(c) = \emptyset$. We define

$$D = \{v \in x: (\forall c \in M) (c \neq 0 \rightarrow (\exists w \in E(c)) (v \leq w \vee w \leq v))\}.$$

We shall show that D is a $\subseteq$ -maximal linearly ordered subset of x . The proof depends on the following three lemmas:

- (1) $D \subseteq M$,
- (2) D is a linearly ordered initial segment of x ,
- (3) $M \subseteq DR$.

By (1) and (3), $M = DR$. If D is not a $\subseteq$ -maximal linearly ordered subset of x , then, by (2), there is a $v \in x$ which is $\leq$ -larger than every $u \in D$. Thus, $DR \subset \overleftarrow{v}R$, which contradicts the maximality of M .

We shall sketch the proofs of (1), (2) and (3).

PROOF OF (1): $D \subseteq M$.

Suppose $v \in D$ and $v \notin M$. Since M is maximal there is a $u \in R$ and a $c \in M$ such that $u \cdot v + c = 1$. Moreover,

$v \notin S$ because $\{v\} \in L$ and $v \in \{v\}F[x]$. Therefore, v is not a unit in R ($u \cdot v \neq 1$), so $c \neq 0$. By the definition of D , there is a $w \in E(c)$ which is comparable with u and $c \in \overleftarrow{w}R$. If $w \leq v$, then $c, v \in \overleftarrow{v}R$, which implies the contradiction that $1 = u \cdot v + c \in \overleftarrow{v}R$. If $v \leq w$, then $c, v \in \overleftarrow{w}R$ so again we have the contradiction $1 = u \cdot v + c \in \overleftarrow{w}R$.

PROOF OF (2): D is a linearly ordered initial segment of x .

Suppose $u, v \in D \subseteq M$. Then $u + v \in M$, so $u + v$ is not a unit element. Therefore, there is a $w \in x$ such that $u + v \in \overleftarrow{w}F[x]$. This implies $u, v \in \overleftarrow{w}$ so u and v are comparable.

Suppose $u \in D, v \leq u$ and $v \in x$. If c is a non-zero element of M then there is a $w \in E(c)$ such that u and w are comparable. If $u \leq w$ then $v \leq w$ so $v \in D$. If $w \leq u$, then $v, w \in \overleftarrow{u}$, so v and w are comparable. This again implies that $v \in D$.

PROOF OF (3): $M \subseteq DR$.

Suppose $c \in M \sim DR$ and suppose $c \neq 0$. No element $u \in D$ is in $E(c)$; for otherwise $c \in \overleftarrow{u}R \subseteq DR$ by (2). Let $v_1, v_2, \dots, v_k$ be the distinct elements of $E(c)$. Since the $v_i \notin D$, there is a non-zero element $w_i \in M$ such that for all $u \in E(w_i)$, u is incomparable with v_i . Since F is an infinite field, we may choose scalars $a_1, a_2, \dots, a_k \in F$ such that in

$$d = c + a_1 w_1 + a_2 w_2 + \dots + a_k w_k \in M$$

no monomial which occurs with a non-zero coefficient in c or some w_i , vanishes in R . Suppose $w \in E(d)$. Then there is some v_i such that $v_i \leq w$. Also, there is a $v \in E(w_i)$ such that $v \leq w$. Therefore, $v, v_i \in \overleftarrow{w}$, which implies that v and v_i are comparable. This contradicts the choice of the v_i , q.e.d.

It follows as a simple corollary that if in AL 15, "unique factorization domain" is replaced by either "commutative

ring with multiplicative identity" or "integral domain with multiplicative identity" the resulting statement remains equivalent to the axiom of choice.

We continue with a form which was given originally by Klimovsky [1962] and later extended and generalized by Felgner [1976].

AL 16: Every group contains a $\subseteq$ -maximal abelian subgroup.

The proof that $M 3 \rightarrow AL 16$ is similar to the proof of 4.3. Alternatively, it is easy to show that $M 7 \rightarrow AL 16$ because being abelian is a property of finite character and a $\subseteq$ -maximal abelian subset of a group is a subgroup.

DEFINITION 6.28: The group $\langle G, \{\cdot, e, {}^{-1}\} \rangle$ is said to be free if there is a non-empty $X \subseteq G$ such that:

- (1) $e \notin X$,
- (2) If $x \in X$, then $x^{-1} \notin X$,
- (3) If $y \in G$, then either $y = e$ or there is a unique $n \in \omega \sim \{0\}$ and a unique sequence $\langle y_1, \dots, y_n \rangle$ such that each $y_i \in X \cup X^{-1}$, $y_i \cdot y_{i+1} \neq e$ for all i , and $y = y_1 \cdot y_2 \cdot \dots \cdot y_n$, where $X^{-1} = \{x^{-1} : x \in X\}$.

The set X is said to generate G and the representation specified in (3) is called the reduced form of y .

Given any non-empty set X there is a free group which is generated by X , and any two such groups are isomorphic, with the isomorphism being the identity on X . Hence, we call such a group the free group generated by X .

THEOREM 6.29: $AL 16 \rightarrow AC 13(2)$.

PROOF: Let x be a set of non-empty sets. For each $u \in x$, let F_u be the free group generated by u . Let 1_u be the identity element of F_u . Let G be the weak direct product of the F_u 's.

$$G = \{f \in \prod_{u \in x} F_u : \{u \in x : f(u) \neq 1_u\} \text{ is finite}\}.$$

If $f, g \in G$, $f \cdot g(u) = f(u) \cdot g(u)$ and $f^{-1}(u) = (f(u))^{-1}$ for all $u \in x$. The identity element 1 is the function f such that for all $u \in x$, $f(u) = 1_u$. Then $\langle G, \{\cdot, 1, {}^{-1}\} \rangle$ is a group. Let A be a $\subseteq$ -maximal abelian subgroup of G . For each $u \in x$, let A_u be the set of u^{th} coordinates of elements of A . Each A_u is an abelian subgroup of F_u and no $A_u = \{1_u\}$. For if $A_u = \{1_u\}$ and $a \in u$ we could extend A_u to $A_u \cup [a]$, where $[a]$ is the cyclic group generated by a , (all powers of a and a^{-1}) thereby contradicting the maximality of A .

If $w \in A_u \sim \{1_u\}$ and w is in reduced form then w must be a power, positive or negative, of a product of elements of $u \cup u^{-1}$. Otherwise, A_u would not be abelian. Let $f(w)$ be the first factor of w , $\ell(w)$ be the last factor of w and $C(w) = \{f(w), \ell(w), (f(w))^{-1}, (\ell(w))^{-1}\}$. Then for all $w_1, w_2 \in A_u \sim \{1_u\}$, $C(w_1) = C(w_2)$ and $1 \leq |C(w_1) \cap u| \leq 2$. Therefore, for each $u \in x$, define $g(u) = C(w) \cap u$ where w is any element of $A_u \sim \{1_u\}$, then g is the required choice function.

DEFINITION 6.30: Suppose $\langle G, \{\cdot, e, {}^{-1}\} \rangle$ is a group.

(a) $(\forall a, b \in G) (a, b) = a^{-1}b^{-1}ab$ is called the commutator of a and b .

(b) If H is the group generated by $\{(a, b) : a, b \in G\}$, H is called the commutator subgroup of G .

(c) If $(a_1, a_2, \dots, a_{n-1}, a_n) = ((a_1, a_2, \dots, a_{n-1}), a_n)$, and G has the property that $(a_1, a_2, \dots, a_k, a_{k+1}) = e$ for all $a_i \in G$, then G is called nil-k. (Nil-1 is abelian.)

(d) G is called metabelian if $((a, b), (c, d)) = e$ for all $a, b, c, d \in G$.

(e) If $k \geq 1$ is a natural number and G' is the commutator subgroup of G , define

$$G^{(0)} = G, G^{(1)} = G', \text{ and } G^{(k+1)} = (G^{(k)})'.$$

We say the solvability degree of G is k if $G^{(k)} = \{e\}$ and $G^{(k-1)} \neq \{e\}$. (Solvability of degree 1 is abelian and solvability of degree ≤ 2 is metabelian.)

Felgner [1976] extended Klimovsky's results and showed that each of the following are equivalent to the axiom of choice.

Let $k \geq 1$ be a natural number.

AL 17(k): Every group contains a $\subseteq$ -maximal nil-k subgroup.

AL 18(k): Every group contains a $\subseteq$ -maximal subgroup with solvability degree $\leq k$.

The proof that each implies the axiom of choice is similar to the proof of 6.29.

Our next forms are statements about vector spaces.

DEFINITION 6.31: Let $\langle F, \{+, \cdot, 0, 1\} \rangle$ be a field. The algebra $\langle V, \{+, 0\} \cup \{O_a : a \in F\} \rangle$ is called a vector space over F iff

- (1) $\langle V, \{+, 0\} \rangle$ is an abelian group.
- (2) $(\forall a, b \in F) (\forall v \in V) O_{a+b}(v) = O_a(v) + O_b(v)$.
- (3) $(\forall a \in F) (\forall u, v \in V) O_a(u + v) = O_a(u) + O_a(v)$.
- (4) $(\forall a, b \in F) (\forall v \in V) O_{a \cdot b}(v) = O_a(O_b(v))$.
- (5) $(\forall v \in V) O_1(v) = v$.

We shall use the notation av instead of $O_a(v)$, and we shall use the same symbols $+$ and 0 for V as for F .

The notions of independent, basis, etc. which were defined earlier (6.12) for algebras in general can be defined more simply for vector spaces.

DEFINITION 6.32: Suppose F is a field and $\langle V, \{+, 0\} \cup \{O_a : a \in F\} \rangle$ is a vector space over F .

- (a) $W \subseteq V$ is called a subspace if W is a subalgebra.
- (b) $A \subseteq V$ is said to be linearly dependent if

$$(\exists n \in \omega) (\exists v_1, \dots, v_n \in A) (\exists a_1, \dots, a_n \in F) 0 = a_1 v_1 + \dots + a_n v_n,$$

and for some i , $a_i \neq 0$. Otherwise, A is said to be linearly independent.

(c) $A \subseteq V$ is said to generate V if

$$(\forall v \in V) (\exists n \in \omega) (\exists u_1, \dots, u_n \in A) (\exists a_1, \dots, a_n \in F)$$

$$v = a_1 u_1 + \dots + a_n u_n.$$

(V is said to be a linear combination of the u_i 's.)

(d) $B \subseteq V$ is said to be a basis of V if B is linearly independent and generates V . (A basis is a $\subseteq$ -maximal linearly independent subset of V .)

(e) If W is also a vector space over F and $T: V \rightarrow W$, T is called linear iff

$$(i) \quad T(v_1 + v_2) = T(v_1) + T(v_2) \quad \text{for all } v_1, v_2 \in V.$$

$$(ii) \quad T(av) = aT(v) \quad \text{for all } v \in V \text{ and for all } a \in F.$$

Let B be the following statement.

B : Every vector space has a basis.

AL 19: For every field F and every vector space V over F , every subset of V which generates V contains a basis.

Let F be a field:

AL 20(F): Every vector space V over F has the property that for all $A \subseteq V$, if A is linearly independent then A can be extended to a basis of V .

AL 21(F): Every vector space V over F has the property that if $S \subseteq V$ is a subspace of V then there is a subspace $S' \subseteq V$ such that $S \cap S' = \{0\}$ and $S \cup S'$ generates V .

It is clear that the axiom of choice implies B , for example AL 12 $\rightarrow B$, but it was unknown up to the present time whether B implies the axiom of choice. However, Andreas Blass sent us a proof that B implies the multiple choice axiom. (Blass credits Dr. R. Harting for asking the question

which Theorem 6.33 answers and for calling attention to the construction of the ground field, which was a key step in the proof.) We shall include his proof here (Theorem 6.33). It is still unknown whether B implies the axiom of choice in NBC° . Bleicher [1964] proved that $AL\ 20(F)$ and $AL\ 21(F)$ each imply the multiple choice axiom and had some partial results involving $AL\ 19$. Halpern [1966] proved that $AL\ 19$ implies the axiom of choice. Armbrust [1972] proved that if F is a field of characteristic zero, then the multiple choice axiom implies $AL\ 21(F)$.

Clearly $AL\ 13 \rightarrow AL\ 19 \rightarrow B$ and $AL'12 \rightarrow AL\ 20(F) \rightarrow B$. We shall show that $AL\ 19 \rightarrow AC\ 2$, $AL\ 20(F) \rightarrow AL\ 21(F) \rightarrow MC\ 1$, $B \rightarrow MC\ 1$, and if F is a field of characteristic zero then $MC\ 1 \rightarrow AL\ 21(F)$.

First, it is easy to see that if F is a field, $AL\ 20(F) \rightarrow AL\ 21(F)$. For suppose V is a vector space over F and $S \subseteq V$ is a subspace. $AL\ 20(F)$ implies that S has a basis B' and that B' can be extended to a basis B of F . Take S' to be the subspace generated by the elements in $B \sim B'$. Then $S \cap S' = \{0\}$ and $S \cup S'$ generates V .

THEOREM 6.33: $B \rightarrow MC\ 1$.

PROOF: Let $x = \{u_i : i \in I\}$ be a non-empty set of pairwise disjoint non-empty sets. Let R be any field such that $R \cap (Ux) = \emptyset$. Let F be the field obtained from R by adding Ux as a set of indeterminates. That is, each $f \in F$ is a rational function of the "variables" in Ux . For each $i \in I$, we define the i -degree of a monomial to be the sum of the exponents of elements of u_i in that monomial. A rational function $f \in F$ is called i -homogeneous of degree d if it is the quotient of two polynomials such that all monomials in the numerator have the same i -degree, n , while all monomials in the denominator have the same i -degree, $n - d$. The rational functions that are i -homogeneous of degree 0 for all $i \in I$ form a subfield $K \subseteq F$. Thus, F is a vector space over the field K and we let V be the subspace which is generated by Ux .

By hypotheses V has a basis, B . For each $i \in I$ and $t \in u_i$ we can express t as a finite linear combination of elements of B with coefficients in K :

$$(1) \quad t = \sum_{b \in B(t)} k_b(t) \cdot b ,$$

where $B(t)$ is a finite subset of B and for each $b \in B(t)$, $k_b(t)$ is a non-zero element of K . If s is another element of the same u_i as t then

$$(2) \quad s = \sum_{b \in B(s)} k_b(s) \cdot b .$$

If we multiply (1) by the element $s/t \in K$ we obtain

$$(3) \quad s = \sum_{b \in B(t)} (s/t) k_b(t) \cdot b .$$

Since B is a basis each element in V is uniquely expressible as a linear combination of elements of B . Thus, it follows from (2) and (3) that $B(s) = B(t)$ and $k_b(s) = (s/t)k_b(t)$. This means that the finite subset $B(t) \subseteq B$ and the elements $k_b(t)/t$ depend only on i , not on the particular $t \in u_i$; we therefore call them B_i and k_{bi} , respectively. Since $k_b(t) \in K$, k_{bi} is i -homogeneous of degree -1 (and j -homogeneous of degree 0 if $j \neq i$). This means that when k_{bi} is written as a quotient of polynomials in reduced form, some variables from u_i must occur in the denominator. We define a multiple choice function g on x such that for each $i \in I$, $g(u_i)$ = the set of those members of u_i which occur in the denominator of k_{bi} (in reduced form) for some $b \in B_i$. Then $g(u_i)$ is a non-empty finite subset of u_i , q.e.d.

If a vector space V has a basis and all bases of V have the same cardinal number κ , then κ is called the dimension of V . Let B' be the following statement:

B' : Every vector space has a dimension.

Then we have $AC \rightarrow B' \rightarrow B \rightarrow MC\ 1$. It is not known if B implies AC in NBG° or if B' implies AC in NBG° .

(The statement B' was suggested by Norbert Brunner.)

THEOREM 6.34: $AL\ 19 \rightarrow AC\ 2$.

PROOF: Let x be a set of pairwise disjoint non-empty sets. Let R be any field such that $R \cap (Ux) = \emptyset$ and let F be the field obtained from R by adding Ux as a set of indeterminates. (The same construction which was used in Theorem 6.33.)

$$F^{[x]} = \{f: f \text{ is a function, } f: x \rightarrow F \text{ and } \{u \in x: f(u) \neq 0\} \text{ is finite}\}.$$

Let V be the vector space of $F^{[x]}$ over the field F .

For each $s \in Ux$ and $u \in x$, define

$$\bar{s}(u) = \begin{cases} s & \text{if } s \in u, \\ 0 & \text{otherwise,} \end{cases}$$

and let $A = \{\bar{s}: s \in Ux\}$. Then it is easy to see that A generates V because each $f \in F^{[x]}$ is a linear combination of $\bar{s}$'s. Thus, $AL\ 19$ implies that there is a $B \subseteq A$ such that B is a basis of V . Suppose $\bar{s}_1, \bar{s}_2 \in B$ and there is a $u \in x$ such that $\bar{s}_1(u) = s_1 \in u$ and $\bar{s}_2(u) = s_2 \in u$. Then $(s_1^{-1})\bar{s}_1 - (s_2^{-1})\bar{s}_2 = 0$, contradicting the fact that B is linearly independent. Thus, for each $u \in x$ there is at most one $\bar{s} \in B$ such that $\bar{s}(u) \neq 0$. On the other hand, since B generates V , for each $u \in x$ there is at least one $\bar{s} \in B$ such that $\bar{s}(u) \neq 0$. Thus, $\{s: \bar{s} \in B\}$ is a choice set for x .

THEOREM 6.35: If F is a field, $AL\ 21(F) \rightarrow MC\ 1$.

PROOF: Let x be a set of pairwise disjoint non-empty sets. Let F be any field. Let

$$L(u, F) = \{a_1 s_1 + \dots + a_n s_n: n \in \omega, a_i \in F \text{ and } s_i \in u\}$$

$L(u, F)$ is the set of all finite formal sums with indeterminates from u and coefficients in F . It can be considered to be a vector space over F , if we assume the usual conventions about addition and scalar multiplication. The set u is then a basis of $L(u, F)$. Let $L_0(u, F)$ be the subspace of $L(u, F)$

in which the sum of the coefficients of each finite sum is zero. That is,

$$L_0(u, F) = \{a_1 s_1 + \dots + a_n s_n \in L(u, F) : a_1 + \dots + a_n = 0\}.$$

Let V be the weak direct product of the $L(u, F)$ for $u \in x$ and V_0 be the weak direct product of the $L_0(u, F)$ for $u \in x$.

$$V = \{f \in \prod_{u \in x} L(u, F) : \{u \in x : f(u) \neq 0\} \text{ is finite}\}.$$

$$V_0 = \{f \in \prod_{u \in x} L_0(u, F) : \{u \in x : f(u) \neq 0\} \text{ is finite}\}.$$

Then V_0 is a subspace of V , and AL 21(F) implies that there is a subspace $V'_0 \subseteq V$ such that

$$(i) \quad V_0 \cap V'_0 = \{0\} \quad \text{and} \quad (ii) \quad V_0 \cup V'_0 \text{ generates } V.$$

In what follows we shall identify each $s \in u \times$ with the function $f_s \in V$ such that

$$f_s(u) = \begin{cases} s & \text{if } s \in u, \\ 0 & \text{otherwise.} \end{cases}$$

For each $v \in V$ there is a unique $w \in V_0$ and a unique $w' \in V'_0$ such that $v = w + w'$. Suppose $s_1, s_2 \in u \in x$, $s_1 = v_1 + v'_1$ and $s_2 = v_2 + v'_2$, where $v_1, v_2 \in V_0$ and $v'_1, v'_2 \in V'_0$. Then

$$s_1 - s_2 = (v_1 - v_2) + (v'_1 - v'_2).$$

Thus, by the definition of V_0 , we have $s_1 - s_2, v_1 - v_2 \in V_0$, so $v'_1 - v'_2 \in V_0 \cap V'_0 = \{0\}$. Therefore, $v'_1 = v'_2$.

Consequently, for each $u \in x$ there is a unique $v_u \in V'_0$ such that for each $s \in u$ there is a $v_0 \in V_0$ with $s = v_0 + v_u$. For each $u \in x$, let $g(u)$ be the finite set of elements in u which generate the u^{th} coordinate of v_u . Then g is a multiple choice function on x .

THEOREM 6.36: If F is a field of characteristic zero then MC 1 $\rightarrow$ AL 21(F).

PROOF: Let F be a field of characteristic zero and V a vector space over F . Let S be a subspace of V . We use

MC 1 to prove that there is a linear mapping T from V to S such that $T|_S$ is the identity function. Then, if $S' = \{v \in V : T(v) = 0\}$, the kernel of T , it is easy to prove that S' is the required subspace. (S' is clearly a subspace. If $u \in S$ then $T(u) = u$ and if $u \in S'$ then $T(u) = 0$ so $S \cap S' = \{0\}$. For any $v \in V$, $T(v) = s \in S$ and $T(s) = s$. Consequently $v = s + (v-s)$, where $s = T(v) \in S$ and $v - s \in S'$.)

It remains to be shown that such a linear mapping exists. Let f be a multiple choice function on $\mathcal{P}(V) \sim \{\emptyset\}$. Let L be the set of all linear mappings from a subspace of V into S and let g be a multiple choice function on $\mathcal{P}(L) \sim \{\emptyset\}$. Define $f(\emptyset) = g(\emptyset) = \emptyset$. For each ordinal α we define a linear function h_α and a set B_α as follows:

$h_0 = \text{identity map on } S, B_0 = S.$

If α is a limit ordinal, $h_\alpha = \bigcup_{\beta < \alpha} h_\beta$ and $B_\alpha = \bigcup_{\beta < \alpha} B_\beta$.

If $\alpha = \beta + 1$, let H_β be the set of all linear extensions of h_β to the subspace S_α which is generated by $B_\alpha = B_\beta \cup f(V \sim B_\beta)$. Suppose $g(H_\beta) = \{k_1, k_2, \dots, k_n\}$. Then

$$h_\alpha = \frac{1}{n} (k_1 + k_2 + \dots + k_n),$$

and $h_\alpha : S_\alpha \rightarrow S$ is a linear mapping. Let α_0 be the smallest element of $\{\alpha : V \sim S_\alpha = \emptyset\}$. Let $T = h_{\alpha_0}$. Then T is the required linear mapping, q.e.d.

As a corollary to Armbrust's [1972] Theorem 6.36, we have

THEOREM 6.37: If F is a field of characteristic p then $\text{MC } 4(p) \rightarrow \text{AL } 21(F)$.

PROOF: The proof is similar to the proof of 6.36. If n is relatively prime to p then $1/n$ exists in a field of characteristic p .

THEOREM 6.38: If F is a field of characteristic p and there is a choice function on the set of all non-empty finite subsets of F then $\text{AL } 21(F) \rightarrow \text{MC } 4(p)$.

PROOF: Follow the proof of 6.35 and for each $u \in x$, construct $v_u \in V'_0$, where $0 \neq v_u = a_1 s_1 + a_2 s_2 + \dots + a_n s_n$, $a_i \in F$, $s_i \in u$, $\sum_{i=1}^n a_i \neq 0$ and no finite subsum of the $a_i s_i$'s is zero. Subdivide the s_i 's into subsets, $[s_i]$, such that $s_j \in [s_i]$ iff $a_j = a_i$. For some $i = 1, 2, \dots, n$, the number of elements in $[s_i]$ is relatively prime to p . Use the choice function to choose one of the a_i 's from the finite set of coefficients of v_u ; then the corresponding set $[s_i]$ is the required finite subset of u , q.e.d.

If AL 20 is ($\exists$ field F) AL 20(F) and similarly AL 21 is ($\exists$ field F) AL 21(F) then $AL\ 20 \leftrightarrow AL\ 20(F)$ and $AL\ 21 \leftrightarrow AL\ 21(F)$. The next two forms, which are due to Bleicher [1964], will be stated with the prefix "There exists a field F " but remain equivalent to the axiom of choice if F is a parameter.

AL 22: There is a field F such that for every vector space V over F if B is a basis for V then for each subspace $S \subseteq V$ there is a subset $B' \subseteq B$ such that if S' is the subspace generated by B' then $S \cap S' = \{0\}$ and $S \cup S'$ generates V .

Let F be a field and let $B(F)$ and $BL(F)$ be the following statements:

$B(F)$: Every vector space over F has a basis.

$BL(F)$: If V is a vector space over F , B is a basis of V and X is any linearly independent subset of V then there is a subset $B' \subseteq B$ such that $X \cap B' = \emptyset$ and $X \cup B'$ is a basis of V .

It is not known whether $B(F)$ or $BL(F)$ imply the axiom of choice but we can show AL 23 does. (The statement B above is: For each field F , $B(F)$, and Blass proved $B \rightarrow MC\ 1$.)

AL 23: There is a field F such that $B(F)$ and $BL(F)$

We shall show that $AL\ 12 \wedge M\ 3 \rightarrow AL\ 23 \rightarrow AL\ 22 \rightarrow AC\ 2$. We noted earlier that $B(F)$ is a special case of AL 12 and the proof that $M\ 3 \rightarrow BL(F)$ is similar to the proof of 4.3.

THEOREM 6.38: AL 23 $\rightarrow$ AL 22.

PROOF: Let V be a vector space over a field F . Let B be a basis of V and S a subspace of V . AL 23 implies S has a basis X . Then since X is linearly independent, AL 23 implies that there is a subset $B' \subseteq B$ such that $X \cap B' = \emptyset$ and $X \cup B'$ is a basis of V . The set B' satisfies the conclusion of AL 22.

THEOREM 6.39: AL 22 $\rightarrow$ AC 2.

PROOF: Let x be a set of pairwise disjoint non-empty sets and let F be a field. Let V and V_0 be the vector spaces constructed in the proof of 6.35. Then V_0 is a subspace of V and $B = \{f_s : s \in \cup x\}$ is a basis for V , if f_s is the function such that

$$f_s(u) = \begin{cases} s & \text{if } s \in u, \\ 0 & \text{otherwise.} \end{cases}$$

We shall identify B with $\cup x$. Now apply AL 22 taking V_0 as the subspace of V and B as the basis, obtaining a subset $B' \subseteq B$ such that if S' is the subspace generated by B' then $V_0 \cap S' = \{0\}$ and $V_0 \cup S'$ generates V .

Let $u \in x$ and let $w = u \cap B'$. We claim w has at most one element, for suppose $s, t \in w$. Then $s - t \in V_0 \cap S' = \{0\}$, which implies $s = t$. On the other hand $w \neq \emptyset$ because for any $s \in u \in x$ there is a $v_0 \in V_0$ and $v' \in S'$ such that $s = v_0 + v'$. The sum of the coefficients of the terms in v_0 is 0, thus the sum of the coefficients of the terms in v' is 1. It follows that B' must contain terms from u for each $u \in x$. Therefore, w is a singleton, so B' is a choice set for x .

If F is a field of characteristic zero, then AL 21(F) can be stated in terms of groups.

DEFINITION 6.40: Suppose $\langle G, \{+, 0\} \rangle$ is a group.

(a) $a \in G$ has order n if $na = 0$ and if $ma = 0$, $n \leq m$. (na is an abbreviation of $a + a + \dots + a$, n times.)

(b) G is said to be torsion-free if the only element in G of finite order is 0 .

(c) A group G is said to be divisible if $(\forall a \in G) (\forall n \in \omega \sim \{0\}) (\exists b \in G) a = nb$.

(d) A group G is said to be uniquely divisible if $(\forall a \in G) (\forall n \in \omega \sim \{0\}) (\exists! b \in G) a = nb$. (For abelian groups, "torsion-free divisible" is the same as "uniquely divisible".)

(e) The group $\langle G, \{+, 0\} \rangle$ is said to be a free abelian group if there is a non-empty $X \subseteq G$ such that

(1) $0 \notin X$,

(2) $x \in X \rightarrow x^{-1} \notin X$. (Let $X^{-1} = \{x^{-1} : x \in X\}$)

(3) $y \in G \rightarrow y = 0$ or there is a unique $n \in \omega \sim \{0\}$ and a sequence $\langle y_1, y_2, \dots, y_n \rangle$ unique up to permutations such that each $y_i \in X \cup X^{-1}$, no y_i is the inverse of any y_j , and $y = y_1 + y_2 + \dots + y_n$. The set X is said to generate G . (See 6.28).

(f) If $\langle G_1, \{+_1, 0_1\} \rangle$ and $\langle G_2, \{+_2, 0_2\} \rangle$ are groups, $f: G_1 \rightarrow G_2$, f is called a homomorphism if $(\forall a, b \in G_1) (f(a +_1 b) = f(a) +_2 f(b))$ and $f(0_1) = 0_2$. If in addition f is 1-1, f is called an isomorphism.

Since a vector space over a field of characteristic zero is an abelian, torsion-free, divisible group it follows from 6.36 that the following statement about groups is equivalent to MC 1.

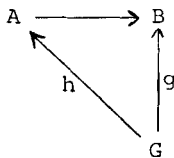
AL 24: If A is an abelian group and B is a torsion-free divisible subgroup of A then there exists a subgroup $B' \subseteq A$ such that $B \cap B' = \{0\}$ and $B \cup B'$ generates A .

(It follows from the proof of 6.36 that it is not necessary for A to be torsion-free or divisible.)

Armbrust [1972] asked if "torsion-free" is omitted from AL 24, is the resulting statement AL 24T implied by MC 1?

Blass [1979] partially answers the question by proving that AL 25 is equivalent to the multiple choice axiom, and AL 26 and AL 27 are equivalent to the axiom of choice.

AL 25: If G is a free abelian group, A is a uniquely divisible abelian group, B is an abelian group, $f:A \rightarrow B$ is a homomorphism mapping A onto B , and $g:G \rightarrow B$ is a homomorphism, then there is a homomorphism $h:G \rightarrow A$ such that $f \circ h = g$. (See Figure 6.1).

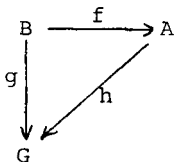


$$f \circ h = g$$

FIGURE 6.1

AL 26: If G is a free abelian group, A and B are abelian groups, $f:A \rightarrow B$ is a homomorphism mapping A onto B , and $g:G \rightarrow B$ is a homomorphism, then there is a homomorphism $h:G \rightarrow A$ such that $f \circ h = g$. (Every free abelian group is projective.) (See Figure 6.1).

AL 27: If G is a divisible abelian group, A and B are abelian groups, $f:B \rightarrow A$ is an isomorphism, and $g:B \rightarrow G$ is a homomorphism, then there is a homomorphism $h:A \rightarrow G$ such that $h \circ f = g$. (Every divisible abelian group is injective.) (See Figure 6.2).



$$h \circ f = g$$

FIGURE 6.2

Clearly, $AL\ 27 \rightarrow AL\ 24T$, for if A is an abelian group, G is a divisible subgroup of A , $B = G$, and the functions f and g are identity functions then AL 27 implies there is

a homomorphism $h:A \rightarrow G$ such that $h|_G$ is the identity. Then the kernel of $h = \{a \in A:h(a) = 0\}$ is the required subgroup of A .

THEOREM 6.41:

(a) AL 25 $\rightarrow$ MC 1.

(b) MC 1 $\rightarrow$ AL 25.

PROOF: (a) Let x be a set of non-empty pairwise disjoint sets and let $w = \cup x$. Let G be the free abelian group generated by x . Let A be the vector space over the rationals generated by the elements of w as indeterminates and let B be the vector space over the rationals generated by the elements of x as indeterminates. Define functions $f:A \rightarrow B$ and $g:G \rightarrow B$ as follows:

If $a \in u \in x$, $f(a) = u$

If $u \in x$, $g(u) = u$.

Extend f and g to all of A and G in the natural manner. Then G, A, B, f and g satisfy the hypothesis of AL 25.

There exists a homomorphism h mapping G into A such that $f \circ h = g$. Thus, if $u \in x$, $h(u) = r_1 a_1 + r_2 a_2 + \dots + r_n a_n$, where the a_i 's are distinct elements of w . However, $f(h(u)) = g(u) = u$, so $\sum_{i=1}^n r_i = 1$ and at least one $a_i \in u$.

(Some a_i 's may not be elements of u , but then the sum of their coefficients must be zero.) Thus, we define a multiple choice function F on x such that for each $u \in x$, $F(u) = \{a_1, a_2, \dots, a_n\} \cap u$.

(b) Let G, A, B, f and g satisfy the hypothesis of AL 25. Let F be a multiple choice function on the non-empty subsets of A . For each generator x of G , let $H(x) = F(f^{-1}\{g(x)\})$. If $H(x)$ has n elements define $h(x) = \frac{1}{n} \sum_{a \in H(x)} a$, q.e.d.

THEOREM 6.42:

(a) AC 1 $\rightarrow$ AL 26.

(b) AL 26 $\rightarrow$ AC 2.

PROOF: (a) Similar to the proof of 6.41(b), but take F to be a choice function. Then unique divisibility is not required.

(b) We shall prove that AL 26 implies both MC 1 and the axiom of choice for finite sets. The proof that AL 26 implies MC 1 is the same as the proof of 6.41(a).

Suppose x is a set of finite sets such that for each $u \in x$, $2 \leq |u| < \omega$. For each $u \in x$, let $q_u = \{v \subseteq u : 2 \leq |v|\}$ and let $y = \bigcup_{u \in x} q_u$. Let G be the free abelian group generated by y . For each $a \in u \in y$ let $z_a = z_u$ be the additive group of integers modulo $|u|$. Let $w = \bigcup_{u \in y} (\{u\} \times u)$. (Then $\langle u, a \rangle \in w$ iff $a \in u \in y$.) Let A be the weak direct product of the z_a 's for $\langle u, a \rangle \in w$ and let B be the weak direct product of the z_u 's for $u \in y$.

$$A = \{\psi \in \prod_{\langle u, a \rangle \in w} z_a : \{\langle u, a \rangle \in w : \psi(u, a) \neq 0\} \text{ is finite}\}$$

$$B = \{\phi \in \prod_{u \in y} z_u : \{u \in y : \phi(u) \neq 0\} \text{ is finite}\}.$$

A and B are abelian groups if addition of functions is defined in the natural way. For $\phi_1, \phi_2 \in B$, $(\phi_1 + \phi_2)(u) = \phi_1(u) + \phi_2(u)$, and the identity function is the zero function; similarly for A .

For each $a \in u \in y$, let $\psi_{\langle u, a \rangle}$ and ϕ_u be the functions defined as follows

$$(\forall \langle v, b \rangle \in w) \psi_{\langle u, a \rangle}(\langle v, b \rangle) = \begin{cases} 0 & \text{if } \langle u, a \rangle \neq \langle v, b \rangle, \\ 1 & \text{if } \langle u, a \rangle = \langle v, b \rangle. \end{cases}$$

$$(\forall v \in y) \phi_u(v) = \begin{cases} 0 & \text{if } u \neq v, \\ 1 & \text{if } u = v. \end{cases}$$

Define functions f and g as follows: For $a \in u \in y$, $f(\psi_{\langle u, a \rangle}) = \phi_u$ and $g(u) = \phi_u$. Extend f to all of A and g to all of G in the natural manner. Then G, A, B, f and g satisfy the hypothesis of AL 26. Therefore, there is a homomorphism $h: G \rightarrow A$ such that $f \circ h = g$.

Let $u \in y$, then $h(u) = \psi = \sum_{a_i \in u} n_i \psi_{\langle u, a_i \rangle}$ where $n_i \in \mathbb{Z}_u$ and $\sum_i n_i \equiv 1 \pmod{|u|}$. This follows from the fact that

$$f(h(u)) = f(\psi) = g(u) = \phi_u.$$

Since $|u| \geq 2$, there must be at least two distinct n_i 's. Let $m = \min\{n_i : a_i \in u\}$. Then $\{a \in u : \text{coefficient of } \psi_{\langle u, a \rangle} \text{ in } \psi \text{ is } m\}$ is a non-empty proper subset of u . Thus, this process reduces the order of the elements of x simultaneously. For each $u \in x$, if $|u| = n$, we will have to apply this process at most $n - 1$ times to choose an element from u . Consequently, we have a choice function on x .

THEOREM 6.43:

- (a) $M 3 \rightarrow AL 27$.
- (b) $AL 27 \rightarrow AC 2$.

PROOF: (a) Let G, A, B, f , and g satisfy the hypothesis of AL 27. Let

$$H = \{\phi : \phi \text{ is a homomorphism from a subgroup of } A \text{ to } G \\ \text{such that } \mathcal{J}(\phi) \supseteq \mathcal{K}(f) \text{ and } (\forall b \in B) g(b) = \phi(f(b))\}.$$

$H \neq \emptyset$ because $g \circ f^{-1} \in H$. H satisfies the hypothesis of M 3 so H has a $\subseteq$ -maximal element, h . We claim that $\mathcal{J}(h) = A$. For suppose $a \in A \sim \mathcal{J}(h)$.

CASE 1. For some positive integer n , $na \in \mathcal{J}(h)$. Let n_0 be the smallest such n . Suppose $h(n_0 a) = x \in G$. Choose $y \in G$ such that $y = x/n_0$. Let

$$A^* = \{b + ka : b \in \mathcal{J}(h) \wedge 0 \leq k < n_0\} \supset \mathcal{J}(h).$$

We have in A^* :

$$b_1 + k_1 a = b_2 + k_2 a \rightarrow b_1 = b_2 \wedge k_1 = k_2$$

$$(b_1 + k_1 a) + (b_2 + k_2 a) = \begin{cases} (b_1 + b_2) + (k_1 + k_2) a, & \text{if } 0 \leq k_1 + k_2 < n_0, \\ (b_1 + b_2 + n_0 a) + (k_1 + k_2 - n_0) a, & \text{if } n_0 \leq k_1 + k_2 < 2n_0. \end{cases}$$

A^* is a subgroup of A . Define $h^*: A^* \rightarrow G$ such that $h^*(b + ka) = h(b) + ky$. Then $h \subset h^*$ and $h^* \in H$. This contradicts the maximality of h .

CASE 2. $na \notin \mathcal{J}(h)$ for all $n \in \omega \sim \{0\}$. Then, since h is a homomorphism, $-na \notin \mathcal{J}(h)$ for all $n \in \omega \sim \{0\}$. Let

$$A^* = \{b \pm na : b \in \mathcal{J}(h) \wedge n \in \omega\}.$$

Define $h^*(b \pm na) = h(b)$. Then $h^* \in H$ and $h \subset h^*$, a contradiction.

(b) We shall first prove $AL\ 27 \rightarrow MC\ 1$, and then prove that $AL\ 27$ implies the axiom of choice for finite sets.

Let x be a set of pairwise disjoint non-empty sets and let $w = \bigcup x$. Let $A = B =$ the free abelian group generated by x . Let Z be the set of integers and Q the set of rational numbers. For each $u \in x$, let

$$R_u = \{\psi \in Q^u : \{a \in u : \psi(a) \neq 0\} \text{ is finite}\}.$$

For $\psi \in R_u$, let $[\psi] = \{\phi \in R_u : \mathcal{K}(\psi - \phi) \subseteq Z \wedge \sum_{a \in u} (\psi - \phi)(a) = 0\}$. Let $S_u = \{[\psi] : \psi \in R_u\}$ and let G be the weak direct product of the S_u 's for $u \in x$.

$$G = \{\psi \in \prod_{u \in x} S_u : \{u \in x : \psi(u) \neq 0\} \text{ is finite}\}.$$

Define $f: B \rightarrow A$ and $g: B \rightarrow G$ as follows:

$$(\forall u \in B) f(u) = 2u = u + u.$$

$$(\forall u \in x) g(u) = l_u, \text{ where } l_u \in G \text{ such that } (\forall v \in x) (v \neq u \rightarrow l_u(v) = [0]) \text{ and } l_u(u) = [\psi_u], \text{ where } \psi_u \in R_u, \mathcal{K}(\psi_u) \subseteq Z \text{ and } \sum_{a \in u} \psi_u(a) = 1.$$

All such functions ψ_u belong to the same equivalence class. Extend g to all of B in the natural way. Then G, A, B, f and g satisfy the hypothesis of AL 27. There is a homomorphism $h:A \rightarrow G$ such that $h \circ f = g$. Then, for each $u \in x$, $h(u) = 1_u/2$. Therefore, if $\phi_u \in h(u)(u)$, then $\sum_{a \in u} \phi_u(a) = 1/2$. Define $F(u) = \{a \in u: \phi_u(a) \in Q \sim Z\}$. Since all $\psi \in h(u)(u)$ have the same values on $Q \sim Z$, for each $u \in x$, $F(u)$ is a uniquely determined finite subset of u , so F is a multiple choice function on x .

To prove the finite axiom of choice let x be a set of finite sets such that for all $u \in x$, $2 \leq |u|$ and if $v \subset u$ with $2 \leq |v|$ then $v \in x$. (We shall give a similar proof to that of 6.42 (b).) Let $A = B =$ the free abelian group generated by x , Z the set of integers, and Q the set of rational numbers. For each $u \in x$, let

$$R_u = \{\psi \in Q^{\{u\} \times u} : \{a \in u : \psi(u,a) \neq 0\} \text{ is finite}\}.$$

Then, continue as in the first part of the proof to define the group G and the function $g:B \rightarrow G$. But here, we define $f:B \rightarrow A$ as follows

$$(\forall u \in x) f(u) = |u|u.$$

Extend f to all of B in the natural way. AL 27 implies there is a homomorphism $h:A \rightarrow G$ such that $h \circ f = g$.

Let $\phi_u \in h(u)(u)$; then $\sum_{a \in u} \phi_u(u,a) = 1/|u|$. Let $\psi_u(u,a) = \phi_u(u,a) - (\text{greatest integer} \leq \phi_u(u,a))$. For each $a \in u$, let $m(a) = |u|\psi_u(u,a) \in Z$. Then $\sum_{a \in u} m(a) \equiv 1 \pmod{|u|}$. For each $j < |u|$ let $n_j = \{a \in u : m(a) = j\}$. Then

$$\sum_{a \in u} m(a) = \sum_{1 \leq j < |u|} j|n_j| \equiv 1 \pmod{|u|}.$$

There must exist a j , $1 \leq j < |u|$, such that $|u|$ is not a factor of $|n_j|$. Let j_0 be the smallest such j . Then $\{a \in u : m(a) = j_0\}$ is a non-empty proper subset of u . This way we can continue, as in the proof of 6.42(b), to construct

a choice function on x , q.e.d.

Grätzer [1962], and Höft & Howard [1981] have studied a representation theorem for multi-algebras which is equivalent to the axiom of choice.

DEFINITION 6.44: $\langle A, \{O_k : k \in K\} \rangle$ is called a multi-algebra if each O_k is a function from A^{n_k} to $\mathcal{P}(A) \sim \{\emptyset\}$.

We shall assume that the multi-algebras we are considering have no constants.

Suppose $\langle B, \{O_k : k \in K\} \rangle$ is an algebra. If $E \subseteq B \times B$ is an equivalence relation then B/E is the set of equivalence classes of B with respect to E . That is:

$$(\forall a \in B) [a] = \{b \in B : \langle a, b \rangle \in E\},$$

$$B/E = \{[a] : a \in B\}.$$

An operation O_k on B can be extended to an operation O_k/E on B/E as follows:

$$O_k/E([a_1], \dots, [a_n]) = \{[b] : (\exists a'_1, \dots, a'_n \in A) (\langle a_i, a'_i \rangle \in E$$

$$\wedge b = O_k(a'_1, \dots, a'_n))\}.$$

Then $\langle B/E, \{O_k/E : k \in K\} \rangle$ is a multi-algebra.

AL 28: Let $\langle A, \{O_k : k \in K\} \rangle$ be a multi-algebra. Then there is an algebra $\langle B, \{O'_k : k \in K\} \rangle$ of the same type as A and an equivalence relation E on B such that $\langle A, \{O_k : k \in K\} \rangle \cong \langle B/E, \{O'_k/E : k \in K\} \rangle$.

AL 29 will be AL 28 with the added restriction that all the multi-operations O_k for $k \in K$ are unary operations. Clearly, AL 28 $\rightarrow$ AL 29. We shall show AC 1 $\rightarrow$ AL 28 and AL 29 $\rightarrow$ AC 1.

Grätzer [1962] proved that the axiom of choice implies AL 28, but the proof we give here is due to Höft & Howard [1981], as is the proof of 6.46.

THEOREM 6.45: AC 1 $\rightarrow$ AL 28.

PROOF: Let $\langle A, \{O_k : k \in K\} \rangle$ be a multi-algebra. Let

$$X = \{O_k(d) : k \in K \wedge d \in \mathcal{D}(O_k)\}.$$

Let F be a choice function on X . Let $B = A \times A$ and define an equivalence relation E on B by

$$\langle a_1, b_1 \rangle E \langle a_2, b_2 \rangle \text{ iff } a_1 = a_2.$$

The operations O'_k are defined on B so that if

$\langle \langle d, e \rangle \rangle = \langle \langle d_1, e_1 \rangle, \dots, \langle d_n, e_n \rangle \rangle$ where $d, e \in \mathcal{D}(O_k)$, then

$$O'_k(\langle \langle d, e \rangle \rangle) = \begin{cases} \langle e_0, e_0 \rangle & \text{if } e_0 = e_1 = \dots = e_n \text{ and } e_0 \in O_k(d) \\ \langle d_0, d_0 \rangle & \text{otherwise, where } d_0 = F(O_k(d)). \end{cases}$$

With these definitions, the multi-operations on B/E will be

$$O'_k/E([\langle \langle d, e \rangle \rangle]) = \{[\langle d', e' \rangle] : (\exists b \in \mathcal{D}(O_k)) \langle d', e' \rangle = O'_k(d, b)\}.$$

If we define $g: B/E \rightarrow A$ such that $g([\langle a, b \rangle]) = a$ then it is easy to prove that g is an isomorphism.

THEOREM 6.46: AL 29 $\rightarrow$ AC 1.

PROOF: Suppose x is a set of non-empty sets. Let $A = \{x\} \cup (\cup x)$. For each $u \in x$, define a unary multi-operation O_u on A as follows:

$$O_u(a) = \begin{cases} u & \text{if } a = x, \\ \{x\} & \text{otherwise.} \end{cases}$$

By AL 29 there is an algebra $\langle B, \{O'_u : u \in x\} \rangle$, of the same type as A , an equivalence relation $E \subseteq B \times B$, and an isomorphism f such that

$$f: \langle B/E, \{O'_u/E : u \in x\} \rangle \cong \langle A, \{O_u : u \in x\} \rangle.$$

Let $b \in B$ such that $f([b]) = x$ and define for each $u \in x$,

$g(u) = f([O'_u(b)])$. Then g is a choice function for x because

$$g(u) \in f(O'_u/E([b])) = O_u(f([b])) = O_u(x) = u.$$

This completes the section on algebra.

§7. CARDINAL NUMBER FORMS

All the statements we make about cardinals could just as well be statements about sets. However, the statements are simpler when we use the notion of a cardinal number, so we shall follow common practice and use cardinal numbers. For the sake of convenience, we shall repeat here some of the properties of cardinal numbers we will use. (Also, see 0.13 and 0.14 and the preceding discussion).

7.1: If x is a set, $|x|$ is the cardinal number of x . We shall use the Greek letters, κ , μ , and ν , and these letters with subscripts, for cardinal numbers.

7.2: For every cardinal κ , there is a set x such that $|x| = \kappa$; for all sets x and y , $|x| = |y|$ iff $x \approx y$; and $|x| \leq |y|$ iff $x \leq y$.

7.3: For every set x , $\Gamma(x) = \{\alpha \in \text{On} : \alpha \leq x\}$ (Γ is Hartogs' function). For each set x , $\Gamma(x)$ is a set. If $|x| = \kappa$, then $|\Gamma(x)| = \kappa^*$. κ^* is the smallest cardinal of a well ordered set which is not $\leq \kappa$.

7.4: $\aleph_0 = |\omega|$ and for each ordinal α , $\aleph_\alpha = |\omega_\alpha|$. An aleph is the cardinal number of an infinite well-ordered set. For each aleph, $\aleph$, there is an ordinal α , such that $\aleph = \aleph_\alpha$. If $\kappa = \aleph_\alpha$ then $\kappa^* = \aleph_{\alpha+1}$.

7.5: For every aleph, $\aleph$, $\aleph^2 = \aleph$.

7.6: For every two alephs, $\aleph$ and $\aleph'$, $\aleph + \aleph' = \aleph \aleph' = \max(\aleph, \aleph')$.

7.7: A cardinal number, κ , is said to be transfinite if $\kappa \geq \aleph_0$.

7.8: If $\kappa_1, \kappa_2, \mu_1$ and μ_2 are transfinite cardinals such that at least one is an aleph, then $\kappa_1 \cdot \kappa_2 \leq \mu_1 + \mu_2$ implies $\kappa_1 \leq \mu_1$ or $\kappa_2 \leq \mu_2$.

7.9: If κ and μ are transfinite cardinals and $\aleph$ is an aleph such that $\aleph \leq \kappa\mu$ then either $\aleph \leq \kappa$ or $\aleph \leq \mu$.

7.10: For each transfinite cardinal κ , κ^* is an aleph and $(\kappa^2)^* = \kappa^*$. (In fact, if $\kappa, \nu \neq 0$ and at least one is infinite then $(\kappa \cdot \mu)^* = \kappa^* \cdot \mu^*$.)

7.11: For each transfinite cardinal κ , $\kappa < \kappa + \kappa^*$ and there are no cardinals between κ and $\kappa + \kappa^*$.

7.12: For all cardinals κ , $\kappa^* \leq 2^{2^{\kappa^2}}$.

7.13: For all cardinals κ , $\kappa^* \leq 2^{2^{2^\kappa}}$.

7.14: For all cardinals κ, μ and for all $n \in \omega \sim \{0\}$. $n\kappa = n\mu \rightarrow \kappa = \mu$.

7.15: For all transfinite cardinals κ , $2^\kappa - \kappa$ exists and is equal to 2^κ .

Georg Cantor may be considered by many to be the founder of set theory, but without any doubt, the work of Alfred Tarski is the foundation for this section. Almost all of the preliminary set theory stated above in 7.8-7.15 is due to Tarski and almost all the ideas in the proofs below are due to Tarski. Most of the forms of the axiom of choice given below were either due directly to Tarski or inspired by his work.

Many of the forms listed below can be weakened, for example, by requiring one of the cardinals to be an aleph or by requiring two of the cardinals to be equal. In some instances these special cases also imply the axiom of choice. When this happens we shall list the special cases as primed forms. Then we shall prove the axiom of choice implies the standard form and the primed form implies the axiom of choice.

Another variation can be obtained for forms which contain " $\kappa \geq \aleph_0$ " in the hypotheses by replacing " $\kappa \geq \aleph_0$ " by " κ is infinite". Since

$$\omega \leq 1 \wedge \kappa \text{ is infinite} \rightarrow \kappa \geq \aleph_0,$$

the resulting statement is equivalent to the axiom of choice. We shall not explicitly state these forms here.

(The greek letters κ , μ , ν , and these letters with subscripts, represent cardinal numbers. The greek letters α , β , γ , ..., and these letters with subscripts represent ordinal numbers.)

CN 1: $(\forall \kappa > 0) (\forall \mu \geq \aleph_0) (\kappa \cdot \mu = \kappa + \mu)$.

CN'1: $(\forall \kappa \geq \aleph_0) (\kappa \cdot \kappa^* = \kappa + \kappa^*)$.

CN 2: $(\forall \kappa, \mu, \nu \geq \aleph_0) [\kappa \leq \mu + \nu \rightarrow (\kappa \leq \mu \vee \kappa \leq \nu)]$.

CN'2: $(\forall \kappa, \mu \geq \aleph_0) (\forall \alpha) [\kappa \leq \mu + \aleph_\alpha \rightarrow (\kappa \leq \mu \vee \kappa \leq \aleph_\alpha)]$.

CN 3: $(\forall \kappa, \mu, \nu \geq \aleph_0) [\kappa \leq \mu \cdot \nu \rightarrow (\kappa \leq \mu \vee \kappa \leq \nu)]$.

CN'3: $(\forall \kappa, \mu \geq \aleph_0) (\forall \alpha) [\kappa \leq \mu \cdot \aleph_\alpha \rightarrow (\kappa \leq \mu \vee \kappa \leq \aleph_\alpha)]$.

CN 4: $(\forall \kappa) (\forall \mu \geq \aleph_0) (\kappa + \mu = \kappa \vee \kappa + \mu = \mu)$.

CN'4: $(\forall \kappa \geq \aleph_0) (\kappa + \kappa^* = \kappa^*)$.

$(\kappa + \kappa^* \neq \kappa \text{ because } \kappa^* \not\leq \kappa.)$

(Tarski [1938a] noted that we can obtain an interesting equivalent formulation of CN 4 in the following way: It is easy to prove in NBG° that

(*) If x is finite then either x has at most one element or there exist disjoint sets u and v such that $|u| < |x|$ and $|v| < |x|$, and $u \cup v = x$.

The converse of (*) is equivalent to CN 4, since as noted earlier, " $\mu \geq \aleph_0$ " in the hypothesis of CN 4 can be replaced by " μ is infinite". Thus, the following statement is equivalent to CN 4.

CN 4F: A set x is finite iff x has at most one element or there exist disjoint sets u and v such that $|u| < |x|$ and $|v| < |x|$, and $u \cup v = x$.

We can obtain forms equivalent to some of the other forms in this section in a similar way. For example,

CN 1F: A set x is finite iff x has at most one element or there exists a non-empty set u disjoint from x such that $u \cup x < u \times x$.

CN 6F: A set x is finite iff x has at most one element or $x < x \times x$. See Tarski [1938a].)

CN 5: $(\forall \kappa)(\forall \mu \geq \aleph_0)(\kappa \cdot \mu = \kappa \vee \kappa \cdot \mu = \mu)$.

CN'5: $(\forall \kappa \geq \aleph_0)(\kappa \cdot \kappa^* = \kappa^*)$.
 $(\kappa \cdot \kappa^* \neq \kappa \text{ because } \kappa^* \not\leq \kappa.)$

CN 6: $(\forall \kappa \geq \aleph_0)(\kappa = \kappa^2)$.

The statement: $(\forall \kappa \geq \aleph_0)(\kappa = 2\kappa)$ has been shown to be weaker than the axiom of choice by Halpern & Howard [1974] (using a Fraenkel-Mostowski model) and by Sageev [1975] (using a Cohen model).

CN 7: $(\forall \kappa, \mu)(\kappa^2 = \mu^2 \rightarrow \kappa = \mu)$.

CN 8: $(\forall \kappa, \mu)(\kappa^* = \mu^* \rightarrow \kappa = \mu)$.

CN 9: $(\forall \kappa \geq \aleph_0) \neg (\exists \mu)(\kappa^* < \mu < \kappa + \kappa^*)$.

CN 10: $(\forall \kappa, \mu)(\kappa^* < \mu^* \rightarrow \kappa < \mu)$.

CN'10: $(\forall \kappa)(\forall \alpha)(\kappa^* < \aleph_{\alpha+1} \rightarrow \kappa < \aleph_\alpha)$.

CN 11: $(\forall \kappa, \mu)(\kappa < \mu \rightarrow \kappa^* < \mu^*)$.

CN'11: $(\forall \alpha)(\forall \mu)(\aleph_\alpha < \mu \rightarrow \aleph_{\alpha+1} < \mu^*)$.

CN 12: $(\forall \kappa \geq \aleph_0)(\forall \mu, \nu)[\kappa + \mu = \kappa + \nu \rightarrow (\mu = \nu \vee (\mu \leq \kappa \wedge \nu \leq \kappa))]$.

CN'12: $(\forall \alpha)(\forall \mu, \nu)[\aleph_\alpha + \mu = \aleph_\alpha + \nu \rightarrow (\mu = \nu \vee (\mu \leq \aleph_\alpha \wedge \nu \leq \aleph_\alpha))]$.

(The following special case of CN 12 is true if $\kappa = 2\kappa$ and is therefore weaker than the axiom of choice:

$$(\forall \kappa \geq \aleph_0)(\forall \mu)(\kappa + \kappa = \kappa + \mu \rightarrow \mu \leq \kappa).$$

CN 13: $(\forall \kappa > 0)(\forall \mu, \nu)[\kappa \cdot \mu = \kappa \cdot \nu \rightarrow (\mu = \nu \vee (\mu \leq \kappa \wedge \nu \leq \kappa))]$.

CN'13: $(\forall \alpha)(\forall \mu, \nu)[\aleph_\alpha \cdot \mu = \aleph_\alpha \cdot \nu \rightarrow (\mu = \nu \vee (\mu \leq \aleph_\alpha \wedge \nu \leq \aleph_\alpha))]$.

CN 14: $(\forall \kappa \geq \aleph_0)(\forall \mu)(\kappa^2 = \kappa \cdot \mu \rightarrow \mu \leq \kappa)$.

CN 15: $(\forall \kappa_1, \kappa_2, \mu_1, \mu_2) (\kappa_1 < \kappa_2 \wedge \mu_1 < \mu_2 \rightarrow \kappa_1 + \mu_1 < \kappa_2 + \mu_2).$

CN'15: $(\forall \alpha) (\forall \kappa, \mu_1, \mu_2) (\aleph_\alpha < \kappa \wedge \mu_1 < \mu_2 \rightarrow \aleph_\alpha + \mu_1 < \kappa + \mu_2).$

CN 16: $(\forall \kappa_1, \kappa_2, \mu_1, \mu_2) (\kappa_1 < \kappa_2 \wedge \mu_1 < \mu_2 \rightarrow \kappa_1 \cdot \mu_1 < \kappa_2 \cdot \mu_2).$

CN'16: $(\forall \alpha) (\forall \kappa, \mu_1, \mu_2) (\aleph_\alpha < \kappa \wedge \mu_1 < \mu_2 \rightarrow \aleph_\alpha \cdot \mu_1 < \kappa \cdot \mu_2).$

CN 17: $(\forall \kappa, \mu, \nu) (\kappa + \mu < \kappa + \nu \rightarrow \mu < \nu).$

CN'17: $(\forall \alpha) (\forall \nu) (\aleph_\alpha < \aleph_\alpha + \nu \rightarrow \aleph_\alpha < \nu).$

(The statement: $(\forall \kappa, \mu) (\kappa + \mu \leq \kappa + \kappa \rightarrow \mu \leq \kappa)$, is true if $\kappa = 2\kappa$ for all $\kappa \geq \aleph_0$, and is therefore weaker than the axiom of choice.)

CN 18: $(\forall \kappa, \mu, \nu) (\kappa \cdot \mu < \kappa \cdot \nu \rightarrow \mu < \nu).$

CN'18: $(\forall \alpha) (\forall \nu) (\aleph_\alpha < \aleph_\alpha \cdot \nu \rightarrow \aleph_\alpha < \nu).$

CN 19: $(\forall \kappa, \mu) (\kappa \cdot \mu \leq \kappa^2 \rightarrow \mu \leq \kappa).$

CN 20: $(\forall \kappa \geq \aleph_0) (\forall \mu, \nu) (\kappa < \nu \wedge \mu < \nu \rightarrow \kappa + \mu \neq \nu).$

CN'20: $(\forall \alpha) (\forall \mu, \nu) (\aleph_\alpha < \nu \wedge \mu < \nu \rightarrow \aleph_\alpha + \mu \neq \nu).$

CN 21: $(\forall \kappa \geq \aleph_0) (\forall \mu, \nu) (\kappa < \nu \wedge \mu < \nu \rightarrow \kappa \cdot \mu \neq \nu).$

CN'21: $(\forall \alpha) (\forall \mu, \nu) (\aleph_\alpha < \nu \wedge \mu < \nu \rightarrow \aleph_\alpha \cdot \mu \neq \nu).$

CN 1, CN 6-7 and CN 15-18 are due to Tarski [1924]. CN 19 is a special case of CN 18. M. S. Lesniewski proved that CN 4 implies the trichotomy and the proof for CN 5 is similar. CN 2-3 follow from the work of Tarski. CN 8 and CN 10-11 are due to Sobocinski [1961b], and CN 9 to Sobocinski [1961c]. CN 12 and CN 20 were given by Tarski in Lindenbaum & Tarski [1926] and CN 13-14 are similar to CN 12. CN 21 is due to Sierpinski [1946].

Each of CN 1-21 follows from the well-ordering theorem and properties of alephs and finite cardinals. The well-ordering theorem implies that each infinite cardinal is an aleph. Then using 7.4-7.6 we easily obtain each of CN 1-21.

We shall now prove the converse, that is each of CN 1-21 implies some form of the axiom of choice.

THEOREM 7.16: $CN'1 \rightarrow WO\ 3$.

PROOF: Suppose $\kappa \geq \aleph_0$. Then by 7.8, $\kappa \cdot \kappa^* = \kappa + \kappa^*$ implies $\kappa \leq \kappa^*$ or $\kappa^* \leq \kappa$. The second alternative is impossible.

THEOREM 7.17: $CN'2 \rightarrow WO\ 3$.

PROOF: Suppose $\kappa \geq \aleph_0$. Since $\kappa + \kappa^* \leq \kappa + \kappa^*$, $CN'2$ implies $\kappa + \kappa^* \leq \kappa$ or $\kappa + \kappa^* \leq \kappa^*$. The first alternative implies $\kappa^* \leq \kappa$, which cannot happen, and the second implies $\kappa \leq \kappa^*$.

The proof that $CN'3 \rightarrow WO\ 3$ is similar to the proof of 7.17 with "+" replaced by ".". Each of $CN'4$ and $CN'5$ imply $WO\ 3$ because they each imply $\kappa \leq \kappa^*$, for $\kappa \geq \aleph_0$.

Even though $CN\ 6$ obviously implies $CN\ 7$, we shall include a direct proof that $CN\ 6 \rightarrow WO\ 3$.

THEOREM 7.18: $CN\ 6 \rightarrow WO\ 3$.

PROOF: Suppose κ is transfinite. Then $CN\ 6$ implies

$$\kappa + \kappa^* = (\kappa + \kappa^*)^2 = \kappa^2 + 2\kappa\kappa^* + (\kappa^*)^2 \geq \kappa \cdot \kappa^*.$$

Thus, 7.8 implies $\kappa \leq \kappa^*$ or $\kappa^* \leq \kappa$. The second alternative is impossible, q.e.d.

THEOREM 7.19: $CN\ 7 \rightarrow WO\ 3$.

PROOF: Let $\kappa \geq \aleph_0$ and let $\mu = \kappa^{\aleph_0}$; then $\kappa \leq \mu$ and $\mu^2 = \mu$. We shall show $(\mu + \mu^*)^2 = (\mu \cdot \mu^*)^2$.

$$\begin{aligned} (\mu + \mu^*)^2 &= \mu^2 + 2\mu \cdot \mu^* + (\mu^*)^2 \\ &= \mu + \mu \cdot \mu^* + (\mu^*)^2 && \text{by definition of } \mu, \\ &= \mu + \mu \cdot \mu^* + \mu^* && \text{by 7.5,} \\ &= \mu \cdot \mu^* && \text{because } \mu^* \text{ is an aleph.} \end{aligned}$$

But since $\mu = \mu^2$ and $\mu^* = (\mu^*)^2$, we obtain the desired result. Therefore, $CN\ 7$ implies $\mu + \mu^* = \mu \cdot \mu^*$. Since $\mu^* \not\leq \mu$ we obtain (by 7.8) $\mu \leq \mu^*$. But $\kappa \leq \mu$, so κ is an aleph.

THEOREM 7.20: CN 8 $\rightarrow$ CN 9.

PROOF: Suppose $\kappa \geq \aleph_0$. Then $(\kappa^*)^* = (\kappa + \kappa^*)^*$. CN 8 implies that $\kappa^* = \kappa + \kappa^*$ so there is nothing between κ^* and $\kappa + \kappa^*$.

THEOREM 7.21: CN 9 $\rightarrow$ CN 6.

PROOF: Suppose $\kappa \geq \aleph_0$. By 7.10,

$$\kappa^* = (\kappa^2)^* \leq \kappa + \kappa^* \leq \kappa^2 + \kappa^*.$$

CN 9 implies $\kappa^* = \kappa + \kappa^*$ or $\kappa + \kappa^* = \kappa^2 + \kappa^*$. The first alternative implies κ is an aleph, so by 7.5, $\kappa = \kappa^2$. The second alternative implies $\kappa \leq \kappa^2 \leq \kappa + \kappa^*$. By 7.11, $\kappa = \kappa^2$ or $\kappa^2 = \kappa + \kappa^*$. The second equality implies $\kappa^* \leq \kappa^2$ which is impossible because $(\kappa^2)^* = \kappa^*$ (7.10), q.e.d.

THEOREM 7.22: CN'10 $\rightarrow$ WO 3.

PROOF: Suppose $\kappa \geq \aleph_0$; then $\kappa^* = \aleph_\alpha$ and $\kappa^{**} = \aleph_{\alpha+1}$ for some α . Since $\kappa^* < \kappa^{**}$, CN'10 implies $\kappa < \kappa^*$.

THEOREM 7.23: CN'11 $\rightarrow$ WO 3.

PROOF: Suppose $\kappa \geq \aleph_0$. If $\kappa^* < \kappa + \kappa^*$, then, by CN'11, $\kappa^{**} < (\kappa + \kappa^*)^* = \kappa^{**}$, contradiction. Therefore, $\kappa^* = \kappa + \kappa^*$, which implies $\kappa \leq \kappa^*$.

THEOREM 7.24: CN'12 $\rightarrow$ WO 3.

PROOF: Let $\kappa \geq \aleph_0$. Then

$$\kappa^* + \kappa = \kappa^* + (\kappa^* + \kappa).$$

CN'12 implies $\kappa = \kappa^* + \kappa$, which is impossible, or $\kappa \leq \kappa^*$ and $\kappa^* + \kappa \leq \kappa^*$. Each implies $\kappa \leq \kappa^*$.

Replacing "+" by "." in the proof of 7.24 gives a proof that CN'13 $\rightarrow$ WO 3.

THEOREM 7.25: CN 14 $\rightarrow$ WO 3.

PROOF: Let κ be any cardinal and let $\mu = \kappa^{\aleph_0}$. Then $\kappa \leq \mu$ and $\mu^2 = \mu$.

$$(\mu + \mu^*)^2 = \mu \cdot \mu^* = (\mu + \mu^*)(\mu \cdot \mu^*).$$

CN 14 implies that $\mu \cdot \mu^* \leq \mu + \mu^*$. Thus, using 7.8, we obtain $\mu^* \leq \mu$, which is impossible, or $\mu \leq \mu^*$, q.e.d.

THEOREM 7.26: CN'15 $\rightarrow$ WO 3.

PROOF: Let κ be any cardinal and let $\mu = \aleph_0 \cdot \kappa$. Then $\kappa \leq \mu$ and $2\mu = \mu$. We have $\mu < \mu + \mu^*$ and $\mu^* \leq \mu + \mu^*$. If $\mu^* = \mu + \mu^*$ then $\mu \leq \mu^*$. Suppose $\mu^* < \mu + \mu^*$. Then, by CN'15,

$$\mu + \mu^* < (\mu + \mu^*) + (\mu + \mu^*) = \mu + \mu^*,$$

clearly a contradiction, q.e.d.

The proof that CN'16 $\rightarrow$ WO 3 is similar to the proof of 7.26 with $\mu = \kappa^{\aleph_0}$ and "+" replaced by ".".

THEOREM 7.27: CN'17 $\rightarrow$ WO 3.

PROOF: Let $\kappa \geq \aleph_0$; then $\kappa^* \leq \kappa^* + \kappa$. If $\kappa^* = \kappa^* + \kappa$, then $\kappa \leq \kappa^*$; otherwise CN'17 implies the contradiction $\kappa^* < \kappa$.

If "+" is replaced by "." in the proof of 7.27, we obtain a proof that CN'18 $\rightarrow$ WO 3. Also, it is clear that CN 19 $\rightarrow$ CN 14.

THEOREM 7.28: CN'20 $\rightarrow$ WO 3.

PROOF: Suppose $\kappa \geq \aleph_0$. We have $\kappa \leq \kappa + \kappa^*$ and $\kappa^* \leq \kappa + \kappa^*$. If strict inequality held for both then CN'20 implies a contradiction. Thus we must have either $\kappa = \kappa + \kappa^*$, which cannot happen, or $\kappa^* = \kappa + \kappa^*$ which implies $\kappa \leq \kappa^*$, q.e.d.

The proof that CN'21 implies WO 3 is obtained from the proof of 7.28 by replacing "+" by ".".

The next set of statements we consider deal with subtraction and division of cardinal numbers.

DEFINITION 7.29: SUBTRACTION AND DIVISION

- (i) $[\kappa \leq \mu \wedge \kappa + \nu = \mu \wedge (\forall \nu_1) (\kappa + \nu_1 = \mu \rightarrow \nu_1 = \nu)] \rightarrow \mu - \kappa = \nu$.
- (ii) $[\kappa \leq \mu \wedge \kappa \cdot \mu = \mu \wedge (\forall \nu_1) (\kappa \cdot \nu_1 = \mu \rightarrow \nu_1 = \nu)] \rightarrow \mu \div \kappa = \nu$.

CN 22: $(\forall \kappa, \mu) (\kappa < \mu \rightarrow \mu - \kappa \text{ exists})$.

(That is, $(\exists! v)(\mu = \kappa + v)$. The statement,
 $\kappa < \mu \rightarrow (\exists v)(\mu = \kappa + v)$ is provable in NBG°.)

CN'22: $(\forall \alpha)(\forall \mu)(\aleph_\alpha < \mu \rightarrow \mu - \aleph_\alpha \text{ exists})$.

CN 23: $(\forall \kappa)(\forall \mu \geq \aleph_0)(\kappa < \mu \rightarrow \mu - \kappa = \mu)$.

CN'23: $(\forall \alpha)(\forall \mu)(\aleph_\alpha < \mu \rightarrow \mu - \aleph_\alpha = \mu)$.

CN 24: $(\forall \kappa_1, \kappa_2)(\forall \mu_1, \mu_2 \geq \aleph_0)[(\kappa_1 < \mu_1 \wedge \kappa_2 < \mu_2 \wedge \kappa_1 + \mu_2 = \kappa_2 + \mu_1) \rightarrow \mu_1 = \mu_2]$.

(If in CN 24 we replace " $\kappa_1 + \mu_2 = \kappa_2 + \mu_1$ " by " $\kappa_1 + \mu_1 = \kappa_2 + \mu_2$ ", the resulting statement holds if $\kappa = 2\kappa$, for all infinite κ . Thus, it is weaker than the axiom of choice.)

CN'24: $(\forall \alpha)(\forall \kappa)(\forall \mu_1, \mu_2 \geq \aleph_0)[(\aleph_\alpha < \mu_1 \wedge \kappa < \mu_2 \wedge \aleph_\alpha + \mu_2 = \kappa + \mu_1) \rightarrow \mu_1 = \mu_2]$.

CN 25: $(\forall \kappa > 0)(\forall \mu \geq \aleph_0)[\kappa < \mu \rightarrow (\exists v)(\mu = \kappa \cdot v)]$.

CN'25: $(\forall \alpha)(\forall \mu)[\aleph_\alpha < \mu \rightarrow (\exists v)(\mu = \aleph_\alpha \cdot v)]$.

CN 26: $(\forall \kappa > 0)(\forall \mu \geq \aleph_0)(\kappa < \mu \rightarrow \mu = \kappa \cdot \mu)$.

CN'26: $(\forall \alpha)(\forall \mu)(\aleph_\alpha < \mu \rightarrow \mu = \aleph_\alpha \cdot \mu)$.

CN 27: $(\forall \kappa > 0)(\forall \mu \geq \aleph_0)(\kappa < \mu \rightarrow \mu \div \kappa \text{ exists})$.

(That is, $(\exists! v)(\mu = \kappa \cdot v)$.)

CN'27: $(\forall \alpha)(\forall \mu)(\aleph_\alpha < \mu \rightarrow \mu \div \aleph_\alpha \text{ exists})$.

CN 28: $(\forall \kappa > 0)(\forall \mu \geq \aleph_0)(\kappa < \mu \rightarrow \mu \div \kappa = \mu)$.

CN'28: $(\forall \alpha)(\forall \mu)(\aleph_\alpha < \mu \rightarrow \mu \div \aleph_\alpha = \mu)$.

CN 29: $(\forall \kappa_1, \kappa_2)(\forall \mu_1, \mu_2 \geq \aleph_0)[(0 < \kappa_1 < \mu_1 \wedge \kappa_2 < \mu_2 \wedge \kappa_1 \cdot \mu_1 = \kappa_2 \cdot \mu_2) \rightarrow \mu_1 = \mu_2]$.

CN'29: $(\forall \alpha)(\forall \kappa)(\forall \mu_1, \mu_2 \geq \aleph_0)[(\aleph_\alpha < \mu_1 \wedge \kappa < \mu_2 \wedge \aleph_\alpha \cdot \mu_1 = \kappa \cdot \mu_2) \rightarrow \mu_1 = \mu_2]$.

CN 30: $(\forall \kappa_1, \kappa_2) (\forall \mu_1, \mu_2 \geq \aleph_0) [(0 < \kappa_1 < \mu_1 \wedge \kappa_2 < \mu_2 \wedge \kappa_1 \cdot \mu_2 = \kappa_2 \cdot \mu_1) \rightarrow \mu_1 = \mu_2]$.

CN'30: $(\forall \alpha) (\forall \kappa) (\forall \mu_1, \mu_2 \geq \aleph_0) [(\aleph_\alpha < \mu_1 \wedge \kappa < \mu_2 \wedge \aleph_\alpha \cdot \mu_2 = \kappa \cdot \mu_1) \rightarrow \mu_1 = \mu_2]$.

CN 22-23 are due to Tarski, Lindenbaum & Tarski [1926]. Later, Sierpinski [1947b], published a proof of CN 22. CN 24 and CN 29-30 were given in Sobocinski [1965]. CN 25-28 are related to the work of Tarski and Sierpinski.

Just like the first twenty one statements, these statements follow easily from the well-ordering theorem. Below we give the proofs that each implies some form of the axiom of choice.

THEOREM 7.30: CN'22 $\rightarrow$ WO 3.

PROOF: Suppose $\kappa \geq \aleph_0$; then $\kappa^* \leq \kappa^* + \kappa$. If $\kappa^* < \kappa^* + \kappa$ then CN'22 implies that there is a unique cardinal μ such that $\kappa^* + \mu = \kappa^* + \kappa$. But $\kappa^* + (\kappa^* + \kappa) = \kappa^* + \kappa$, so we must have $\kappa = \kappa^* + \kappa$ which is a contradiction. Therefore, $\kappa^* = \kappa^* + \kappa$, which implies $\kappa \leq \kappa^*$.

Clearly, CN'23 $\rightarrow$ CN'22.

THEOREM 7.31: CN'24 $\rightarrow$ CN'23.

PROOF: Suppose $\aleph_\alpha < \mu$. Then there is a cardinal κ such that $\aleph_\alpha + \kappa = \mu$. Since $0 < \kappa$ we can apply CN'24 and obtain $\mu = \kappa$. Thus, $\mu - \aleph_\alpha = \mu$.

THEOREM 7.32: CN'25 $\rightarrow$ WO 3.

PROOF: Let $\kappa \geq \aleph_0$; then $\kappa^* \leq \kappa + \kappa^*$. If $\kappa^* < \kappa + \kappa^*$ then CN'25 implies that there is a ν such that $\kappa + \kappa^* = \kappa^* \cdot \nu$. Then, using 7.8, $\kappa^* \leq \kappa$, which is impossible, or $\nu \leq \kappa^*$. The latter inequality implies

$$\kappa \leq \kappa + \kappa^* = \kappa^* \cdot \nu = \kappa^*.$$

Since CN'28 $\rightarrow$ CN'27 $\rightarrow$ CN'25 and CN'26 $\rightarrow$ CN'25, it remains to be shown that each of CN'29 and CN'30 implies the axiom of choice.

THEOREM 7.33: CN'29 $\rightarrow$ CN'26.

PROOF: Suppose $\aleph_\alpha < \mu$. Since $1 < \aleph_\alpha \cdot \mu$ and $\aleph_\alpha \cdot \mu = 1 \cdot (\aleph_\alpha \cdot \mu)$, CN'29 implies $\mu = \aleph_\alpha \cdot \mu$.

THEOREM 7.34: CN'30 $\rightarrow$ CN'26.

PROOF: Suppose $\aleph_\alpha < \mu$; then $\aleph_\alpha < \aleph_\alpha \cdot \mu$, $1 < \mu$ and $\aleph_\alpha \cdot \mu = 1 \cdot (\aleph_\alpha \cdot \mu)$. So, CN'30 implies $\mu = \aleph_\alpha \cdot \mu$.

The following group of statements are, for the most part, weakened forms of CN 6.

CN 31: $(\forall \kappa \geq \aleph_0) (\exists \mu) (\kappa = \mu^2)$.

CN 32($\aleph_\alpha$): $(\forall \kappa) (\aleph_\alpha \cdot \kappa = \aleph_\alpha \cdot \kappa^2)$.

CN 33: $(\exists \alpha) (\forall \kappa \geq \aleph_0)$ (There is no $\leq$ -well-ordered set M of cardinal numbers between κ and κ^2 such that $M \cong \alpha$.)

CN 34: $(\exists \alpha) (\forall \kappa, \mu) (\kappa^* = \mu^* \rightarrow$ There is no $\leq$ -well-ordered set M of cardinal numbers between κ and μ such that $M \cong \alpha$.)

CN 35: $(\forall \kappa, \mu \geq \aleph_0) [\kappa \leq \mu < 2^\kappa \wedge (\exists v_1) (\kappa = v_1^2) \rightarrow (\exists v_2) (v = v_2^2)]$.

CN 36: $(\forall \kappa, \mu \geq \aleph_0) [\kappa \leq \mu < 2^\kappa \rightarrow (\exists v) (\mu = \kappa \cdot v)]$.

CN 31, while not explicitly stated in the literature, follows from the work of Tarski and Sierpinski; CN 32($\aleph_\alpha$) is due to Alas [1968]; CN 33 first appeared in Rubin & Rubin [1963]; CN 34 is a generalization of a form given in Sobocinski [1962a]; and CN 35-36 are due to Seki [1961].

Clearly, CN 6 implies each of CN 31-33; CN 8 $\rightarrow$ CN 34; CN 25 $\rightarrow$ CN 36; and CN 31 $\rightarrow$ CN 35. It remains to be shown that each of CN 32-36 imply the axiom of choice.

THEOREM 7.35: CN 32($\aleph_\alpha$) $\rightarrow$ WO 3.

PROOF: Suppose $\kappa \geq \aleph_0$. Let $\mu = \aleph_\alpha \cdot \kappa + (\aleph_\alpha \cdot \kappa)^*$. Then $\aleph_\alpha \cdot \mu = \mu$. CN 32($\aleph_\alpha$) implies

$$\begin{aligned} \aleph_\alpha \cdot \kappa + (\aleph_\alpha \cdot \kappa)^* &= [\aleph_\alpha \cdot \kappa + (\aleph_\alpha \cdot \kappa)^*]^2 \\ &\geq (\aleph_\alpha \cdot \kappa) \cdot (\aleph_\alpha \cdot \kappa)^*. \end{aligned}$$

Using 7.8, we obtain $\aleph_\alpha \cdot \kappa \leq (\aleph_\alpha \cdot \kappa)^*$ or $(\aleph_\alpha \cdot \kappa)^* \leq \aleph_\alpha \cdot \kappa$. The second alternative is impossible and the first implies κ is an aleph.

THEOREM 7.36: CN 33 $\rightarrow$ WO 3.

PROOF: Suppose $\mu \geq \aleph_0$, but suppose μ is not an aleph. We shall show that CN 33 is false. Let α be an arbitrary ordinal, let $\mu^* = \aleph_\delta$ and let $\kappa = \mu + \aleph_{\delta+\alpha}$. We shall show that there is a $\leq$ -well-ordered set M of cardinal numbers between κ and κ^2 such that $M \cong \alpha$.

For $\beta < \alpha$, let

$$v_\beta = \mu \cdot \aleph_{\delta+\beta} + \aleph_{\delta+\alpha}.$$

If $\beta < \gamma < \alpha$, $v_\beta < v_\gamma$; for suppose $v_\beta = v_\gamma$. Then

$$\mu \cdot \aleph_{\delta+\beta} + \aleph_{\delta+\alpha} = \mu \cdot \aleph_{\delta+\gamma} + \aleph_{\delta+\alpha}.$$

Therefore, $\mu \cdot \aleph_{\delta+\gamma} \leq \mu \cdot \aleph_{\delta+\beta} + \aleph_{\delta+\alpha}$. It follows from 7.8 that $\mu \leq \aleph_{\delta+\alpha}$ or $\aleph_{\delta+\gamma} \leq \mu \cdot \aleph_{\delta+\beta}$. The first alternative is impossible because we assumed that μ was not an aleph. The second alternative, by 7.9, implies $\aleph_{\delta+\gamma} \leq \mu$ or $\aleph_{\delta+\gamma} \leq \aleph_{\delta+\beta}$. The first alternative cannot happen because $\mu^* = \aleph_\delta$ and the second cannot happen either because $\beta < \gamma$. Thus, $\beta < \gamma$ implies $v_\beta < v_\gamma$.

Let $M = \{v_\beta : \beta < \alpha\}$; then $M \cong \alpha$. It is easy to see that for each $\beta < \alpha$, $\kappa \leq v_\beta \leq \kappa^2$. We shall show that equality cannot occur. Suppose $\kappa = v_\beta$ for some $\beta < \alpha$. Then $\mu + \aleph_{\delta+\alpha} = \mu \cdot \aleph_{\delta+\beta} + \aleph_{\delta+\alpha}$. But this equality implies

$$\mu \cdot \aleph_{\delta+\beta} \leq \mu + \aleph_{\delta+\alpha}.$$

Using 7.8, we obtain the contradiction $\mu \leq \aleph_{\delta+\alpha}$ or $\aleph_{\delta+\beta} \leq \mu$. On the other hand, if $\kappa^2 = v_\beta$ for some β , then $\mu^2 + \mu \cdot \aleph_{\delta+\alpha} = \mu \cdot \aleph_{\delta+\beta} + \aleph_{\delta+\alpha}$. This implies

$$\mu \cdot \aleph_{\delta+\alpha} \leq \mu \cdot \aleph_{\delta+\beta} + \aleph_{\delta+\alpha}.$$

Using 7.8 and 7.9 we obtain the contradiction:

$$\mu \leq \aleph_{\delta+\alpha} \vee \aleph_{\delta+\alpha} \leq \mu \vee \aleph_{\delta+\alpha} \leq \aleph_{\delta+\beta}.$$

Thus, we have shown that for each $\beta < \alpha$, $\kappa < v_\beta < \kappa^2$, q.e.d.

The proof of 7.36 is also a proof that $CN\ 34 \rightarrow WO\ 3$, because, by 7.10, for all $\kappa \geq \aleph_0$, $\kappa^* = (\kappa^2)^*$.

THEOREM 7.37: $CN\ 35 \rightarrow WO\ 3$.

PROOF: Let $\kappa \geq \aleph_0$ and let $\mu = \kappa^{\aleph_0}$. Then $\kappa \leq \mu$ and $\mu^2 = \mu$. By 7.12,

$$\mu^* \leq 2^{2^{\mu^2}} = 2^{2^\mu}.$$

Therefore,

$$2^\mu \leq 2^\mu + \mu^* \leq 2^\mu + 2^{2^\mu} = 2^{2^\mu}.$$

If $2^\mu + \mu^* = 2^{2^\mu}$ then, by 7.15, $\mu^* = 2^{2^\mu}$ which implies that μ , and also κ , is an aleph. Suppose

$$2^\mu \leq 2^\mu + \mu^* < 2^{2^\mu}.$$

Since $2^\mu = (2^\mu)^2$, it follows from CN 35 that there is a v such that $2^\mu + \mu^* = v^2$. By 7.8, $v \leq \mu^*$ or $v \leq 2^\mu$. The first inequality implies that 2^μ , and therefore κ , is an aleph. If $v \leq 2^\mu$, then

$$\mu^* \leq 2^\mu + \mu^* = v^2 \leq (2^\mu)^2 = 2^\mu.$$

This implies,

$$\mu \leq \mu + \mu^* \leq 2^\mu.$$

If $\mu + \mu^* = 2^\mu$, then, by 7.15, $\mu^* = 2^\mu$, implying κ is an aleph. Suppose

$$\mu \leq \mu + \mu^* < 2^\mu.$$

Since $\mu^2 = \mu$, we can apply CN 35 again. Therefore, there is a v_1 such that $\mu + \mu^* = (v_1)^2$. Now we are finished because by 7.8, $v_1 \leq \mu$ or $v_1 \leq \mu^*$. If the first inequality holds we would have

$$\mu^* \leq \mu + \mu^* = (v_1)^2 \leq \mu^2 = \mu$$

which is impossible. The second inequality implies that μ , and therefore κ , is an aleph.

The proof that $CN\ 36 \rightarrow WO\ 3$ is similar. Whenever $CN\ 35$ is used in the proof of 7.37, use $CN\ 36$ and the same results follow.

Our next three statements involve the notion of "immediate successor".

DEFINITION 7.38: IMMEDIATE SUCCESSOR

- (a) $\kappa <_1 \mu$ iff $\kappa < \mu \wedge \neg (\exists v)(\kappa < v < \mu)$.
- (b) $\kappa <_2 \mu$ iff $\kappa < \mu \wedge (\forall v)(\kappa < v \rightarrow \mu \leq v)$.
- (c) $\kappa <_3 \mu$ iff $\kappa < \mu \wedge (\forall v)(v < \mu \rightarrow v \leq \kappa)$.

It follows from 7.11 that if κ is transfinite then $\kappa <_1 \kappa + \kappa^*$. Thus,

$$(*) \quad (\forall \kappa \geq \aleph_0)(\exists \mu)(\kappa <_1 \mu)$$

is provable in NBG° . If we replace " $<_1$ " in (*) by " $<_2$ ", the resulting statement, $CN\ 37$, is equivalent to the axiom of choice. (Tarski [1954].) If in (*) we replace " $<_1$ " by " $<_3$ ", we do not know if the resulting statement implies the axiom of choice. However the existence of a 3-successor does imply that $(\forall \kappa \geq \aleph_0)(\kappa = 2\kappa)$. (Tarski [1954] and Truss [1973]). Sageev [1975] has shown that the converse is not true.

$$CN\ 37: (\forall \kappa \geq \aleph_0)(\exists \mu)(\kappa <_2 \mu).$$

$$CN'37: (\forall \alpha)(\exists \mu)(\aleph_\alpha <_2 \mu).$$

$$CN\ 38: (\forall \kappa, \mu \geq \aleph_0)[\kappa <_1 \mu \rightarrow (\forall v)(v \cdot \kappa = v \cdot \mu \vee v \cdot \kappa <_1 v \cdot \mu)].$$

$$CN'38: (\forall \alpha)(\forall v)(v \cdot \aleph_\alpha = v \cdot \aleph_{\alpha+1} \vee v \cdot \aleph_\alpha <_1 v \cdot \aleph_{\alpha+1}).$$

($CN\ 38 \rightarrow CN'38$ because $\aleph_\alpha <_1 \aleph_{\alpha+1}$.)

$$CN\ 39: (\forall \kappa \geq \aleph_0)(\forall \mu, v)(\kappa < \mu \wedge \kappa <_3 v \rightarrow v \leq \mu).$$

$$CN'39: (\forall \kappa \geq \aleph_0)(\kappa^* < \kappa^* + \kappa \rightarrow \kappa^{**} \leq \kappa^* + \kappa).$$

($CN\ 39 \rightarrow CN'39$ because $\kappa^* <_3 \kappa^{**}$.)

$CN\ 37$ is due to Tarski [1954]; $CN\ 38$ to Sudan [1938]; and $CN\ 39$ is related to work given in Sobocinski [1962b].

Using 7.4-7.6, it is easy to see that the well-ordering theorem implies each of CN 37-39.

THEOREM 7.39: CN'37 $\rightarrow$ WO 3.

PROOF: Suppose $\kappa \geq \aleph_0$. CN'37 implies that κ^* has a 2-successor, μ . But $\kappa^* < \kappa^{**}$. Therefore, by 7.38(b), $\mu \leq \kappa^{**}$. Since there is nothing between κ^* and κ^{**} , $\mu = \kappa^{**}$. Thus, $\kappa^* <_2 \kappa^{**}$.

Clearly, $\kappa^* \leq \kappa + \kappa^*$. If $\kappa^* < \kappa + \kappa^*$ then by 7.38(b), $\kappa^{**} \leq \kappa + \kappa^*$. This inequality, by 7.5 and 7.8, implies the contradiction $\kappa^{**} \leq \kappa$ or $\kappa^{**} \leq \kappa^*$. Thus, we must have $\kappa^* = \kappa + \kappa^*$ which implies $\kappa \leq \kappa^*$.

(This proof is credited to A. Levy by Truss [1973].)

THEOREM 7.40: CN'38 $\rightarrow$ WO 3.

PROOF: Let $\kappa \geq \aleph_0$ and suppose $\kappa^* = \aleph_\alpha$. CN'38 implies that either $\kappa \cdot \aleph_\alpha = \kappa \cdot \aleph_{\alpha+1}$ or $\kappa \cdot \aleph_\alpha <_1 \kappa \cdot \aleph_{\alpha+1}$. However,

$$\kappa \cdot \aleph_\alpha \leq \kappa \cdot \aleph_\alpha + \aleph_{\alpha+1} \leq \kappa \cdot \aleph_{\alpha+1}.$$

Therefore, either

$$(1) \quad \kappa \cdot \aleph_\alpha = \kappa \cdot \aleph_\alpha + \aleph_{\alpha+1}$$

or

$$(2) \quad \kappa \cdot \aleph_\alpha + \aleph_{\alpha+1} = \kappa \cdot \aleph_{\alpha+1}.$$

If (1) holds then $\aleph_{\alpha+1} \leq \kappa \cdot \aleph_\alpha$. But $\aleph_{\alpha+1} \not\leq \kappa$ and $\aleph_{\alpha+1} \not\leq \aleph_\alpha$, which contradicts 7.9. If (2) holds, then by 7.8, either $\aleph_{\alpha+1} \leq \kappa \cdot \aleph_\alpha$, which is impossible, or $\kappa \leq \aleph_{\alpha+1}$, q.e.d.

THEOREM 7.41: CN'39 $\rightarrow$ WO 3.

PROOF: It follows from 7.5 and 7.8 that the conclusion of CN'39 is false. Thus, the hypotheses is false which implies $\kappa^* = \kappa^* + \kappa$. Consequently, $\kappa \leq \kappa^*$.

We continue with statements involving cardinal exponentiation.

Let $\nu > 1$ be a cardinal,

CN 40(ν): $(\forall \kappa, \mu) (\kappa^\nu < \mu^\nu \rightarrow \kappa < \mu)$.

CN 41: $(\exists \mu > 1) (\forall \kappa_1, \kappa_2) [(\forall v) (1 < v \leq \mu \rightarrow \kappa_1^v < \kappa_2^v) \rightarrow \kappa_1 < \kappa_2]$.

CN 42: $(\forall \kappa, \mu, v) (\kappa \neq 0 \wedge \kappa^\mu < \kappa^v \rightarrow \mu < v)$.

CN'42: $(\forall \kappa, \mu) (\forall \alpha) (\kappa \neq 0 \wedge \kappa^\mu < \kappa^{\aleph_\alpha} \rightarrow \mu < \aleph_\alpha)$.

CN 40(v) and CN 42 were given by Tarski in Lindenbaum & Tarski [1926], and CN 41 is a generalization of CN 40.

For $v > 1$, CN 40(v) $\rightarrow$ CN 41. The trichotomy and monotonicity laws for cardinal exponentiation imply CN 40(v) and CN 42. We shall show that CN 41 and CN 42 each imply the well ordering theorem.

THEOREM 7.42: CN 41 $\rightarrow$ WO 3.

PROOF: Suppose $\kappa \geq \aleph_0$. Let $\mu_1 = 2^{\mu \cdot \aleph_0}$; then $\mu \cdot \mu_1 = \mu_1$. Let $\lambda = \kappa^{\mu_1}$, $\kappa_1 = \lambda \cdot \lambda^*$ and $\kappa_2 = \lambda + (\lambda \cdot (\lambda^*)^\mu)^*$. Suppose $1 < v \leq \mu$, then

$$\begin{aligned} \kappa_2^v &= [\lambda + (\lambda \cdot (\lambda^*)^\mu)^*]^v \\ &\geq \lambda \cdot ((\lambda \cdot (\lambda^*)^\mu)^*)^{v-1} \\ &\geq \lambda \cdot (((\lambda^*)^\mu)^*)^{v-1} \\ &\geq \lambda \cdot (\lambda^*)^v && \text{(by 7.5)} \\ &= (\lambda \cdot \lambda^*)^v && \text{(because } \lambda^v = \lambda) \\ &= \kappa_1^v. \end{aligned}$$

We shall show next that $\kappa_1^v < \kappa_2^v$.

$$\begin{aligned} (\kappa_1^v)^* &= ((\lambda \cdot \lambda^*)^v)^* \\ &= (\lambda \cdot (\lambda^*)^v)^* \\ &\leq (\lambda \cdot (\lambda^*)^\mu)^* \\ &\leq \kappa_2 \leq \kappa_2^v. \end{aligned}$$

Therefore, $\kappa_1^v < \kappa_2^v$ because $(\kappa_1^v)^* \neq \kappa_1^v$. CN 41 implies that $\kappa_1 < \kappa_2$. That is,

$$\lambda \cdot \lambda^* < \lambda + (\lambda \cdot (\lambda^*)^\mu)^*.$$

By 7.8, either $\lambda^* \leq \lambda$, which is impossible, or $\lambda \leq (\lambda \cdot (\lambda^*)^\mu)^*$. Since $\kappa \leq \lambda$, we obtain the desired result.

THEOREM 7.43: $\text{CN}'42 \rightarrow \text{WO } 3$.

PROOF: Let $\mu \geq \aleph_0$ and let $\kappa = 2^\mu$. Then $\kappa^\mu = \kappa \leq \kappa^{\kappa^*}$. If $\kappa = \kappa^{\kappa^*}$, then since $\kappa^* \leq \kappa^{\kappa^*}$, we would have $\kappa^* \leq \kappa$ which is impossible. Therefore $\kappa^\mu < \kappa^{\kappa^*}$ and $\kappa \neq 0$, so $\text{CN}'42$ implies $\mu < \kappa^*$, q.e.d.

Let $\kappa \cap \mu$ be the greatest lower bound of κ and μ , and $\kappa \cup \mu$ their least upper bound. Tarski [1949] has shown that the following statement is provable in NBG° :

(*) If $\kappa \cap \mu$ exists then $\kappa \cup \mu$ exists and $\kappa + \mu = (\kappa \cap \mu) + (\kappa \cup \mu)$.

(If $v = \kappa \cap \mu$ then $\kappa = v + \kappa_1$ and $\mu = v + \mu_1$ for some κ_1 and μ_1 . It can be shown that $\kappa \cup \mu = v + \kappa_1 + \mu_1$. It was pointed out by Arthur L. Rubin that the statement: $(\forall \kappa, \mu) (\kappa \cap \mu \text{ exists})$, is weaker than the axiom of choice.)

We have shown that if "+" is replaced by "·" in (*), the resulting statement is equivalent to the axiom of choice.

CN 43: $(\forall \kappa, \mu) [\text{If } \kappa \cap \mu \text{ exists then } \kappa \cup \mu \text{ exists and } \kappa \cdot \mu = (\kappa \cap \mu) \cdot (\kappa \cup \mu)]$.

CN'43: $(\forall \kappa) (\forall \alpha) [\text{If } \kappa \cap \aleph_\alpha \text{ exists then } \kappa \cup \aleph_\alpha \text{ exists and } \kappa \cdot \aleph_\alpha = (\kappa \cap \aleph_\alpha) \cdot (\kappa \cup \aleph_\alpha)]$.

The trichotomy, T1, implies that for all cardinals κ and μ , $\kappa \leq \mu$ or $\mu \leq \kappa$. Thus, $\kappa \cap \mu = \min(\kappa, \mu)$, $\kappa \cup \mu = \max(\kappa, \mu)$ and CN 43 follows.

THEOREM 7.44: $\text{CN}'43 \rightarrow \text{WO } 3$.

PROOF: Let $v \geq \aleph_0$ and let α be chosen so that $v^* \leq \aleph_{\alpha+1}$. Let $\kappa = v + \aleph_\alpha$; then $\kappa \cap \aleph_{\alpha+1} = \aleph_\alpha$ and $\kappa \cup \aleph_{\alpha+1} = v + \aleph_{\alpha+1}$.

$$(1) \quad \kappa \cdot \aleph_{\alpha+1} = (v + \aleph_\alpha) \cdot \aleph_{\alpha+1} = v \cdot \aleph_{\alpha+1} + \aleph_{\alpha+1}.$$

$$(2) \quad (\kappa \cap \aleph_{\alpha+1}) \cdot (\kappa \cup \aleph_{\alpha+1}) = \aleph_\alpha \cdot (v + \aleph_{\alpha+1}) = v \cdot \aleph_\alpha + \aleph_{\alpha+1}.$$

CN'43 implies $\kappa \cdot \aleph_{\alpha+1} = (\kappa \cap \aleph_{\alpha+1}) \cdot (\kappa \cup \aleph_{\alpha+1})$. So it follows from (1) and (2) that

$$\nu \cdot \aleph_{\alpha+1} \leq \nu \cdot \aleph_{\alpha} + \aleph_{\alpha+1}.$$

Using 7.8, we obtain $\aleph_{\alpha+1} \leq \nu \cdot \aleph_{\alpha}$ or $\nu \leq \aleph_{\alpha+1}$. But $\aleph_{\alpha+1} \not\leq \nu$ and $\aleph_{\alpha+1} \not\leq \aleph_{\alpha}$ so the first inequality contradicts 7.9. Therefore, we are left with $\nu \leq \aleph_{\alpha+1}$.

DEFINITION 7.45:

- (a) $\mathcal{P}_{<\omega}(X)$ is the class of all finite subsets of X .
- (b) If x is a set and $|x| = \kappa$, then $e(\kappa) = |\mathcal{P}_{<\omega}(x)|$.

It is easy to prove the following three facts about e in NBG°:

$$7.46: (\forall \alpha) e(\aleph_{\alpha}) = \aleph_{\alpha}.$$

$$7.47: (\forall \kappa, \mu) e(\kappa + \mu) = e(\kappa) \cdot e(\mu).$$

$$7.48: (\forall \kappa, \alpha) (\kappa = \kappa \cdot \aleph_0 \rightarrow e(\kappa \cdot \aleph_{\alpha}) = e(\kappa) \cdot \aleph_{\alpha}).$$

The following three forms are due to Alas [1968,1970].

$$\text{CN 44: } (\forall \kappa, \mu \geq \aleph_0) [e(\kappa + \mu) = e(\kappa) + e(\mu)].$$

$$\text{CN'44: } (\forall \kappa \geq \aleph_0) (\forall \alpha) [e(\kappa + \aleph_{\alpha}) = e(\kappa) + \aleph_{\alpha}].$$

(For any α , $e(\aleph_{\alpha}) = \aleph_{\alpha}$.)

$$\text{CN 45: } (\forall \kappa, \mu \geq \aleph_0) (e(\kappa) = e(\mu) \wedge \kappa \leq \mu \rightarrow \kappa = \mu).$$

$$\text{CN 46: } (\forall \kappa, \mu \geq \aleph_0) [e(\kappa + \mu) = e(\kappa) \vee e(\kappa + \mu) = e(\mu)].$$

$$\text{CN'46: } (\forall \kappa \geq \aleph_0) (e(\kappa + [e(\kappa)]^*) = [e(\kappa)]^*).$$

(CN'46 is a special case of CN 46 because for any aleph, $\aleph$, $e(\aleph) = \aleph$ and $e(\kappa + [e(\kappa)]^*) \neq e(\kappa)$.)

It follows from the well ordering theorem and properties of alephs that each of CN 44-46 holds.

THEOREM 7.49: CN'44 $\rightarrow$ WO 3.

PROOF: Let $\kappa \geq \aleph_0$ and let $\aleph_{\alpha} = [e(\kappa)]^*$. By CN'44, $e(\kappa + \aleph_{\alpha}) = e(\kappa) + \aleph_{\alpha}$. But, $e(\kappa) \cdot \aleph_{\alpha} = e(\kappa + \aleph_{\alpha})$ (by 7.46

and 7.47). Therefore, $e(\kappa) \cdot \aleph_\alpha = e(\kappa) + \aleph_\alpha$. By 7.8, $\aleph_\alpha \leq e(\kappa)$, which is impossible, or $e(\kappa) \leq \aleph_\alpha$, which implies $\kappa \leq \aleph_\alpha$.

THEOREM 7.50: CN 45 $\rightarrow$ WO 3.

PROOF: Let $\kappa = \mu \cdot \aleph_0$. Then $\kappa + \kappa^* \leq \kappa \cdot \kappa^*$ and by 7.46, 7.47 and 7.48, $e(\kappa + \kappa^*) = e(\kappa \cdot \kappa^*)$. CN 45 implies $\kappa + \kappa^* = \kappa \cdot \kappa^*$. Using 7.8 we obtain $\kappa \leq \kappa^*$, which implies $\mu \leq \kappa^*$.

THEOREM 7.51: CN'46 $\rightarrow$ WO 3.

PROOF: $\kappa \leq e(\kappa + [e(\kappa)]^*) = [e(\kappa)]^*$.

The following three forms, CN 47-49, are due to Tarski [1938,1939].

CN 47: $(\forall s \neq \emptyset)(\exists t)(t = \{x \subseteq t : |s| \not\leq |x|\})$.

CN 48: $(\forall s \neq \emptyset)(\exists t)(t \approx \{x \subseteq t : |s| \not\leq |x|\})$.

CN 49: $(\forall t)(\forall \mu)[|t| \geq \aleph_0 \wedge \mu \leq |t| \rightarrow |\{x \subseteq t : \mu \not\leq |x|\}| = |t|^\mu]$.

We shall prove

AC 1 $\wedge$ T 1 $\rightarrow$ CN 47 $\rightarrow$ CN 48 $\rightarrow$ WO 3.

T 1 $\wedge$ AC 1 $\wedge$ WO 1 $\rightarrow$ CN 49 $\rightarrow$ CN 48.

It is clear that CN 47 $\rightarrow$ CN 48 because CN 48 is a special case of CN 47.

THEOREM 7.52: AC 1 $\wedge$ T 1 $\rightarrow$ CN 47.

PROOF: Let $s \neq \emptyset$ and suppose $|s| = \kappa$. Let α be a successor ordinal such that $(2^\kappa)^* \leq \aleph_\alpha$. Define a function f on ω_α as follows: For $\beta \in \omega_\alpha$,

$$f(\beta) = \{x : x \subseteq \cup f''\beta \wedge |x| < \kappa\}.$$

Then define $t = \cup f''\omega_\alpha$. We shall show that t is the required set. Suppose $x \in t$. Then for some $\beta < \omega_\alpha$, $x \in f(\beta)$. So $|x| < \kappa$ which implies $\kappa \not\leq |x|$. Also, $x \subseteq \cup f''\beta \subseteq \cup f''\omega_\alpha = t$.

Conversely, suppose $x \subseteq t$ and $\kappa \not\leq |x|$. T 1 implies $|x| < \kappa$. For each $u \in x$, let $B_u = \{\beta : u \in f(\beta)\}$ and let

γ_u be the smallest element of B_u . Let $R = \{\gamma_u : u \in x\}$. Then $2^{|R|} \leq 2^{|x|} \leq 2^\kappa$. Thus,

$$(i) \quad |R| < 2^\kappa.$$

Since R is a set of ordinal numbers, R is either finite or $|R|$ is an aleph. Thus, it follows from (i) and the definition of $\aleph_\alpha$ that

$$(ii) \quad |R| < \aleph_\alpha.$$

Let $\theta = \bigcup R$. We shall show $\theta < \omega_\alpha$. If $\beta \in R$ then $\beta < \omega_\alpha$. This implies $|\beta| < \aleph_\alpha$. Since α is a successor ordinal $|\beta| \leq \aleph_{\alpha-1}$. Let A_β be the set of all 1-1 functions mapping β into $\omega_{\alpha-1}$. Let $C = \{A_\beta : \beta \in R\}$. C is a set of non-empty sets. Thus, AC 1 implies that there is a choice function g on C such that for each $\beta \in R$, $g_\beta \in A_\beta$.

It is clear that $\theta \leq \bigcup_{\beta \in R} (\{\beta\} \times \beta)$. For each $\gamma \in \beta \in R$ define a function h such that $h(\beta, \gamma) = \langle \beta, g_\beta(\gamma) \rangle$. Then h is a 1-1 function mapping $\bigcup_{\beta \in R} (\{\beta\} \times \beta)$ into $\bigcup_{\beta \in R} (\{\beta\} \times \omega_{\alpha-1})$. Therefore,

$$(iii) \quad \theta \leq R \times \omega_{\alpha-1}.$$

It follows from (ii) and (iii) that

$$|\theta| \leq |R \times \omega_{\alpha-1}| = |R| \cdot \aleph_{\alpha-1} = \aleph_{\alpha-1}.$$

Consequently, we obtain

$$(iv) \quad \theta < \omega_\alpha.$$

Now we shall show $x \in t$. For every $u \in x$. There is a $\beta \leq \theta$ such that $u \in f(\beta)$. Therefore, $u \in \bigcup f''(\theta+1)$, which implies $x \subseteq \bigcup f''(\theta+1)$. By (iv), $\theta < \omega_\alpha$. Since ω_α is a limit ordinal, $\theta + 1 < \omega_\alpha$. We have shown above that $|x| < \kappa$. So it follows from the definition of t that $x \in t$, q.e.d.

Let P be the statement obtained from CN 47 by replacing " $|s| \not\leq |x|$ " by " $|x| < |s|$ ". Dana Scott noted that the only place the axiom of choice is used in proving P is in proving (iii), and therefore, (iv). That is, for any successor ordinal α , ω_α is not the union of $< \omega_\alpha$ ordinals, each of

which is $< \omega_\alpha$. (ω_α is regular.) It follows from the work of Paul Cohen that this statement does not imply the axiom of choice in NBG. However, we do not know if P is provable in NBG° or NBG.

Before proving $CN\ 48 \rightarrow WO\ 3$, we shall prove the following lemma (Tarski [1939]):

LEMMA 7.53: Let t and v be sets such that $|v| \leq |t|$; then there is a set w with the following properties:

- (i) $w \subseteq t$,
- (ii) $w \not\subseteq v$,
- (iii) w can be well ordered.

PROOF: Since $|v| \leq |t|$, there is a 1-1 function f mapping v into t . Define a function g such that for each ordinal number α ,

$$g(\alpha) = \begin{cases} f(g''\alpha) & \text{if } g''\alpha \in v, \\ u \notin t & \text{otherwise.} \end{cases}$$

As long as $g(\alpha) \in t$, g is 1-1. Therefore, there is an ordinal α such that $g(\alpha) = u \notin t$. Let γ be the smallest such ordinal. Then $w = g''\gamma$ is the required set.

THEOREM 7.54: $CN\ 48 \rightarrow WO\ 3$.

PROOF: Let s be a non-empty set. $CN\ 48$ implies there is a set t such that $t \approx \{x \subseteq t : |s| \not\leq |x|\} = v$. Since $|v| = |t|$, t and v satisfy the hypothesis of 7.53. Let w be a set satisfying (i), (ii) and (iii) of 7.53. Since $w \subseteq t$ and $w \not\subseteq v$ we must have $|s| \leq |w|$. By 7.53 (iii), w can be well-ordered, q.e.d.

THEOREM 7.55: $T\ 1 \wedge AC\ 1 \wedge WO\ 1 \rightarrow CN\ 49$.

PROOF: Let t and μ satisfy the hypothesis of $CN\ 49$, and let $\kappa = |t|$ and $s = \{x \subseteq t : \mu \not\leq |x|\}$. Then $T\ 1$ implies that $s = \{x \subseteq t : |x| \leq \mu\}$. Let u be a set so that $|u| = \mu$. Since $\kappa = \kappa + 1$, $t = a \cup \{t_0\}$, where $t_0 \notin a$ and $|a| = \kappa$. So $s \approx \{x \subseteq a : |x| \leq \mu\}$. For each $x \subseteq a$ such that $|x| \leq \mu$,

there is an injection, f_x , mapping a subset of u onto x . Define g_x on u so that

$$g_x(v) = \begin{cases} f_x(v) & \text{if } v \in \mathcal{D}(f_x), \\ t_0 & \text{otherwise.} \end{cases}$$

Therefore, using AC 1, for each $x \in s$ we can find a function g_x mapping u onto $x \subseteq t$, and $x, y \in s$, $x \neq y$ implies $g_x \neq g_y$. Thus $s \leq t^u$, which implies $|s| \leq \kappa^\mu$.

Conversely, WO 1 implies both κ and μ can be well ordered. Since $\mu \leq \kappa$, it follows from 7.6 that $\mu \cdot \kappa = \kappa$. This implies that there is a function $h: t \times u \approx t$. Let $f \in t^u$. Define

$$\psi(f) = \{h(f(v), v) : v \in u\}.$$

Then $\psi: t^u \hookrightarrow s$. Therefore, $\kappa \leq |s|$, q.e.d.

THEOREM 7.56: CN 49 $\rightarrow$ CN 48.

PROOF: Let s be a non-empty set and let $|s| = \mu$. If $\mu = 0$ take $t = \emptyset$, and if $\mu = 1$ take $t = \{\emptyset\}$. Suppose $\mu \geq 2$. Let $\kappa = 2^{\aleph_0}$; then $\kappa^\mu = \kappa$. Let t be a set such that $|t| = \kappa$. CN 49 implies

$$\begin{aligned} \kappa^\mu &= |\{x \subseteq t : \mu \nless |x|\}| \\ &\geq |\{x \subseteq t : \mu \nless |x|\}| \\ &\geq |t| = \kappa. \end{aligned}$$

Therefore, $t \approx \{x \subseteq t : |s| \nless |x|\}$.

CN 50-52 are the last three forms in this section.

CN 50: $(\forall x) [|x| < \bigcup_{i \in \omega} A_i \rightarrow (\exists n \in \omega) (|x| \leq \bigcup_{i \leq n} A_i)]$.

CN 51: If J is a set and A and B are functions with domain J then

$$(\forall j \in J) (|A_j| < |B_j|) \rightarrow \left(\bigcup_{j \in J} A_j \right) < \prod_{j \in J} B_j.$$

CN 52: If J is a set, A is a function with domain J and for each $j \in J$, $2 \leq |A_j|$ then

$$\left| \bigcup_{j \in J} A_j \right| \leq \left| \prod_{j \in J} A_j \right|.$$

CN 50 was stated without proof in Lindenbaum & Tarski [1926] by Tarski, and later Tarski [1948] published a proof of its equivalence to the axiom of choice. CN 51 is frequently referred to as König's Theorem or Lemma. König [1905] proved it from the axiom of choice for countable J and later Zermelo [1908] proved it for the general case. CN 52 is a variation of CN 51 which is due to Arthur L. Rubin.

We shall prove first that CN 50 is equivalent to the well ordering theorem.

THEOREM 7.57: WO 1 $\rightarrow$ CN 50.

PROOF: Suppose $x < \bigcup_{i \in \omega} A_i$. It is clear that CN 50 holds if x is finite, and we may assume without loss of generality that each A_i is infinite. WO 1 implies that the cardinal number of an infinite set is an aleph. Let $|x| = \aleph_\alpha$, $|A_i| = \aleph_{\beta_i}$ and $\gamma = \bigcup_{i \in \omega} \beta_i$. It follows from properties of alephs that for each $n \in \omega$

$$\left| \bigcup_{i \leq n} A_i \right| = \max\{\aleph_{\beta_i} : i \leq n\}.$$

If $\beta_i < \alpha$ for all i , then $\gamma \leq \alpha$, so $\left| \bigcup_{i \in \omega} A_i \right| \leq \aleph_0 \cdot \aleph_\gamma = \aleph_\gamma \leq \aleph_\alpha$. This contradicts the hypothesis that $\aleph_\alpha < \left| \bigcup_{i \in \omega} A_i \right|$. Thus we must have $\beta_n \geq \alpha$ for some n , which implies $|x| \leq \left| \bigcup_{i \leq n} A_i \right|$.

Before proving that CN 50 implies the well ordering theorem we shall prove a preliminary lemma. It follows from 7.13 that for all transfinite y , $\Gamma(y) < \mathcal{P}(\mathcal{P}(\mathcal{P}(y)))$. However, we need a little stronger result.

LEMMA 7.58: There is a function F such that for each transfinite set y , F_y is a 1-1 function mapping $\Gamma(y)$ into $\mathcal{P}(\mathcal{P}(\mathcal{P}(y)))$.

PROOF: Let $\alpha \in \Gamma(y)$ and let f be a 1-1 function mapping α into y . Let

$$A_f = \{z: (\exists \beta) (\beta \leq \alpha \wedge z = f''\beta)\}.$$

Then $A_f \in \mathcal{P}(\mathcal{P}(y))$. Let

$$Q = \{A_f: (\exists \alpha \in \Gamma(y)) (f: \alpha \rightarrow y)\} \subseteq \mathcal{P}(\mathcal{P}(y)).$$

There is a natural mapping, ψ , from Q onto $\Gamma(y)$, namely, for each $A_f \in Q$, let $\psi(A_f) = \mathcal{P}(f)$. Thus, we can construct a 1-1 function h mapping $\mathcal{P}(\Gamma(y))$ into $\mathcal{P}(Q)$. (For each $u \subseteq \Gamma(y)$, $h(u) = \psi^{-1}''u$.) Define a function g such that for each $\alpha \in \Gamma(y)$, $g(\alpha) = \{\alpha\}$. Then $g: \Gamma(y) \rightarrow \mathcal{P}(\Gamma(y))$. Now, for each $\alpha \in \Gamma(y)$ define $F_y(\alpha) = h(g(\alpha))$, then F_y is the required function.

THEOREM 7.59: CN 50 $\rightarrow$ WO 3.

PROOF: Let x be transfinite. Construct a sequence of sets such that $y_0 = x$ and for each $i \in \omega$, $y_{i+1} = \mathcal{P}(y_i)$.

Let $u = \bigcup_{i \in \omega} \Gamma(y_i)$. It follows from 7.58 that

$u \prec \bigcup_{i \in \omega} (y_i \times \{i\})$. If $u \prec \bigcup_{i \in \omega} (y_i \times \{i\})$, then it follows

from CN 50 that there is an $n \in \omega$ such that

$u \prec \bigcup_{i \leq n} (y_i \times \{i\})$. Let $u_i \subseteq y_i$ such that $u \approx \bigcup_{i \leq n} (u_i \times \{i\})$.

Then

$$|u| = \sum_{i \leq n} |u_i| = \max_{i \leq n} |u_i| = |u_j|.$$

This implies $\Gamma(y_j) \prec u \prec y_j$ which is a contradiction. Thus

$u \approx \bigcup_{i \in \omega} (y_i \times \{i\})$, which implies $x \prec u$, q.e.d.

We shall prove that AC 1 $\rightarrow$ CN 51 $\wedge$ CN 52 and CN 51 $\vee$ CN 52 $\rightarrow$ AC 6.

THEOREM 7.60: AC 1 $\rightarrow$ CN 52.

PROOF: Suppose J and A satisfy the hypothesis of CN 52. We may assume without loss of generality that the A_j 's are pairwise disjoint. For each $j \in J$, let u_j and v_j be distinct elements of A_j . (Use AC 1 to choose these elements.) For each $x \in \bigcup_{j \in J} A_j$, there is a unique $i \in J$

such that $x \in A_i$. Define $F_x \in \prod_{j \in J} A_j$ as follows:

$$F_x(j) = \begin{cases} x & \text{if } j = i, \\ u_j & \text{if } j \neq i \wedge x \neq u_j, \\ v_j & \text{if } j \neq i \wedge x = u_j. \end{cases}$$

Then the function G such that for each $x \in \bigcup_{j \in J} A_j$, $G(x) = F_x$, is an injection.

THEOREM 7.61: AC 1 $\rightarrow$ CN 51.

PROOF: Suppose J , A and B satisfy the hypotheses of CN 51. We may assume, without loss of generality that the A_j 's are pairwise disjoint and $|B_j| \geq 2$ for all $j \in J$. Using a proof similar to the proof of 7.60, it is easy to show

$$|\bigcup_{j \in J} A_j| \leq |\prod_{j \in J} B_j|.$$

Suppose $F: \bigcup_{j \in J} A_j \approx \prod_{j \in J} B_j$. For $j \in J$, let F_j be the projection of F onto B_j . That is, $F_j(x) = F(x)(j)$ and

$$F_j \upharpoonright A_j: A_j \rightarrow B_j.$$

Since $A_j < B_j$, $F_j \upharpoonright A_j$ cannot be surjective. Thus, using AC 1, we can choose $t_j \in B_j \sim F_j \upharpoonright A_j " A_j$. Let

$g = \{\langle j, t_j \rangle : j \in J\}$. Then $g \in \prod_{j \in J} B_j$. Consequently, there is a unique $j \in J$ and $x \in A_j$ such that $F(x) = g$. In particular,

$$t_j = g(j) = F(x)(j) = F_j(x) \in F_j \upharpoonright A_j " A_j,$$

which is absurd.

It is easy to see that each of CN 51 and CN 52 implies AC 6. Let $\{x_j : j \in J\}$ be a set of non-empty sets. Without loss of generality, we may assume $|x_j| \geq 2$ for each $j \in J$. Then CN 51 implies $\prod_{j \in J} x_j \neq \emptyset$. (take $B_j = x_j$ and $A_j = \emptyset$ for each $j \in J$) and CN 52 implies $\prod_{j \in J} x_j \neq \emptyset$ because $\bigcup_{j \in J} x_j \neq \emptyset$.

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§8. FORMS FROM TOPOLOGY, ANALYSIS, AND LOGIC

In preceding sections we tried to make the text complete in itself so the reader would not have to refer to the literature for definitions and proofs. Since this section contains material from several different branches of mathematics, to continue this procedure would make the section unnecessarily long. Here we shall refer the reader to the literature for many of the definitions and subject matter proofs.

We shall first consider some statements from logic. The first two are Löwenheim-Skolem theorems and the third is a compactness theorem. The proofs that these three theorems are implied by the axiom of choice are due to Tarski (Tarski & Vaught [1957]), Malcev [1936] and Henkin [1949]. The proofs given here that they imply the axiom of choice are due to Vaught [1956].

In the following three propositions, "formula" means a "first order formula".

- LOG 1: A formula having a model in a set of cardinality κ also has a model in a set of cardinality μ if $\aleph_0 \leq \mu \leq \kappa$.
- LOG 2: A formula having a model in a set of cardinality $\aleph_0$ also has a model in a set of cardinality κ , for all $\kappa \geq \aleph_0$.
- LOG 3: If Q is a set of formulas in which the set of individual constants has cardinality κ and every finite subset of Q has a model, then Q has a model in a set whose cardinality is not greater than $\kappa + \aleph_0$.

We shall show that $\text{LOG } 3 \rightarrow \text{LOG } 2 \rightarrow \text{LOG } 1 \rightarrow \text{CN } 6$.

THEOREM 8.1: $\text{LOG } 3 \rightarrow \text{LOG } 2$.

PROOF: Let ψ be a formula which has a model in a set x where $|x| = \aleph_0$. Let κ be a cardinal such that $\aleph_0 \leq \kappa$. Let C be a set such that $|C| = \kappa$ and $C \cap x = \emptyset$. For each $a, b \in C$, $a \neq b$, we define the formula ψ_{ab} as $\neg(a = b)$. Let $A = \{\psi_{ab} : a, b \in C \wedge a \neq b\}$ and let $Q = \{\psi\} \cup A$. Since each finite subset of A has a finite subset of C as a model, ψ has x as a model, and $x \cap C = \emptyset$, it follows that each finite subset of Q has a model. Thus, Q satisfies the hypotheses of LOG 3. Therefore, Q has a model in a set whose cardinality is not greater than $\kappa + \aleph_0 = \kappa$. By construction of Q , the cardinal number of the model cannot be less than κ . Hence, ψ has a model in a set of cardinal number κ , q.e.d.

The proof that $\text{LOG } 2 \rightarrow \text{LOG } 1$ uses the Löwenheim-Skolem Theorem (see Skolem [1929]) which is provable in NBG° :

Every consistent set of formulas (with no individual constants) has a model in a set of cardinality $\aleph_0$.

THEOREM 8.2: $\text{LOG } 2 \rightarrow \text{LOG } 1$.

PROOF: Suppose the formula ϕ has a model in a set of cardinality κ . Then ϕ is consistent. Therefore, by the Löwenheim-Skolem Theorem, ϕ has a model in a set of cardinality $\aleph_0$. So by LOG 2, ϕ has a model in a set of any cardinality greater than $\aleph_0$, namely μ where $\aleph_0 \leq \mu \leq \kappa$.

THEOREM 8.3: $\text{LOG } 1 \rightarrow \text{CN } 6$.

PROOF: Let κ be an arbitrary transfinite cardinal number. Let $\mu = 2^{\kappa \cdot \aleph_0}$; then $\mu^2 = (2^{\kappa \cdot \aleph_0})^2 = 2^{\kappa \cdot \aleph_0 \cdot 2} = 2^{\kappa \cdot \aleph_0} = \mu$ and $\aleph_0 \leq \kappa \leq \mu$. Consider the following formula:

S: $(\forall x, y)(\exists z)R(x, y, z) \wedge$

$(\forall x, y, z, x', y', z')[R(x, y, z) \wedge R(x', y', z')] \rightarrow (z = z' \leftrightarrow (x = x' \wedge y = y'))]$.

An algebra $\langle A, R \rangle$, where A is a set and R a ternary relation, is a model for S iff R is a 1-1 function mapping $A \times A$ into A (so that $|A|^2 = |A|$). Clearly, S has a model of cardinality μ . Therefore, by LOG 1, S has a model of cardinality κ . Therefore, $\kappa^2 = \kappa$, q.e.d.

Our next statement is due to Klimovsky [1956].

LOG 4: If A is a set of sentences of first-order logic and $B \subseteq A$ is a consistent subset, then there exists a $\subseteq$ -maximal consistent set C such that $B \subseteq C \subseteq A$.

Since the property of being consistent is a property of finite character, $M'7 \rightarrow$ LOG 4.

THEOREM 8.4: LOG 4 $\rightarrow$ AC 2.

PROOF: Let t be a set of pairwise disjoint non-empty sets. Let A be the following set of sentences:

1. $(\exists! x)\dot{u}x$; for each $u \in t$.
2. $a \neq b$; for each $a, b \in \bigcup t$, $a \neq b$.
3. $\dot{u}a$; for each $a \in u \in t$.

Let B consist of all sentences of the first two types. Any $\subseteq$ -maximal consistent extension of B determines a choice function on t .

Dzik [1981] also has a logical equivalent of the axiom of choice, but since it is a special case of AL'3 we will not include it here.

$\aleph_0$ ' theorem is frequently used in logic. We shall show that $\aleph_0$ ' theorem together with the Boolean prime ideal theorem, BPI, is equivalent to the axiom of choice. As was mentioned earlier in Section 6, in 1967 Halpern & Levy [1971] proved that BPI does not imply the axiom of choice in NBG, and it was shown by Blass [1977] that $\aleph_0$ ' theorem does not imply the axiom of choice in NBG.

DEFINITION 8.5: Let A and X be non-empty sets, R a binary relation on A , $\mathcal{F}$ a filter on $\mathcal{P}(X)$, and $M = \langle A, R \rangle$.

(i) For $f_1, f_2 \in A^X$ define:

$$f_1 S f_2 \text{ iff } \{u \in X: f_1(u) = f_2(u)\} \in \mathcal{F}.$$

Since $X \in \mathcal{F}$ and $\mathcal{F}$ is closed with respect to finite intersections, it is easy to see that S is an equivalence relation on A^X . Let

$$[f] = \{g \in A^X: f S g\}$$

and let $A^X/\mathcal{F} = \{[f]: f \in A^X\}$.

(ii) For $[f], [g] \in A^X/\mathcal{F}$, define R' as follows:

$$[f]R'[g] \text{ iff } \{u \in X: f(u) R g(u)\} \in \mathcal{F}.$$

Then we define the reduced power

$$M^X/\mathcal{F} = \langle A^X/\mathcal{F}, R' \rangle.$$

Using the axiom of choice, Łos [1955] proved the following:

LT: If A, X, R and M are defined as in 8.5 and $\mathcal{F}$ is an ultrafilter on X then $M^X/\mathcal{F}$ and M are elementarily equivalent.

(Actually, Łos proved the theorem for any relational system M which has arbitrarily many finitary relations. But LT is sufficient for our purposes.)

The proof of the following theorem is due to P. E. Howard [1975].

THEOREM 8.6: $LT + BPI \rightarrow AC 2$.

PROOF: Let X be a set of pairwise disjoint non-empty sets such that for each $y \in X$, $y \cap X = \emptyset$. Let $A = X \cup (UX)$ and define R on A so that

$$\begin{aligned} t R y & \quad \text{if } t \in y \in X, \\ t R t & \quad \text{if } t \in UX, \\ \neg u R v & \quad \text{otherwise.} \end{aligned}$$

Suppose X has no choice function. Define $I \subseteq \mathcal{P}(X)$ so that

$$I = \{u \subseteq X: u \text{ has a choice function}\}.$$

Then I is an ideal, and since X has no choice function, I is a proper ideal. Using BPI' we extend I to a maximal (proper) ideal J . (It was shown in section 6 that $\text{BPI} \leftrightarrow \text{BPI}'$.) Let $\mathcal{F} = \{X \sim u: u \in J\}$. Then $\mathcal{F}$ is an ultrafilter.

By LT, $M^{X/\mathcal{F}}$ and M are elementarily equivalent, and by the definition of R , $M \vdash (\forall y)(\exists t)t R y$. Thus $M^{X/\mathcal{F}} \vdash (\forall y)(\exists t)t R y$. In particular, let i_X be the identity function on X and let $y = [i_X]$. Then, it follows that for some function $f \in A^X$, $\{u \in X: f(u) R i_X(u)\} \in \mathcal{F}$. That is, $Z = \{u \in X: f(u) \in u\} \in \mathcal{F}$. This implies that $X \sim Z \in J$. But clearly Z has a choice function, so $Z \in J$. This contradicts the fact that J is a proper ideal. Hence, the assumption that X has no choice function leads to a contradiction, q.e.d.

Blass [1977] proves that the statement

UP: All ultrafilters are principal.

is consistent with NBG. It is clear that $\text{UP} \rightarrow \text{LT}$ and that $\text{AC} \rightarrow \neg \text{UP}$. Thus in Blass' model of NBG, LT is true and AC is false.

Probably the most well-known and important topological equivalent of the axiom of choice is the Tychonoff Compactness Theorem. (See, for example, Kelley [1955] for definitions of topological terms.)

TOP 1: The product of compact spaces is compact in the product topology.

A. Tychonoff [1935] derived TOP 1 from a maximal principle and later J. Kelley [1950] proved the converse. L. E. Ward [1962] showed that a weaker version of TOP 1 is equivalent to AC.

TOP 2: The product of mutually homeomorphic compact spaces is compact in the product topology.

Then O. T. Alas [1969] proved that a weaker version of TOP 2 is still equivalent to AC.

TOP 3: The product of a set of mutually homeomorphic topological spaces $\{(X_i, T_i): i \in I\}$, where for each $i \in I$ T_i has three elements, is compact in the product topology.

Clearly TOP 1 $\rightarrow$ TOP 2 $\rightarrow$ TOP 3. We shall give the proof that TOP 3 $\rightarrow$ AC.

THEOREM 8.7: TOP 3 $\rightarrow$ AC 7.

PROOF: Let $\{X_i: i \in I\}$ be a set of equipollent sets. Let a be an element such that $a \notin \bigcup_{i \in I} X_i$. For each $i \in I$, let $Y_i = X_i \cup \{a\}$ and define $T_i = \{\emptyset, \{a\}, Y_i\}$. Then $\{(Y_i, T_i): i \in I\}$ satisfies the hypothesis of TOP 3. Let $W = \prod_{i \in I} Y_i$ be given the product topology and define $Z_i = \{f: f \in W \wedge f(i) \in X_i\}$. Then $\{Z_i: i \in I\}$ is a family of closed sets with the finite intersection property. Consequently, by TOP 3, $\bigcap_{i \in I} Z_i = \prod_{i \in I} X_i$ is non-empty.

If in TOP 1 and 2 the topological spaces are T 1, the resulting statements are equivalent to the axiom of choice. However, if the topological spaces are T 2, it was shown by Łoś & Ryll-Nardzewski [1954] that the resulting statements are equivalent to BPI. (See also Rubin & Scott [1954].) We next give a result of Franklin & Thomas [1971].

DEFINITION 8.8:

(a) A function f from one topological space to another is called perfect if the image of a closed set is closed and the inverse image of each point is compact.

(b) An onto function is irreducible if the image of a proper closed subset of the domain is a proper subset of the range.

(c) If $X = \bigcup_{i \in I} X_i$ is a union of topological spaces X_i , X is called a free union if $O \subseteq X$ is open iff $O \cap X_i$ is open in X_i for each $i \in I$.

TOP 4: Any perfect onto map from one topological space to another has an irreducible restriction to a closed subspace of its domain.

TOP 4 is provable from a maximal principle.

THEOREM 8.9: TOP 4 $\rightarrow$ AC 2.

PROOF: Let $\{X_i : i \in I\}$ be a family of non-empty pairwise disjoint sets and for each $i \in I$, let T_i be the cofinite topology on X_i (a set is open iff its complement is finite). Let $X = \bigcup_{i \in I} X_i$ be the free union. If the index set I is given the discrete topology then the function $f: X \rightarrow I$ such that for each $u \in X_i$, $f(u) = i$, is perfect and maps X onto I . Thus, TOP 4 implies that there is a subspace $Y \subseteq X$ on which f is irreducible. The intersection of Y with each X_i is a singleton, q.e.d.

N. Brunner [1983] studies statements of the following form:

TOP 5(P): Each T_2 (Hausdorff) space has the property P .

TOP 6(P,Q): Each T_2 space which has the property P also has the property Q .

For P and Q , Brunner considers the following, where X is a space and $T \subseteq \mathcal{P}(X)$ is its topology.

W 1: T can be well ordered.

W 2: X can be well ordered.

D 1: X is covered by a well ordered family of closed, discrete sets.

D 2: X is covered by a well ordered family of discrete sets.

B 1: T has a well orderable base.

B 2: There is a function $f: X \times W \rightarrow T$ such that W can be well ordered and for each $x \in X$, $f''(\{x\} \times W)$ is a neighborhood base at x .

B 3: There is a function $f: X \times W \rightarrow T$ such that W can be well ordered and for each $x \in X$, $\cap f''(\{x\} \times W) = \{x\}$.

L 1: Each open cover of X has a well orderable subcover.

L 2: Each open cover of X has a well orderable refinement.

S: X has a dense well orderable subset.

C: Each family $\mathcal{O} \subseteq T$ of pairwise disjoint sets can be well ordered.

A 1: If $\mathcal{O} \subseteq T$ covers X there is a mapping $f: X \rightarrow \mathcal{O}$ such that for all $x \in X$, $x \in f(x)$.

A 2: If $\mathcal{O} \subseteq T$ covers X there is a mapping $f: X \rightarrow T$ such that for all $x \in X$, $x \in f(x)$ and $f''X$ refines $\mathcal{O}$.

It is easy to show that the following implications hold (proofs are given in Brunner [1983]):

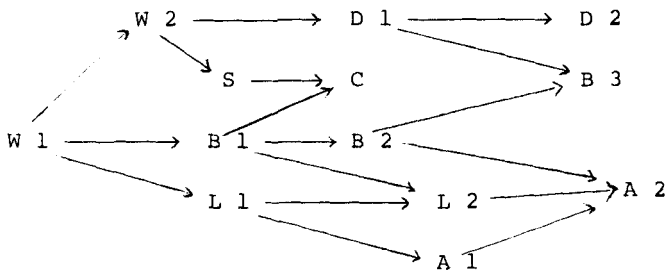


Figure 8.1

Also, if $P' \rightarrow P$ and $Q \rightarrow Q'$ then $\text{TOP } 6(P, Q) \rightarrow \text{TOP } 6(P', Q')$ and $\text{TOP } 5(Q) \rightarrow \text{TOP } 5(Q')$; and if $P \rightarrow Q$ then $\text{NBG}^\circ \vdash \text{TOP } 6(P, Q)$. It is also clear that $\text{AC} \rightarrow \text{TOP } 5(Q) \rightarrow \text{TOP } 6(P, Q)$ for all P and Q .

THEOREM 8.10: Each of the following is equivalent to the axiom of choice.

1. $\text{TOP } 6(P, Q)$, where $P = D\ 1, D\ 2, B\ 2, B\ 3$, or $A\ 2$, and $Q = W\ 1, W\ 2, B\ 1, L\ 1, L\ 2, S, C$, or $A\ 1$.

2. $\text{TOP } 6(P, Q)$, where $P = L\ 1, L\ 2$, or $A\ 1$ and $Q = W\ 1, W\ 2, B\ 1, S$, or C .

3. $\text{TOP } 5(Q)$, where $Q = W\ 1, W\ 2, B\ 1, L\ 1, L\ 2, S, C$, or $A\ 1$.

PROOF: It follows from Figure 8.1 and the remarks preceding the theorem that to prove 1. it is sufficient to prove that each of $\text{TOP } 6(P, Q)$, where $P = D\ 1$ or $B\ 2$, and $Q = L\ 2, C$, or $A\ 1$, implies AC.

Let X_1 be a set and $T_1 = \mathcal{P}(X_1)$; then $\langle X_1, T_1 \rangle$ is a topological space satisfying $D\ 1$ and $B\ 2$. If $\langle X_1, T_1 \rangle$ satisfies $L\ 2$ or C then X_1 can be well ordered.

Let Y be a set of non-empty pairwise disjoint sets such that no one is an element of another, $X_2 = Y \cup (\cup Y)$ and $T_2 = \mathcal{P}(X_2)$ then $\langle X_2, T_2 \rangle$ satisfies $D\ 1$ and $B\ 2$. Suppose $\langle X_2, T_2 \rangle$ satisfies $A\ 1$. The set $O = \{\{a, u\}: a \in u \in Y\}$ is an open cover of X_2 . Suppose $f: X_2 \rightarrow O$ is an $A\ 1$ mapping. For each $u \in Y$, let $g(u) = f(u) \sim \{u\}$. Then g is a choice function on y .

To prove 2. it is sufficient to show that $\text{TOP } 6(L\ 1, C)$ implies AC.

Let X be a set with the discrete topology and $X^* = X \cup \{X\}$ its one point compactification. The topology T^* of X^* is all open subsets of X and all subsets $U \subseteq X^*$ such that $X^* \sim U$ is a closed compact subset of X . In our case $X^* \sim U$ is finite. Thus, $\langle X^*, T^* \rangle$ satisfies $L\ 1$. If $\langle X^*, T^* \rangle$ satisfies C then $\{\{a\}: a \in X\}$ can be well ordered. Therefore, X can be well ordered.

Part 3. follows from 1.

THEOREM 8.11: Each of the following is equivalent to the multiple choice axiom, MC. (See section 5. We shall use the fact that $\text{MC } 1 \leftrightarrow \text{MC } 2$.)

1. $\text{TOP } 6(P, Q)$, where $P = D \ 2, L \ 1, L \ 2, A \ 1$, or $A \ 2$, and $Q = D \ 1, B \ 2$, or $B \ 3$.

2. $\text{TOP } 6(D \ 2, A \ 2)$.

3. $\text{TOP } 5(Q)$, where $Q = D \ 1, B \ 2, B \ 3$, or $A \ 2$.

PROOF: To show that MC implies each one of the above statements it is sufficient to show that $\text{MC } 2 \rightarrow \text{TOP } 5(Q)$ for $Q = D \ 1$ and $B \ 2$.

Let $\langle X, T \rangle$ be a $T \ 2$ space. Since points are closed $\text{MC } 2$ implies $\text{TOP } 5(D \ 1)$.

To prove $\text{TOP } 5(B \ 2)$, we use the fact that $\text{MC } 2$ implies that there is a well-ordered family of finite sets W such that $T = UW$. Define $f: X \times W \rightarrow T$ so that if $\langle a, E \rangle \in X \times W$,

$$f(a, E) = \begin{cases} \bigcap \{u \in E: a \in u\} & \text{if } a \in UE, \\ X & \text{otherwise.} \end{cases}$$

If $a \in O \in T$, then there is an $E \in W$ such that $O \in E$. Thus, $a \in f(a, E) \subseteq O$, thereby proving $\text{TOP } 5(B \ 2)$.

To prove the converse, it is sufficient to prove that each of $\text{TOP } 6(D \ 2, B \ 3)$, $\text{TOP } 6(L \ 1, B \ 3)$ and $\text{TOP } 6(D \ 2, A \ 2)$ implies MC.

Let X be an infinite set with the discrete topology and $X^* = X \cup \{X\}$ with the topology T^* , its one point compactification. (See 8.10). Then $\langle X^*, T^* \rangle$ satisfies $L \ 1$ and $D \ 2$. Suppose W is a well ordered family of sets such that $f: X^* \times W \rightarrow T^*$ is a $B \ 3$ -function. Then

$$\bigcap \{f(X, u): u \in W\} = \{X\}.$$

Therefore, $X = \bigcup \{X^* \sim f(X, u): u \in W\}$ and for each $u \in W$, $X^* \sim f(X, u)$ is finite. This, completes the proof of part 1. of the theorem.

To prove 2. we shall show that $\text{TOP } 6(D \ 2, A \ 2) \rightarrow \text{MC } 1$. Let Y be a family of pairwise disjoint infinite sets, such

that no one is an element of another. For each $u \in Y$, let u^* be the one point compactification of the discrete topology on u . Let $X = \bigcup \{u^* : u \in Y\}$ be the free union (see 8.8(c)), and let T be its topology. The sets Y and $\bigcup Y$ are both discrete subsets of X and $X = Y \cup (\bigcup Y)$, so $\langle X, T \rangle$ satisfies D 2.

Let $\mathcal{O} = \{u^* \sim \{a\} : a \in u \in Y\}$. Then $\mathcal{O}$ is an open cover of X . Suppose f is an $\mathcal{A}$ 2-mapping for $\mathcal{O}$. That is, $f: X \rightarrow T$ such that for all $x \in X$, $x \in f(x)$ and $f''X$ refines $\mathcal{O}$. For each $u \in Y$, let $g(u) = u \sim f(u)$. By the definition of f and T , $g(u)$ is a finite subset of u . To complete the proof we must show that $g(u) \neq \emptyset$. Since $f''X$ refines $\mathcal{O}$, there is a $v \in Y$ and an $a \in v$ such that $f(u) \subseteq v^* \sim \{a\}$. But $u \in f(u)$, so $v = u$. This implies $a \in g(u)$. Thus g is the required multiple choice function.

Since $\text{TOP } 5(Q) \rightarrow \text{TOP } 6(P, Q)$ for all P and Q , this completes the proof of the theorem.

THEOREM 8.12: Each of the following is equivalent to PW.

1. $\text{TOP } 6(P, W 1)$, where $P = W 2, B 1$, or S .
2. $\text{TOP } 6(B 1, Q)$, where $Q = W 2, D 1$, or $D 2$.
3. $\text{TOP } 6(S, W 2)$.

PROOF: First we note that PW implies $\text{TOP } 6(P, Q)$ for $P = B 1, S$, or $W 2$ because if PW holds, spaces with the property P also have the property $W 1$.

To complete the proof of 1., let X be well ordered and let $T = \mathcal{P}(X)$. Then $\langle X, T \rangle$ satisfies $W 2, B 1$ and S . If $\langle X, T \rangle$ satisfies $W 1$, then $\mathcal{P}(X)$ can be well ordered.

Since $W 2 \rightarrow D 1 \rightarrow D 2$, it follows that $\text{TOP } 6(B 1, W 2) \rightarrow \text{TOP } 6(B 1, D 1) \rightarrow \text{TOP } 6(B 1, D 2)$. However, we claim $\text{TOP } 6(B 1, D 2) \rightarrow \text{TOP } 6(B 1, W 2)$ because if T has a well ordered base, then any discrete set D can be well ordered as follows: if $a, b \in D$, $a \neq b$, $a R b$ iff the first base element B such that $B \cap D = \{a\}$, precedes the first

base element B' such that $B' \cap D = \{b\}$. Thus, if X is covered by a well ordered family of discrete sets, X can be well ordered. Therefore, to prove 2. it is sufficient to prove that $\text{TOP } 6(B\ 1, W\ 2) \rightarrow \text{PW}$.

Suppose X is a well ordered set. Let T be the product topology on 2^X . For each $a \in X$ and $f \in 2^X$, let $p_a \circ f = f(a)$. Then $Z = \{p_a^{-1}\{1\} : a \in X\}$ is a subbase for the topology and Z can be well ordered. Therefore, $\langle 2^X, T \rangle$ satisfies $B\ 1$. Thus, $W\ 2$ implies 2^X can be well ordered.

We also note that the above space also satisfies S because if

$$D = \{f \in 2^X : f^{-1}\{1\} \text{ is finite}\}$$

then D is dense and can be well ordered. Consequently, $\text{TOP } 6(S, W\ 2) \rightarrow \text{PW}$, q.e.d.

THEOREM 8.13: Each of the following imply PW .

1. $\text{TOP } 6(P, D\ 1)$, where $P = B\ 2, B\ 3, S$ or C .
2. $\text{TOP } 6(C, Q)$, where $Q = W\ 1$ or $W\ 2$.
3. $\text{TOP } 5(D\ 2)$.

PROOF: To prove 1., it is sufficient to prove $\text{TOP } 6(S, D\ 1) \rightarrow \text{PW}$ and $\text{TOP } 6(B\ 2, D\ 1) \rightarrow \text{PW}$. Suppose X is a well ordered set and T is the lexicographical order topology on 2^X . The space $\langle 2^X, T \rangle$ is compact and it satisfies S because the set $D = \{g \in 2^X : g^{-1}\{1\} \text{ is finite}\}$ is dense and well ordered. Let $\leq$ be the lexicographic ordering relation on 2^X . For each $g \in 2^X$, let $\vec{g} = \{h \in 2^X : g < h\}$ and let $\overleftarrow{g} = \{h \in 2^X : h < g\}$. Define a function f from $2^X \times D$ to T as follows: If $g \in 2^X$ and $h \in D$

$$f(g, h) = \begin{cases} \vec{h} & \text{if } h < g, \\ \overleftarrow{h} & \text{if } g < h, \\ 2^X & \text{if } g = h. \end{cases}$$

Then for each $g \in 2^X$, $f"(\{g\} \times D)$ is a subbase for the neighborhood system at g . Now, it is easy to see how to construct a B_2 function for 2^X .

Since $\langle 2^X, T \rangle$ is compact each closed, discrete subset of 2^X is finite. Moreover, each finite subset of 2^X is well ordered by $\leq$. Thus, if $\langle 2^X, T \rangle$ satisfies D_1 , 2^X can be well ordered. This completes the proof of 1.

Since $W_1 \rightarrow W_2 \rightarrow D_1$, it follows that $TOP_6(C, W_1) \rightarrow TOP_6(C, W_2) \rightarrow TOP_6(C, D_1)$. In part 1, we proved $TOP_6(C, D_1) \rightarrow PW$, therefore, part 2. holds. Part 3. follows from Theorem 8.12 part 2. because $TOP_5(D_2) \rightarrow TOP_6(B_1, D_2)$.

COROLLARY 8.14: $MC_2 \rightarrow TOP_5(D_2) \rightarrow PW$.

PROOF: It follows from 8.11 that $MC_2 \rightarrow TOP_5(D_2)$ and from 8.13 that $TOP_5(D_2) \rightarrow PW$.

It follows from 8.10-8.14 that each of the following statements are equivalent to AC in NBG:

$TOP_5(Q)$, for all Q (in our list).

$TOP_6(P, W_1)$, for $P \neq W_1$.

$TOP_6(P, W_2)$, for $P \neq W_1, W_2$.

$TOP_6(P, D_1)$, for $P \neq W_1, W_2, D_1$.

$TOP_6(B_1, D_2)$.

$TOP_6(P, B_1)$, for $P \neq W_1, W_2, B_1, S, C$.

$TOP_6(P, Q)$ for $P = D_2, L_1, L_2, A_1, A_2$, and $Q = B_2, B_3$.

$TOP_6(P, Q)$, for $P = D_1, D_2, B_2, B_3, A_2$, and $Q = L_1, L_2$.

$TOP_6(P, Q)$, for $P = D_1, D_2, B_2, B_3, A_2$, and $Q = S, C$.

$TOP_6(P, A_1)$, for $P = D_1, D_2, B_2, B_3, A_2$.

$TOP_6(D_2, A_2)$.

There are several unanswered questions. For example, whether $\text{TOP } 6(W 2, Q)$, for $Q = B 1, B 2, L 1, L 2, A 1$, or $A 2$, implies AC. See Brunner [1983] for details.

Howes [1980] considers another property which is useful in topology:

$B 2'$: Each point of X has a neighborhood base which can be well ordered and the well ordering is compatible with set inclusion. (By this we mean that if u and v are elements of the neighborhood base and $u \subseteq v$ then u follows v in the well ordering.)

Howes [1980] and Vaughan [1975] prove that $(\forall X) B 2' \leftrightarrow MC$. (There is an error in Howes proof but it is easily corrected.)

THEOREM 8.15: $MC 2 \leftrightarrow (\forall X) B 2'$.

PROOF: Let $p \in X$ and let T be any neighborhood base at p . Then MC implies that for some ordinal λ , $T = \bigcup_{\alpha < \lambda} S_\alpha$, where each S_α is a finite non-empty set. For each $\alpha < \lambda$, let $u_\alpha = \bigcap S_\alpha$. Then $U = \{u_\alpha : \alpha < \lambda\}$ is a well ordered neighborhood base at p .

Next, we have to find a well ordering which is compatible with $\subseteq$. Towards this end we shall find a subset $U' \subseteq U$ such that if $u_\alpha, u_\beta \in U'$ and $u_\alpha \subseteq u_\beta$ then $\alpha \geq \beta$. Define a function $f: \lambda \rightarrow \lambda$ such that:

$$f(0) = 0;$$

$$\text{if } 0 < \alpha < \lambda, f(\alpha) = \min\{\gamma: (\forall \beta < \alpha) (u_\gamma \not\subseteq u_{f(\beta)})\}.$$

Let $U' = \{u_{f(\alpha)} : \alpha < \lambda\}$. Then it is clear that U' is the required neighborhood base.

Conversely, let X be a set and T the finite complement topology on X . (That is, a set is open if its complement is finite.) Let $\langle P, \leq \rangle$ be a well ordered neighborhood base for some point $p \in X$. Define a function $f: X \setminus \{p\} \rightarrow P$ so that for each $y \in X, y \neq p$, $f(y) = A$, where A is the $\leq$ -first

element of P such that $A \subseteq X \sim \{y\}$. For each $A \in \mathcal{K}(f)$, $f^{-1}\{A\}$ is finite and $X \sim \{p\} = \bigcup_{A \in P} f^{-1}\{A\}$. Thus, X is a well ordered union of finite sets, thereby proving MC 2.

The last part of this section deals with equivalents of the axiom of choice in analysis. There is no sharp dividing line between analysis and topology. There may be some difference of opinion whether, for example, the Krein-Milman theorem is analysis, but these results seemed to fit together so we shall include them here. Most non-constructive principles used in real analysis are weaker than the axiom of choice since they deal with sets of cardinality $\leq 2^{\aleph_0}$. Here we will consider more general statements.

We shall mention several well known non-constructive theorems in analysis. HB is the Hahn-Banach theorem, KM is the Krein-Milman theorem and AL is Alaoglu's theorem.

HB: Let p be a real-valued function defined on the vector space X over the reals, $\mathbb{R}$, such that $p(x+y) \leq p(x) + p(y)$ for all $x, y \in X$ and $p(ax) = ap(x)$ for each $a \geq 0$, $a \in \mathbb{R}$, (p is called a sublinear functional). Suppose that f is a linear functional defined on a subspace $S \subseteq X$ and that $f(s) \leq p(s)$ for all $s \in S$. Then there is a linear functional F defined on X such that $F(x) \leq p(x)$ for all $x \in X$ and $F(s) = f(s)$ for all $s \in S$.

KM: Let K be a compact convex set in a locally convex topological vector space X . Then K has an extreme point. (An extreme point is a point which is not an interior point of any line segment which lies in K).

If X is a vector space, a nonnegative real-valued function $\| \cdot \|$ defined on X is called a norm if for all $x, y \in X$ and all $a \in \mathbb{R}$:

1. $\|x\| = 0$ iff $x = 0$.
2. $\|x + y\| \leq \|x\| + \|y\|$.
3. $\|ax\| = |a| \|x\|$.

If X is a normed vector space, X^* , the dual of X , is the space of bounded linear functionals on X . A base at the point 0^* of X^* for the weak* topology is given by sets of the form $\{f \in X^*: f(x_i) < \varepsilon, i = 1, 2, \dots, n\}$ where $\{x_1, x_2, \dots, x_n\}$ is a finite subset of X . For each $f \in X^*$, $\|f\| = \sup\{f(x) : \|x\| \leq 1\}$.

AL: The unit sphere $S^* = \{f \in X^*: \|f\| \leq 1\}$ in the dual of a normed vector space X over the reals is compact in the weak* topology.

The proofs of HB, KM, and AL in $NBG^\circ + AC$ are given, for example, in Royden [1963]. We obtain variations of KM and AL by replacing "compact" by "convex-compact". (Some authors call convex-compact, quasicompact.) If X is a topological vector space, $Y \subseteq X$ is called convex-compact if whenever C_i , for $i \in I$, are closed convex sets in X , and $\{Y \cap C_i : i \in I\}$ has the finite intersection property, then $\bigcap_{i \in I} (Y \cap C_i) \neq \emptyset$. Let KMC and ALC be the statements obtained from KM and AL, respectively, by replacing "compact" by "convex-compact". Then

$$KMC \rightarrow KM \quad \text{and} \quad AL \rightarrow ALC.$$

(KMC is also provable in $NBG^\circ + AC$.)

The following results are known:

- I $BPI \not\vdash AC$ in NBG . (Halpern & Levy [1971])
 - II $HB \not\vdash BPI$ in NBG .
 - III $HB + KM \not\vdash AC$ in NBG° .
- } (Pincus [1972][1974])
- IV $BPI \rightarrow HB$. (Zos & Ryll-Nardzewski [1951] and Luxemburg [1962])
 - V $ALC \leftrightarrow HB$. (Luxemburg [1969])
 - VI $AL \leftrightarrow BPI$. (Rubin & Scott [1954])

We shall include results due to Bell & Fremlin [1972], Edwards [1975] and Lembcke [1979].

AN 1: The unit sphere in the dual of a normed vector space over the reals has an extreme point.

AN 1 is a well-known theorem about normed vector spaces. Bell & Fremlin [1972] prove that it implies the axiom of choice.

AN 2: Let X be a real vector space, Y a subspace and S a subset of X . Suppose $p: X \rightarrow \mathbb{R}$ is a sublinear functional on X and $f: Y \rightarrow \mathbb{R}$ a linear functional such that $f(y) \leq p(y)$ for all $y \in Y$. Then the set Z of all p -dominated linear extensions of f to X has an element g such that for all $h \in Z$ with $g(s) \leq h(s)$ for all $s \in S$, we have $g(s) = h(s)$ for all $s \in S$. That is, g is S -maximal in Z .

AN 3: Same hypotheses as AN 2 (no subset S is required) but the conclusion is that Z has an extreme point.

Andenaes [1970] proved AN 2 and Bonsall [1954, 1956] proved AN 3, while Lembcke [1979] proved that $\text{AN } 2 \rightarrow \text{MC}$ and $\text{AN } 3 \rightarrow \text{AC}$.

We shall prove first that $\text{AN } 3 \rightarrow \text{AN } 1$.

THEOREM 8.16: $\text{AN } 3 \rightarrow \text{AN } 1$.

PROOF: Let X be a real normed vector space with norm p . Let f be the 0-functional on $Y = \{0\}$. Then the unit ball of X^* is the set Z of AN 3. Thus, $\text{AN } 3 \rightarrow \text{AN } 1$.

The proof of the following theorem is immediate.

THEOREM 8.17:

1. $\text{AL} + \text{KM} \rightarrow \text{AN } 1$.
2. $\text{ALC} + \text{KMC} \rightarrow \text{AN } 1$.

COROLLARY 8.18:

1. $\text{BPI} + \text{KM} \rightarrow \text{AN } 1$.
2. $\text{HB} + \text{KMC} \rightarrow \text{AN } 1$.

PROOF: The proof follows from 8.16 and V and VI above.

The following proof is due to Bell & Fremlin [1972].

THEOREM 8.19: $\text{AN } 1 \rightarrow \text{AC } 1$.

PROOF: Let $\{A_i : i \in I\}$ be a family of pairwise disjoint sets and let $A = \bigcup_{i \in I} A_i$. Define

$$S = \{f \in \mathbb{R}^A : \sup_{i \in I} \sum_{t \in A_i} |f(t)| \leq 1\}$$

$$W = \{f \in \mathbb{R}^A : \sup_{i \in I} \sum_{t \in A_i} |f(t)| < \infty\}$$

$$X = \{f \in \mathbb{R}^A : (\forall \epsilon > 0) (\{t \in A : |f(t)| > \epsilon\} \text{ is finite}) \\ \text{and } \sum_{i \in I} \sup_{t \in A_i} |f(t)| < \infty\}.$$

Then W and X are normed vector spaces with norms defined by:

$$\|f\|_W = \sup_{i \in I} \sum_{t \in A_i} |f(t)|, \text{ for } f \in W,$$

$$\|g\|_X = \sum_{i \in I} \sup_{t \in A_i} |g(t)|, \text{ for } g \in X.$$

Also, S is the unit sphere of W . But W is isometrically isomorphic to the dual X^* of X , and therefore, S may be regarded as the unit sphere of X^* . $\text{AN } 1$ implies that S has an extreme point e . We shall show that for each $i \in I$ there is a unique $t \in A_i$ such that $e(t) \neq 0$.

First suppose there is an $i \in I$ such that $e(t) = 0$ for all $t \in A_i$. Let $u \in A_i$ and choose $f, g \in S$ such that

$$f(u) = 1, \quad g(u) = -1$$

$$f(t) = g(t) = e(t) \text{ for all } t \in A \sim \{u\}.$$

Then $e = (f + g)/2$ and $e \neq f, e \neq g$. This contradicts the fact that e is an extreme point of S .

Next, suppose there is an $i \in I$ and two distinct elements $u, v \in A_i$ such that $e(u) \neq 0$ and $e(v) \neq 0$. Define $f, g \in S$ as follows:

$$f(u) = e(u)(1 + |e(v)|)$$

$$f(v) = e(v)(1 - |e(u)|)$$

$$g(u) = e(u)(1 - |e(v)|)$$

$$g(v) = e(v)(1 + |e(u)|)$$

$$f(t) = g(t) = e(t), \quad \text{for all } t \in A \sim \{u, v\}.$$

Then $e \neq f$, $e \neq g$ and $e = (f + g)/2$, again contradicting the definition of e as an extreme point.

Now we define a choice function, F , on $\{A_i : i \in I\}$, by $F(A_i)$ is the unique $t \in A_i$ such that $e(t) \neq 0$.

COROLLARY 8.20: Each of the following is equivalent to AC:

AN 1, AN 3, AL + KM, ALC + KMC, BPI + KM, and HB + KMC.

Lembcke [1979] gives a proof that $\text{AN } 2 \rightarrow \text{MC } 1$ which is similar to the Bell & Fremlin [1972] proof, Theorem 8.18, that $\text{AN } 1 \rightarrow \text{AC } 1$. Thus it follows that $\text{NBG} \vdash \text{AN } 2 \leftrightarrow \text{AC}$.

In the same paper Lembcke gives the following form:

AN 4: Let X and Y be T_2 spaces, $f: X \rightarrow Y$ a continuous map, and μ an (inner) regular Borel measure on Y . If μ is inner regular with respect to the system of sets $\{f(K) : K \text{ is a compact subset of } X\}$, then there is a regular Borel measure ν on X such that $\mu = \nu f$, where $\nu f = \nu \circ f^{-1}$. (See, for example, Halmos [1950] for terminology.)

Lembcke [1970, 1977] proves that $\text{AC} \rightarrow \text{AN } 4$ and the following proof is given in Lembcke [1979].

THEOREM 8.21: $\text{AN } 4 \rightarrow \text{MC } 1$.

PROOF: Let $\{A_i : i \in I\}$ be a family of non-empty, pairwise disjoint sets and let $X = \bigcup_{i \in I} A_i$. Give X and $Y = I$

the discrete topologies. Define $f: X \rightarrow Y$ such that for each $a \in A_i \subseteq X$, $f(a) = i$, and define μ on Y to be the counting measure, that is, for each $S \subseteq Y$, $\mu(S) = |S|$ if S is finite, otherwise $\mu(S) = +\infty$.

Then μ is a regular Borel measure on Y . A set is compact iff it is finite. All compact subsets of Y are images of compact subsets of X . Thus, AN 4 implies that there is a regular Borel measure ν on X such that $\nu f = \mu$. Therefore, for each $i \in I$, $\nu(A_i) = \mu(\{i\}) = 1$. Since compact subsets are finite and ν is inner regular, for each $i \in I$ there is an $a \in A_i$ such that $\nu(\{a\}) \neq 0$. Let $s_i = \sup\{\nu(\{a\}) : a \in A_i\}$. Then for each $i \in I$, there is a finite number of $a \in A_i$ such that $\nu(\{a\}) = s_i$. Define g on $\{A_i : i \in I\}$ such that for each $i \in I$,

$$g(A_i) = \{a \in A_i : \nu(\{a\}) = s_i\}.$$

Then g is a multiple choice function.

Thus, we have $\text{NBG} \vdash \text{AN } 4 \leftrightarrow \text{AC}$.

The next form, AN 5, is a theorem of classical analysis. Edwards [1975] proved it implies the axiom of choice. Before stating AN 5 we give some definitions.

DEFINITION 8.22: $\langle X, \leq \rangle$ is a real vector lattice if X is a vector space over $\mathbb{R}$, $\langle X, \leq \rangle$ is a lattice, and the vector operations preserve $\leq$. (That is, for all $x, y, z \in X$ and for all positive real numbers a if $x \leq y$ then $x + z \leq y + z$ and $ax \leq ay$.) An element $e \in X$ is said to be an order-unit if for each $x \in X$ there is a positive integer n such that $-ne \leq x \leq ne$. (Alternatively, if X is a normed lattice, $e \in X$ is called an order-unit if for all $x \in X$ such that $\|x\| \leq 1$, $x \leq e$. For our purposes it doesn't matter which definition is used.)

Our interest will be in real vector lattices of bounded real valued functions. If f and g are real valued functions with domain A ,

$$\begin{aligned}
f &\leq g \quad \text{if} \quad f(a) \leq g(a), \quad \forall a \in A, \\
(f \vee g)(a) &= \max(f(a), g(a)), \quad \forall a \in A, \\
(f \wedge g)(a) &= \min(f(a), g(a)), \quad \forall a \in A, \\
(f + g)(a) &= f(a) + g(a), \quad \forall a \in A, \\
(bf)(a) &= b(f(a)), \quad \forall b \in \mathbb{R}, \quad \forall a \in A.
\end{aligned}$$

The characteristic function on A , χ_A , is an example of an order-unit in both senses.

AN 5: Let X be a real vector lattice with a distinguished order-unit (denoted by 1), Y a sublattice of X , and Z the set of all positive linear functionals F on X such that $F(1) = 1$. Then there exists an extreme point M of the convex set Z which is Y -minimal in Z . (That is, if M' is any other extreme point of Z such that $M'(y) \leq M(y)$ for all $y \in Y$, then $M'(y) = M(y)$ for all $y \in Y$.)

The next theorem is due to Edwards [1975].

THEOREM 8.23: AN 5 $\rightarrow$ AL'1.

PROOF: Let $\langle L, \leq \rangle$ be a lattice with identity and at least one other element. Then there is a set A and a collection of subsets of A , $\mathcal{L}$, such that $\langle L, \leq \rangle \cong \langle \mathcal{L}, \subseteq \rangle$. For each $S \subseteq A$ let χ_S be the characteristic function for S . Then for $a \in A$,

$$\chi_S(a) = \begin{cases} 1 & \text{if } a \in S, \\ 0 & \text{if } a \in A \sim S. \end{cases}$$

We shall denote $A \sim S$ by S' .

Let J be a proper ideal in L and let $\mathcal{J}$ be its image in $\mathcal{L}$. Then there is a Boolean algebra ideal $\mathcal{K} \subseteq \mathcal{P}(A)$ such that $\mathcal{J} = \mathcal{K} \cap \mathcal{L}$. (Take $\mathcal{K} = \{S \subseteq A : (\exists T \in \mathcal{J}) S \subseteq T\}$.)

Define an equivalence relation R on $\mathbb{R}^A$ as follows:
For $f, g \in \mathbb{R}^A$

$$f R g \quad \text{iff} \quad (\exists K \in \mathcal{K}) (\forall a \in K') (f(a) = g(a)).$$

Let $[f] = \{g \in \mathbb{R}^A : f \leq g\}$ and let X be the vector space generated by $\{[\chi_S] : S \subseteq A\}$. Then X is a real vector lattice with order-unit $1 = [\chi_A]$. Let $Y = \{[\chi_S] : S \in \mathcal{L}\}$ and let Z be the set of all positive linear functionals F on X with $F(1) = 1$. AN 5 implies that Z has a Y -minimal extreme point, M .

First we note that for each characteristic function χ_S for $S \subseteq A$, $M([\chi_S])$ is 0 or 1. Since $[\chi_S] + [\chi_{S^c}] = 1$, $0 \leq M([\chi_S]) \leq 1$. Suppose $0 < M([\chi_S]) < 1$ for some $S \subseteq A$. For $[f] \in X$, let $g_S = f|_S + 0|_{S^c}$ and $g_{S^c} = f|_{S^c} + 0|_S$. Then $f = g_S + g_{S^c}$. Therefore,

$$\begin{aligned} M([f]) &= M([g_S]) + M([g_{S^c}]) \\ &= M([\chi_S]) \left(\frac{M([g_S])}{M([\chi_S])} \right) + M([\chi_{S^c}]) \left(\frac{M([g_{S^c}])}{M([\chi_{S^c}])} \right). \end{aligned}$$

This contradicts the fact that M is an extreme point of Z .

Let $\mathcal{U} = \{S \subseteq A : M([\chi_S]) = 1\}$. Then $\mathcal{U}$ is a Boolean algebra ultrafilter in $\mathcal{P}(A)$. Moreover, $\mathcal{K} \cap \mathcal{U} = \emptyset$ because if $K \in \mathcal{K}$ then $[\chi_K] = [\chi_\emptyset]$, so $M([\chi_K]) = 0$.

Let $\mathcal{K} = \mathcal{L} \sim \mathcal{U}$. Then $\mathcal{K}$ is an ideal in the lattice $(\mathcal{L}, \subseteq)$ such that $\mathcal{J} \subseteq \mathcal{K}$. We claim that $\mathcal{K}$ is a $\subseteq$ -maximal proper ideal containing $\mathcal{J}$. Suppose not. Suppose there is an ideal $\mathcal{G} \supset \mathcal{K}$. Then if we repeat the above argument with $\mathcal{G}$ instead of $\mathcal{J}$ we would obtain an extreme point M' of Z such that $M'([\chi_S]) = 0$ for all $S \in \mathcal{G}$, contradicting the fact that M is a Y -minimal extreme point of Z , q.e.d.

In the above proof, the function M is a two-valued measure. Thus, AN 5 is equivalent to the following:

AN 6: If $(L, \leq)$ is a Boolean algebra with identity 1, and K is a sublattice of L , then there exists a two-valued K -minimal measure μ on L such that $\mu(1) = 1$.

If "K-minimal" is omitted from AN 6 then the resulting statement is equivalent to BPI.

PART II, CLASS FORMS

INTRODUCTION

We now turn our attention to the class or global forms of the axiom of choice. J. von Neumann [1925] listed CWO 10 as an axiom of set theory (Axiom IV 2) and stated that the well ordering theorem can be derived from it. Then in 1928 (von Neumann [1928]), he proved that CWO 10 implies that the universe can be well ordered. Gödel [1940] used axiom E, which is equivalent to our CAC 1, as an axiom of set theory and Bernays [1941] used CAC 4 as an axiom. Bernays also proved that CAC 4 was equivalent to CAC 5. Since Rubin & Rubin [1963], other people who have worked with the class form of the axiom choice include Easton [1964], Felgner [1971, 1976a], Felgner & Flannagan [1978], Flannagan [1975], Fleisher [1973], Gaifman [1968], Grišen [1972], Harper & Rubin [1977], Howard, Rubin & Rubin [1978], and Isbell & Wright [1966].

For each set form of the axiom of choice in Part I which has the form:

For every x there exists a y such that $\phi(x,y)$,

we can obtain a class form,

There is a function F such that for every x , $\phi(x,F(x))$.

For example, see CWO 1, Gödel's axiom E, and the following form of the trichotomy:

There is a function F such that for all sets x and y , $F(x,y)$ is a 1-1 function of a subset of x into a subset of y such that either $\mathcal{P}(F(x,y)) = x$ or $\mathcal{P}(F(x,y)) = y$.

Other ways of obtaining class forms are illustrated by CMS 1-4, CMR 8(A), CH, and CWO 12. However, by far the most common method of obtaining a class form from a set form is by changing the word "set" to "class". Most of the class forms we consider are obtained in this way.

The only class forms we are considering are class forms of the well ordering theorem, the axiom of choice, and the maximal principles. The letter "C" will precede the name of of each proposition (AC becomes CAC, WO becomes CWO and M becomes CM) and the letter "C" will also follow the number of each theorem and definition in Part II.

The axiom of foundation plays a more important role in proving equivalences of class forms than it did for set forms. To prove equivalences for set forms, we frequently had to construct collections of sets which had certain properties. In the corresponding class forms these sets may become classes and cannot be elements. Another type of proof which is used for the set forms but only works for the class forms with the axiom of foundation is to deduce from the fact that there is a choice function on the class of all non-empty subsets of a class, that the class can be well ordered.

However, we give here two forms of the well ordering theorem, CWO 5 (Each proper class is equipollent to On) and CWO 12($\mathcal{P}(On)$ can be well ordered) such that for every other class form S that we give here we can prove in NBG^o that $CWO\ 5 \rightarrow S \rightarrow CWO\ 12$. Even though the axiom of foundation is frequently needed to prove equivalences, the only proof that we have to give in NBG is that $CWO\ 12 \rightarrow CWO\ 5$.

§1. THE WELL ORDERING THEOREM

CWO 1-4 are class forms of WO 1, CWO 5 is a class form of WO 2, CWO 6(m)-8 are class forms of WO 4(m)-6 and CWO 9 is a class form of MC 2.

CWO 1: There is a function F such that for each set x , $F(x)$ well orders x .

CWO 2: There is a relation R such that every set is well ordered by R .

CWO 3: Every class can be well ordered. (Equivalently, the universe, V , can be well ordered.)

CWO 4: There is a relation R which well orders V in such a way that each proper initial R -segment of V is a set.

CWO 5: Each proper class is equipollent to On . (Equivalently, $V \approx On$.)

Let $m \in \omega$ such that $m \geq 1$.

CWO 6(m): For each class X there is a function F defined on On such that for each $\alpha \in On$, $F(\alpha) \leq m$ and $\bigcup_{\alpha \in On} F(\alpha) = X$.

CWO 7: There is an $m \in \omega \sim \{0\}$ such that CWO 6(m).

CWO 8: For each class X there exists an $m \in \omega \sim \{0\}$ and a function F defined on On such that for each $\alpha \in On$, $F(\alpha) \leq m$ and $\bigcup_{\alpha \in On} F(\alpha) = X$.

CWO 9: Every class X is the union of a well ordered class of finite sets.

In each of CWO 6(m)-9, we obtain an equivalent statement by replacing "X" by "V". Thus, each implies that the universe is a well ordered union of finite sets.

The next form, CWO 10, is a class form of PROJ given in section 5. It was given as an axiom of set theory in von Neumann [1925, Axiom IV 2] as a replacement for the well ordering theorem and in von Neumann [1928] it was proved that $\text{CWO } 10 \rightarrow \text{CWO } 3$.

CWO 10: Every proper class can be mapped onto V. (For every proper class X , $V \prec^* X$.)

The following statement CT is a class form of the trichotomy, but it is easily seen to be equivalent to a form of the well ordering theorem.

CT: Each pair of proper classes is equipollent.

It is not known whether the statement:

Each pair of proper classes is comparable.

implies a class form of the axiom of choice. Feigner & Flannagan [1978] have shown that the statement:

On can be injected into every proper class.

does not imply the set form of the axiom of choice in NBG.

CWO 11, CWO 12, and CPW are class forms of PW, CLW is a class form of LW, and CCW is a class form of WO 8.

CWO 11: For each proper class X , $\mathcal{P}(X) \prec X$.

CWO 12: $\mathcal{P}(\text{On})$ can be well ordered.

CPW: The power class of a well ordered class can be well ordered.

CLW: Every linearly ordered class can be well ordered.

CCW: Every class on which there is a choice function can be well ordered.

The following implications are immediate.

$CT \leftrightarrow CWO\ 5 \rightarrow CWO\ 3 \rightarrow CWO\ 2 \rightarrow CWO\ 1$; $CWO\ 3 \leftrightarrow CCW$;
 $CWO\ 5 \leftrightarrow CWO\ 4 \leftrightarrow CWO\ 10$; $CWO\ 6(1) \leftrightarrow CWO\ 5$;
 $CWO\ 6(m) \rightarrow CWO\ 6(n)$ if $m \leq n$;
 $CWO\ 6(m) \rightarrow CWO\ 7 \rightarrow CWO\ 8 \rightarrow CWO\ 9$;
 $CWO\ 5 \rightarrow CWO\ 11 \rightarrow CPW \rightarrow CWO\ 12$.

The proof that $CWO\ 8 \rightarrow CWO\ 5$ is similar to the proof of Theorem 1.1. In fact, the proof here is a little easier because it is sufficient to take the "X" of $CWO\ 8$ to be "V" and we automatically have $V \times V \subseteq V$. The proofs that $CLW \rightarrow CPW$, and $CWO\ 12 \rightarrow CWO\ 4$ are similar to the proofs for the corresponding set forms. (See Theorems 5.5 and 5.7 and notice that $CWO\ 12 \rightarrow PW$. The proof that $CWO\ 12 \rightarrow CWO\ 4$ is in NBG.) To complete the proof of equivalences we give the following two theorems.

THEOREM 1.1C: $CWO\ 1 \rightarrow CWO\ 12$.

PROOF: For each $u \subseteq On$ let $\rho(u)$ be the smallest ordinal number α such that for all $\beta \in u$, $\beta < \alpha$. For each α , let $x_\alpha = \{u \subseteq On: \rho(u) = \alpha\}$. Then since $x \subseteq \mathcal{P}(\alpha)$, x_α is a set. Let F be the function guaranteed by $CWO\ 1$ such that for each $\alpha \in On$, $F(x_\alpha)$ well orders x_α . Then we can well order $\mathcal{P}(On)$ by a relation R as follows: for $u, v \subseteq On$,

$$u R v \text{ iff } (\rho(u) < \rho(v)) \vee (\rho(u) = \rho(v) = \alpha \wedge u F(x_\alpha) v).$$

THEOREM 1.2C: $CWO\ 9 \rightarrow CLW$.

PROOF: Suppose X is a class which is linearly ordered by R . $CWO\ 9$ implies that X is a well ordered union of finite sets. A linear ordering on a finite set is a well ordering. Thus, using R we can well order X .

In Figure 1.1C, we summarize the results of this section. All implications were proved in NBG°.

All the forms in the set $\textcircled{A}$ are equivalent in NBG°, where $\textcircled{A} = \{CT, CWO\ 4, CWO\ 5, CWO\ 6(m), CWO\ 7, CWO\ 8, CWO\ 10\}$.

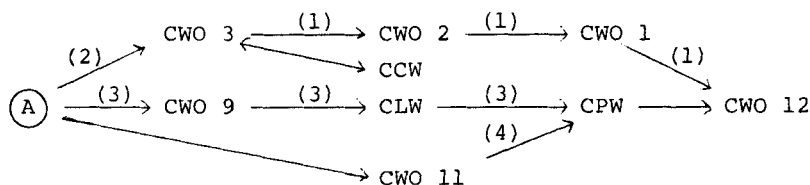


FIGURE 1.1C

We have shown that $\text{CWO } 12 \rightarrow \textcircled{A}$ in NBG; thus all the statements in Figure 1.1C are equivalent in NBG. (It was shown by Howard, Rubin & Rubin [1978] that the implications (1) and (2) are not reversible in NBG° , but a strongly inaccessible cardinal was used to prove (2). Felgner & Jech [1973] proved that the implications (3) are not reversible in NBG° . The proofs in Felgner & Jech were for the corresponding set forms, but it is easy to see that they hold for the class forms also. It follows from model (e) in Felgner & Jech [1973] that (4) is not reversible in NBG°).

To conclude this section we shall show that $\text{CWO } 2$ along with DC, the Principle of Dependent Choices (see Part I, Section 2) implies $\text{CWO } 3$. Clearly, $\text{CWO } 3 \rightarrow \text{DC}$.

THEOREM 1.3C: $\text{CWO } 2 \wedge \text{DC} \rightarrow \text{CWO } 3$.

PROOF: Let W be an irreflexive relation which well orders each set. Suppose W does not well order V . Then there is a non-empty class A such that A has no W -first element. Define a relation R as follows:

$$R = \{\langle x, y \rangle \in A \times A : \langle y, x \rangle \in W\}.$$

Since A has no W -first element, $\mathcal{D}(R) = A$. Therefore, $\mathcal{R}(R) \subseteq \mathcal{D}(R)$ so DC implies that there is a function f such that for each $n \in \omega$, $\langle f(n), f(n+1) \rangle \in R$. But, by the definition of R , $\langle f(n+1), f(n) \rangle \in W$ for each $n \in \omega$. This implies the contradiction that the set $\{f(n) : n \in \omega\}$ has no W -first element, q.e.d.

§2. THE AXIOM OF CHOICE

We can obtain class forms of the axiom of choice, CAC k , from the corresponding set forms, AC k , for $1 \leq k \leq 15$, but $k \neq 6, 7, 9$ by substituting the word "class" for "set" in the appropriate places. AC 20 and 21 are class forms of AC 4 and 18, respectively, which were proved to be equivalent to a set form in NBG (see Part I, §5). AC 6 and 7 have no class analogues because the direct product of a proper class of non-empty sets is empty. AC 17 has no class analogue for similar reasons - there does not exist a class which has a proper class as an element. AC 9, 18, and 19 do have class analogues, but they are written in a different form. We do not know if AC 16(n, k) has a class form. We shall list the CAC k forms below.

CAC 1: If S is a class of non-empty sets, there is a function F such that for each $x \in S$, $F(x) \in x$.

Gödel's [1940] Axiom E was the statement:

E: There exists a function F such that for each non-empty set x , $F(x) \in x$.

Clearly, $E \rightarrow \text{CAC } 1$, and $\text{CAC } 1 \rightarrow E$ because the class S in CAC 1 may be taken to be $V \sim \{\emptyset\}$.

CAC 2: If T is a pairwise disjoint class of non-empty sets, there is a class C which consists of one and only one element from each set in T .

CAC 3: For every function F there is a function G such that for every x , if $x \in \mathcal{P}(F)$ and $F(x) \neq \emptyset$ then $G(x) \in F(x)$.

CAC 4: For every relation R there is a function F such that $\mathcal{I}(F) = \mathcal{I}(R)$ and $F \subseteq R$. (See AC 20 in Section 5).

CAC 5: For every function F there is a function G such that $\mathcal{I}(G) = \mathcal{K}(F)$ and for every $x \in \mathcal{I}(G)$, $F(G(x)) = x$.

CAC 8: If S is a class of pairs of equipollent sets then there is a function which associates with each pair a bijection mapping one onto the other.

CAC 9: Let R be a relation such that for each $x, y \in \mathcal{I}(R)$, $R''\{x\} \approx R''\{y\}$; then there exists a relation S such that for each $x, y \in \mathcal{I}(R)$, $S''\{\{x, y\}\}$ is a bijection mapping $R''\{x\}$ onto $R''\{y\}$.

We do not know if the statement obtained from AC 9 by changing the word "set" to "class" is equivalent to a class form of the axiom of choice.

Let $n \geq 2$ be a natural number.

CAC 10(n): If S is a class of infinite sets, there is a function F such that for each $x \in S$, $F(x)$ is a decomposition of x into a set of disjoint sets of size between 2 and n .

CAC 11: There exists a natural number $n \geq 2$ such that CAC 10(n).

CAC 12: If S is a class of infinite sets there is a natural number $n \geq 2$ and a function F such that for each $x \in S$, $F(x)$ is a decomposition of x into a set of disjoint sets of size between 2 and n .

Each of the set forms AC 10(n), 11 and 12 remain equivalent to the set form of the axiom of choice if the size of each of the sets in the partition is exactly n . However, for the class forms, we were not able to prove equivalence in NBG^o, so we shall state the latter forms explicitly.

Let $n \geq 2$ be a natural number.

CAC 10E(n): If S is a class of infinite sets, there is a function F such that for each $x \in S$, $F(x)$ is a decomposition of x into a set of disjoint sets of size exactly n .

Similarly, for CAC 11E and CAC 12E. We have CAC 10E(2) is the same as CAC 10(2).

Let $m \geq 1$ be a natural number.

CAC 13(m): If S is a class of non-empty sets there is a function F such that for each $x \in S$, $F(x)$ is a non-empty subset of x with $F(x) \prec m$.

CAC 14: There is a natural number $m \geq 1$ such that CAC 13(m).

CAC 15: If S is a class of non-empty sets then there is a natural number $m \geq 1$ and a function F such that for each $x \in S$, $F(x)$ is a non-empty subset of x with $F(x) \prec m$.

CAC 18 and 18C are class forms of AC 18 (See AC 21 in Part I, Section 5.), CAC 19 is a class form of AC 19.

CAC 18: For any non-empty class A , and for every function B whose domain is A and B_a is a non-empty set for each $a \in A$

$$x \in \bigcap_{a \in A} \bigcup_{b \in B_a} X_{a,b} \leftrightarrow \text{There is a function } F \text{ with domain } A$$

$$\text{such that } (\forall a \in A) [F(a) \in B_a \wedge x \in X_{a,F(a)}].$$

CAC 18C: For any non-empty class A , for every class-valued function B whose domain is A

$$x \in \bigcap_{a \in A} \bigcup_{b \in B[a]} X_{a,b} \leftrightarrow \text{There is a function } F \text{ with domain } A$$

$$\text{such that } (\forall a \in A) [F(a) \in B[a] \wedge x \in X_{a,F(a)}].$$

(See Definition 0.4(k) for the definition of a class-valued function).

CAC 19: For any non-empty class A , each of whose elements is non-empty,

$$x \in \bigcap_{a \in A} \bigcup_{b \in a} b \leftrightarrow \text{There exists a function } F \text{ with domain}$$

A such that $(\forall a \in A)[F(a) \in a \wedge x \in F(a)]$.

CAC 22 and 23 are class forms of the multiple choice axiom, MC 1.

CAC 22: If S is a class of non-empty sets then there is a function F such that for each $x \in S$, $F(x)$ is a non-empty finite subset of x .

CAC 23: There exists a function F such that for each non-empty set x , $F(x)$ is a non-empty finite subset of x .

CAC 24 is a class form of MC 3.

CAC 24: If S is a class of infinite sets, then there is a function F such that for each $x \in S$, $F(x)$ is a partition of x into pairwise disjoint finite sets, each of which has at least 2 elements.

The proofs that CAC 1-5 are equivalent are the same as the corresponding proofs for the set forms with one exception. The proof that AC 3 $\rightarrow$ AC 4 (see 2.7) does not hold for the class forms because for some $x \in \mathcal{P}(R)$ (where R is the relation described in CAC 4) $\{y: \langle x, y \rangle \in R\}$ may be a proper class. Therefore, if we could find a relation $S \subseteq R$ such that $\mathcal{P}(R) = \mathcal{P}(S)$ and for all x , $S''\{x\}$ is a set, the proof that CAC 3 $\rightarrow$ CAC 4 would follow. Let PR be the following statement:

PR: For every relation R there is a relation $S \subseteq R$ such that $\mathcal{P}(S) = \mathcal{P}(R)$ and for all x , $S''\{x\}$ is a set.

LEMMA 2.1C: $\text{NBG} \vdash \text{PR}$.

PROOF: Let R be an arbitrary relation; define S so that $x S y \leftrightarrow x R y \wedge (\forall z)[x R z \rightarrow \rho(y) \leq \rho(z)]$. It follows from 0.8 and 0.9 that the class of all elements with a

given rank is a set; so S is the required relation.

It is also clear that $CAC\ 4 \rightarrow PR$.

THEOREM 2.2C: $CAC\ 3 \wedge PR \rightarrow CAC\ 4$.

PROOF: Let R be an arbitrary relation and let S be the associated relation whose existence is guaranteed by PR . For each $x \in \mathcal{D}(S)$ define a function F so that $F(x) = S''\{x\}$. $CAC\ 3$ implies there is a function G such that for each $x \in \mathcal{D}(F)$, if $F(x) \neq \emptyset$ then $G(x) \in F(x)$. G is the required function, q.e.d.

It was shown in Howard, Rubin & Rubin [1978] that $CAC\ 3$ does not imply $CAC\ 4$ in NBG° .

We shall show next that each of $CAC\ 1-5$ is equivalent to the class form of the well ordering theorem.

Clearly $CWO\ 1 \rightarrow CAC\ 1$.

THEOREM 2.3C: $CAC\ 1 \rightarrow CWO\ 1$.

PROOF: First, we have $CAC\ 1 \rightarrow WO\ 1$. For each non-empty set x define

$$S_x = \{R \subseteq x \times x : R \text{ well orders } x\}.$$

$WO\ 1$ implies that $S_x \neq \emptyset$ for each set x . Define $S = \{S_x : x \in V\}$. $CAC\ 1$ implies there is a function G such that for each $S_x \in S$, $G(S_x) \in S_x$. For each $x \in V$ define $F(x) = G(S_x)$. F is the required function.

We can also prove the following theorem in NBG° .

THEOREM 2.4C: $CWO\ 3 \rightarrow CAC\ 4$.

PROOF: Let R be an arbitrary relation. $CWO\ 3$ implies that there is a relation S which well orders V . Define a function F so that for each $x \in \mathcal{D}(R)$,

$$F(x) = S\text{-first element of } R''\{x\}.$$

F is the required function.

The proofs of the following are the same as for the corresponding set forms. (See Part 1 Section 2 and Theorem 5.23.)

$CAC\ 1 \leftrightarrow CAC\ 8; \quad CAC\ 10(n) \rightarrow CAC\ 13(n-1), \quad n \geq 2;$
 $CAC\ 1 \rightarrow CAC\ 10E(n) \rightarrow CAC\ 10(n) \rightarrow CAC\ 11 \rightarrow CAC\ 12 \rightarrow CAC\ 15;$
 $CAC\ 1 \rightarrow CAC\ 13(n) \rightarrow CAC\ 14 \rightarrow CAC\ 15 \rightarrow CAC\ 22 \rightarrow CLW;$
 $CAC\ 4 \leftrightarrow CAC\ 18C \rightarrow CAC\ 18 \leftrightarrow CAC\ 19 \leftrightarrow CAC\ 1;$
 $NBG \vdash CLW \rightarrow CWO\ 4.$

The following implications are easy to prove:

$CAC\ 11 \leftrightarrow CAC\ 12 \rightarrow CAC\ 14 \leftrightarrow CAC\ 15;$
 $CAC\ 22 \leftrightarrow CAC\ 23; \quad CAC\ 1 \leftrightarrow CAC\ 10(2) (= CAC\ 10E(2));$
 $CAC\ 24 \rightarrow CAC\ 22 \quad (\text{similar to the proof of 2.12});$
 $CAC\ 10E(n) \rightarrow CAC\ 11E \leftrightarrow CAC\ 12E \rightarrow CAC\ 11.$

We can also prove the following implications in NBG° :

THEOREM 2.5C:

- (a) If $n \geq 2$, $CAC\ 13(n) \rightarrow CAC\ 10(n+1).$
- (b) $CAC\ 14 \rightarrow CAC\ 11.$
- (c) $CAC\ 22 \rightarrow CAC\ 24.$

PROOF: (a) Let S be the class of all infinite sets. $CAC\ 13(n)$ implies that there is a function F such that for each $x \in S$, $F(x)$ is a non-empty finite subset of x such that $F(x) \leq n$. For each ordinal α and $x \in S$, let

$$G(x, \alpha) = \begin{cases} F(x \sim \bigcup_{\beta < \alpha} G(x, \beta)) & \text{if } x \neq \bigcup_{\beta < \alpha} G(x, \beta), \\ \emptyset, & \text{otherwise.} \end{cases}$$

Then $x_p = \{G(x, \alpha) : G(x, \alpha) \neq \emptyset \wedge \alpha \in On\}$ is a well ordered partition of x into non-empty sets each of which has $\leq n$ elements, but some elements of x_p may only have one element. If there are an infinite number of $G(x, \alpha)$'s with one element, combine them in pairs. If there are a finite number, k of them and k is even, again combine them in pairs. If k is odd, $k \geq 3$ and $n \geq 3$, combine the first three $G(x, \alpha)$'s which are singletons and then combine the remaining ones in pairs. Finally, if $k = 1$ or $n = 2$, we have to combine the

$G(x, \alpha)$'s which are singletons with other sets in the partition and in this case we may end up with a set in the partition which has $n + 1$ elements. (We can carry out this procedure simultaneously for each set in S because each partition has a defined well ordering).

This same procedure may be used to prove both (b) and (c), q.e.d.

Thus it follows from 2.5C and preceding results that for $n \geq 2$, $CAC\ 13(n) \leftrightarrow CAC\ 10(n+1)$; $CAC\ 14 \leftrightarrow CAC\ 11$; and $CAC\ 22 \leftrightarrow CAC\ 24$.

We shall show next that $CWO\ 5 \rightarrow CAC\ 9 \rightarrow CAC\ 4$.

THEOREM 2.6C: $CWO\ 5 \rightarrow CAC\ 9$.

PROOF: Let R be a relation satisfying the hypotheses of $CAC\ 9$. $CWO\ 5$ implies that there is a bijection F mapping

$\bigcup_{x \in \mathcal{D}(R)} (R''\{x\} \times \{x\})$ onto On . For each $x \in \mathcal{D}(R)$ let

$F_x = F \upharpoonright (R''\{x\} \times \{x\})$. By hypothesis, for all $x, y \in \mathcal{D}(R)$, $\mathcal{R}(F_x) \approx \mathcal{R}(F_y)$, and by definition of F , $\mathcal{R}(F_x), \mathcal{R}(F_y) \subseteq On$ and $\mathcal{R}(F_x) \cap \mathcal{R}(F_y) = \emptyset$. Let G_{xy} be the unique bijection mapping $\mathcal{R}(F_x)$ onto $\mathcal{R}(F_y)$ which preserves order. Then

$F_y^{-1} \circ G_{yx} \circ F_x$ is a bijection from $R''\{x\} \times \{x\}$ onto $R''\{y\} \times \{y\}$, q.e.d.

THEOREM 2.7C: $CAC\ 9 \rightarrow CAC\ 4$.

PROOF: Let R be a relation. Define $Z = [\bigcup_{x \in \mathcal{D}(R)} R''\{x\}]^\omega$. Then, just as in the proof of 2.8, we show that for each $x \in \mathcal{D}(R)$, $R''\{x\} \times Z \approx Z$. Applying $CAC\ 9$, we construct a relation S such that for all $x, y \in \mathcal{D}(R)$,

$$S''\{\langle x, y \rangle\}: (R''\{x\} \times Z) \approx (R''\{y\} \times Z).$$

Choose $x \in \mathcal{D}(R)$, $a \in R''\{x\}$ and $f \in Z$; then define a function F so that

$$F = \{\langle y, b \rangle: b \in y \wedge (\exists g \in Z) S''\{\langle x, y \rangle\}(a, f) = \langle b, g \rangle\}.$$

F is the required function, q.e.d.

The last class form of the axiom of choice we are considering in this section is due to Hilbert [1923]. It is a metamathematical statement and has no set counterpart. Hilbert stated it in the following form:

There is a function F such that for each property P if $F(P)$ has the property P then every x has the property P.

However, such a statement is meaningless in the system of set theory in which we are working. F is some sort of metamathematical function which has not been defined. Through discussions with A. Levy and R. Vaught we were able to come up with a more meaningful formulation. To do this we introduce the metamathematical notion of the iota operator.

DEFINITION 2.8C: If ϕ is a formula with at least one free variable then:

$$X = (\iota Y)\phi(Y) \leftrightarrow [(\exists! Z)\phi(Z) \wedge \phi(X)] \vee [\neg (\exists! Z)\phi(Z) \wedge X = V].$$

The statement CH below is the contrapositive of Hilbert's statement.

CH: There is a formula ϕ with at least three free variable such that

$$(\exists X)(\forall Y)(Y \neq \emptyset \rightarrow (\iota Z)\phi(Z, X, Y) \in Y).$$

We shall prove $CWO\ 3 \rightarrow CH \rightarrow CAC\ 4$.

THEOREM 2.9C: $CWO\ 3 \rightarrow CH$.

PROOF: Let $\phi(Z, X, Y)$ mean Z is the X-first element of Y. $CWO\ 3$ implies that there is a relation X which well orders the universe. Therefore, for each non-empty class Y, $(\iota Z)\phi(Z, X, Y) \in Y$, q.e.d.

THEOREM 2.10C: $CH \rightarrow CAC\ 4$.

PROOF: Let R be an arbitrary relation, and let ϕ be a formula and X a class which satisfies CH. We define a function F such that for all $x \in \mathcal{D}(R)$,

$$F(x) = \{z \mid \phi(z, x, R''\{x\})\}.$$

Then it follows from CH that F is a function which satisfies CAC 4.

We shall summarize the results from Section 1 and 2, Part 2, in Figure 2.1C.

All the statements in each of the following sets are equivalent.

- (A) = {CT, CWO 4, 5, 6(n), 7, 8, 10}, $n \geq 1$.
- (B) = {CAC 4, 5, 18C}.
- (C) = {CWO 1, E, CAC 1, 2, 3, 8, 10(2), 18, 19}.
- (D_m) = {CAC 10(m), 13(m-1)}, $m \geq 3$.
- (D_E) = {CAC 11E, 12E}.
- (D) = {CAC 11, 12, 14, 15}.
- (E) = {CAC 22, 23, 24}.

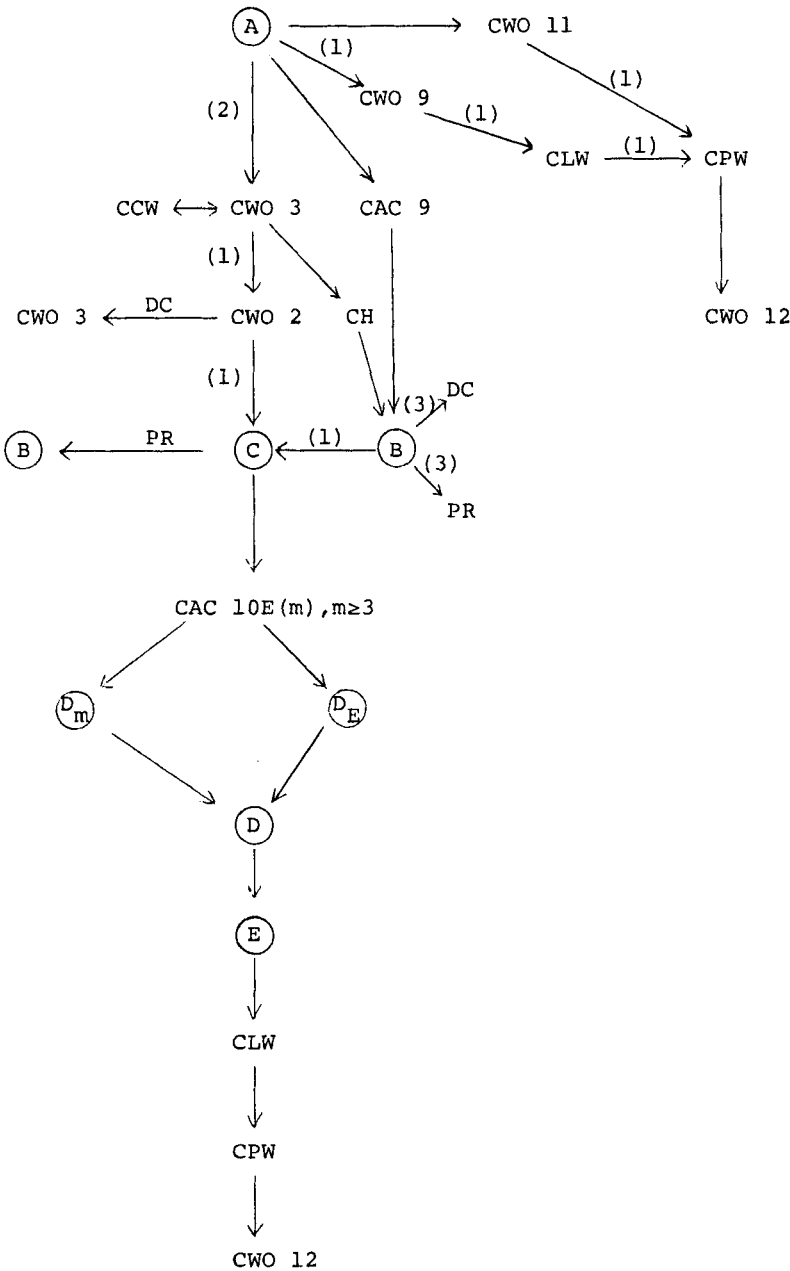


FIGURE 2.1C

The implications (1) and (2) in Figure 2.1C are not reversible in NBG° , but (2) used the existence of a strongly inaccessible cardinal. The implications (3) are not reversible in NBG . We have shown in Part II, section 1, that $\text{NBG} \vdash \text{CWO } 12 \rightarrow \text{CWO } 4$. Thus, all the class forms of the axiom of choice given in Figure 2.1C are equivalent in NBG .

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§3. MAXIMAL PRINCIPLES

The first four maximal principles we give here are obtained from M1-4 by changing the word "set" to "class".

CM 1: If R is a partial ordering relation on a non-empty class X and if every subclass of X which is linearly ordered by R has an R -upper bound, then X has an R -maximal element.

CM 2: If R is a partial ordering relation on a non-empty class X and if every subclass of X which is well ordered by R has an R -upper bound, then X has an R -maximal element.

CM 3: If every non-empty nest which is a subclass of a non-empty class has its union an element of X , then X has a $\subseteq$ -maximal element.

CM 4: If every non-empty well ordered nest which is a subclass of a non-empty class X has its union an element of X , then X has a $\subseteq$ -maximal element.

The following four statements are also class forms of M1-4.

CMS 1: If R is a partial ordering relation on a non-empty class X and if every subset of X which is linearly ordered by R has an R -upper bound then either X has an R -maximal element or X has a subclass which is a proper class and which is linearly ordered by R .

CMS 2 is obtained from CMS 1 by replacing "linearly ordered" by "well ordered".

CMS 3: If every non-empty nest which is a subset of a non-empty class X has its union an element of X , then either X has a $\subseteq$ -maximal element or there is a subclass of X which is a nest and a proper class.

CMS 4 is obtained from CMS 3 by replacing "nest" by "well ordered nest".

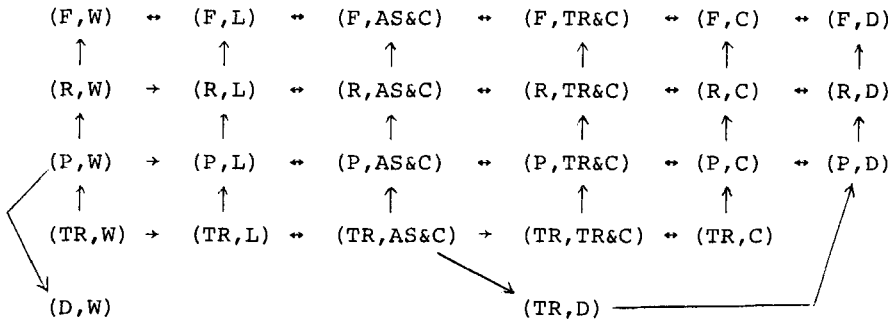
Clearly $CM\ n \rightarrow CMS\ n$ for $n = 1, 2, 3, 4$. It is also easy to see that $CMS\ n \rightarrow CM\ n$ for $n = 3, 4$ because if a proper class is linearly ordered by $\subseteq$, (that is, a nest) then its union is a proper class and cannot be an element. (Howard, Rubin & Rubin [1978] have shown that if $n = 1$ or 2 , CMS n does not imply $CM\ n$ in NBG° if there exists a strongly inaccessible cardinal.) Later we shall show that they are all equivalent in NBG .

There are also the two corresponding types of class forms for $M\ 1(Q, U)$. (See pp. 37-8 for the possibilities for Q and U .)

$CM\ 1(Q, U)$: Every non-empty class which is Q -ordered by R in which each U -ordered subclass has an R -upper bound, has an R -maximal element.

$CMS\ 1(Q, U)$: Every non-empty class which is Q -ordered by R in which each U -ordered subset has an R -upper bound, either has an R -maximal element or contains a U -ordered subclass which is a proper class.

$CM\ 1(P, L)$ is $CM\ 1$ and $CM\ 1(P, W)$ is $CM\ 2$. Similarly for $CMS\ 1(P, L)$ and $CMS\ 1(P, W)$. It is clear that $CM\ 1(Q, U) \rightarrow CMS\ 1(Q, U)$ for all Q and U . We proved in Part I, Section 3 that the $M\ 1(Q, U)$ are equivalent if $Q = TR, P, RA$, or F and $U = C, AS\&C, TR\&C, L, D$, or W ; or if $Q = D$ and $U = W$. (See pp. 37-41.) These same proofs may be used to prove that the corresponding forms of $CMS\ 1(Q, U)$ are all equivalent. However, these proofs do not all hold for the stronger forms, $CM\ 1(Q, U)$. For these forms we were only able to prove the following in NBG° : (In figure 3.1C, " (Q, U) " means " $CM\ 1(Q, U)$ ".)



CM 1(Q,U)

FIGURE 3.1C

(See Harper & Rubin [1977].)

We shall also consider class forms of M5, M 5(Q,U), M 6-7, M 8(A), M 9(A), M 10-11, M 15(A:B), M 16(A:B), M 17, and A, the anti-chain axiom. (See pp 276-278.)

The following implications are easy to prove: CM 5 → CM 1, CWO 5 → CM 1(TR,W) → CM 2 → CM 1, and CM 1 → CMS 1.

THEOREM 3.1C: The following statements are equivalent: CMS 1, CMS 2, CMS 3, CMS 4, CM 3, CM 4, CM 6 and CM 17. (CM 8(K) is the same as CM 6.)

PROOF: The following implications are easily seen to hold:

$$\begin{aligned} \text{CMS 2} &\rightarrow \text{CMS 1} \rightarrow \text{CMS 3} \\ \text{CMS 2} &\rightarrow \text{CMS 4} \rightarrow \text{CMS 3} . \end{aligned}$$

First, we will prove that CMS 3 → CMS 2. Let L be a non-empty class and R a partial ordering relation on X . Define Y as follows:

$$\begin{aligned} Y = \{ &s \subseteq \mathcal{P}(X) : (\forall x \in s) (R \text{ well orders } x) \wedge \\ &(\forall x, y \in s) (x \text{ is an } R\text{-initial segment of } y \vee \\ &y \text{ is an } R\text{-initial segment of } x) \} . \end{aligned}$$

Let $z \subseteq Y$ be a nest, then $Uz \in Y$. Therefore, Y satisfies the hypothesis of CMS 3. It follows that either Y has an R -maximal element or Y has a subclass which is a nest and a proper class.

Suppose the first alternative holds. Then Y has an R -maximal element t . It follows that Ut can be well ordered by R . The hypothesis of CMS 2 implies that Ut has an R -upper bound, b . Thus, b is an R -maximal element of X . If the second alternative holds then there exists a nest $Z \subseteq Y$ such that Z is a proper class. Then, UUZ is a subclass of X which is a proper class and well ordered by R .

As was announced earlier $CM\ 3 \leftrightarrow CMS\ 3$ and $CM\ 4 \leftrightarrow CMS\ 4$. Thus, the first six statements given in the theorem are equivalent.

We shall show next that $CM\ 6 \rightarrow CMS\ 3$ and $CMS\ 1 \rightarrow CM\ 6$.

Assume $CM\ 6$ holds and let X be a non-empty class. Then $CM\ 6$ implies that there exists a $\subseteq$ -maximal subclass $Z \subseteq X$ such that Z is a nest. If Z is a proper class then X has a subclass which is a nest and a proper class. If Z is a set, the hypotheses of CMS 3 implies that $UZ \in X$. In this case, UZ is a $\subseteq$ -maximal element of X .

Next, assume $CMS\ 1$ holds and let X be a non-empty class. Define Y as follows:

$$Y = \{x \subseteq X: x \text{ is a } \subseteq\text{-maximal nest in } X \cap \mathcal{P}(Ux)\}$$

Y is partially ordered by inclusion. Let $s \subseteq Y$ be a nest. Clearly, $CMS\ 1 \rightarrow M\ 1 \rightarrow M'6$ and $M'6$ implies that there is a $\subseteq$ -maximal nest y such that

$$(i) \quad Us \subseteq y \subseteq \mathcal{P}(UUs).$$

It is clear that y is a $\subseteq$ -upper bound of s . We claim that $y \in Y$. Since $\mathcal{P}$ and U are monotone, it follows from (i) that

$$(ii) \quad \mathcal{P}(UU s) \subseteq \mathcal{P}(Uy) \subseteq \mathcal{P}(U\mathcal{P}(UU s)).$$

For any set z , $U\mathcal{P}(z) = z$. Therefore, from (ii) we obtain

$$(iii) \quad \mathcal{P}(UU s) \subseteq \mathcal{P}(Uy) \subseteq \mathcal{P}(UU s).$$

Thus, $\mathcal{P}(Uy) = \mathcal{P}(UU s)$, so it follows from (i) that $y \in Y$.

CMS 1 implies that either Y has a $\subseteq$ -maximal element which is clearly a $\subseteq$ -maximal nest in X , or Y contains a subclass S which is a nest and a proper class. US is clearly a nest, and we claim it is a $\subseteq$ -maximal nest contained in X . For suppose not. Suppose there exists a $u \in X$ such that $US \cup \{u\}$ is a nest. Then, either $v \subseteq u$ for all $v \in US$ or there exists a $v \in US$ such that $u \subset v$. In the first case $UU s \subseteq u$, which is a contradiction because u is a set and S is a proper class. In the second case, $v \in US$ implies there is an $x \in S$ such that $v \in x$. Since $x \cup \{u\} \subseteq US \cup \{u\}$, $x \cup \{u\}$ is a nest. Moreover, since $u \subset v \in x$, $x \cup \{u\} \subseteq X \cap \mathcal{P}(Ux)$. But this is a contradiction because $x \in Y$, and therefore x is supposed to be a $\subseteq$ -maximal nest in $X \cap \mathcal{P}(Ux)$.

Finally, we shall show that $CM 17 \leftrightarrow CMS 3$.

Suppose $CM 17$ holds and X is a non-empty class. We define a relation R on $X \times On$ as follows: If $s, t \in X$

$$\langle s, \alpha \rangle R \langle t, \beta \rangle \leftrightarrow (s \subset t \wedge \alpha < \beta) \vee$$

$$(s = t, s \text{ is a } \subseteq\text{-maximal element of } X \wedge \alpha < \beta).$$

Define a function F on $X \times On$ as follows:

$$F(s, \alpha) = \{ \{ \langle s, \alpha \rangle \} \} \cup \{ F(s', \alpha') : \langle s', \alpha' \rangle \in X \times On \\ \wedge \langle s', \alpha' \rangle R \langle s, \alpha \rangle \}.$$

It is easily shown that

$$F(s, \alpha) = F(t, \beta) \leftrightarrow \langle s, \alpha \rangle = \langle t, \beta \rangle,$$

$$F(s, \alpha) \in F(t, \beta) \leftrightarrow \langle s, \alpha \rangle R \langle t, \beta \rangle \quad (\text{see 4.46}).$$

CM 17 implies that $\{F(s, \alpha) : \langle s, \alpha \rangle \in X \times \text{On}\}$ has a $\subseteq$ -maximal subclass Q such that for all distinct $s, t \in Q$, $s \in t$ or $t \in s$. Therefore, $X \times \text{On}$ has a $\subseteq$ -maximal subclass Z which is linearly ordered by $R \cup I$, where I is the identity relation on $X \times \text{On}$.

Suppose X has no $\subseteq$ -maximal element. $\mathcal{J}(Z)$ is a nest. If $\mathcal{J}(Z)$ is a set, then by the hypothesis of CMS 3 $\bigcup \mathcal{J}(Z) \in X$, and therefore, $\bigcup \mathcal{J}(Z)$ would be a $\subseteq$ -maximal element of X . Consequently, if X has no $\subseteq$ -maximal element, $\mathcal{J}(Z)$ is a subclass of X which is a nest and a proper class.

Conversely, suppose CMS 3 holds. Let X be a class. For each set x , let $\text{TC}(x)$ be the transitive closure of x . (That is, $\text{TC}(x) = x \cup \bigcup x \cup \bigcup \bigcup x \cup \dots$) Define $Y = \{y \subseteq X : y \text{ is a } \subseteq\text{-maximal subset of } X \cap \text{TC}(y) \text{ such that } \in \text{ is connected on } y\}$. CMS 3 $\rightarrow$ M 3 $\rightarrow$ M 17 which implies $Y \neq \emptyset$. If Y has a $\subseteq$ -maximal element then CM 17 holds. If Y has no $\subseteq$ -maximal element then CMS 3 implies that there is a nest $Z \subseteq Y$ such that Z is a proper class. We claim that $\bigcup Z$ is a $\subseteq$ -maximal subclass of X such that $\in$ is connected on $\bigcup Z$.

It is clear that $\bigcup Z \subseteq X$ and that $\in$ is connected on $\bigcup Z$. Suppose $\bigcup Z$ is not $\subseteq$ -maximal. Then there is an $x \in X \sim \bigcup Z$ such that $\in$ is connected on $\bigcup Z \cup \{x\}$. Therefore, for all $y \in \bigcup Z$ either $y \in x$ or $x \in y$. Since $\bigcup Z$ is a proper class $\bigcup Z \not\subseteq x$; therefore, there is a $y \in \bigcup Z$ such that $x \in y$. Consequently, there is a $t \in Z$ such that $y \in t$. We have $Z \subseteq Y$; thus $t \in Z$ implies $t \in Y$. Since $x \in y \in t$, $x \in \text{TC}(t)$. Since $\in$ is connected on $\bigcup Z \cup \{x\}$ and $t \subseteq \bigcup Z$, it follows that $t \cup \{x\}$ is $\in$ -connected. However, t is a maximal subset of $X \cap \text{TC}(t)$ which is $\in$ -connected; therefore, $x \in t$. This implies the contradiction that $x \in \bigcup Z$.

Now we have completed the proof of Theorem 3.1C.

Our next concern are the forms CM 5(Q, U) where the possibilities for Q and U are A, TR, AS, C, TR&C, AS&C, D (directed), RA, and F (see pp 37-8).

We note first that facts (1)-(10) on pp 42-43 all hold for $CM\ 5(Q,U)$. Moreover, the proof that $M\ 5(TR,AS) \rightarrow AC\ 2$ (pp 43-4) also gives a proof that $CM\ 5(TR,AS) \rightarrow CAC\ 2$. The equivalences given on p 78 for $M\ 5(TR\&C,U)$ also hold for $CM\ 5(TR\&C,U)$ where $U = AS, AS\&C, P, L, D$ and RA . Since each of the properties $AS, TR, C, AS\&C, TR\&C, P, L, D$ and RA are properties of finite character it follows that $CM\ 7$ implies $CM\ 5(Q,U)$ for each of the 50 pairs of values for Q and U given in Theorem 4.11 and for the 6 pairs of values given on p 78. Thus, it remains to show that each of $CM\ 5(D,L)$, $CM\ 5(F,L)$ and $CM\ 5(TR\&C,L)$ implies some form of the axiom of choice. It is easy to see that a modification of the proofs on p 43 of $M\ 5(D,L) \rightarrow M\ 5$ and $M\ 5(F,L) \rightarrow M\ 2$ give us $CM\ 5(D,L) \rightarrow CMS\ 1$ and $CM\ 5(F,L) \rightarrow CMS\ 2$, respectively.

THEOREM 3.2C: $CM\ 5(TR\&C,L) \rightarrow CWO\ 12$.

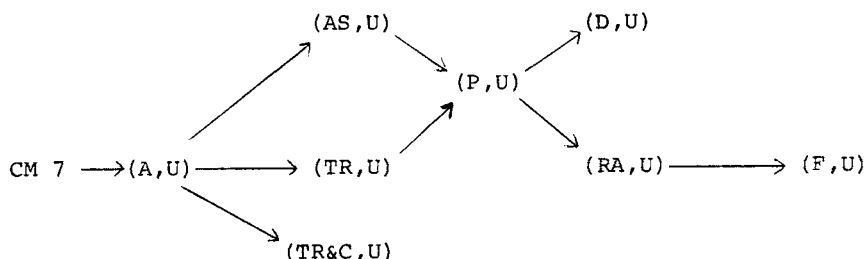
PROOF: First, we can prove using the first part of the proof of Theorem 5.8, that $CM\ 5(TR\&C,L)$ implies that a linearly ordered class of non-empty sets has a choice function, CAC^{LO} .

For each set $u \subseteq On$, we define the rank of u , $\rho(u)$ to be the smallest ordinal number α such that for all $\beta \in u$, $\beta < \alpha$. Every subset of On has a rank and the class of all subsets with a given rank is a set. For each $\alpha \in On$, let $x_\alpha = \{u \subseteq On : \rho(u) = \alpha\}$. Each x_α can be linearly ordered by a relation R such that if $u \neq v$, $u R v$ iff the ϵ -first element of $(u \cup v) \setminus (u \cap v)$ is an element of u . As mentioned above, $CM\ 5(TR\&C,L) \rightarrow CAC^{LO}$, and clearly, $CAC^{LO} \rightarrow AC^{LO} \rightarrow LW$. LW implies each x_α can be well ordered. Let R_α be the set of all well orderings of x_α and let $X = \{R_\alpha : \alpha \in On\}$. CAC^{LO} implies that X has a choice function, F . Then $\mathcal{P}(On)$ can be well ordered by a relation S defined as follows:

$$u S v \text{ iff } \rho(u) < \rho(v) \vee (\rho(u) = \rho(v) = \alpha \wedge u F(R_\alpha) v).$$

In Figure 3.2C below we summarize the results for the 56 statements of the form $CM\ 5(Q,U)$. All implications are

provable in NBG° . (In Figure 3.2C, " (Q,U) " means " $\text{CM } 5(Q,U)$ ".)



where $U = \text{TR}, \text{AS}, \text{C}, \text{TR}\&\text{C}, \text{AS}\&\text{C}, \text{P}, \text{L}, \text{D}$, or RA .

$$(\text{TR}, \text{AS}) \leftrightarrow (\text{TR}, \text{P}) \rightarrow \text{CAC } 2.$$

$$(\text{D}, \text{RA}) \leftrightarrow (\text{D}, \text{L}); \quad (\text{D}, \text{U}) \rightarrow \text{CMS } 1, \quad \text{for } U = \text{C}, \text{AS}\&\text{C}, \text{TR}\&\text{C}, \text{ or } \text{L}.$$

$$(\text{F}, \text{U}) \rightarrow \text{CMS } 2, \quad \text{for } U = \text{C}, \text{AS}\&\text{C}, \text{TR}\&\text{C}, \text{L}, \text{D}, \text{ or } \text{W}.$$

$$(\text{TR}\&\text{C}, \text{U}) \rightarrow \text{CWO } 12, \quad \text{for } U = \text{AS}, \text{AS}\&\text{C}, \text{P}, \text{L}, \text{D}, \text{ or } \text{RA}.$$

$$(\text{C}, \text{U}) \leftrightarrow (\text{AS}, \text{L}), \quad \text{for } U = \text{P}, \text{D}, \text{L}, \text{ or } \text{RA}.$$

$$(\text{C}, \text{AS}) \leftrightarrow (\text{AS}, \text{C}); \quad (\text{C}, \text{AS}\&\text{C}) \leftrightarrow (\text{AS}, \text{AS}\&\text{C}).$$

$$\text{CM } 5 \leftrightarrow (\text{P}, \text{U}), \quad \text{for } U = \text{C}, \text{AS}\&\text{C}, \text{TR}\&\text{C}, \text{ or } \text{L}.$$

$$\text{CM } 5(Q, U)$$

FIGURE 3.2C

We have shown in Section 1 that $\text{CWO } 12 \rightarrow \text{CWO } 5$ in NBG and in Section 2 that $\text{CAC } 2 \rightarrow \text{CWO } 5$ in NBG . We have shown in Theorem 3.1C that $\text{CMS } 1 \leftrightarrow \text{CMS } 2$ in NBG° . Next we shall show that $\text{CAC } 4 \rightarrow \text{CMS } 3$, $\text{CMS } 1 \rightarrow \text{CWO } 12$ and $\text{CWO } 5 \rightarrow \text{CM } 7$.

THEOREM 3.3C: $\text{CAC } 4 \rightarrow \text{CMS } 3$.

PROOF: Let X be a class which satisfies the hypothesis of $\text{CMS } 3$ and let R be a relation defined as follows:

$$R = \{ \langle x, s \rangle : x \subseteq X, s \in X, \text{ and } (\forall t) (t \in x \rightarrow t \subset s) \}.$$

CAC 4 implies there is a function $F \subseteq R$ such that $\mathcal{I}(F) = \mathcal{I}(R)$. If there is a nest $n \subseteq X$ such that n has no strict $\subseteq$ -upper bound, then On , which by the hypothesis of CMS 3 is an element of X , is a $\subseteq$ -maximal element of X . So suppose every nest n has a strict $\subseteq$ -upper bound; then $F(n)$ is a strict $\subseteq$ -upper bound of n . Define a function G on On such that $G(\alpha) = F(G''\alpha)$ for all $\alpha \in On$. Then it is easy to see that $\mathcal{K}(G) \subseteq X$, $\mathcal{K}(G)$ is a nest and a proper class, q.e.d.

THEOREM 3.4C: CMS 1 $\rightarrow$ CWO 12.

PROOF: For each $u \subseteq On$, $\rho(u)$ is the smallest ordinal α such that for each $\beta \in u$, $\beta < \alpha$. As we mentioned in the proof of 3.2C, each subset of On has a rank, and the class of all subsets of On of a given rank is a set. For each $\alpha \in On$, let $y_\alpha = \{u \subseteq On : \rho(u) \leq \alpha\}$. CMS 1 $\rightarrow$ M 1 $\rightarrow$ WO 1, so each y_α can be well-ordered. Let

$$R_\alpha = \{R \subseteq y_\alpha \times y_\alpha : R \text{ well orders } y_\alpha\},$$

and let $X = \bigcup_{\alpha \in On} R_\alpha$. Let S be a relation defined on X so that if $R, R' \in X$, $R S R'$ iff R' is an extension of R . Then X and S satisfy the hypothesis of CMS 1. It is easy to see that X has no S -maximal element. Therefore, CMS 1 implies that there is a proper class $Y \subseteq X$ such that Y is linearly ordered by S . Then UY is a relation which well orders $\bigcup_{\alpha \in On} y_\alpha = \mathcal{P}(On)$, q.e.d.

THEOREM 3.5C: CWO 5 $\rightarrow$ CM 7.

PROOF: Let X be a class and P a property of finite character. CWO 5 implies that there is a bijection F such that $F: On \approx X$. Let α_0 be the smallest ordinal number such that $\{\alpha_0\}$ has the property P . (If no such ordinal number exists then $\emptyset$ is the $\subseteq$ -maximal subclass of X with the property P .) Define a function G on On such that:

$$G(\alpha) = \begin{cases} F(\alpha) & \text{if } \{F(\alpha)\} \cup \{G(\beta) : \beta < \alpha\} \text{ has the property } P, \\ F(\alpha_0) & \text{otherwise.} \end{cases}$$

Then $\mathcal{K}(G)$ is a $\subseteq$ -maximal subclass of X with the property P , q.e.d.

To complete the discussion of the CM 5(Q,U) forms, we shall look at CM 5(A,UN) and CM 5(A,UN(n)). Since these statements come up again later and this notation is rather cumbersome, we shall give these statements new numbers, CM 20(n).

Let $n \in \omega$ be such that $n \geq 2$.

CM 20(n): If X is a class and R an n -ary relation there exists a $\subseteq$ -maximal subclass $Y \subseteq X$ such that $Y^n \subseteq R$.

CM 5(A,UN) is CM 20(2) which we shall denote by CM 20. The following implications are proved in a similar manner to the corresponding implications for the set forms:

$$\text{CM 7} \rightarrow \text{CM 20}(n) \rightarrow \text{CM 20}(n-1), \quad \text{for } n \geq 3,$$

$$\text{CM 20}(n) \rightarrow \text{CM 20} \rightarrow \text{CM 5(A,C)} \rightarrow \text{CM 5}.$$

(CM 20(n) $\rightarrow$ CM 20(n-1) because if R is an $(n-1)$ -ary relation then $R \times V$ is an n -ary relation. A $\subseteq$ -maximal subclass Y such that $Y^n \subseteq R \times V$ induces a $\subseteq$ -maximal subclass $Z (= Y)$, such that $Z^{n-1} \subseteq R$.)

The class form of the anti-chain axiom, CA, is as follows:

CA: For each class X and partial ordering relation R there is a $\subseteq$ -maximal class $Y \subseteq X$ which is an R -anti-chain.

The property of being an anti-chain is a property of finite character so CM 7 $\rightarrow$ CA. CM 8($\bar{K}$) is a special case of CA

(R is the $\subseteq$ relation), so $CA \rightarrow CM\ 8(\bar{K})$. We shall show $CM\ 8(\bar{K}) \rightarrow CWO\ 12$.

THEOREM 3.6C: $CM\ 8(\bar{K}) \rightarrow CWO\ 12$.

PROOF: Just as in the proof of Theorem 3.2C, let R be a relation which linearly orders $\mathcal{P}(On)$ and let $x_\alpha = \{u \subseteq On : \rho(u) = \alpha\}$. Let $Y = \{\langle x, u \rangle : \emptyset \neq x \subseteq \mathcal{P}(On) \wedge u \in x\}$. Define a relation S on Y so that

$$\langle x, u \rangle S \langle y, v \rangle \quad \text{iff} \quad x = y \wedge u R v \wedge \rho(u) \leq \rho(v).$$

Then $\overleftarrow{\langle x, u \rangle} = \{\langle y, v \rangle \in Y : \langle y, v \rangle S \langle x, u \rangle\}$ is a set for each $\langle x, u \rangle \in Y$, and $\langle x, u \rangle S \langle y, v \rangle$ iff $\overleftarrow{\langle x, u \rangle} \subseteq \overleftarrow{\langle y, v \rangle}$ for each $\langle x, u \rangle, \langle y, v \rangle \in Y$. Let $Z = \{\overleftarrow{\langle x, u \rangle} : \langle x, u \rangle \in Y\}$. $CM\ 8(\bar{K})$ implies that Z contains a $\subseteq$ -maximal subclass G with the property $\bar{K}$. Then we can define a choice function F on $\mathcal{P}(\mathcal{P}(On)) \sim \{\emptyset\}$ so that if $\emptyset \neq x \subseteq \mathcal{P}(On)$, $F(x) = u$, where u is the unique element of x such that $\overleftarrow{\langle x, u \rangle} \in G$. The function F can be used to well order each x_α , for $\alpha \in On$. Since $\mathcal{P}(On) = \bigcup_{\alpha \in On} x_\alpha$, this implies $\mathcal{P}(On)$ can be well ordered, q.e.d.

Our next consideration will be the class forms $CM\ 8(A)$, $9(A)$, 10 , 11 , 12 , 13 , and 14 . The choices for A are D , K , J , D_m , and their complements $\bar{D}$, $\bar{K}$, $\bar{J}$, and $\bar{D}_m$ where

- (1) $X D Y$ iff $X \cap Y = \emptyset$,
- (2) $X K Y$ iff $X \subseteq Y \vee Y \subseteq X$,
- (3) $X J Y$ iff $X \not\subseteq Y \wedge Y \not\subseteq X \wedge X \cap Y \neq \emptyset \wedge X \cup Y \neq V$,
- (4) A class X has the property D_m iff for all $x_1, x_2, \dots, x_m \in X$, x_i 's distinct, $\bigcap_{i=1}^m x_i = \emptyset$.

We noted above that $CM\ 8(K)$ is the same as $CM\ 6$ and that $CA \rightarrow CM\ 8(\bar{K}) \rightarrow CWO\ 12$. (Theorem 3.6C). The proof that $CM\ 8(J) \rightarrow CM\ 8(\bar{K})$ is similar to the proof of 4.24. (Substitute J for $\bar{J}$ and $\bar{K}$ for K in 4.24.) $CM\ 8(\bar{D})$ is $M\ 19$ and is

equivalent to a set form of the axiom of choice in NBG. Similarly for $\text{CM } 8(\bar{D}_m)$, $\text{CM } 12$, and $\text{CM } 13$. (See 5.21). $\text{CM } 14$ is equivalent to a set form of the axiom of choice in NBG° . (See 4.31) The proof that $\text{CM } 8(D_m) \rightarrow \text{CAC } 13(m-1)$ for $m \geq 2$ is similar to the proof of 4.27. Clearly, $\text{CM } 8(D_m) \rightarrow \text{CM } 10 \rightarrow \text{CM } 11$. The $\text{CM } 8(A)$ statements which remain are those in which $A = D$ or $\bar{J}$. We shall show that $\text{CM } 11 \rightarrow \text{CAC } 4$ and $\text{CM } 8(\bar{J}) \rightarrow \text{CM } 8(D) \rightarrow \text{CAC } 4$.

The proof that $\text{CM } 11 \rightarrow \text{CAC } 4$ is due to A. L. Rubin.

THEOREM 3.7: $\text{CM } 11 \rightarrow \text{CAC } 4$.

PROOF: Let R be any relation and let

$$X_1 = \{ \{ \langle 0, a \rangle, \langle 1, a, b \rangle \} : \langle a, b \rangle \in R \}.$$

For any set u and for any $m \in \omega$, $m \geq 2$, let $w_u = \omega \times u$

and let $\binom{w_u}{m}$ be the set of all m -element subsets of w_u . For each such u and m define the following three sets

$$z_{m,u} = \left\{ t \in \binom{w_u}{m} : \begin{array}{l} \text{the element of } t \text{ with the largest} \\ \text{first coordinate is unique} \end{array} \right\},$$

$$f_{m,u}(v) = \{ t \in \binom{w_u}{m} : v \in t \},$$

$$y_{m,u} = \{ f_{m,u}(v) \sim z_{m,u} : v \in w_u \}.$$

Then we define the classes X_2 and X as follows:

$$X_2 = \{ \{ \langle m, u \rangle \} \times s : 2 \leq m \in \omega, 2 \leq |u| < m, s \in y_{m,u} \},$$

$$X = X_1 \cup X_2.$$

$\text{CM } 11$ implies that there is a natural number $m \geq 2$ and a $\subseteq$ -maximal subclass $Y \subseteq X$ such that Y has the property D_m . For each $a \in \mathcal{J}(R)$, let

$$u = \{b: \{\langle 0, a \rangle, \langle 1, a, b \rangle\} \in Y\}.$$

By definition, $u \subseteq R^{\omega}\{a\}$. It follows from the maximality of Y that $u \neq \emptyset$, and since Y satisfies D_m , $|u| < m$. If $|u| = 1$ define $F(a)$ to be the unique element of u . Otherwise, define

$$v = \{\langle n, b \rangle \in w_u: \{\langle m, u \rangle\} \times (f_{m,u}(n, b) \sim z_{m,u}) \in Y\}.$$

We claim that each m -element subset of v is an element of $z_{m,u}$. For suppose not. Suppose $t = \{\langle n_1, b_1 \rangle, \langle n_2, b_2 \rangle, \dots, \langle n_m, b_m \rangle\}$ is an m -element subset of v such that $n_1 = n_2$ and for all $i = 1, 2, \dots, m$, $n_i \leq n_1$. Then for each $i = 1, 2, \dots, m$,

$$y_i = \{\langle m, u \rangle\} \times \{f_{m,u}(n_i, b_i) \sim z_{m,u}\} \in Y,$$

and $t \in f_{m,u}(n_i, b_i) \sim z_{m,u}$. This implies $\bigcap_{i=1}^m y_i \neq \emptyset$ which contradicts the fact that Y satisfies D_m . This proves the claim.

Now, we claim that there is an $n \in \omega$ such that there is a unique $b \in u$ with $\langle n, b \rangle \in v$. To prove this claim we note that because of the maximality of Y , $|v| \geq m$. (Otherwise we could add an element to v with a larger first coordinate.) If $|v| \geq m$, then, for example, $\{b \in u: \langle m, b \rangle \in v\}$ has one element. (Otherwise there is an m -element subset of v which is not an element of $z_{m,u}$.) This proves the second claim.

Let n_0 be the smallest n which satisfies the second claim. Then we define $F(a)$ to be the unique b such that $\langle n_0, b \rangle \in v$. This proves CAC 4, because we have given a constructive procedure which gives us $F(a)$ for each $a \in \mathcal{D}(R)$, q.e.d.

THEOREM 3.8C: $CM\ 8(\bar{J}) \rightarrow CM\ 8(D)$.

PROOF: Let X be a class. For each $s \in X$, define $s_u = s \cup \{\langle s, u \rangle\}$ where for each $t \in X$, if $t \neq s$ then $\langle s, u \rangle \notin t$. (We can always find such an element u since we

could replace X by $X \times \{a\}$ for any a .) If $s \neq t$, where $s, t \in X$, then $s_u \not\subseteq t_u$. Thus, $s_u D t_u$ iff $s_u \bar{J} t_u$. Also, $s \neq t$ implies $s_u D t_u$ iff $s D t$. $CM\ 8(\bar{J})$ implies that $\{s_u : s \in X\}$ has a $\subseteq$ -maximal subclass with the property $\bar{J}$. Hence, X has a $\subseteq$ -maximal subclass with the property D , q.e.d.

THEOREM 3.9C: $CM\ 8(D) \rightarrow CAC\ 4$.

PROOF: Let R be a relation. Define $X = \{\langle 0, u \rangle, \langle u, v \rangle : u R v\}$. Two elements in X , $\langle 0, u \rangle, \langle u, v \rangle$ and $\langle 0, u' \rangle, \langle u', v' \rangle$ are disjoint iff $u \neq u'$. Thus, a $\subseteq$ -maximal subclass of X with the property D produces a function satisfying $CAC\ 4$.

It is also true that $CM\ 9(J) \rightarrow CM\ 9(\bar{K}) \rightarrow CAC\ 2$. The proof that $CM\ 9(J) \rightarrow CM\ 9(\bar{K})$ is similar to the proof of 4.24, and the proof that $CM\ 9(\bar{K}) \rightarrow CAC\ 2$ is similar to the proof of 4.25. Clearly, $CM\ 9(A) \rightarrow CM\ 8(A)$, for all A .

We shall show that $CM\ 7 \rightarrow CM\ 9(A)$, for all $A = D, \bar{D}, J, \bar{J}, K, \bar{K}$. Since each of these properties are properties of finite character, to show that $CM\ 7 \rightarrow CM\ 9(A)$, it is sufficient to prove that $CM\ 7$ implies that the universe can be linearly ordered.

LV: V can be linearly ordered.

LEMMA 3.10C: $CM\ 7 \rightarrow LV$.

PROOF: If X is a class of ordered pairs we shall say X is circular if there exists a finite sequence of length $n > 1$ of ordered pairs in X , $\langle x_0, x_1 \rangle, \langle x_1, x_2 \rangle, \langle x_2, x_3 \rangle, \dots, \langle x_{n-1}, x_n \rangle$, such that $x_0 = x_n$ and $\{x_0, x_1, \dots, x_n\} \approx n$. If no such sequence exists X is called non-circular. Let P be a property defined for each class X of ordered pairs so that X has the property P iff X is non-circular. P is a property of finite character.

$CM\ 7$ implies that there is a $\subseteq$ -maximal class, R , with the property P . We claim that R is a reflexive, anti-

symmetric, transitive, and connected relation on V . Clearly R is reflexive and anti-symmetric. To show R is transitive, suppose $\langle x, y \rangle, \langle y, z \rangle \in R$ and $\langle x, z \rangle \notin R$. This implies, by the maximality of R , that $R \cup \{\langle x, z \rangle\}$ is circular. Since $\langle x, y \rangle$ and $\langle y, z \rangle$ are elements of R , it follows that R must have been circular. This is a contradiction. Thus $\langle x, z \rangle \in R$ and R is transitive.

Suppose $\langle x, y \rangle \notin R$, $x \neq y$; then $R \cup \{\langle x, y \rangle\}$ is circular. There exists a finite sequence of ordered pairs $\langle z_0, z_1 \rangle, \langle z_1, z_2 \rangle, \dots, \langle z_{n-2}, z_{n-1} \rangle, \langle z_{n-1}, z_0 \rangle$ in $R \cup \{\langle x, y \rangle\}$ such that $n > 1$, the z_i 's are distinct and one of the ordered pairs in the sequence is $\langle x, y \rangle$. There is no loss of generality in assuming $\langle z_0, z_1 \rangle$ is $\langle x, y \rangle$. Then the following ordered pairs are in R : $\langle y, z_2 \rangle, \langle z_2, z_3 \rangle, \dots, \langle z_{n-2}, z_{n-1} \rangle, \langle z_{n-1}, x \rangle$. Since R is transitive, we must have $\langle y, x \rangle \in R$. Thus, R is connected, so R linearly orders V , q.e.d.

Figure 3.3C below illustrates the relationships between the statements CM 8(A), 9(A), 10, and 11 in NBG° .

$\text{CM } 7 \rightarrow \text{CM } 9(A) \leftrightarrow \text{LV} \wedge \text{CM } 8(A)$, for all A ;

$\text{CM } 9(J) \rightarrow \text{CM } 9(\bar{K}) \rightarrow \text{CAC } 2$;

$\text{CM } 8(\bar{J}) \rightarrow \text{CM } 8(D) \rightarrow \text{CAC } 4 \rightarrow \text{CM } 8(K) \leftrightarrow \text{CM } 6 \rightarrow \text{CWO } 12$;

$\text{CM } 8(J) \rightarrow \text{CM } 8(\bar{K}) \rightarrow \text{CWO } 12$;

$\text{CM } 8(D_m) \rightarrow \text{CM } 10 \rightarrow \text{CM } 11 \rightarrow \text{CAC } 4$.

FIGURE 3.3C

Since $\text{CWO } 12 \rightarrow \text{CWO } 4$ in NBG , and $\text{CWO } 4 \rightarrow \text{CM } 7$ in NBG° , all the statements in Figure 3.3C are equivalent to $\text{CWO } 4$ in NBG . Moreover we have $\text{CM } 14 \leftrightarrow \text{M } 14$ in NBG° and each of $\text{CM } 8(\bar{D})$, $\text{CM } 8(\bar{D}_m)$, $\text{CM } 12$ and $\text{CM } 13$ is equivalent to a set form of the axiom of choice in NBG .

If X and Y are distinct, non-empty sets exactly one of D , K or J holds between them. However, if X and Y

are classes there is another possibility, namely $X \cup Y = V$. Consequently, we define another relation, U .

DEFINITION 3.11C: Let X and Y be classes.

- (i) $X \cup Y$ iff $X \cup Y = V$.
- (ii) $X \bar{U} Y$ iff $X \cup Y \neq V$.

The property U is dual to D . That is, $X D Y$ iff $(V \sim X) \cup (V \sim Y)$. If X and Y are distinct, non-empty, non-universal and non-complementary classes, exactly one of the properties D , K , J , or U holds between them. $\bar{D}$, $\bar{K}$, $\bar{J}$ and $\bar{U}$ are the complements of D , K , J and U , so $\bar{D} = K \vee J \vee U$, $\bar{K} = D \vee J \vee U$, $\bar{J} = D \vee K \vee U$, and $\bar{U} = D \vee K \vee J$. For the sake of completeness we shall also consider the six properties, $D \vee K$ (DK), $D \vee J$ (DJ), $D \vee U$ (DU), $K \vee J$ (KJ), $K \vee U$ (KU), and $J \vee U$ (JU).

It would be impossible to state the form CM 8(A) where U or $\bar{U}$ is substituted for A because it would be necessary to construct a class whose elements are classes. To avoid this difficulty we restate CM 8(A) as follows:

CMR 8(A): For every relation R and class X there exists a $\subseteq$ -maximal subclass $Y \subseteq X$ such that for all $x, y \in Y$, if $x \neq y$ then $R\{x\} A R\{y\}$.

CMR 8(A) is equivalent to a class form of the axiom of choice if $A = D, \bar{D}, J, \bar{J}, K, \bar{K}, D_m, \bar{D}_m, \bar{D}!_m, U, \bar{U}, DK, DJ, DU, KJ, KU, \text{ or } JU$.

(The property $D!_m$ is excluded because it is the same as D .)

The following propositions are also equivalent to a class form of the axiom of choice.

CMR 9(A): The universe can be linearly ordered and for every relation R and class X there exists a $\subseteq$ -maximal subclass $Y \subseteq X$ such that for all $x, y \in Y$, if $x \neq y$ then $R\{x\} A R\{y\}$, where $A = D, \bar{D}, J, \bar{J}, K, \bar{K}, D_m, \bar{D}_m, \bar{D}!_m, U, \bar{U}, DK, DJ, DU, KJ, KU, \text{ or } JU$.

We shall not spend much time on the CMR 9(A) statements because CMR 9(A) is the same as $LV \wedge \text{CMR } 8(A)$. However, CMR 9($\bar{K}$) is of some interest because of historical reasons.

CMR 10: There exists a natural number $m \geq 2$ such that CMR 8(D_m).

CMR 11: For every relation R and class X there is a natural number $m \geq 2$ and a $\subseteq$ -maximal subclass $Y \subseteq X$ such that for all $y_1, y_2, \dots, y_m \in Y$, y_i 's distinct, $\bigcap_{i=1}^m R\{y_i\} = \emptyset$.

The statement:

There exists a natural number $m \geq 2$ such that CMR 8($\bar{D}_m$).

is also equivalent to a class form of the axiom of choice.

(We shall show later $\text{CMR } 8(D_m) \leftrightarrow \text{CMR } 8(\bar{D}_m)$.) However, as we mentioned earlier for the set forms, if in CMR 11 we replace " $\bigcap_{i=1}^m R\{y_i\} = \emptyset$ " by " $\bigcap_{i=1}^m R\{y_i\} \neq \emptyset$ ", the resulting statement is provable in NBG°. (See the discussion preceding definition 4.28.) This was the reason for introducing $\bar{D}_m!$.

CMR 12: There exists a natural number $m \geq 2$ such that CMR 8($\bar{D}_m!$).

CMR 13: For every relation R and class X there exists a natural number $m \geq 2$ and a $\subseteq$ -maximal subclass $Y \subseteq X$ such that for all $y_1, y_2, \dots, y_m \in Y$, $\bigcap_{i=1}^m R\{y_i\} \neq \emptyset$.

CMR 14: If X is a class and R a relation there exists a $\subseteq$ -maximal subclass $Y \subseteq X$ such that for all natural numbers $m \geq 2$ and for all $y_1, y_2, \dots, y_m \in Y$, $\bigcap_{i=1}^m R\{y_i\} \neq \emptyset$.

Even though CM 14 is equivalent to M 14 in NBG°, it was proved by A. L. Rubin that CMR 14 $\leftrightarrow$ CM 7 in NBG°.

There is no advantage to introducing "MR" forms for the set forms of the axiom of choice, because they are clearly

equivalent to the corresponding "M" forms. However, for the class forms it gives us a method for handling collections of proper classes.

Each CMR 8 form implies the corresponding CM 8 form. (Define R so that $u R v$ iff $v \in u$. Then, for each x , $R''\{x\} = x$.) We shall show that each of CMR 8(A), for $A = D, \bar{D}, J, \bar{J}, U, \bar{U}, DK, DJ, DU, KJ, KU$ or JU is equivalent to CM 20, and that $CMR\ 8(K) \leftrightarrow CM\ 5$, and $CMR\ 8(\bar{K}) \leftrightarrow CA$. The proof that $CMR\ 8(J) \rightarrow CMR\ 8(\bar{K})$ is similar to the proof of 4.24. (Substitute J for $\bar{J}$ and $\bar{K}$ for K in 4.24.)

THEOREM 3.12C:

- (a) $CM\ 5 \leftrightarrow CMR\ 8(K)$
- (b) $CA \leftrightarrow CMR\ 8(\bar{K})$.

PROOF: (a) Suppose CM 5 holds and let R be an arbitrary relation. Define a relation S for all $x, y \in \mathcal{J}(R)$ such that

$$x S y \leftrightarrow R''\{x\} \subseteq R''\{y\}$$

S is a partial ordering so by CM 5 there exists a $\subseteq$ -maximal subclass $Y \subseteq \mathcal{J}(R)$ such that Y is linearly ordered by R . Y is the required class.

Conversely, let R be a partial ordering relation on a class X . Then we claim that for all $x, y \in X$,

$$(1) \quad x R y \leftrightarrow R''\{y\} \subseteq R''\{x\}.$$

For suppose $x R y$ and $z \in R''\{y\}$. Then $z R y$ holds. Since R is transitive, $z R x$ holds, which implies $R''\{y\} \subseteq R''\{x\}$. Now suppose $R''\{y\} \subseteq R''\{x\}$. Since R is reflexive, $y \in R''\{y\}$ which implies that $y \in R''\{x\}$. Thus, $x R y$ holds.

Therefore, it follows from (1) that $CMR\ 8(K)$ implies CM 5.

The proof of (b) is similar.

THEOREM 3.13C: $CM\ 20 \rightarrow CMR\ 8(A)$, where $A = D, \bar{D}, J, \bar{J}, U, \bar{U}, DK, DJ, DU, KJ, KU, \text{ or } JU$.

PROOF: Let R be a relation and X a class. Define a relation S such that for each $x, y \in \mathcal{D}(R)$,

$$x S y \leftrightarrow x = y \vee (x \neq y \wedge R''\{x\} A R''\{y\})$$

$CM\ 20$ implies X has a $\subseteq$ -maximal subclass Y such that $Y \times Y \subseteq S$. This is an equivalent way of saying that Y is a $\subseteq$ -maximal subclass of X such that for all $x, y \in Y$, if $x \neq y$ then $R''\{x\} A R''\{y\}$.

To prove the converse we use the fact that each of these relations are universal. That is, given any symmetric relation R and any of the above relations A , there is a relation S such that if $x \neq y$, $x R y$ iff $S''\{x\} A S''\{y\}$. (A similar remark holds for the set forms and was used by Kurepa [1952]. For D and $\bar{D}$ this was previously observed by Szpilrajn-Marczewski [1945].)

THEOREM 3.14C:

(a) $CMR\ 8(D) \rightarrow CM\ 20$. (The proof is similar if D is replaced by DK, DU or $\bar{J}$. By duality, an analogous proof will hold for U and KU .)

(b) $CMR\ 8(\bar{D}) \rightarrow CM\ 20$. (The proof is similar if $\bar{D}$ is replaced by J, JU or JK . By duality, an analogous proof will hold for $\bar{U}$ and DJ .)

PROOF: (a) Let R be a relation. Define a relation S such that for each $x \in \mathcal{D}(R)$,

$$x S y \leftrightarrow (\exists z)[y = \{x, z\} \wedge \langle x, z \rangle \in \bar{R} \cup \bar{R}^{-1} \cup I].$$

(That is, $\langle x, z \rangle \notin R$ or $\langle x, z \rangle \notin R^{-1}$ or $x = z$.)

If $x, y \in \mathcal{D}(S)$ and $x \neq y$ then $S''\{x\} \not\subseteq S''\{y\}$ because $\{x\} \in S''\{x\}$, but $\{x\} \notin S''\{y\}$. If $x \neq y$ then $S''\{x\} \cap S''\{y\} \subseteq \{\{x, y\}\}$.

If $X \subseteq \mathcal{D}(R) = \mathcal{D}(S)$ then the following statements are equivalent:

- (i) $S''\{x\} \supseteq S''\{y\}$, for all $x, y \in X, x \neq y$.
- (ii) $S''\{x\} \cap S''\{y\} = \emptyset$, for all $x, y \in X, x \neq y$.
- (iii) $\langle x, y \rangle \notin \bar{R} \cup \bar{R}^{-1}$, for all $x, y \in X, x \neq y$.
- (iv) $\langle x, y \rangle \in R \cap R^{-1}$, for all $x, y \in X, x \neq y$.
- (v) $X \times X \subseteq (R \cap R^{-1}) \cup I$.

If X is a $\subseteq$ -maximal subclass of the $\mathcal{J}(R)$ such that (i) holds and

$$Z = \{x \in X : \langle x, x \rangle \notin R\} \cup \{x \in X : (\exists y \in X) (\langle x, y \rangle \in R \wedge \langle y, x \rangle \notin R)\}$$

then $Y = X \sim Z$ is a $\subseteq$ -maximal subclass of $\mathcal{J}(R)$ such that $Y \times Y \subseteq R$.

(b) Let R be a relation and define S such that for each $x \in \mathcal{J}(R)$,

$$x S y \leftrightarrow (\exists z) [y = \{x, z\} \wedge \langle x, z \rangle \in (R \cap R^{-1}) \cup I].$$

Just as in part (a) we have that if $x, y \in \mathcal{J}(S)$ and $x \neq y$, then $S''\{x\} \not\subseteq S''\{y\}$ and $S''\{x\} \cap S''\{y\} \subseteq \{\{x, y\}\}$.

If $X \subseteq \mathcal{J}(R) = \mathcal{J}(S)$, the following statements are equivalent:

- (i) $S''\{x\} \bar{\supseteq} S''\{y\}$, for all $x, y \in X, x \neq y$.
- (ii) $S''\{x\} \cap S''\{y\} \neq \emptyset$, for all $x, y \in X, x \neq y$.
- (iii) $\langle x, y \rangle \in R \cap R^{-1}$, for all $x, y \in X, x \neq y$.
- (iv) $X \times X \subseteq (R \cap R^{-1}) \cup I$.

Then, just as in part (a), we construct a $\subseteq$ -maximal subclass $Y \subseteq \mathcal{J}(R)$ such that $Y \times Y \subseteq R$, q.e.d.

The following corollary follows from Theorems 3.13C and 3.14C.

COROLLARY 3.15C: $CM\ 20 \leftrightarrow CMR\ 8(A)$, for $A = D, \bar{D}, J, \bar{J}, U, \bar{U}, DK, DJ, DU, KJ, KU$, and JU .

THEOREM 3.16C: $LV \wedge CM\ 20 \rightarrow CMR\ 9(\bar{K})$.

PROOF: Substitute $\bar{K}$ for D in 3.13C.

The proof that $\text{CMR } 9(\bar{K}) \rightarrow \text{CAC } 4$ is similar to the proof of 4.25.

THEOREM 3.17C: $\text{CMR } 9(\bar{K}) \rightarrow \text{CAC } 4$.

PROOF: Let L be a relation which linearly orders V . Define a relation S as follows:

$$S = \{ \langle \langle x, y \rangle, \langle x, z \rangle \rangle : \langle x, y \rangle, \langle x, z \rangle \in R \wedge \langle y, z \rangle \in L \}.$$

If $\langle x, y \rangle, \langle z, w \rangle \in \mathcal{I}(S)$ then

$$S''\{\langle x, y \rangle\} K S''\{\langle z, w \rangle\} \quad \text{iff } x = z.$$

$\text{CMR } 9(\bar{K})$ implies that there exists a $\subseteq$ -maximal subclass $F \subseteq \mathcal{I}(S)$ such that if $s, t \in F$ and $s \neq t$ then $S''\{s\} \bar{K} S''\{t\}$. Clearly, $\mathcal{I}(F) = \mathcal{I}(R)$ and $F \subseteq R$. If $\langle x, y \rangle, \langle x, z \rangle \in F$ then $S''\{\langle x, y \rangle\} K S''\{\langle x, z \rangle\}$. But, it follows from the definition of F that this cannot happen unless $y = z$. Thus, F is a function which satisfies CAC 4.

Clearly, $\text{CMR } 9(A) \leftrightarrow \text{LV} \wedge \text{CMR } 8(A)$, for all A . Since $\text{CM } 7 \rightarrow \text{LV}$, it follows that $\text{CM } 7 \rightarrow \text{CMR } 9(A)$ for each A which is a property of finite character. Even though each of $\text{CM } 8(\bar{D}_m)$, $\text{CM } 12$ and $\text{CM } 13$ is equivalent to a set form of the axiom of choice in NBG, we shall prove that each of $\text{CMR } 8(\bar{D}_m)$, $\text{CMR } 12$, and $\text{CMR } 13$ is equivalent to a class form. We shall show that each of $\text{CMR } 8(D_m)$, $\text{CMR } 8(\bar{D}_m)$, $\text{CMR } 8(\bar{D}!_m)$, is equivalent to $\text{CM } 20(m)$; and each of $\text{CMR } 10, 11, 12, 13$ is equivalent to $\text{CM } 20$.

The following implications hold for logical reasons.

$$\text{CMR } 8(D_m) \rightarrow \text{CMR } 10 \rightarrow \text{CMR } 11,$$

$$\text{CMR } 8(\bar{D}!_m) \rightarrow \text{CMR } 12 \rightarrow \text{CMR } 13.$$

The proof that $\text{CMR } 8(\bar{D}_m) \rightarrow \text{CMR } 8(D_m)$ is similar to the proof of Theorem 4.21. A proof similar to the proof of 3.13C gives us that $\text{CM } 20(m)$ implies each of $\text{CMR } 8(A)$, for $A = D_m, \bar{D}_m$ and $\bar{D}!_m$.

We shall show next that $\text{CMR } 8(D_m) \rightarrow \text{CMR } 8(\bar{D}_m)$, but first a preliminary definition and lemma.

DEFINITION 3.18C: If P is a property of m classes, R is a relation and X is a class then

$$P_R[X] \text{ iff } P[R\{x_1\}, R\{x_2\}, \dots, R\{x_m\}]$$

for all $x_1, x_2, \dots, x_m \in X$ such that the x_i 's are distinct.

Using this notation we could have stated $\text{CMR } 8(A)$ as follows: For every relation R and class X there exists a $\subseteq$ -maximal subclass Y such that $A_R[Y]$.

LEMMA 3.19C: For every relation R and natural number $m \geq 2$ there exists a relation S such that

$$\bar{D}_{mR}[X] \text{ iff } D_{mS}[X],$$

for every class X .

PROOF: Let R and m satisfy the hypothesis. Define S as follows.

$$S = \{ \langle x, y \rangle : x \in \mathcal{D}(R), y \subseteq \mathcal{D}(R), x \in y, \text{ and} \\ (y = \{x\} \vee (y \approx m \wedge D_{mR}[y])) \}.$$

Thus, we have,

$$\begin{aligned} D_{mS}[X] &\leftrightarrow (\forall y \subseteq X) (y \approx m \rightarrow \bigcap_{x \in y} S\{x\} = \emptyset) \\ &\leftrightarrow (\forall y \subseteq X) (y \approx m \rightarrow \bar{D}_{mR}[y]) \\ &\leftrightarrow \bar{D}_{mR}[X]. \end{aligned}$$

It follows immediately from 3.19C that $\text{CMR } 8(D_m) \rightarrow \text{CMR } 8(\bar{D}_m)$.

COROLLARY 3.20C: For each natural number $m \geq 2$,

$$\text{CMR } 8(D_m) \leftrightarrow \text{CMR } 8(\bar{D}_m).$$

Theorems 3.21C-3.25C are all due to A. L. Rubin.

THEOREM 3.21C: For each natural number $m \geq 2$,
 $CM\ 20(m) \leftrightarrow CMR\ 8(\bar{D}!_m)$.

PROOF: The proof that $CM\ 20(m) \rightarrow CMR\ 8(\bar{D}!_m)$ is similar to the proof of 3.13C.

Conversely, suppose $CMR\ 8(\bar{D}!_m)$ holds. Let X be any class and R any relation. Define a relation S so that

$$x\ S\ y \text{ iff } x \in y, y \prec m, \text{ and } y^m \subseteq R.$$

$CMR\ 8(\bar{D}!_m)$ implies that there is a $\subseteq$ -maximal subclass $Y \subseteq X$ such that for all $x_1, x_2, \dots, x_m \in Y$, $\bigcap_{i=1}^m S''\{x_i\} \neq \emptyset$. Since

$$S''\{x_i\} = \{y \subseteq X: x_i \in y, y \prec m, \text{ and } y^m \subseteq R\},$$

it follows that $\bigcap_{i=1}^m S''\{x_i\} = \{\{x_1, x_2, \dots, x_m\}\}$. Therefore, for all $x_1, x_2, \dots, x_m \in Y$, $\{x_1, x_2, \dots, x_m\}^m \subseteq R$. This implies that Y is a $\subseteq$ -maximal subclass of X such that $Y^m \subseteq R$, q.e.d.

The next result shows that the factorial, $!$, in $\bar{D}!_m$ behaves like a factorial.

THEOREM 3.22C: For all natural numbers $m \geq 2$,
 $CMR\ 8(\bar{D}!_m) \leftrightarrow \bigwedge_{i=2}^m CMR\ 8(\bar{D}_i)$.

$$\left(\bigwedge_{i=2}^m CMR\ 8(\bar{D}_i) \right) = CMR\ 8(\bar{D}) \wedge CMR\ 8(\bar{D}_3) \wedge \dots \wedge CMR\ 8(\bar{D}_m).$$

PROOF: To prove $CMR\ 8(\bar{D}!_m) \rightarrow \bigwedge_{i=2}^m CMR\ 8(\bar{D}_i)$, it follows from 3.21C that it is sufficient to prove that $CM\ 20(m) \rightarrow CMR\ 8(\bar{D}_k)$ for each k , $2 \leq k \leq m$. Since $CM\ 20(m) \rightarrow CM\ 20(k)$, use a proof similar to the proof of Theorem 3.13C to prove that $CM\ 20(k) \rightarrow CMR\ 8(\bar{D}_k)$.

Conversely, let X be a class, R a relation, and assume $\bigwedge_{i=2}^m CMR\ 8(\bar{D}_i)$. Define $Y_1 = X \cap \mathcal{D}(R)$. For each k ,

$2 \leq k < m$, define Y_{k+1} as a $\subseteq$ -maximal subclass of Y_k such that $\bar{D}_{k+1,R}[Y_{k+1}]$. Y_m is the required class, q.e.d.

The next theorem is the basis for proving the remaining equivalences and it is the most difficult of these theorems to prove.

THEOREM 3.23C: For all natural numbers $m \geq 3$,
 $\text{CMR } 8(\bar{D}_m) \rightarrow \text{CMR } 8(\bar{D}_{m-1})$.

PROOF: Let X be a class and R a relation. For any class Y and natural number $n \geq 1$, let

$$\binom{Y}{n} = \{y \subseteq Y : y \approx n\}.$$

We define a class Y as follows:

$$Y_0 = \{\emptyset\} \times X,$$

$$Y_{n+1} = Y_n \cup \left[\binom{Y_n}{m-1} \times X \right],$$

$$Y = \bigcup_{n \in \omega} Y_n.$$

$$\text{Then, } Y = \left[\{\emptyset\} \cup \binom{Y}{m-1} \right] \times X.$$

(Note that Y is definable in NBG° since it can be expressed as a union of sets). For each $u \in Y$, the rank of u is the smallest $n \in \omega$ such that $u \in Y_n$. Next, we define a class Z such that

$$Z = \{u \in \binom{Y}{m} : (\exists w)((w = \emptyset \vee w \in \binom{Y}{m-1}))$$

$$\wedge u \subseteq w \cup (\{w\} \times X) \wedge \bar{D}_{m-1,R}[\{t : \langle w, t \rangle \in u\}]\}.$$

Also, define a relation S such that $\mathcal{J}(S) = Y$ and

$$\langle y, u \rangle \in S \text{ iff } (y \in u \in Z) \vee u = \{y\}.$$

Let W be a $\subseteq$ -maximal subclass of Y such that $\bar{D}_{m,S}[W]$.

We see first that $\bar{D}_{m,S}[W] \rightarrow \binom{W}{m} \subseteq Z$. This is because $u \in \bigcap_{i=1}^m S''\{y_i\} \leftrightarrow u = \{y_1, y_2, \dots, y_m\}$.

The next claim is that W has at least m elements. Suppose not. Let w be any $m-1$ element subset of Y containing W . Let $u = w \cup \{\langle w, a \rangle\}$ such that $a \in X$. Then $\bar{D}_{m,S}[u]$ and $W \subset u$. This contradicts the maximality of W .

Suppose $\langle w_1, x_1 \rangle, \langle w_2, x_2 \rangle, \dots, \langle w_m, x_m \rangle \in W$, where $\langle w_i, x_i \rangle$ are distinct. Let t be a w_i of maximal rank and let $v = \{\langle w_i, x_i \rangle : 1 \leq i \leq m\}$. Since $v \in Z$, it follows that for all $i = 1, 2, \dots, m$, either $w_i = t$ or $\langle w_i, x_i \rangle \in t$. Since $|t| \leq m-1$, we must have that $|\mathcal{D}(W)| \leq m$ and $W = t \cup (\{t\} \times T)$. Since $\binom{W}{m} \subseteq Z$, it follows that T is a $\subseteq$ -maximal subclass of X such that $\bar{D}_{m-1,R}[T]$, q.e.d.

COROLLARY 3.24C: The following statements are equivalent for each $m \geq 3$: CM 20(m), CMR 8(D_m), CMR 8($\bar{D}_m$), and CMR 8($\bar{D}!_m$).

PROOF: By 3.20C, $\text{CMR } 8(D_m) \leftrightarrow \text{CMR } 8(\bar{D}_m)$. By 3.22C, $\text{CMR } 8(\bar{D}!_m) \rightarrow \text{CMR } 8(\bar{D}_m)$. It follows from 3.23C that $\text{CMR } 8(\bar{D}_m) \rightarrow \bigwedge_{i=2}^m \text{CMR } 8(\bar{D}_i)$, and from 3.22C that $\bigwedge_{i=2}^m \text{CMR } 8(\bar{D}_i) \rightarrow \text{CMR } 8(\bar{D}!_m)$. Thus, $\text{CMR } 8(D_m) \leftrightarrow \text{CMR } 8(\bar{D}_m) \leftrightarrow \text{CMR } 8(\bar{D}!_m)$. To complete the proof, in 3.21c we proved $\text{CM } 20(m) \leftrightarrow \text{CMR } 8(\bar{D}!_m)$, q.e.d.

THEOREM 3.25C: The following statements are equivalent: CM 20, CMR 10, 12, 13.

PROOF: It follows from 3.24C that $\text{CMR } 10 \leftrightarrow \text{CMR } 12$. To complete the proof we shall show that $\text{CM } 20 \rightarrow \text{CMR } 12 \rightarrow \text{CMR } 13 \rightarrow \text{CM } 20$.

$\text{CM } 20 \rightarrow \text{CMR } 12$. It follows from Theorem 3.13C that $\text{CM } 20 \rightarrow \text{CMR } 8(\bar{D})$, and clearly $\text{CMR } 8(\bar{D}) \rightarrow \text{CMR } 12$ (take $m = 2$).

Clearly $\text{CMR } 12 \rightarrow \text{CMR } 13$.

$\text{CMR } 13 \rightarrow \text{CM } 20$. Let X be a class and R a relation. There is no loss of generality in assuming R is symmetric

and reflexive. Define a relation S so that if $x \in X$,

$$\langle x, y \rangle \in S \text{ iff } x \in y, y < \omega, \text{ and } y \times y \subseteq R.$$

CMR 13 implies that there is a natural number $m \geq 2$ and a $\subseteq$ -maximal $Y \subseteq X$ such that for all $y_1, y_2, \dots, y_m \in Y$,

$$\bigcap_{i=1}^m S''\{y_i\} \neq \emptyset. \text{ This implies that for all } x, y \in Y,$$

$$S''\{x\} \cap S''\{y\} \neq \emptyset, \text{ which in turn implies } \{x, y\} \times \{x, y\} \subseteq R.$$

Therefore, $Y \times Y \subseteq R$. We claim Y is $\subseteq$ -maximal. For suppose $x \in X \sim Y$ and $(Y \cup \{x\}) \times (Y \cup \{x\}) \subseteq R$. Then, using the definition of S , if $y_1, y_2, \dots, y_{m-1} \in Y$,

$$S''\{x\} \cap \bigcap_{i=1}^{m-1} S''\{y_i\} \supseteq \{y_1, y_2, \dots, y_{m-1}, x\} \neq \emptyset. \text{ Since } Y \text{ was}$$

$\subseteq$ -maximal with this property, $x \in Y$, contradiction. Thus, CMR 13 $\rightarrow$ CM 20, q.e.d.

THEOREM 3.26C: CMR 11 $\rightarrow$ CMR 8($\bar{D}$).

PROOF: Let X be a class and S a relation. We define a sequence of classes X_n , and a sequence of relations S_n using the construction in Theorem 3.23C. Let $X_2 = X$ and $S_2 = S$. Let X_{n+1} be a class so that if $Z_{n+1} \subseteq X_{n+1}$ is $\subseteq$ -maximal for the property $\bar{D}_{n+1, S_{n+1}}[Z_{n+1}]$ then $Z_n \subseteq X_n$ is $\subseteq$ -maximal for the property $\bar{D}_{n, S_n}[Z_n]$. Using 3.19C, we construct a relation T_n such that $\bar{D}_{n, S_n}[Z]$ iff $D_{n, T_n}[Z]$ for all classes Z . Now define a relation R :

$$R = \{\langle x, y \rangle : (\exists n \in \omega) (\exists z, w) [x = \langle n, z \rangle, y = \langle n, w \rangle, \text{ and } \langle z, w \rangle \in T_n]\}.$$

(The definition of R is constructive and, since R can be expressed as a union of sets, it is definable in NBG $^\circ$.)

CMR 11 implies there is an $m \geq 2$ and a $\subseteq$ -maximal $Y \subseteq \mathcal{L}(R)$ such that $D_{m, R}[Y]$. Let $Z_m = \{z : \langle m, z \rangle \in Y\}$. Then $Z_m \subseteq \mathcal{L}(T_m)$ is $\subseteq$ -maximal such that $D_{m, T_m}[Z_m]$. By the definition of T_m , $Z_m \subseteq \mathcal{L}(S_m)$ is $\subseteq$ -maximal such that $\bar{D}_{m, S_m}[Z_m]$. Using the definition of S_m , we now construct a $Z_2 \subseteq X$ which is $\subseteq$ -maximal with the property $\bar{D}_S[Z_2]$, q.e.d.

COROLLARY 3.27C: The following statements are equivalent: CM 20, CMR 10, 11, 12, 13.

PROOF: $\text{CMR } 10 \rightarrow \text{CMR } 11$ for logical reasons. By 3.15C, $\text{CM } 20 \leftrightarrow \text{CMR } 8(\bar{D})$. The corollary is a consequence of 3.25C and 3.26C.

To complete our discussion of these CMR statements, we shall prove that $\text{CMR } 14 \leftrightarrow \text{CM } 7$. (This proof is also due to A. L. Rubin.)

THEOREM 3.28C: $\text{CM } 7 \leftrightarrow \text{CMR } 14$.

PROOF: Since the finite intersection property is a property of finite character it follows that $\text{CM } 7 \rightarrow \text{CMR } 14$.

Conversely, let X be a class and P a property of finite character. Define a relation R on X so that

$$\langle x, y \rangle \in R \text{ iff } x \in y, y < \omega, \text{ and } P(y).$$

CMR 14 gives us a $\subseteq_m$ -maximal subclass $Y \subseteq X$ such that for all $x_1, x_2, \dots, x_m \in Y$, $\bigcap_{i=1}^m R''\{x_i\} \neq \emptyset$. This implies that $\{x_1, x_2, \dots, x_m\}$ has the property P , so that every finite subset of Y has the property P . Thus, since P is a property of finite character, Y has the property P . It follows from the definition of Y that Y is a $\subseteq$ -maximal subclass of X with the property P , q.e.d.

We summarize our results for the CMR statements in Figure 3.4C. (Also see Theorems 3.1C, 3.4C and Figure 3.3C.)

The statements in each of the following sets are equivalent:

$$(F) = \{\text{CM } 7, \text{CMR } 14\}.$$

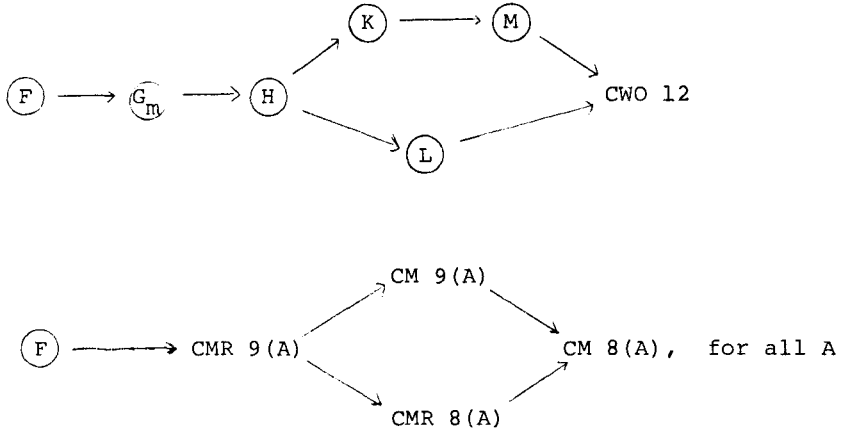
$$(G_m) = \{\text{CM } 20(m), \text{CMR } 8(A)\}, \text{ where } A = D_m, \bar{D}_m, \bar{D}_m! \text{ and } m \geq 3.$$

$$(H) = \{\text{CM } 20, \text{CMR } 10, 11, 12, 13, \text{CMR } 8(A)\}, \text{ where } A = D, \bar{D}, J, \bar{J}, U, \bar{U}, DK, DJ, DU, KJ, KU, JU.$$

$$(K) = \{\text{CM } 5, \text{CMR } 8(K)\}.$$

$$(L) = \{\text{CA}, \text{CMR } 8(\bar{K})\}.$$

$(M) = \{CMR\ 1, 2, 3, 4, CM\ 3, 4, CM\ 6\ (CM\ 8(K)), CM\ 17, CMS\ 1(D,W), CMS\ 1(Q,U)\}$, where $Q = TR, P, RA, F$, and $U = C, AS\&C, L, D, W$.



$CMR\ 9(A) \leftrightarrow LV \wedge CMR\ 8(A)$, for all A

$CM\ 9(A) \leftrightarrow LV \wedge CM\ 8(A)$, for all A

FIGURE 3.4C

To conclude this section we shall consider the forms $CM\ 15(A:B)$ and $CMR\ 15(A:B)$.

$CM\ 15(A:B)$: Any class which contains a $\subseteq$ -maximal subclass with the property A contains a $\subseteq$ -maximal subclass with the property B .

We shall use the notation introduced in Definition 3.18C to state $CMR\ 15(A:B)$.

$CMR\ 15(A:B)$: For any relation R and class X , if there exists a $\subseteq$ -maximal subclass $Y \subseteq X$ such that $A_R[Y]$ then there exists a $\subseteq$ -maximal subclass $Z \subseteq X$ such that $B_R[Z]$.

Also, $CM\ 16(A:B)$ is defined to be $LV \wedge CM\ 15(A:B)$, and $CMR\ 16(A:B)$ is defined to be $LV \wedge CMR\ 15(A:B)$. It is easy to see that the following implications hold for all A and B .

$$\begin{array}{ll}
 \text{CMR } 15(A:B) \rightarrow \text{CM } 15(A:B) \\
 \text{CMR } 16(A:B) \rightarrow \text{CM } 16(A:B) \\
 (*) \quad \text{CM } 8(B) \rightarrow \text{CM } 15(A:B) \\
 \text{CMR } 8(B) \rightarrow \text{CMR } 15(A:B) \\
 \text{CM } 9(B) \rightarrow \text{CM } 16(A:B) \\
 \text{CMR } 9(B) \rightarrow \text{CMR } 16(A:B)
 \end{array}$$

First, we shall consider the forms $\text{CM } 15(A:B)$, where the choices for A and B , $A \neq B$, are $D, \bar{D}, J, \bar{J}, K, \bar{K}$. $\text{CM } 8(\bar{D})$ is $M 19$ and is equivalent to a set form of the axiom of choice in NBG. Thus, by $(*)$ above, $\text{CM } 8(A:\bar{D})$ cannot be stronger than a set form of the axiom of choice, and since $\text{CM } 8(A:\bar{D})$ implies $M 8(A:\bar{D})$, $\text{CM } 8(A:\bar{D})$ is equivalent to a set form in NBG.

By using simple modifications in Theorems 4.35, 4.36, 4.38, 4.40, and 4.41 (with $A = K$ or $\bar{K}$), we obtain the corresponding results for $\text{CM } 15(A:B)$. (We may always assume that any class X in any one of these theorems is different from V , for X could be replaced, for example, by $X \times \{0\}$.) Using these results we obtain:

$$\text{CM } 15(A:B) \rightarrow \text{CM } 8(B), \text{ for } A = K, J, \text{ or } \bar{D}; B = \bar{K}, \bar{J}, \text{ or } D. \quad (4.35)$$

$$\text{CM } 15(\bar{J}:\bar{K}) \rightarrow \text{CM } 8(\bar{K}). \quad (4.36)$$

$$\text{CM } 15(A:B) \rightarrow \text{CM } 8(B), \text{ for } A = D \text{ or } \bar{D}; B = K \text{ or } \bar{K}. \quad (4.38)$$

$$\text{CM } 15(D:\bar{J}) \rightarrow \text{CM } 8(K). \quad (4.38)$$

$$\text{CM } 15(A:J) \rightarrow \text{CM } 8(\bar{K}), \text{ for } A = D \text{ or } \bar{D}. \quad (4.38)$$

$$\text{CM } 15(A:K) \rightarrow \text{CM } 8(K), \text{ for } A = \bar{K} \text{ or } J. \quad (4.40)$$

$$\text{CM } 15(\bar{K}:B) \rightarrow \text{CM } 8(D), \text{ for } B = D \text{ or } \bar{J}. \quad (4.41)$$

The following theorems give us the results for the remaining statements: $\text{CM } 15(\bar{J}:B)$, where $B = D, J, \text{ or } K$; and $\text{CM } 15(A:J)$, where $A = K \text{ or } \bar{K}$.

THEOREM 3.29C:

$$(a) \quad \text{CM } 15(\bar{J}:K) \rightarrow \text{CM } 8(K),$$

$$(b) \quad \text{CM } 15(A:J) \rightarrow \text{CM } 8(\bar{K}), \quad \text{for } A = \bar{J} \text{ or } \bar{K}.$$

PROOF: Let X be a class. Let a, b, c and d be distinct elements. For each $u \in X$, define

$$u_0 = (\{0\} \times u) \cup \{a, b, c\},$$

$$u_1 = (\{1\} \times u) \cup \{a, c, d\},$$

and define $y_0 = \{a, b\}$, $y_1 = \{c, d\}$. Let

$$Y = \{u_0 : u \in X\} \cup \{u_1 : u \in X\} \cup \{y_0, y_1\}.$$

Each of the following is easy to verify for all $u, v \in X$:

- (1) $u_0 J v_1$,
- (2) $y_0 K u_0$ and $y_1 K u_1$,
- (3) $y_0 J u_1$ and $y_1 J u_0$,
- (4) $y_0 D y_1$,
- (5) $u_i K v_i$ iff $u K v$, $i = 0$ or 1 ,
- (6) $u_i J v_i$ iff $u_i \bar{K} v_i$ iff $u \bar{K} v$, $i = 0$ or 1 .

It follows from (1)-(4) that $\{y_0, y_1\}$ is a $\subseteq$ -maximal subclass of Y with the property $\bar{J}$. Using CM 15($\bar{J}:K$) we obtain that Y has a $\subseteq$ -maximal subclass, Z , with the property K . It follows from (1) that if $u_i, v_j \in Z$ then $i = j$. Therefore, using (5) we see that $\{u \in X : u_i \in Z\}$ is a $\subseteq$ -maximal subclass of X with the property K . This proves (a).

To prove (b) with $A = \bar{J}$, we let Z' be a $\subseteq$ -maximal subclass of Y with the property J . By (4) there is at most one $y_i \in Z'$. If $y_0 \in Z'$ then by (2) $u_0 \notin Z'$ for any $u \in X$. Then using (6), $W = \{u \in X : u_1 \in Z'\}$ is a $\subseteq$ -maximal subclass of X with the property $\bar{K}$. Similarly, if $y_1 \in Z'$. If no $y_i \in Z'$, then by (1), $u_0 \in Z'$ iff $u_1 \in Z'$

so W is again a $\subseteq$ -maximal subclass with the property $\bar{K}$.

If $A = \bar{K}$, then we note that $\{y_0, y_1\}$ is a $\subseteq$ -maximal subclass of Y with the property $\bar{K}$; thus the proof is the same as that for $A = \bar{J}$, q.e.d.

THEOREM 3.30C: $CM\ 15(K:J) \rightarrow CM\ 8(\bar{K})$.

PROOF: Let X be a class and suppose $a, b, c \notin U X$. Let $u_a = u \cup \{a\}$, $y_0 = \{a, b\}$, $y_1 = \{a, b, c\}$ and $Y = \{u_a : u \in X\} \cup \{y_0, y_1\}$. We claim $\{y_0, y_1\}$ is a $\subseteq$ -maximal subclass of Y with the property K . Clearly $y_0 K y_1$. It is also easy to see that $y_i J u_a$ for all $u \in X$ and $i = 0, 1$. Thus, $CM\ 15(K:J)$ implies that Y has a $\subseteq$ -maximal subclass Z with the property J . For all $u, v \in X$, $u_a J v_a$ iff $u \bar{K} v$. Therefore, $\{u \in X : u_a \in Z\}$ is a $\subseteq$ -maximal subclass of X with the property $\bar{K}$.

THEOREM 3.31C: $CM\ 15(\bar{J}:D) \rightarrow CWO\ 12$.

PROOF: The proof is similar to that of 3.6C, because the class Z has the property $\bar{J}$. Thus, a $\subseteq$ -maximal subclass of Z with the property D can be used to construct the choice function F of 3.6C.

We illustrate our results with the table in Figure 3.5C.

A \ B	D	J	$\bar{J}$	K	$\bar{K}$
D	X	$\bar{K}$	K	K	$\bar{K}$
$\bar{D}$	D	$\bar{K}$	$\bar{J}$	K	$\bar{K}$
J	D	X	$\bar{J}$	K	$\bar{K}$
$\bar{J}$	CWO 12	$\bar{K}$	X	K	$\bar{K}$
K	D	$\bar{K}$	$\bar{J}$	X	$\bar{K}$
$\bar{K}$	D	$\bar{K}$	D	K	X

CM 15(A:B)

FIGURE 3.5C

The letters in the body of the table indicate the CM 8 form which is implied by the CM 15(A:B) form. For example, there is a K in the first row, third column. This means that $\text{CM } 15(D, \bar{J}) \rightarrow \text{CM } 8(K)$. The X's are for the CM 15(A:A) statements which are provable in NBG° and the CWO 12 means $\text{CM } 15(\bar{J}:D) \rightarrow \text{CWO } 12$.

Combining Figure 3.5C with the fact that $\text{CM } 8(B) \rightarrow \text{CM } 15(A:B)$ for all A and B, we obtain

THEOREM 3.32C:

$\text{CM } 8(\bar{K}) \leftrightarrow \text{CM } 15(A:\bar{K})$, for all $A \neq \bar{K}$.

$\text{CM } 8(K) \leftrightarrow \text{CM } 15(A:K)$, for all $A \neq K$.

$\text{CM } 8(D) \leftrightarrow \text{CM } 15(A:D)$, for all $A \neq D, \bar{J}$.

$\text{CM } 8(D) \rightarrow \text{CM } 15(\bar{J}:D) \rightarrow \text{CWO } 12$.

$\text{CM } 8(J) \rightarrow \text{CM } 15(A:J) \rightarrow \text{CM } 8(\bar{K})$, for all $A \neq J$.

$\text{CM } 8(\bar{J}) \rightarrow \text{CM } 15(A:\bar{J})$, for $A = \bar{D}, J, K$.

$\text{CM } 8(\bar{J}) \rightarrow \text{CM } 15(D:\bar{J}) \rightarrow \text{CM } 8(K)$.

$\text{CM } 8(\bar{J}) \rightarrow \text{CM } 15(\bar{K}:\bar{J}) \rightarrow \text{CM } 8(D)$.

(The possible values for A are $D, \bar{D}, J, \bar{J}, K$ or $\bar{K}$).

Thus it follows that all CM 15(A:B) forms with $A \neq B$ and $B \neq \bar{D}$ are equivalent to a class form of the axiom of choice in NBG . (See Figure 3.3C.)

Finally, we turn our attention to the CMR 15(A:B) statements. Clearly $\text{CMR } 8(B) \rightarrow \text{CMR } 15(A:B)$ for all A and B. We have shown $\text{CMR } 8(K) \leftrightarrow \text{CM } 5(3.12C(a))$, $\text{CMR } 8(\bar{K}) \leftrightarrow \text{CA}(3.12C(b))$ and $\text{CMR } 8(A) \leftrightarrow \text{CM } 20$ for $A = D, \bar{D}, J, \bar{J}, U, \bar{U}, DK, DJ, DU, KJ, KU$, and $JU(3.15C)$. We shall show below that for all A and B, $A \neq B$, there is a B' such that $\text{CMR } 15(A:B) \rightarrow \text{CMR } 8(B')$, except for $\text{CMR } 15(JK:B)$, $B = \bar{D}, \bar{U}$, which are provable in NBG° .

To accomplish this monumental task, for any class X and relation R we shall construct 9 relations, S_i , $i = 1, 2, \dots, 9$, such that when $\text{CMR } 15(A:B)$ is applied with

S_1 we obtain a $\subseteq$ -maximal subclass $Y_1 \subseteq X$ such that $B'_R[Y_1]$, for some B' .

In the 9 constructions which follow we assume X is a given class and R is a given relation. We shall also assume without loss of generality that for all $x \in X$, $R"\{x\}$ is neither empty nor the universe. Under these assumptions, if $R"\{x\}$ and $R"\{y\}$ are not complementary, exactly one of D , K , J , or U holds between them. If $R"\{x\}$ and $R"\{y\}$ are complementary then both D and U hold.

In what follows we shall replace the statement " Y is a $\subseteq$ -maximal subclass such that $A_S[Y]$ " by " Y is $A(S)$ -maximal".

1. For each $x \in X$,

$$S_1"\{x\} = (\{0\} \times R"\{x\}) \cup \{\langle 1, x \rangle\}.$$

If $x, y \in X$, $x \neq y$, then $S_1"\{x\} A S_1"\{y\}$, for $A = \bar{K}, \bar{U}$, and $\bar{K}\bar{U}$ ($= DJ$). Thus, for each $x \in X$, $\{x\}$ is $A(S_1)$ -maximal, for $A = K, U$, or KU ; and X itself is $A(S_1)$ -maximal for $A = \bar{K}, \bar{U}$ or DJ . We also have that for $x \neq y$,

(a) $S_1"\{x\} B S_1"\{y\}$ iff $R"\{x\} D R"\{y\}$, for $B = D, DK, DU$, or $\bar{J}$.

(b) $S_1"\{x\} B S_1"\{y\}$ iff $R"\{x\} \bar{D} R"\{y\}$, for $B = \bar{D}, JU, JK$ or J .

If $A = K, U, KU, \bar{K}, \bar{U}$ or DJ , the hypothesis of CMR 15(A:B) is satisfied. Thus, there is a subclass Y which is $B(S_1)$ -maximal. It follows from (a) that Y is $D(R)$ -maximal, if $B = D, DK, DU$ or $\bar{J}$; and it follows from (b) that Y is $\bar{D}(R)$ -maximal if $B = \bar{D}, JU, JK$, or J . Thus, we have proved the following:

If $A = K, U, KU, \bar{K}, \bar{U}$ or DJ ,

CMR 15(A:B) $\rightarrow$ CMR 8(D), for $B = D, DK, DU$, or $\bar{J}$,

CMR 15(A:B) $\rightarrow$ CMR 8($\bar{D}$), for $B = \bar{D}, JU, JK$, or J .

Using the fact that D and U are duals of each other (i.e., $X D Y$ iff $(V \sim X) U (V \sim Y)$) and K and J are self-dual, we obtain, in addition:

If $A = K, D, DK, \bar{K}, \bar{D}, JU,$

CMR 15(A:B) $\rightarrow$ CMR 8(U), for $B = U, KU,$

CMR 15(A:B) $\rightarrow$ CMR 8($\bar{U}$), for $B = \bar{U}, DJ.$

If $A = D, DK, \bar{D}, JU,$

CMR 15(A:B) $\rightarrow$ CMR 8(U), for $B = \bar{J}, DU,$

CMR 15(A:B) $\rightarrow$ CMR 8($\bar{U}$), for $B = J, JK.$

2. Let F be any 1-1 function mapping X into V such that $\mathcal{K}(F) \neq V$. For each $x \in X,$

$$S_2''\{x\} = F''(R''\{x\}).$$

Then $S_2''\{x\} \bar{U} S_2''\{y\}$ for all $x, y \in X$. Thus, for each $x \in X,$ $\{x\}$ is $U(S_2)$ -maximal, and X is $\bar{U}(S_2)$ -maximal. Moreover,

$$S_2''\{x\} A S_2''\{y\} \text{ iff } R''\{x\} A R''\{y\}, \text{ for } A = K \text{ or } \bar{K}.$$

It follows that

$$\text{CMR } 15(A:B) \rightarrow \text{CMR } 8(B), \text{ for } A = U \text{ or } \bar{U}, \text{ and } B = K \text{ or } \bar{K};$$

and by duality,

$$\text{CMR } 15(A:B) \rightarrow \text{CMR } 8(B), \text{ for } A = D \text{ or } \bar{D}, \text{ and } B = K \text{ or } \bar{K}.$$

3. Let $x \in X$ and let z_0, z_1 be distinct sets such that for all $x \in X, z_i \neq \langle x, j \rangle$, for $i, j = 0, 1$. Then, for $i = 0, 1,$ we define

$$S_3''\{\langle x, i \rangle\} = R''\{x\} \times \{i\}, S_3''\{z_i\} = V \sim (R''X \times \{i\}).$$

Then the following seven statements are easy to verify, if $x, y \in X, i \neq j, i, j = 0, 1$.

$$(a) S_3''\{z_0\} \bar{U} S_3''\{z_1\}.$$

$$(b) S_3''\{z_i\} D S_3''\{\langle x, i \rangle\}.$$

$$(c) S_3''\{z_i\} K S_3''\{\langle x, j \rangle\}.$$

$$(d) S_3''\{\langle x, 0 \rangle\} D S_3''\{\langle y, 1 \rangle\}.$$

$$(e) S_3''\{\langle x, i \rangle\} D S_3''\{\langle y, i \rangle\} \text{ iff } R''\{x\} D R''\{y\}.$$

$$(f) S_3''\{\langle x, i \rangle\} K S_3''\{\langle y, i \rangle\} \text{ iff } R''\{x\} K R''\{y\}.$$

$$(g) \quad S_3''\{\langle x, i \rangle\} \bar{U} S_3''\{\langle y, i \rangle\}.$$

Therefore, $\{z_0, z_1\}$ is $A(S_3)$ -maximal for $A = U, \bar{D}, \bar{K}, DU, KU$ and JU ; $\mathcal{D}(S_3) \sim \{z_i\}$ for $i = 0, 1$, are $\bar{U}(S_3)$ -maximal; and each $\{z_i\}$, $i = 0, 1$, is $J(S_3)$ -maximal.

For $B = D, DJ, K, DK, JK$, and $\bar{J}$, let Y_B be $B(S_3)$ -maximal.

(i) $B = D$. Since $S_3''\{z_0\} \cap S_3''\{z_1\} \neq \emptyset$, there is at most one $z_i \in Y_D$. If $z_i \in Y_D$, then it follows from (c) that no $\langle x, j \rangle \in Y_D$, with $j \neq i$. It follows from (b) and (e) that $\{x \in X: \langle x, i \rangle \in Y_D\}$ is $D(R)$ -maximal.

If no $z_i \in Y_D$, then by (d), Y_D is a union of disjoint classes, Y_0 and Y_1 , such that $D_{S_3}[Y_0]$ and $D_{S_3}[Y_1]$. Thus, each of $\{x \in X: \langle x, i \rangle \in Y_D\}$ for $i = 0, 1$, is $D(R)$ -maximal.

(ii) $B = DJ$. The proof is similar to (i), but instead of finishing with a $DJ(R)$ -maximal class, we finish with a $\bar{K}(R)$ -maximal class. (This follows from (g).)

(iii) $B = K$. By (d), if there is a $\langle x, i \rangle \in Y_K$, then no $\langle y, j \rangle \in Y_K$ if $i \neq j$. By (a), (b) and (d), if $\langle x, i \rangle \in Y_K$ then $z_j \in Y_K$ if $i \neq j$, but $z_i \notin Y_K$. Therefore, Y_K contains exactly one z_i and a class of $\langle x, j \rangle$'s, with $j \neq i$. It follows from (f) that $\{x \in X: \langle x, j \rangle \in Y_K\}$ is a $K(R)$ -maximal class.

(iv) $B = DK$. In this case Y_{DK} contains exactly one z_i and two classes Y_0 and Y_1 as in part (i). Thus, $\{x \in X: \langle x, i \rangle \in Y_{DK}\}$ for $i = 0, 1$, are each $DK(R)$ -maximal.

(v) $B = JK$. This is similar to (i). There is at most one $z_i \in Y_{JK}$, but, because of (g), we end with a $\bar{D}(R)$ -maximal subclass instead of a $JK(R)$ -maximal subclass.

(vi) $B = \bar{J}$. In this case $Y_{\bar{J}} = \{z_0, z_1\} \cup Y_0 \cup Y_1$, where Y_0 and Y_1 are defined in (i). Again, because of

(g) , $\{x \in X: \langle x, i \rangle \in Y_{\bar{J}}\}$, $i = 0, 1$ are each $DK(R)$ -maximal subclasses.

Using the results obtained in (i)-(vi) and the duality of D and U , we list below the new results obtained from S_3 .

CMR 15(A:D) $\rightarrow$ CMR 8(D), $A = \bar{D}, J, JU, DU$.

CMR 15(A:U) $\rightarrow$ CMR 8(U), $A = \bar{U}, J, DJ, DU$.

CMR 15(A:DJ) $\rightarrow$ CMR 8($\bar{K}$), $A = J, U, \bar{U}, KU, DU$.

CMR 15(A:JU) $\rightarrow$ CMR 8($\bar{K}$), $A = J, D, \bar{D}, DK, DU$.

CMR 15(A:K) $\rightarrow$ CMR 8(K), $A = J, \bar{K}, DJ, KU, DK, JU, DU$.

CMR 15(A:DK) $\rightarrow$ CMR 8(DK), $A = \bar{D}, J, JU, DU$.

CMR 15(A:KU) $\rightarrow$ CMR 8(KU), $A = \bar{U}, J, DJ, DU$.

CMR 15(A:JK) $\rightarrow$ CMR 8($\bar{D}$), $A = J, DU$.

CMR 15(A: $\bar{J}$) $\rightarrow$ CMR 8(DK), $A = J, DU$.

4. Let z be any set such that $z \notin X$. For each $x \in X$, define

$$S_4''\{x\} = R''\{x\} \times \{0\}, S_4''\{z\} = \{\langle z, 1 \rangle\}.$$

Then $\{z\}$ is $A(S_4)$ -maximal for $A = \bar{D}, J, K, KU, JU$ or JK . Suppose Y_B is a $B(S_4)$ -maximal subclass.

Then the following are true.

- (i) $\{x \in X: x \in Y_{\bar{D}}\}$ is $D(R)$ -maximal.
- (ii) $\{x \in X: x \in Y_{DJ}\}$ is $\bar{K}(R)$ -maximal.
- (iii) $\{x \in X: x \in Y_{DK}\}$ is $DK(R)$ -maximal.
- (iv) $\{x \in X: x \in Y_{DU}\}$ is $D(R)$ -maximal.
- (v) $\{x \in X: x \in Y_{\bar{J}}\}$ is $DK(R)$ -maximal.
- (vi) $\{x \in X: x \in Y_{\bar{K}}\}$ is $\bar{K}(R)$ -maximal.

Using (i)-(vi) and duality, the additional results we obtain from this example are as follows:

CMR 15(JK:B) $\rightarrow$ CMR 8(B), $B = D, U, DK, KU$.

CMR 15(A:DU) $\rightarrow$ CMR 8(D), $A = J, JK$.

CMR 15(JK: $\bar{J}$) $\rightarrow$ CMR 8(DK).

CMR 15(A: $\bar{K}$) $\rightarrow$ CMR 8($\bar{K}$), $A = J, K, KU, JU, JK, DJ, DK$.

5. Let Z be a non-empty, non-universal class and suppose $y \notin X$. Then we define,

$$S_5''\{x\} = (R''\{x\} \times Z) \cup (V \sim (V \times Z)),$$

$$S_5''\{y\} = V \sim (V \times Z).$$

It is easy to see that $\{y\}$ is $A(S_5)$ -maximal for $A = J, DJ$, and DU . (We are assuming $R''\{x\} \neq V$ for all $x \in X$.) Suppose Y is $\bar{U}(S_5)$ -maximal. Since

$$S_5''\{x\} \bar{U} S_5''\{y\} \quad \text{iff} \quad R''\{x\} \bar{U} R''\{y\},$$

$Y \sim \{y\}$ is $\bar{U}(R)$ -maximal.

Using example 5 and duality we obtain the following additional results:

CMR 15(A: $\bar{U}$) $\rightarrow$ CMR 8($\bar{U}$), $A = J, DJ, DU$.

CMR 15(A: $\bar{D}$) $\rightarrow$ CMR 8($\bar{D}$), $A = J, JU, DU$.

6. Suppose $\emptyset \neq Z \subset V$ and $z \notin X$. Define

$$S''\{x\} = R''\{x\} \times V, \quad S''\{z\} = V \times Z.$$

The range of S is $V \times V$. We are going to modify S so that its range is V . Let F be a function mapping V onto $\mathcal{K}(S)$. Then for each $u \in \mathcal{D}(S)$,

$$S_6''\{u\} = F^{-1}S''\{u\}.$$

Since F is a function, if $Y_1, Y_2 \subseteq \mathcal{K}(S)$,

$$F^{-1}Y_1 \cap F^{-1}Y_2 \quad \text{iff} \quad Y_1 \cap Y_2, \quad \text{for } A = D \text{ or } K.$$

$$F^{-1}Y_1 \cup F^{-1}Y_2 \quad \text{iff} \quad Y_1 \cup Y_2 = V \times V.$$

For all $x \in X$, $S_6''\{x\} \cap S_6''\{z\}$. Thus $\{z\}$ is $A(S_6)$ -maximal for $A = D, K, DK, DU, KU$ and $\bar{J}$. Suppose Y_B is $B(S_6)$ -maximal. We have for all $x, y \in X$, $x \neq y$,

$$S_6''\{x\} \cap S_6''\{y\} \text{ iff } R''\{x\} \cap R''\{y\}, \text{ for } B = D, K, J, U.$$

Therefore, for $B = \bar{D}, \bar{K}, \bar{U}, J, DJ, JU$ and JK ,

$$\{x \in X: x \in Y_B\} \text{ is } B(R)\text{-maximal.}$$

The additional results using S_6 and duality are listed below.

$$\text{CMR } 15(\bar{J}:B) \rightarrow \text{CMR } 8(B), \quad B = \bar{D}, \bar{U}, \bar{K}, J, DJ, JU, JK.$$

$$\text{CMR } 15(A:\bar{D}) \rightarrow \text{CMR } 8(\bar{D}), \quad A = D, DK.$$

$$\text{CMR } 15(A:\bar{U}) \rightarrow \text{CMR } 8(\bar{U}), \quad A = U, KU.$$

$$\text{CMR } 15(DU:B) \rightarrow \text{CMR } 8(B), \quad B = J, \bar{K}.$$

7. Let In be the set of all integers, $T = \bigcup_{x \in X} R''\{x\}$, and $z \notin X$. Then for each $x \in X$ and $i \in \text{In}$, we define S_7 as follows:

$$S_7''\{\langle x, i \rangle\} = R''\{x\} \times \{i\} \times \{1, 2\},$$

$$S_7''\{\langle z, i \rangle\} = T \times \{j \in \text{In}: j \leq i\} \times \{1\}.$$

(See 4.43) The following are easy to verify for $x, y \in X$, $i, j \in \text{In}$.

$$(a) \quad S_7''\{\langle x, i \rangle\} \cap S_7''\{\langle y, i \rangle\} \text{ iff } R''\{x\} \cap R''\{y\}, \text{ for } A = D \text{ or } K.$$

$$(b) \quad S_7''\{\langle x, i \rangle\} \cap S_7''\{\langle y, j \rangle\}.$$

$$(c) \quad S_7''\{\langle x, i \rangle\} \cap S_7''\{\langle y, j \rangle\}, \quad \text{if } i \neq j.$$

$$(d) \quad S_7''\{\langle z, i \rangle\} \subseteq S_7''\{\langle z, j \rangle\}, \quad \text{if } i \leq j.$$

$$(e) \quad S_7''\{\langle x, i \rangle\} \cap S_7''\{\langle z, j \rangle\}, \quad \text{if } i \leq j.$$

$$(f) \quad S_7''\{\langle x, i \rangle\} \cap S_7''\{\langle z, j \rangle\}, \quad \text{if } i > j.$$

If $Z = \{\langle z, i \rangle : i \in \text{In}\}$ then Z is $A(S_7)$ -maximal for $A = K, DK, JK, KU, \bar{D}$, and $\bar{J}$. Suppose Y_B is $B(S_7)$ -maximal, $B = D, DU, J$ or JU .

i) $B = D$. It follows from (d) that there is at most one $\langle z, i \rangle \in Y_D$. If no $\langle z, i \rangle$ is an element of Y_D then $\{x \in X : \langle x, i \rangle \in Y_D\}$ is $D(R)$ -maximal for each $i \in \text{In}$. If $\langle z, i \rangle \in Y_D$ then for each $j > i$, $\{x \in X : \langle x, j \rangle \in Y_D\}$ is $D(R)$ -maximal.

The proof is analogous if "D" is replaced by "DU".

(ii) $B = J$. Using (c), (d) and (e), we see that there is exactly one $\langle z, i \rangle \in Y_J$. It also follows from (e) that if $\langle x, i \rangle, \langle y, j \rangle \in Y_J$, then $i = j$. Thus, there is a $j \in \text{In}$, $j \leq i$, such that if $\langle x, k \rangle \in Y_J$ then $k = j$. Then, by (a) and (b), $\{x \in X : \langle x, j \rangle \in Y_J\}$ is $JU(R)$ -maximal. Similarly, if "J" is replaced by "JU" we obtain a $JU(R)$ -maximal class.

Using (i), (ii) and duality, we obtain the following results:

CMR 15($\bar{J}:B$) $\rightarrow$ CMR 8(B), $B = D, U, DU$.

CMR 15($KU:U$) $\rightarrow$ CMR 8(U).

CMR 15($DK:D$) $\rightarrow$ CMR 8(D).

CMR 15($JK:B$) $\rightarrow$ CMR 8(JU), $B = J, JU$.

CMR 15($JK:DJ$) $\rightarrow$ CMR 8(DJ).

8. For each $x \in X$, let

$$S_8^*\{x\} = (R^*\{x\} \times \{0\}) \cup \{\langle 0, 1 \rangle\}.$$

Then, X is $JK(S_8)$ -maximal. Let Y be $K(S_8)$ -maximal. Then Y is $K(R)$ -maximal. Therefore,

$$\text{CMR } 15(JK:K) \rightarrow \text{CMR } 8(K).$$

9. Let $\emptyset \neq Z \subset V$ and $z \notin X$. For each $x \in X$,

$$S''\{\langle x, 0 \rangle\} = R''\{x\} \times Z \times V,$$

$$S''\{\langle x, 1 \rangle\} = R''\{x\} \times (V \sim Z) \times V,$$

$$S''\{\langle z, 0 \rangle\} = (V \times Z \times Z) \cup (V \times (V \sim Z) \times V),$$

$$S''\{\langle z, 1 \rangle\} = (V \times Z \times V) \cup (V \times (V \sim Z) \times Z).$$

To obtain S_9 from S , we modify S as we did in example 6. We have $\mathcal{R}(S) = V \times V \times V$ and $\mathcal{R}(S_9) = V$.

If $x, y \in X$, $i \neq j$, $i, j = 0, 1$, the following relations hold:

$$S_9''\{\langle x, i \rangle\} D S_9''\{\langle y, j \rangle\},$$

$$S_9''\{\langle x, i \rangle\} K S_9''\{\langle z, j \rangle\},$$

$$S_9''\{\langle x, i \rangle\} J S_9''\{\langle z, i \rangle\},$$

$$S_9''\{\langle z, i \rangle\} U S_9''\{\langle z, j \rangle\}.$$

We see that $\{\langle z, 0 \rangle, \langle z, 1 \rangle\}$ is a $\bar{J}(S_9)$ -maximal subclass and for $i = 0, 1, \{\langle z, i \rangle\}$ are each $D(S_9)$ -maximal. Suppose Y_B is $B(S_9)$ -maximal.

If $B = K$, then there is exactly one $\langle z, i \rangle \in Y_K$, so if $i \neq j$, $\{x \in X: \langle x, j \rangle \in Y_K\}$ is $K(R)$ -maximal.

If $B = DK$, there is at most one $\langle z, i \rangle \in Y_{DK}$. If $\langle z, i \rangle \in Y_{DK}$, then $\{x \in X: \langle x, j \rangle \in Y_{DK}\}$ is $DK(R)$ -maximal if $j \neq i$. If no $\langle z, i \rangle \in Y_{DK}$, then $\{x \in X: \langle x, i \rangle \in Y_{DK}\}$ is $DK(R)$ -maximal for $i = 0, 1$.

Using this example and duality we obtain the following results:

$$\text{CMR } 15(\bar{J}:B) \rightarrow \text{CMR } 8(B), \quad B = K, DK, KU.$$

$$\text{CMR } 15(D:DK) \rightarrow \text{CMR } 8(DK).$$

$$\text{CMR } 15(U:KU) \rightarrow \text{CMR } 8(KU).$$

We summarize our results in Theorem 3.33C.

THEOREM 3.33C:

CMR 15(A:B) $\leftrightarrow$ CMR 8(B), for all A and B, $A \neq B$,
except $A = JK$ and $B = \bar{D}$ or $\bar{U}$ and those listed below.

- (i) CMR 8(DJ) $\rightarrow$ CMR 15(A:DJ) $\rightarrow$ CMR 8($\bar{K}$), for
 $A = J, U, \bar{U}, KU$, and DU .
- (ii) CMR 8(JU) $\rightarrow$ CMR 15(A:JU) $\rightarrow$ CMR 8($\bar{K}$), for
 $A = J, D, \bar{D}, DK$, and DU .

(Possible values for A and B are $D, \bar{D}, K, \bar{K}, J, \bar{J}, U, \bar{U}, DK, DJ, DU, KJ, KU, JU$.)

PROOF: Corollary 3.15C and examples 1-9 above.

Thus, it follows from 3.33C and preceding results that CMR 15(A:B) is equivalent to a class form of the axiom of choice for all A and B, $A \neq B$, except for the two statements CMR 15(JK: $\bar{D}$) and its dual, CMR 15(JK: $\bar{U}$). A. L. Rubin has shown that these two statements are provable in NBG°.

LEMMA 3.34C: CMR 15(JK: $\bar{D}$) $\wedge$ CMR 15(JK: $\bar{U}$).

PROOF: Let X be any class and suppose $Y \subseteq X$ is a $\subseteq$ -maximal subclass such that $JK_R[Y]$. Let

$$Z = \{z \in X: (\forall y \in Y) R''\{z\} \bar{D} R''\{y\}\}.$$

Then, we claim that $W = Y \cup Z$ is a $\subseteq$ -maximal subclass of X such that $\bar{D}_R[W]$.

To show $\bar{D}_R[W]$, it is sufficient to show that for all $z_1, z_2 \in Z \sim Y$, $R''\{z_1\} \bar{D} R''\{z_2\}$. Suppose $z_1, z_2 \in Z \sim Y$. Since $z_1, z_2 \in Z$ and $z_1, z_2 \notin Y$, there must be $y_1, y_2 \in Y$ such that

$$(1) \quad R''\{y_1\} \cup R''\{z_1\} \quad \text{and} \quad R''\{y_2\} \cup R''\{z_2\}.$$

Moreover, since $y_1, y_2 \in Y$,

$$(2) \quad R''\{y_1\} \bar{U} R''\{y_2\}.$$

It follows from (1) and (2) that there is a $v \in V \sim (R''\{y_1\} \cup R''\{y_2\})$ such that $v \in R''\{z_1\} \cap R''\{z_2\}$.

Therefore, $R\{z_1\} \bar{D} R\{z_2\}$ and $\bar{D}_R[W]$.

It is clear from the definition of W that it is a $\subseteq$ -maximal subclass of X such that $\bar{D}_R[W]$. This proves CMR 15(JK: $\bar{D}$). We obtain CMR 15(JK: $\bar{U}$) by duality.

We summarize the results of Part II below.

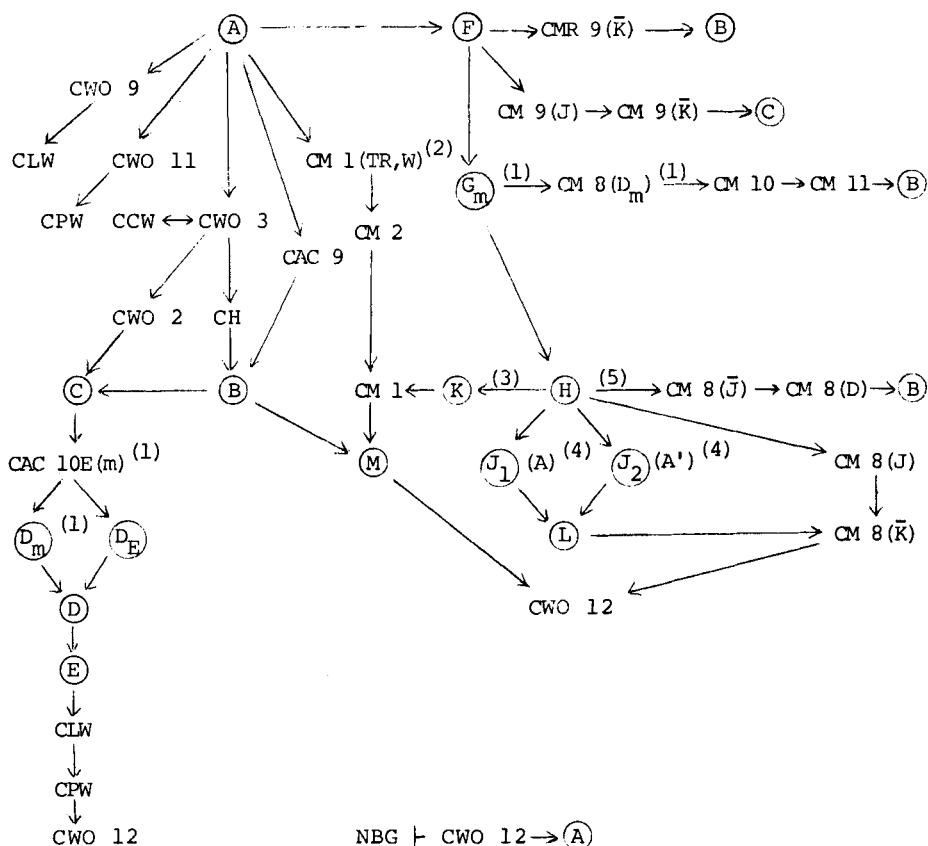
The statements in each of the following sets are equivalent:

- (A) = {CT, CWO 4, 5, 6(n), 7, 8, 10}, $n \geq 1$.
- (B) = {CAC 4, 5, 18C}.
- (C) = {CWO 1, E, CAC 1, 2, 3, 8, 10(2), 18, 19}.
- (D_m) = {CAC 10(m), 13(m-1)}, $m \geq 3$.
- (D_E) = {CAC 11E, 12E}.
- (D) = {CAC 11, 12, 14, 15}.
- (E) = {CAC 22, 23, 24}.
- (F) = {CM 7, CMR 14}.
- (G_m) = {CM 20(m), CMR 8(A)}, where $A = D_m, \bar{D}_m, \bar{D}_m^!$, and $m \geq 3$.
- (H) = {CM 20, CMR 10, 11, 12, 13, CMR 8(B_1), CMR 15($A_2:B_2$),
 $A_2 \neq B_2$, CMR 15($A_3:DJ$), CMR 15($A_4:JU$)}, where
 $B_1 = D, \bar{D}, J, \bar{J}, U, \bar{U}, DJ, DK, DU, JK, JU, KU$;
 $A_2 = D, \bar{D}, J, \bar{J}, K, \bar{K}, U, \bar{U}, DJ, DK, DU, JK, JU, KU$;
 $B_2 = D, \bar{D}, J, \bar{J}, U, \bar{U}, DK, DU, JK, KU$, but $A_2 = JK$
and $B_2 = \bar{D}, \bar{U}$ are omitted; $A_3 = D, \bar{D}, \bar{J}, K, \bar{K}, DK,$
 KJ, JU ; $A_4 = \bar{J}, K, \bar{K}, U, \bar{U}, DJ, KJ, KU$.
- (J_1) = CMR 15(A:DJ).
- (J_2) = CMR 15(A:JU).
- (K) = {CM 5, CMR 8(K), CMR 15(A:K)}, where $A \neq K$.

(L) = {CA, CMR 8($\bar{K}$), CMR 15(A: $\bar{K}$)}, where $A \neq \bar{K}$.

(Possible values of A in (K) and (L) are D, $\bar{D}$, J, $\bar{J}$, K, $\bar{K}$, U, $\bar{U}$, DJ, DK, DU, JK, JU, KU.)

(M) = {CMS 1, 2, 3, 4, CM 3, 4, 6(CM 8(K)), CM 17, CMS 1(D,W), CMS 1(Q,U)}, where $Q = TR, P, RA, F$ and $U = C, AS\&C, TR\&C, L, D, W$.



(1) $m \geq 3$.

(2) See Figure 3.1C for the remaining CM 1(Q,U) forms.

(3) CM 5 is CM 5(P,L). See Figure 3.2C for the remaining CM 5(Q,U) forms.

(4) $\langle A, A' \rangle = \langle J, J \rangle, \langle U, D \rangle, \langle \bar{U}, \bar{D} \rangle, \langle KU, DK \rangle, \langle DU, DU \rangle$.

(5) See Theorem 3.32C for the CM 15(A:B) forms.

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LIST OF THE SET FORMS

In parentheses following each of the statements we shall give the name of the person who is given credit for proving it equivalent to the axiom of choice (when applicable) and the date followed by the page number where it occurs in the text. Rubin 1963 refers to the 1963 edition of "Equivalents" and Rubin 1985 refers to the present work.

THE WELL ORDERING THEOREM

WO 1: Every set can be well-ordered. (1)

WO 2: Every set is equipollent to an ordinal number. (1)

WO 3: Every set is equipollent to a subset of an ordinal number. (1)

Let m be a natural number, $m \geq 1$.

WO 4(m): For every set x there exists an ordinal number α and a function f defined on α such that $f(\beta) \leq m$ for every $\beta < \alpha$ and $\bigcup_{\beta < \alpha} f(\beta) = x$. (Levy 1962; 1)

WO 5: There exists a natural number $m \geq 1$ such that WO 4(m). (Levy 1962; 1)

WO 6: For every set x there exists a natural number $m \geq 1$, an ordinal number α , and a function f defined on α such that $f(\beta) \leq m$ for every $\beta < \alpha$ and $\bigcup_{\beta < \alpha} f(\beta) = x$. (Levy 1962, Rubin 1963; 1)

WO 7: For every set x , x is finite iff for each well ordering R of x , R^{-1} also well orders x . (Tarski 1924a, 1938a; 5)

WO 8: Every set on which there is a choice function can be well ordered. (Brunner, Rubin 1985; 6)

THE AXIOM OF CHOICE

- AC 1: If s is a set of non-empty sets, there is a function f such that for every $x \in s$, $f(x) \in x$. (Zermelo 1904; 7)
- AC 2: If t is a pairwise disjoint set of non-empty sets, there is a set c which consists of one and only one element from each set in t . (Russell 1906; 7)
- AC 3: For every function f there is a function g such that for every s , if $x \in \mathcal{D}(f)$ and $f(x) \neq \emptyset$, then $g(x) \in f(x)$. (Zermelo 1904; 7)
- AC 4: For every relation r there is a function f such that $\mathcal{D}(f) = \mathcal{D}(r)$ and $f \subseteq r$. (Bernays 1941; 7)
- AC 5: For every function f there is a function g such that $\mathcal{D}(g) = \mathcal{R}(f)$ and for every $x \in \mathcal{D}(g)$, $f(g(x)) = x$. (Bernays 1941; 7)
- AC 6: The Cartesian product of a set of non-empty sets is non-empty. (Zermelo 1904; 8)
- AC 7: The Cartesian product of a set of non-empty sets of the same cardinality is non-empty. (Ward 1962; 8)
- AC 8: If s is a set of pairs of equipollent sets then there is a function which associates with each pair a bijection mapping one onto the other. (Pincus 1974; 8)
- AC 9: If s is a set of sets of the same cardinality then there is a function which associates with each pair a bijection mapping one onto the other. (Pincus 1974; 8)

Let $n \geq 2$ be a natural number.

- AC 10(n): If s is a set of infinite sets, there is a function f such that for each $x \in s$, $f(x)$ is a decomposition of x into a set of disjoint sets of size between 2 and n . (Sierpinski 1965, Levy 1962;8)

AC 11: There exists a natural number $n \geq 2$ such that
AC 10(n). (Sierpinski 1965, Levy 1962; 8)

AC 12: If s is a set of infinite sets there is a natural
number $n \geq 2$ and a function f such that for each
 $x \in s$, $f(x)$ is a decomposition of x into a set of
disjoint sets of size between 2 and n .
(Sierpinski 1965, Levy 1962, Rubin 1963; 8)

Let $m \geq 1$ be a natural number.

AC 13(m): If s is a set of non-empty sets there is a
function f such that for each $x \in s$, $f(x)$ is
a non-empty subset of x with $f(x) \prec m$. (Levy
1962; 8)

AC 14: There is a natural number $m \geq 1$ such that AC 13(m).
(Levy 1962; 8)

AC 15: If s is a set of non-empty sets then there is a
natural number $m \geq 1$ and a function f such that
for each $x \in s$, $f(x)$ is a non-empty subset of x
with $f(x) \prec m$. (Levy 1962, Rubin 1963; 8)

Let n and k be natural numbers with $n > 2$ and $1 < k < n$.

AC 16(n,k): If s is an infinite set then there is a set
 t_n of n -element subsets of s such that each
 k -element subset of s is a subset of exactly
one element of t_n . (Frascella 1965, 1966, Rubin
& Rubin 1967; 9)

AC 17: If s is a set, $t = \mathcal{P}(s) \sim \{\emptyset\}$, F is the set of
all functions from t to s and g is a function
from F to t then there is an $f \in F$ such that
 $f(g(f)) \in g(f)$. (Tarski 1951; 9)

AC 18: For any non-empty set A , and for every function B
whose domain is A and B_a is a non-empty set for
each $a \in A$,

$$\bigcap_{a \in A} \bigcup_{b \in B_a} x_{a,b} = \bigcup_{\substack{f \in \prod_{a \in A} B_a}} \bigcap_{a \in A} x_{a,f(a)}.$$

(Collins 1954, Farrah 1955; 9)

AC 19: For any non-empty set A , each of whose elements is non-empty,

$$\bigcap_{a \in A} \bigcup_{b \in a} b = \bigcup_{f \in C(A)} \bigcap_{a \in A} f(a) ,$$

where $C(A)$ is the set of all choice functions on A .
(Collins 1954, Farrah 1955, Rubin 1963; 9)

$(\forall \alpha)DC(\alpha)$: For each ordinal α , if R is a relation between subsets and elements of a set x such that
 $(\forall y \subseteq x)(y < \alpha \rightarrow (\exists u \in x)\langle y, u \rangle \in R)$ then there
 is a function f mapping α into x such that
 $(\forall \beta < \alpha)\langle f''\beta, f(\beta) \rangle \in R$. (Levy 1964; 19)

THE TRICHOTOMY

T 1: For all sets x and y , $x < y$, $x \approx y$ or $y < x$.
(Hartogs 1915; 21)

T 2: For every two non-empty sets, there is a mapping of one onto the other. (Lindenbaum 1926, Sierpinski 1948; 21)

If $n \geq 2$ is a natural number,

T 3(n): If x is a set of cardinal numbers such that $|x| = n$ then there exists $\kappa, \mu \in x$ such that $\kappa < \mu$ or $\mu < \kappa$. (Tarski 1964; 22)

T 4: For all cardinals μ , if $\mu \leq \kappa_1, \kappa_2 \leq 2^\mu$ then either $\kappa_1 \leq \kappa_2$ or $\kappa_2 \leq \kappa_1$. (Kruse 1962; 24)

T 5: For all cardinals κ and μ , if $\mu < 2^\kappa$ then $\kappa \leq \mu$ or $\mu \leq \kappa$. (Sobocinski 1961a; 27)

T 6: For all cardinals κ , and alephs $\aleph$, if $\aleph < 2^\kappa$ then $\kappa \leq \aleph$ or $\aleph \leq \kappa$. (Sobocinski 1961a; 27)

T 7: For all cardinals κ_1, κ_2 , and μ , if $\aleph_0 \leq \kappa_1, \kappa_2 < \mu$ and if for all v_1 and v_2 , $\kappa_1 + v_1 = \kappa_2 + v_2 = \mu$ implies $v_1 = v_2$, then $\kappa_1 \leq \kappa_2$ or $\kappa_2 \leq \kappa_1$.
(Sobocinski 1961b; 28)

- T 8: For all sets x and y , $\mathcal{P}_{<\omega}(x) \prec \mathcal{P}_{<\omega}(y)$ or $\mathcal{P}_{<\omega}(y) \prec \mathcal{P}_{<\omega}(x)$. (Alas 1968; 29)
- T 9: For all sets x and y , $x \prec \mathcal{P}_{<\omega}(y)$ or $\mathcal{P}_{<\omega}(y) \prec x$. (Alas 1968; 29)
- T 10: For all sets x and y , $x \prec \mathcal{P}_{<\omega}(y)$ or $y \prec \mathcal{P}_{<\omega}(x)$. (Alas 1968; 29)

MAXIMAL PRINCIPLES

- M 1: If R is a partial ordering relation on a non-empty set x and if every subset of x which is linearly ordered by R has an R -upper bound, then x has an R -maximal element. (Bourbaki 1939, Tukey 1940; 32)
- M 2: If R is a partial ordering relation on a non-empty set x and if every subset of x which is well-ordered by R has an R -upper bound, then x has an R -maximal element. (Kneser 1950, Szele 1950; 33)
- M 3: If every non-empty nest which is a subset of a non-empty set x has its union an element of x , then x has a $\subseteq$ -maximal element. (Kuratowski 1922; Zorn 1935; 33)
- M 4: If every non-empty well-ordered nest which is a subset of a non-empty set x has its union an element of x , then x has a $\subseteq$ -maximal element. (Kuratowski 1922, Hausdorff 1927; 33)
- M 5: If R is a partial ordering relation on x , then there exists a $\subseteq$ -maximal subset of x which is linearly ordered by R . (Hausdorff 1914; 33)
- M 6: For every set x , there exists a $\subseteq$ -maximal subset of x which is a nest. (Hausdorff 1907, 1909; 33)
- M 7: For every set x and every property P of finite character, there exists a $\subseteq$ -maximal subset of x which has the property P . (Teichmüller 1939, Tukey 1940, Bourbaki 1940; 34)

- M'1: If R is a partial ordering on x and if every subset of x which is linearly ordered by R has an R -upper bound and if $y \in x$ then there is an R -maximal element $z \in x$ such that $y R z$. (Bourbaki 1939, Tukey 1940, Kuratowski 1922; 36)
- M'2: If R is a partial ordering on x and if every subset of x which is well ordered by R has an R -upper bound and if $y \in x$ then there is an R -maximal element $z \in x$ such that $y R z$. (Kneser 1950, Szele 1950, Kuratowski 1922; 36)
- M'3: If every non-empty nest which is a subset of a non-empty set x has its union an element of x and if $y \in x$ then there is a $\subseteq$ -maximal element $z \in x$ such that $y \subseteq z$. (Kuratowski 1922, Zorn 1935; 36)
- M'4: If every non-empty well ordered nest which is a subset of a non-empty set x has its union in x and if $y \in x$ then there is a $\subseteq$ -maximal element $z \in x$ such that $y \subseteq z$. (Kuratowski 1922, Hausdorff 1927; 36)
- M'5: If R is a partial ordering on x and $y \subseteq x$ such that y is linearly ordered by R , then there exists a $\subseteq$ -maximal set $z \subseteq x$ such that $y \subseteq z$. (Hausdorff 1914, Kuratowski 1922; 36)
- M'6: For every set x and every nest $y \subseteq x$ there is a $\subseteq$ -maximal nest $z \subseteq x$ such that $y \subseteq z$. (Hausdorff 1907, 1909, Kuratowski 1922; 36)
- M'7: For every set x , for every property P of finite character, and for every set $y \subseteq x$ such that y has the property P , there is a $\subseteq$ -maximal set $z \subseteq x$ such that $y \subseteq z$ and z has the property P . (Teichmüller 1939, Tukey 1940, Bourbaki 1940; 37)
- M 1(Q,U): Every non-empty set which is Q -ordered by R in which each U -ordered subset has an R -upper bound, has an R -maximal element, where $Q = TR, P, RA$ or F and $U = C, AS\&C, TR\&C, L, D$, or W ; and $Q = D$ and $U = W$. (Harper & Rubin 1976, Felgner 1967; 37)

- M 5(Q,U): Every set which is Q-ordered by R contains a $\subseteq$ -maximal subset which is U-ordered by R, where
 $Q = A$ or TR and $U = AS, C, AS\&C, TR\&C, P, L, D,$
or RA ; $Q = AS$ or P and $U = C, AS\&C, TR\&C, L,$
 $D,$ or RA ; $Q = C$ and $U = AS, AS\&C, P, L, D,$ or
 RA ; $Q = D, RA$ or F and $U = C, AS\&C, TR\&C,$ or L ;
 $Q = D$ and $U = RA$; $Q = RA$ and $U = D$; $Q = F$
and $U = D$ or W ; and $Q = A$ and $U = UN$ or
 $UN(n), n \geq 2$. (Wallace 1944, Gottschalk 1952,
Höft & Howard 1973, Harper & Rubin 1976; 41,45)
- M'5(Q,U): In every Q-ordered set, every U-ordered subset can
be extended to a $\subseteq$ -maximal U-ordered set, where Q
and U are the same as in M 5(Q,U), and also
 $Q = A$ and $U = TR$. (Harper & Rubin 1976; 44)
- M 8(A): Every set has a $\subseteq$ -maximal subset with the property A;
where $A = D$ (Vaught 1952; 68); $A = \bar{D}, J$ or $\bar{J}$
(Kurepa 1952; 68); $A = D_m$ or $\bar{D}_m$ (Chang 1960,
Levy 1962, Rubin 1963; 71); $A = D \vee \bar{D} \vee J \vee \bar{J} \vee K \vee K^*$
(A. Rubin & H. Rubin 1985; 54)
- M 9($\bar{K}$): Every set can be linearly ordered, and every set has
a $\subseteq$ -maximal subset with the property $\bar{K}$. (Kurepa
1952; 54)
- M 10: There exists a natural number $m \geq 2$ such that $M 8(D_m)$.
(Chang 1960, Levy 1962; 57)
- M 11: For every set x there exists a natural number $m \geq 2$
such that x has a $\subseteq$ -maximal subset which has the
property D_m . (Chang 1960, Levy 1962, Rubin 1963; 57)
- M 12: There exists a natural number $m \geq 2$ such that
 $M 8(\bar{D}_m)$. (Rubin 1963; 58)
- M 13: For every set x, there exists a natural number $m \geq 2$
such that x has a $\subseteq$ -maximal subset which has the
property $\bar{D}_m$. (Rubin 1963; 58)
- M 14: Every set contains a $\subseteq$ -maximal subset with the finite
intersection property. (Klimovski 1956; 60)

- M 14S: Every class contains a $\subseteq$ -maximal subclass with the finite intersection property. (Klimovski 1956, Rubin 1985; 61)
- M'14: If $y \subseteq x$ and y has the finite intersection property then there is a $\subseteq$ -maximal set z with the finite intersection property such that $y \subseteq z \subseteq x$. (Klimovski 1956; 61)
- M 15(A:B): Any set which contains a $\subseteq$ -maximal subset with the property A contains a $\subseteq$ -maximal subset with the property B , where $A \neq B$, $A = D, \bar{D}, J, \bar{J}, K$ or $\bar{K}$ and $B = D, \bar{D}, J, \bar{J}$, or K . (Rubin 1963, 1985; 65)
- M 16(A:B): Every set can be linearly ordered and any set which contains a $\subseteq$ -maximal subset with the property A contains a $\subseteq$ -maximal subset with the property B , where $A \neq B$, $A, B = D, \bar{D}, J, \bar{J}, K$ or $\bar{K}$. (Rubin 1963; 65).
- M 17: Every set x contains a $\subseteq$ -maximal subset y such that for every $s, t \in y$, $s \neq t$, either $s \in t$ or $t \in s$. (Rubin 1963; 69)
- M 18: Suppose x is partially ordered by R . For each $t \in x$, let

$$\vec{t} = \{s \in x: t R s \wedge t \neq s\}.$$
 Suppose every R -linearly ordered subset of x has an R -upper bound. Then there is a $t \in x$ such that $\vec{t}$ has no R -first element. (Blair & Tomber 1960, Rubin 1963; 71)
- M 19(Q,U): Every set x such that x is Q -ordered by R and every U -ordered subset has an R -upper bound has the property that every function f from x to x such that for all $u \in x$, $u R f(u)$, has a fixed point, where Q and U have the same values as in M 1(Q,U). (Abian 1983, Rubin 1985; 72)

FORMS EQUIVALENT TO THE AXIOM OF CHOICE UNDER THE AXIOMS OF
EXTENSIONALITY AND FOUNDATION

- PW: The power set of a well-ordered set can be well-ordered.
(H. Rubin 1960; 73)
- LW: Every linearly ordered set can be well-ordered. (H. Rubin
1960; 73)
- INJ: If x is a non-empty set and A is a proper class then
 $x < A$. (von Neumann 1925; 73)
- PROJ: If x is a non-empty set and A is a proper class
then A can be mapped onto x . (von Neumann 1925; 74)
- MC 1: If s is a set of non-empty sets then there is a
function f such that for each $x \in s$, $f(x)$ is a
non-empty finite subset of x . (Levy 1962, Felgner &
Jech 1973; 74)
- MC 2: Every set is the union of a well-ordered set of finite
sets. (Levy 1962, Rubin 1985; 74)
- MC 3: If s is a set of infinite sets, then there is a
function f such that for each $x \in s$, $f(x)$ is a
partition of x into pairwise disjoint finite sets,
each of which has at least 2 elements. (Sierpinski
1965, Levy 1962, Rubin 1985; 74)
- A: Every partially ordered set contains a $\subseteq$ -maximal antichain.
(Kurepa 1952; 74)
- AC_Q^P : For each set x , if x is a P -ordered set of non-empty
sets such that each $u \in x$ can be Q -ordered, then there
is a function f such that for each $u \in x$, $f(u) \in u$,
where $P = LO$ and $Q = \emptyset$, LO or DLO ; and $P = \emptyset$ and
 $Q = LO$ or DLO . (Howard & Rubin 1977; 78).
- MC_Q^P : For each set x , if x is a P -ordered set of non-empty
sets such that each $u \in x$ can be Q -ordered, then there
is a function f such that for each $u \in x$, $f(u)$ is
a non-empty, finite subset of u , where $P = LO$ and
 $Q = \emptyset$, LO or DLO ; and $P = \emptyset$ and $Q = LO$ or DLO .
(Howard & Rubin 1977; 78)

- ORD: If x is a set and f is a function such that for each $u \in x$ $f(u)$ is a non-empty set of linear orderings of u , then $\{f(u) : u \in x\}$ has a choice function. (Truss 1972; 78)
- W-ORD: If x is a set and f is a function such that for each $u \in x$, $f(u)$ is a non-empty set of well-orderings of u , then $\{f(u) : u \in x\}$ has a choice function. (Truss 1972; 78)
- TCL: Every transitive and connected set contains a $\subseteq$ -maximal linear ordered set. (Howard & Rubin 1977; 78)
- MC 4(n): If s is a set of non-empty sets then there is a function f such that for all $x \in s$, $f(x)$ is a non-empty finite subset of x and $(|f(x)|, n) = 1$. (Bleicher 1963, 1965; 83)
- INF 2: For every set x , if every infinite subset of x is structured, then x can be well ordered. (Brunner 1982; 84)
- U_Q^P : The union of a P -ordered family of sets, each of which can be Q -ordered, can be well-ordered, where $P = \emptyset$ or LO and $Q = \text{LO, DLO, WO or DWO}$; $P = \text{WO}$ and $Q = \emptyset, \text{LO, or DLO}$; and $P = \text{LO}$ and $Q = \emptyset$. (J. E. Rubin 1982; 85)
- M 5(Q,U): Every set which is Q -ordered by R contains a $\subseteq$ -maximal subset which is U ordered by R , where $Q = \text{TR, AS, AS\&TR, D or RA}$ and $U = \text{VC}$. (Rubin 1985; 86)
- M 19: Every class has a $\subseteq$ -maximal subclass with the property $\bar{D}$. (Kurepa 1952, Rubin 1963; 87)
- AC 20: For every relation R whose domain is a set (its range may be a proper class) there is a function f such that $\mathcal{J}(f) = \mathcal{J}(R)$ and $f \subseteq R$. (Bernays 1941, Rubin 1963; 88)

AC 21: For any non-empty set A , and for each relation B whose domain is A ,

$$\bigcap_{a \in A} \bigcup_{b \in B_a} x_{a,b} = \bigcup_{\substack{f \in \prod_{a \in A} B_a}} x_{a,f(a)}$$

where B_a is defined to be $B''\{a\}$ for each $a \in A$.
(Collins 1954, Farrah 1955, Rubin 1963; 88)

($\forall \alpha$)DCC(α): For each ordinal α , if R is a relation between subsets and elements of a class X such that
($\forall y \subseteq X$) ($y < \alpha \rightarrow (\exists u \in X) \langle y, u \rangle \in R$) then there is a function f mapping α into x such that
($\forall \beta < \alpha$) ($f''\beta, f(\beta) \rangle \in R$. (Felgner 1976a; 89)

T 11: $\{\alpha \in \text{On} : T(N_\alpha) \wedge N_{\alpha+1} \leq 2^{N_\alpha}\}$ is unbounded. (Kruse 1963; 90)

T 12: $\{\alpha \in \text{On} : T(N_\alpha) \wedge T(2^{N_\alpha})\}$ is unbounded. (Kruse 1963; 90)

T 13: For all cardinals κ and μ , if there is an ordinal α such that $\kappa, \mu \leq 2^{|\alpha|}$ then $\kappa \leq \mu$ or $\mu \leq \kappa$.
(Kruse 1963; 90)

T 14: $\{\alpha \in \text{On} : T(N_\alpha) \wedge \text{cof}(\omega_{\alpha+1}) \geq \beta\}$ is unbounded for each $\beta \in \text{On}$. (Kruse 1963; 91)

ALGEBRAIC FORMS

AL 1: Every lattice with an identity element and at least one other element contains a $\subseteq$ -maximal proper ideal.
(Scott 1953; 95)

AL'1: Every proper ideal of a lattice with an identity element can be extended to a $\subseteq$ -maximal proper ideal. (Scott 1953; 95)

AL'2: If A is an algebraic system, B a subalgebra, and $a \in A$, but $a \notin B$, then there exists a $\subseteq$ -maximal subalgebra which contains B as a subset, but does not contain a as an element. (Blair 1957; 95)

- AL'3: If x is an arbitrary set, C a finitary closure operator, P a property of finite character, and y a C -closed subset of x which has the property P , then there is a $\subseteq$ -maximal C -closed subset of x which contains y as a subset and has the property P .
(Scott 1953, Rubin 1963; 95)
- AL'4: If x is an arbitrary set, $y \subseteq x$, and $R : y$ then there is a $\subseteq$ -maximal subset z of x such that $y \subseteq z$ and $R : z$. (Beth 1959; 97)
- AL 5: If B is a Boolean algebra and $S \subseteq B$ such that $0 \notin S$, then there exists a $\subseteq$ -maximal proper ideal disjoint from S . (Mrowka 1955; 98)
- AL 5($\vee$): If B is a Boolean algebra and $S \subseteq B$ such that $0 \notin S$ and S is closed with respect to $\vee$ (join), then there exists a $\subseteq$ -maximal ideal disjoint from S .
(Howard 1977; 99)
- AL 5($\sim$): If B is a Boolean algebra and $S \subseteq B$ such that $0 \notin S$ and S is closed with respect to $\sim$ (complementation), then there exists a $\subseteq$ -maximal ideal disjoint from S . (Howard 1977; 100)
- AL 1D: Every distributive lattice with an identity element and at least one other element contains a $\subseteq$ -maximal proper ideal. (Klimovski 1958; 101)
- DAL 1D: Every distributive lattice with a zero element and at least one other element contains an ultrafilter.
(Klimovski 1958; 101)
- AL 6: If $\mathcal{L} = \langle L, \{\vee, \wedge\} \rangle$ is a complete lattice, $a, b, c \in L$, $a \leq b$, c is compact, $c \leq b$ and $c \not\leq a$ then there exists an element $m \in L$ such that $a \leq m \leq b$, $c \not\leq m$ and if there is a $d \in L$ such that $m < d \leq b$ then $c \leq d$. (Diener 1956; 102)
- AL 7: Every subalgebra is the intersection of all meet-irreducible subalgebras containing it as a subset.
(McCoy 1938, Birkhoff & Frink 1948, Fuchs 1949, Diener 1956, Schmidt 1953; 103)

- AL 8: Every proper subalgebra has a proper meet-irreducible subalgebra containing it as a subset. (Diener 1956; 103)
- AL 9: The set of absolutely dispensible elements is equal to the intersection of the set of all $\subseteq$ -maximal proper subalgebras. (Frattini 1885, Neumann 1937, Schmidt 1953; 103)
- AL 10: For every element in a basis there exists a $\subseteq$ -maximal subalgebra which does not contain it as an element. (Schmidt 1953; 103)
- AL 11: If A has a finite basis and $B \subset A$ is a subalgebra then there exists a $\subseteq$ -maximal proper subalgebra which contains B as a subset. (Krull 1929; Schmidt 1953; 103)
- AL 12: Every algebra has a basis. (Schmidt 1962; 103)
- AL'12: Any algebraically independent subset of an algebra can be extended to a basis. (Schmidt 1962; Rubin 1985; 109)
- AL 13: Any subset of an algebra which generates the algebra contains a basis. (Schmidt 1962; Rubin 1985; 109)
- AL 14: On every non-empty set there exists a cancellative groupoid. (Hajnal & Kertesz 1972; 110)
- AL 15: Every unique factorization domain contains a $\subseteq$ -maximal proper ideal. (Hodges 1979; 112)
- AL 16: Every group contains a $\subseteq$ -maximal abelian subgroup. (Klimovsky 1962, Felgner 1976; 116)
- Let $k \geq 1$ be a natural number,
- AL 17(k): Every group contains a $\subseteq$ -maximal nil- k subgroup. (Felgner 1976; 118)
- AL 18(k): Every group contains a $\subseteq$ -maximal subgroup with solvability degree $\leq k$. (Felgner 1976; 118)
- B: Every vector space has a basis. (Blass 1984; 119)

AL 19: For every field F and every vector space V over F , every subset of V which generates V contains a basis. (Bleicher 1964, Halpern 1966; 119)

Let F be a field,

AL 20(F): Every vector space V over F has the property that for all $A \subseteq V$, if A is linearly independent then A can be extended to a basis of V . (Bleicher 1964; 119)

B': Every vector space has a dimension. (Brunner; 121)

AL 20: There exists a field F such that AL 20(F). (Bleicher 1964; 125)

Let F be a field,

AL 21(F): Every vector space V over F has the property that if $S \subseteq V$ is a subspace of V then there is a subspace $S' \subseteq V$ such that $S \cap S' = \{0\}$ and $S \cup S'$ generates V . (Bleicher 1964; Armbrust 1972; 119)

AL 21: There exists a field F such that AL 21(F). (Bleicher 1964; 125)

AL 22: There is a field F such that for every vector space V over F if B is a basis for V then for each subspace $S \subseteq V$ there is a subset $B' \subseteq B$ such that if S' is the subspace generated by B' then $S \cap S' = \{0\}$ and $S \cup S'$ generates V . (Bleicher 1964; 125)

AL 23: There is a field F such that every vector space over F has a basis, and if V is a vector space over F , B is a basis of V and X is any linearly independent subset of V , then there is a subset $B' \subseteq B$ such that $X \cap B' = \emptyset$ and $X \cup B'$ is a basis of V . (Bleicher 1964; 125)

- AL 24: If A is an abelian group and B is a torsion-free divisible subgroup of A then there exists a subgroup $B' \subseteq A$ such that $B \cap B' = \{0\}$ and $B \cup B'$ generates A . (Bleicher 1964, Armbrust 1972; 127)
- AL 25: If G is a free abelian group, A is a uniquely divisible abelian group, B is an abelian group, $f:A \rightarrow B$ is a homomorphism mapping A onto B , and $g:G \rightarrow B$ is a homomorphism then there is a homomorphism $h:G \rightarrow A$ such that $f \circ h = g$. (Blass 1979; 128)
- AL 26: If G is a free abelian group, A and B are abelian groups, $f:A \rightarrow B$ is a homomorphism mapping A onto B , and $g:G \rightarrow B$ is a homomorphism then there is a homomorphism $h:G \rightarrow A$ such that $f \circ h = g$. (Blass 1979; 128)
- AL 27: If G is a divisible abelian group, A and B are abelian groups, $f:B \rightarrow A$ is an isomorphism, and $g:B \rightarrow G$ is a homomorphism then there is a homomorphism $h:A \rightarrow G$ such that $h \circ f = g$. (Blass 1979; 128)
- AL 28: Let $\langle A, \{O_k : k \in K\} \rangle$ be a multi-algebra. Then there is an algebra $\langle B, \{O'_k : k \in K\} \rangle$ of the same type as A and an equivalence relation E on B such that $\langle A, \{O_k : k \in K\} \rangle \cong \langle B/E, \{O'_k/E : k \in K\} \rangle$. (Grätzer 1962, Höft & Howard 1980; 134)
- AL 29: Let $\langle A, \{O_k : k \in K\} \rangle$ be a multi-algebra such that all the multi-operations O_k for $k \in K$ are unary operations. Then there is an algebra $\langle B, \{O'_k : k \in K\} \rangle$ of the same type as A and an equivalence relation E on B such that $\langle A, \{O_k : k \in K\} \rangle \cong \langle B/E, \{O'_k/E : k \in K\} \rangle$. (Grätzer 1962, Höft & Howard 1980; 134)

CARDINAL NUMBER FORMS

- CN 1: $(\forall \kappa > 0)(\forall \mu \geq \aleph_0)(\kappa \cdot \mu = \kappa + \mu)$. (Tarski 1924; 139)
- CN'1: $(\forall \kappa \geq \aleph_0)(\kappa \cdot \kappa^* = \kappa + \kappa^*)$. (Tarski 1924; 139)

- CN 1F: A set x is finite iff x has at most one element or there exists a non-empty set u disjoint from x such that $u \cup x < u \times x$. (Tarski 1938a; 140)
- CN 2: $(\forall \kappa, \mu, \nu \geq \aleph_0) [\kappa \leq \mu + \nu \rightarrow [\kappa \leq \mu \vee \kappa \leq \nu]]$. (Tarski; 139)
- CN'2: $(\forall \kappa, \mu \geq \aleph_0) (\forall \alpha) [\kappa \leq \mu + \aleph_\alpha \rightarrow (\kappa \leq \mu \vee \kappa \leq \aleph_\alpha)]$. (Tarski; 139)
- CN 3: $(\forall \kappa, \mu, \nu \geq \aleph_0) [\kappa \leq \mu \cdot \nu \rightarrow [\kappa \leq \mu \vee \kappa \leq \nu]]$. (Tarski; 139)
- CN'3: $(\forall \kappa, \mu \geq \aleph_0) (\forall \alpha) [\kappa \leq \mu \cdot \aleph_\alpha \rightarrow (\kappa \leq \mu \vee \kappa \leq \aleph_\alpha)]$. (Tarski; 139)
- CN 4: $(\forall \kappa) (\forall \mu \geq \aleph_0) (\kappa + \mu = \kappa \vee \kappa + \mu = \mu)$. (Lesniewski, Sierpinski 1947; 139)
- CN'4: $(\forall \kappa \geq \aleph_0) (\kappa + \kappa^* = \kappa^*)$. (Lesniewski, Sierpinski 1947; 139)
- CN 4F: A set x is finite iff x has at most one element or there exist disjoint sets u and v such that $|u| < |x|$ and $|v| < |x|$, and $u \cup v = x$. (Tarski 1938a; 139)
- CN 5: $(\forall \kappa) (\forall \mu \geq \aleph_0) (\kappa \cdot \mu = \kappa \vee \kappa \cdot \mu = \mu)$. (Lesniewski, Sierpinski 1947; 140)
- CN'5: $(\forall \kappa \geq \aleph_0) (\kappa \cdot \kappa^* = \kappa^*)$. (Lesniewski, Sierpinski 1947; 140)
- CN 6: $(\forall \kappa \geq \aleph_0) (\kappa = \kappa^2)$. (Tarski 1924; 140)
- CN 6F: A set x is finite iff x has at most one element or $x < x \times x$. (Tarski 1938a; 140)
- CN 7: $(\forall \kappa, \mu) (\kappa^2 = \mu^2 \rightarrow \kappa = \mu)$. (Tarski 1924; 140)
- CN 8: $(\forall \kappa, \mu) (\kappa^* = \mu^* \rightarrow \kappa = \mu)$. (Sobocinski 1961b; 140)
- CN 9: $(\forall \kappa \geq \aleph_0) \neg (\exists \mu) (\kappa^* < \mu < \kappa + \kappa^*)$. (Sobocinski 1961c; 140)
- CN 10: $(\forall \kappa, \mu) (\kappa^* < \mu^* \rightarrow \kappa < \mu)$. (Sobocinski 1961b; 140)
- CN'10: $(\forall \kappa) (\forall \alpha) (\kappa^* < \aleph_{\alpha+1} \rightarrow \kappa < \aleph_\alpha)$. (Sobocinski 1961b; 140)

- CN 11: $(\forall \kappa, \mu) (\kappa < \mu \rightarrow \kappa^* < \mu^*)$. (Sobocinski 1961b; 140)
- CN'11: $(\forall \alpha) (\forall \mu) (\aleph_\alpha < \mu \rightarrow \aleph_{\alpha+1} < \mu^*)$. (Sobocinski 1961b; 140)
- CN 12: $(\forall \kappa \geq \aleph_0) (\forall \mu, \nu) [\kappa + \mu = \kappa + \nu \rightarrow (\mu = \nu \vee (\mu \leq \kappa \wedge \nu \leq \kappa))]$. (Tarski 1926; 140)
- CN'12: $(\forall \alpha) (\forall \mu, \nu) [\aleph_\alpha + \mu = \aleph_\alpha + \nu \rightarrow (\mu = \nu \vee (\mu \leq \aleph_\alpha \wedge \nu \leq \aleph_\alpha))]$. (Tarski 1926; 140)
- CN 13: $(\forall \kappa > 0) (\forall \mu, \nu) [\kappa \cdot \mu = \kappa \cdot \nu \rightarrow (\mu = \nu \vee (\mu \leq \kappa \wedge \nu \leq \kappa))]$. (Tarski 1926; 140)
- CN'13: $(\forall \alpha) (\forall \mu, \nu) [\aleph_\alpha \cdot \mu = \aleph_\alpha \cdot \nu \rightarrow (\mu = \nu \vee (\mu \leq \aleph_\alpha \wedge \nu \leq \aleph_\alpha))]$. (Tarski 1926; 140)
- CN 14: $(\forall \kappa \geq \aleph_0) (\forall \mu) (\kappa^2 = \kappa \cdot \mu \rightarrow \mu \leq \kappa)$. (Tarski 1926; 140)
- CN 15: $(\forall \kappa_1, \kappa_2, \mu_1, \mu_2) (\kappa_1 < \kappa_2 \wedge \mu_1 < \mu_2 \rightarrow \kappa_1 + \mu_1 < \kappa_2 + \mu_2)$ (Tarski 1924; 141)
- CN'15: $(\forall \alpha) (\forall \kappa, \mu_1, \mu_2) (\aleph_\alpha < \kappa \wedge \mu_1 < \mu_2 \rightarrow \aleph_\alpha + \mu_1 < \kappa + \mu_2)$. (Tarski 1924; 141)
- CN 16: $(\forall \kappa_1, \kappa_2, \mu_1, \mu_2) (\kappa_1 < \kappa_2 \wedge \mu_1 < \mu_2 \rightarrow \kappa_1 \cdot \mu_1 < \kappa_2 \cdot \mu_2)$. (Tarski 1924; 141)
- CN'16: $(\forall \alpha) (\forall \kappa, \mu_1, \mu_2) (\aleph_\alpha < \kappa \wedge \mu_1 < \mu_2 \rightarrow \aleph_\alpha \cdot \mu_1 < \kappa \cdot \mu_2)$. (Tarski 1924; 141)
- CN 17: $(\forall \kappa, \mu, \nu) (\kappa + \mu < \kappa + \nu \rightarrow \mu < \nu)$. (Tarski 1924; 141)
- CN'17: $(\forall \alpha) (\forall \nu) (\aleph_\alpha < \aleph_\alpha + \nu \rightarrow \aleph_\alpha < \nu)$. (Tarski 1924; 141)
- CN 18: $(\forall \kappa, \mu, \nu) (\kappa \cdot \mu < \kappa \cdot \nu \rightarrow \mu < \nu)$. (Tarski 1924; 141)
- CN'18: $(\forall \alpha) (\forall \nu) (\aleph_\alpha < \aleph_\alpha \cdot \nu \rightarrow \aleph_\alpha < \nu)$. (Tarski 1924; 141)
- CN 19: $(\forall \kappa, \mu) (\kappa \cdot \mu \leq \kappa^2 \rightarrow \mu \leq \kappa)$. (Tarski 1924; 141)
- CN 20: $(\forall \kappa \geq \aleph_0) (\forall \mu, \nu) (\kappa < \nu \wedge \mu < \nu \rightarrow \kappa + \mu \neq \nu)$. (Tarski 1926; 141)
- CN'20: $(\forall \alpha) (\forall \mu, \nu) (\aleph_\alpha < \nu \wedge \mu < \nu \rightarrow \aleph_\alpha + \mu \neq \nu)$. (Tarski 1926; 141)
- CN 21: $(\forall \kappa \geq \aleph_0) (\forall \mu, \nu) (\kappa < \nu \wedge \mu < \nu \rightarrow \kappa \cdot \mu \neq \nu)$. (Sierpinski 1946; 141)

- CN'21: $(\forall \alpha) (\forall \mu, \nu) (\aleph_\alpha < \nu \wedge \mu < \nu \rightarrow \aleph_\alpha \cdot \mu \neq \nu)$. (Sierpinski 1946; 141)
- CN 22: $(\forall \kappa, \mu) (\kappa < \mu \rightarrow \mu - \kappa \text{ exists})$. (Tarski 1926, Sierpinski 1947; 145)
- CN'22: $(\forall \alpha) (\forall \mu) (\aleph_\alpha < \mu \rightarrow \mu - \aleph_\alpha \text{ exists})$. (Tarski 1926; 145)
- CN 23: $(\forall \kappa) (\forall \mu \geq \aleph_0) (\kappa < \mu \rightarrow \mu - \kappa = \mu)$. (Tarski 1926; 145)
- CN'23: $(\forall \alpha) (\forall \mu) (\aleph_\alpha < \mu \rightarrow \mu - \aleph_\alpha = \mu)$. (Tarski 1926; 145)
- CN 24: $(\forall \kappa_1, \kappa_2) (\forall \mu_1, \mu_2 \geq \aleph_0) [(\kappa_1 < \mu_1 \wedge \kappa_2 < \mu_2 \wedge \kappa_1 + \mu_2 = \kappa_2 + \mu_1) \rightarrow \mu_1 = \mu_2]$. (Sobocinski 1965; 145)
- CN'24: $(\forall \alpha) (\forall \kappa) (\forall \mu_1, \mu_2 \geq \aleph_0) [(\aleph_\alpha < \mu_1 \wedge \kappa < \mu_2 \wedge \aleph_\alpha + \mu_2 = \kappa + \mu_1) \rightarrow \mu_1 = \mu_2]$. (Sobocinski 1965; 145)
- CN 25: $(\forall \kappa > 0) (\forall \mu \geq \aleph_0) [\kappa < \mu \rightarrow (\exists \nu) (\mu = \kappa \cdot \nu)]$. (Tarski, Sierpinski; 145)
- CN'25: $(\forall \alpha) (\forall \mu) [\aleph_\alpha < \mu \rightarrow (\exists \nu) (\mu = \aleph_\alpha \cdot \nu)]$. (Tarski, Sierpinski; 145)
- CN 26: $(\forall \kappa > 0) (\forall \mu \geq \aleph_0) (\kappa < \mu \rightarrow \mu = \kappa \cdot \mu)$. (Tarski, Sierpinski; 145)
- CN'26: $(\forall \alpha) (\forall \mu) (\aleph_\alpha < \mu \rightarrow \mu = \aleph_\alpha \cdot \mu)$. (Tarski, Sierpinski; 145)
- CN 27: $(\forall \kappa > 0) (\forall \mu \geq \aleph_0) (\kappa < \mu \rightarrow \mu \div \kappa \text{ exists})$. (Tarski, Sierpinski; 145)
- CN'27: $(\forall \alpha) (\forall \mu) (\aleph_\alpha < \mu \rightarrow \mu \div \aleph_\alpha \text{ exists})$. (Tarski, Sierpinski; 145)
- CN 28: $(\forall \kappa > 0) (\forall \mu \geq \aleph_0) (\kappa < \mu \rightarrow \mu \div \kappa = \mu)$. (Tarski, Sierpinski; 145)
- CN'28: $(\forall \alpha) (\forall \mu) (\aleph_\alpha < \mu \rightarrow \mu \div \aleph_\alpha = \mu)$. (Tarski, Sierpinski; 145)
- CN 29: $(\forall \kappa_1, \kappa_2) (\forall \mu_1, \mu_2 \geq \aleph_0) [(0 < \kappa_1 < \mu_1 \wedge \kappa_2 < \mu_2 \wedge \kappa_1 \cdot \mu_1 = \kappa_2 \cdot \mu_2) \rightarrow \mu_1 = \mu_2]$. (Sobocinski 1965; 145)
- CN'29: $(\forall \alpha) (\forall \kappa) (\forall \mu_1, \mu_2 \geq \aleph_0) [(\aleph_\alpha < \mu_1 \wedge \kappa < \mu_2 \wedge \aleph_\alpha \cdot \mu_1 = \kappa \cdot \mu_2) \rightarrow \mu_1 = \mu_2]$. (Sobocinski 1965; 145)

- CN 30: $(\forall \kappa_1, \kappa_2) (\forall \mu_1, \mu_2 \geq \aleph_0) [(0 < \kappa_1 < \mu_1 \wedge \kappa_2 < \mu_2 \wedge \kappa_1 \cdot \mu_2 = \kappa_2 \cdot \mu_1) \rightarrow \mu_1 = \mu_2]$. (Sobocinski 1965; 146)
- CN'30: $(\forall \alpha) (\forall \kappa) (\forall \mu_1, \mu_2 \geq \aleph_0) [(\aleph_\alpha < \mu_1 \wedge \kappa < \mu_2 \wedge \aleph_\alpha \cdot \mu_2 = \kappa \cdot \mu_1) \rightarrow \mu_1 = \mu_2]$. (Sobocinski 1965; 146)
- CN 31: $(\forall \kappa \geq \aleph_0) (\exists \mu) (\kappa = \mu^2)$. (Tarski, Sierpinski; 147)
- CN 32($\aleph_\alpha$): $(\forall \kappa) (\aleph_\alpha \cdot \kappa = \aleph_\alpha \cdot \kappa^2)$. (Alas 1968; 147)
- CN 33: $(\exists \alpha) (\forall \kappa \geq \aleph_0)$ (There is no $\leq$ -well ordered set M of cardinal numbers between κ and κ^2 such that $M \cong \alpha$.) (Tarski 1924; Rubin 1963; 147)
- CN 34: $(\exists \alpha) (\forall \kappa, \mu) (\kappa^* = \mu^* \rightarrow$ There is no $\leq$ -well ordered set M of cardinal numbers between κ and μ such that $M \cong \alpha$.) (Sobocinski 1962a, Rubin 1985; 147)
- CN 35: $(\forall \kappa, \mu \geq \aleph_0) [\kappa \leq \mu < 2^\kappa \wedge (\exists v_1) (\kappa = v_1^2) \rightarrow (\exists v_2) (\mu = v_2^2)]$. (Seki 1961; 147)
- CN 36: $(\forall \kappa, \mu \geq \aleph_0) [\kappa \leq \mu < 2^\kappa \rightarrow (\exists v) (\mu = \kappa \cdot v)]$. (Seki 1961; 147)
- CN 37: $(\forall \kappa \geq \aleph_0) (\exists \mu) (\kappa <_2 \mu)$. (Tarski 1954; 150)
- CN'37: $(\forall \alpha) (\exists \mu) (\aleph_\alpha <_2 \mu)$. (Tarski 1954; 150)
- CN 38: $(\forall \kappa, \mu \geq \aleph_0) [\kappa <_1 \mu \rightarrow (\forall v) (v \cdot \kappa = v \cdot \mu \vee v \cdot \kappa <_1 v \cdot \mu)]$. (Sudan 1938; 150)
- CN'38: $(\forall \alpha) (\forall v) (v \cdot \aleph_\alpha = v \cdot \aleph_{\alpha+1} \vee v \cdot \aleph_\alpha <_1 v \cdot \aleph_{\alpha+1})$. (Sudan 1938; 150)
- CN 39: $(\forall \kappa \geq \aleph_0) (\forall \mu, v) (\kappa < \mu \wedge \kappa <_3 v + v \leq \mu)$. (Sobocinski 1962b; 150)
- CN'39: $(\forall \kappa \geq \aleph_0) (\kappa^* < \kappa^* + \kappa \rightarrow \kappa^{**} \leq \kappa^* + \kappa)$. (Sobocinski 1962b; 150)

Let $v > 1$ be a cardinal,

- CN 40(v): $(\forall \kappa, \mu) (\kappa^v < \mu^v \rightarrow \kappa < \mu)$. (Tarski 1926; 151)
- CN 41: $(\exists \mu > 1) (\forall \kappa_1, \kappa_2) [(\forall v) (1 < v \leq \mu \rightarrow \kappa_1^v < \kappa_2^v) \rightarrow \kappa_1 < \kappa_2]$. (Tarski 1926; 151)

- CN 42: $(\forall \kappa, \mu, \nu) (\kappa \neq 0 \wedge \kappa^\mu < \kappa^\nu \rightarrow \mu < \nu)$. (Tarski 1926; 151)
- CN'42: $(\forall \kappa, \mu) (\forall \alpha) (\kappa \neq 0 \wedge \kappa^\mu < \kappa^{\aleph_\alpha} \rightarrow \mu < \aleph_\alpha)$. (Tarski 1926; 151)
- CN 43: $(\forall \kappa, \mu)$ [If $\kappa \cap \mu$ exists then $\kappa \cup \mu$ exists and $\kappa \cdot \mu = (\kappa \cap \mu) \cdot (\kappa \cup \mu)$]. (Tarski 1949, Rubin 1963; 153)
- CN'43: $(\forall \kappa) (\forall \alpha)$ [If $\kappa \cap \aleph_\alpha$ exists then $\kappa \cup \aleph_\alpha$ exists and $\kappa \cdot \aleph_\alpha = (\kappa \cap \aleph_\alpha) \cdot (\kappa \cup \aleph_\alpha)$]. (Tarski 1949, Rubin 1963; 153)
- CN 44: $(\forall \kappa, \mu \geq \aleph_0) [e(\kappa + \mu) = e(\kappa) + e(\mu)]$. (Alas 1968, 1970; 154)
- CN'44: $(\forall \kappa \geq \aleph_0) (\forall \alpha) [e(\kappa + \aleph_\alpha) = e(\kappa) + \aleph_\alpha]$. (Alas 1968, 1970; 154)
- CN 45: $(\forall \kappa, \mu \geq \aleph_0) (e(\kappa) = e(\mu) \wedge \kappa \leq \mu \rightarrow \kappa = \mu)$. (Alas 1968, 1970; 154)
- CN 46: $(\forall \kappa, \mu \geq \aleph_0) [e(\kappa + \mu) = e(\kappa) \vee e(\kappa + \mu) = e(\mu)]$. (Alas 1968, 1970; 154)
- CN'46: $(\forall \kappa \geq \aleph_0) (e(\kappa + [e(\kappa)]^*) = [e(\kappa)]^*)$. (Alas 1968, 1970; 154)
- CN 47: $(\forall s \neq \emptyset) (\exists t) (t = \{x \subseteq t: |s| \not\leq |x|\})$. (Tarski 1938, 1939; 155)
- CN 48: $(\forall s \neq \emptyset) (\exists t) (t \approx \{x \subseteq t: |s| \not\leq |x|\})$. (Tarski 1938, 1939; 155)
- CN 49: $(\forall t) (\forall \mu) [|t| \geq \aleph_0 \wedge \mu \leq |t| \rightarrow |\{x \subseteq t: \mu \not\leq |x|\}| = |t|^\mu]$. (Tarski 1938, 1939; 155)
- CN 50: $(\forall x) [|x| < |\bigcup_{i \in \omega} A_i| \rightarrow (\exists n \in \omega) (|x| \leq |\bigcup_{i \leq n} A_i|)]$. (Tarski 1926, 1948; 158)
- CN 51: If J is a set and A and B are functions with domain J then
 $(\forall j \in J) (|A_j| < |B_j|) \rightarrow (|\bigcup_{j \in J} A_j| < |\bigcup_{j \in J} B_j|)$.
 (König 1905, Zermelo 1908; 158)
- CN 52: If J is a set, A is a function with domain J and for each $j \in J$, $2 \leq |A_j|$ then
 $|\bigcup_{j \in J} A_j| \leq |\bigcup_{j \in J} A_j|$ (König 1905, A.L. Rubin; 159)

FORMS FROM LOGIC

- LOG 1: A formula having a model in a set of cardinality κ also has a model in a set of cardinality μ if $\aleph_0 \leq \mu \leq \kappa$. (Vaught 1956, Tarski 1957, Malcev 1936, Henkin 1949; 163)
- LOG 2: A formula having a model in a set of cardinality $\aleph_0$ also has a model in a set of cardinality κ , for all $\kappa \geq \aleph_0$. (Vaught 1956, Tarski 1957, Malcev 1936, Henkin 1949; 163)
- LOG 3: If Q is a set of formulas in which the set of individual constants has cardinality κ and every finite subset of Q has a model, then Q has a model in a set whose cardinality is not greater than $\kappa + \aleph_0$. (Vaught 1956, Tarski 1957, Malcev 1936, Henken 1949; 163)
- LOG 4: If A is a set of sentences of first-order logic and $B \subseteq A$ is a consistent subset, then there exists a $\subseteq$ -maximal consistent set C such that $B \subseteq C \subseteq A$. (Klimovski 1956; 165)

FORMS FROM TOPOLOGY

- TOP 1: The product of compact spaces is compact in the product topology. (Tychonoff 1935, Kelley 1950; 167)
- TOP 2: The product of mutually homeomorphic compact spaces is compact in the product topology. (Tychonoff 1935, Ward 1962; 167)
- TOP 3: The product of a set of mutually homeomorphic topological spaces $\{\langle X_i, T_i \rangle : i \in I\}$, where for each $i \in I$, T_i has three elements, is compact in the product topology. (Tychonoff 1935, Alas 1969; 168)
- TOP 4: Any perfect onto map from one topological space to another has an irreducible restriction to a closed subspace of its domain. (Franklin & Thomas 1971; 169)

- TOP 5(P): Each T_2 (Hausdorff) space has the property P , where P is any one of the properties given in the list on pp 169-170. (Brunner 1983; 169)
- TOP 6(P,Q): Each T_2 space which has the property P also has the property Q , where P and Q are chosen from the list on pp 169-70 and (i) $Q = W_1$, $P \neq W_1$; (ii) $Q = W_2$, $P \neq W_1, W_2$; (iii) $Q = D_1$, $P \neq W_1, W_2, D_1$; (iv) $Q = D_2$, $P = B_1, B_2, B_3, L_2, A_2$; (v) $Q = B_1$, $P \neq W_1, W_2, B_1, S, C$; (vi) $Q = B_2, B_3$, $P = D_2, L_1, L_2, A_1, A_2$; (vii) $Q = L_1, L_2$, $P = D_1, D_2, B_2, B_3, A_2$; (viii) $Q = S, C$, $P = D_1, D_2, B_2, B_3, A_2$; (ix) $A = A_1$, $P = D_1, D_2, B_2, B_3, A_2$; and (x) $Q = A_2$, $P = D_2$. (Brunner 1983; 169)
- B 2': Each topological space X has the property that each point of X has a neighborhood base which can be well ordered and the well ordering is compatible with set inclusion. (Howes 1980, Vaughan 1975; 176)

FORMS FROM ANALYSIS

- AN 1: The unit sphere in the dual of a normed vector space over the reals has an extreme point. (Bell & Fremlin 1972; 179)
- AN 2: Let X be a real vector space, Y a subspace and S a subset of X . Suppose $p: X \rightarrow \mathbb{R}$ is a sublinear functional on X and $f: Y \rightarrow \mathbb{R}$ a linear functional such that $f(y) \leq p(y)$ for all $y \in Y$. Then the set Z of all p -dominated linear extensions of f to X has an element g such that for all $h \in Z$ and $g(s) \leq h(s)$ for all $s \in S$, we have $g(s) = h(s)$ for all $s \in S$. That is, g is S -maximal in Z . (Andenaes 1970, Lembcke 1979; 179)

- AN 3: Let X be a real vector space and Y a subspace. Suppose $p: X \rightarrow \mathbb{R}$ is a sublinear functional on X and $f: Y \rightarrow \mathbb{R}$ a linear functional such that $f(y) \leq p(y)$ for all $y \in Y$. Then the set Z of all p -dominated linear extensions of f to X has an extreme point. (Bonsall 1954, 1956, Lembcke 1979; 179)
- AN 4: Let X and Y be T_2 spaces, $f: X \rightarrow Y$ a continuous map, and μ an (inner) regular Borel measure on Y . If μ is inner regular with respect to the system of sets $\{f^{-1}K: K \text{ is a compact subset of } Y\}$, then there is a regular Borel measure ν on X such that $\mu = \nu f$. (Lembcke 1970, 1977, 1979; 181)
- AN 5: Let X be a real vector lattice with a distinguished order-unit (denoted by 1), Y a sublattice of X , and Z the set of all positive linear functionals F on X such that $F(1) = 1$. Then there exists an extreme point M of the convex set Z which is Y -minimal in Z . (Edwards 1975; 183)
- AN 6: If $\langle L, \leq \rangle$ is a Boolean algebra with identity 1 , and K is a sublattice of L then there exists a two-valued K -minimal measure μ on L such that $\mu(1) = 1$. (Lembcke 1970, 1977, 1979; 184)

Each of the following is equivalent to the axiom of choice:

AL + KM, ALC + KMC, BPI + KM, and HB + KMC. (Bell & Fremlin 1972; 181)

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LIST OF THE CLASS FORMS

In this list, in parentheses following each statement we shall give the page number where it occurs in the text.

THE WELL ORDERING THEOREM

- CWO 1: There is a function F such that for each set x , $F(x)$ well orders x . (187)
- CWO 2: There is a relation R such that every set is well ordered by R . (187)
- CWO 3: Every class can be well ordered. (Equivalently, the universe, V , can be well ordered.) (187)
- CWO 4: There is a relation R which well orders V in such a way that each proper initial R -segment of V is a set. (187)
- CWO 5: Each proper class is equipollent to On . (Equivalently, $V \approx On$.) (187)

Let $m \in \omega$ such that $m \geq 1$.

- CWO 6(m): For each class X there is a function F defined on On such that for each $\alpha \in On$, $F(\alpha) \prec m$ and $\bigcup_{\alpha \in On} F(\alpha) = X$. (187)
- CWO 7: There is an $m \in \omega \sim \{0\}$ such that CWO 6(m). (187)
- CWO 8: For each class X there exists an $m \in \omega \sim \{0\}$ and a function F defined on On such that for each $\alpha \in On$, $F(\alpha) \prec m$ and $\bigcup_{\alpha \in On} F(\alpha) = X$. (187)

- CWO 9: Every class X is the union of a well ordered class of finite sets. (187)
- CWO 10: Every proper class can be mapped onto V . (For every proper class X , $V \prec^* X$.) (188)
- CT: Each pair of proper classes are equipollent. (188)
- CWO 11: For each proper class X , $\mathcal{P}(X) \prec X$. (188)
- CWO 12: $\mathcal{P}(\text{On})$ can be well ordered. (188)
- CPW: The power class of a well ordered class can be well ordered. (188)
- CLW: Every linearly ordered class can be well ordered. (188)
- CCW: Every class on which there is a choice function can be well ordered. (188)

THE AXIOM OF CHOICE

- CAC 1: If S is a class of non-empty sets, there is a function F such that for each $x \in S$, $F(x) \in x$. (191)
- E: There exists a function F such that for each non-empty set x , $F(x) \in x$. (191)
- CAC 2: If T is a pairwise disjoint class of non-empty sets, there is a class C which consists of one and only one element from each set in T . (191)
- CAC 3: For every function F there is a function G such that for every x , if $x \in \mathcal{D}(F)$ and $F(x) \neq \emptyset$ then $G(x) \in F(x)$. (191)
- CAC 4: For every relation R there is a function F such that $\mathcal{D}(F) = \mathcal{D}(R)$ and $F \subseteq R$. (192)
- CAC 5: For every function F there is a function G such that $\mathcal{D}(G) = \mathcal{R}(F)$ and for every $x \in \mathcal{D}(G)$, $F(G(x)) = x$. (192)

CAC 8: If S is a class of pairs of equipollent sets then there is a function which associates with each pair a bijection mapping one onto the other. (192)

CAC 9: Let R be a relation such that for each $x, y \in \mathcal{D}(R)$, $R''\{x\} \approx R''\{y\}$; then there exists a relation S such that for each $x, y \in \mathcal{D}(R)$, $S''\{\langle x, y \rangle\}$ is a bijection mapping $R''\{x\}$ onto $R''\{y\}$. (192)

Let $n \geq 2$ be a natural number.

CAC 10(n): If S is a class of infinite sets, there is a function F such that for each $x \in S$, $F(x)$ is a decomposition of x into a set of disjoint sets of size between 2 and n . (192)

CAC 11: There exists a natural number $n \geq 2$ such that CAC 10(n). (192)

CAC 12: If S is a class of infinite sets there is a natural number $n \geq 2$ and a function F such that for each $x \in S$, $F(x)$ is a decomposition of x into a set of disjoint sets of size between 2 and n . (192)

Let $n \geq 2$ be a natural number.

CAC 10E(n): If S is a class of infinite sets, there is a function F such that for each $x \in S$, $F(x)$ is a decomposition of x into a set of disjoint sets of size exactly n . (193)

CAC 11E: There exists a natural number $n \geq 2$ such that CAC 10E(n). (193)

CAC 12E: If S is a class of infinite sets there is a natural number $n \geq 2$ and a function F such that for each $x \in S$, $F(x)$ is a decomposition of x into a set of disjoint sets of size exactly n . (193)

Let $m \geq 1$ be a natural number.

CAC 13(m): If S is a class of non-empty sets there is a function F such that for each $x \in S$, $F(x)$ is a non-empty subset of x with $F(x) \prec m$. (193)

- CAC 14: There is a natural number $m \geq 1$ such that CAC 13(m).
(193)
- CAC 15: If S is a class of non-empty sets then there is a natural number $m \geq 1$ and a function F such that for each $x \in S$, $F(x)$ is a non-empty subset of x with $F(x) \leq m$. (193)
- CAC 18: For any non-empty class A , and for every function B whose domain is A and B_a is a non-empty set for each $a \in A$,
- $$x \in \bigcap_{a \in A} \bigcup_{b \in B_a} X_{a,b} \leftrightarrow \text{There is a function } F \text{ with}$$
- domain A such that $(\forall a \in A)[F(a) \in B_a \wedge x \in X_{a,F(a)}]$.
(193)
- CAC 18C: For any non-empty class A , for every class-valued function B whose domain is A ,
- $$x \in \bigcap_{a \in A} \bigcup_{b \in B[a]} X_{a,b} \leftrightarrow \text{There is a function } F \text{ with}$$
- domain A such that $(\forall a \in A)[F(a) \in B[a] \wedge$
 $x \in X_{a,F(a)}]$. (193)
- CAC 19: For any non-empty class A , each of whose elements is non-empty,
- $$x \in \bigcap_{a \in A} \bigcup_{b \in a} b \leftrightarrow \text{There exists a function } F \text{ with domain}$$
- A such that $(\forall a \in A)[F(a) \in a \wedge x \in F(a)]$. (194)
- CAC 22: If S is a class of non-empty sets then there is a function F such that for each $x \in S$, $F(x)$ is a non-empty finite subset of x . (194)
- CAC 23: There exists a function F such that for each non-empty set x , $F(x)$ is a non-empty finite subset of x . (194)

CAC 24: If S is a class of infinite sets, then there is a function F such that for each $x \in S$, $F(x)$ is a partition of x into pairwise disjoint finite sets, each of which has at least 2 elements. (194)

CH: There is a formula ψ with at least three free variables such that

$$(\exists X)(\forall Y)(Y \neq \emptyset \rightarrow (\exists Z)\psi(Z, X, Y) \in Y). \quad (198)$$

MAXIMAL PRINCIPLES

CM 1: If R is a partial ordering relation on a non-empty class X and if every subclass of X which is linearly ordered by R has an R -upper bound, then X has an R -maximal element. (203)

CM 2: If R is a partial ordering relation on a non-empty class X and if every subclass of X which is well ordered by R has an R -upper bound, then X has an R -maximal element. (203)

CM 3: If every non-empty nest which is a subclass of a non-empty class X has its union an element of X , then X has a $\subseteq$ -maximal element. (203)

CM 4: If every non-empty well ordered nest which is a subclass of a non-empty class X has its union an element of X , then X has a $\subseteq$ -maximal element. (203)

CMS 1(2): If R is a partial ordering relation on a non-empty class X and if every subset of X which is linearly ordered (well ordered) by R has an R -upper bound then either X has an R -maximal element or X has a subclass which is a proper class and which is linearly ordered (well ordered) by R . (203)

CMS 3(4): If every non-empty nest (well ordered nest) which is a subset of a non-empty class X has its union an element of X , then either X has a $\subseteq$ -maximal element or there is a subclass of X which is a nest (well ordered nest) and a proper class. (204)

CM 1(Q,U): Every non-empty class which is Q-ordered by R in which each U-ordered subclass has an R-upper bound, has an R-maximal element, where $Q = F, R, P,$ or TR , and $U = W, L, AS\&C, TR\&C, C,$ or D ; $Q = D$ and $U = W$. (204)

CMS 1(Q,U): Every non-empty class which is Q-ordered by R in which each U-ordered subset has an R-upper bound, either has an R-maximal element or contains a U-ordered subclass which is a proper class, where $Q = TR, P, RA,$ or F and $U = C, AS\&C, TR\&C, L, D,$ or W ; $Q = D$ and $U = W$. (204)

CM 5: If R is a partial ordering on X, then there exists a $\subseteq$ -maximal subclass of X which is linearly ordered by R. (205)

CM 5(Q,U): Every class which is Q-ordered by R contains a $\subseteq$ -maximal subclass which is U-ordered by R, where $Q = A$ or TR and $U = AS, C, AS\&C, TR\&C, P, L, D,$ or RA ; $Q = AS$ or P and $U = C, AS\&C, TR\&C, L, D,$ or RA ; $Q = C$ and $U = AS, AS\&C, P, L, D,$ or RA ; $Q = D$ and $U = C, AS\&C, TR\&C, L,$ or RA ; $Q = RA$ and $U = C, AS\&C, TR\&C, L,$ or D ; or $Q = F$ and $U = C, AS\&C, TR\&C, L, D,$ or W . (205)

CM 6: For every class X, there exists a $\subseteq$ -maximal subclass of X which is a nest. (205)

CM 7: For every class X and every property P of finite character, there exists a $\subseteq$ -maximal subclass of X which has the property P. (205)

CM 8(A): Every class has a $\subseteq$ -maximal subclass with the property A, where $A = D, K, J, \bar{K}, \bar{J}$ or $D_m, m \geq 2$. (205)

CM 9(A): Every class can be linearly ordered and CM 8(A), where $A = D, K, J, \bar{K}, \bar{J},$ or $D_m, m \geq 2$. (205)

CM 10: There exists a natural number $m \geq 2$ such that CM 8(D_m). (205)

CMR 11: For every class X there exists a natural number $m \geq 2$ such that X has a $\subseteq$ -maximal subclass which has the property D_m . (205)

CMR 8(A): For every relation R and class X there exists a $\subseteq$ -maximal subclass $Y \subseteq X$ such that for all $x, y \in Y$, if $x \neq y$ then $R\{x\} A R\{y\}$, where $A = D, \bar{D}, J, \bar{J}, K, \bar{K}, D_m, \bar{D}_m, \bar{D}_m!, U, \bar{U}, DK, DJ, DU, KJ, KU, \text{ or } JU$. (218)

CMR 9(A): The universe can be linearly ordered and for every relation R and class X there exists a $\subseteq$ -maximal subclass $Y \subseteq X$ such that for all $x, y \in Y$, if $x \neq y$ then $R\{x\} A R\{y\}$, where $A = D, \bar{D}, J, \bar{J}, K, \bar{K}, D_m, \bar{D}_m, \bar{D}_m!, U, \bar{U}, DK, DJ, DU, KJ, KU, \text{ or } JU$. (218)

CMR 10: There exists a natural number $m \geq 2$ such that $CMR\ 8(D_m)$. (219)

CMR 11: For every relation R and class X there is a natural number $m \geq 2$ and a $\subseteq$ -maximal subclass $Y \subseteq X$ such that for all $y_1, y_2, \dots, y_m \in Y$, y_i 's distinct,

$$\bigcap_{i=1}^m R\{y_i\} = \emptyset. \quad (219)$$

CMR 12: There exists a natural number $m \geq 2$ such that $CMR\ 8(\bar{D}_m!)$. (219)

CMR 13: For every relation R and class X there exists a natural number $m \geq 2$ and a $\subseteq$ -maximal subclass $Y \subseteq X$ such that for all $y_1, y_2, \dots, y_m \in Y$,

$$\bigcap_{i=1}^m R\{y_i\} \neq \emptyset. \quad (219)$$

CMR 14: If X is a class and R a relation there exists a $\subseteq$ -maximal subclass $Y \subseteq X$ such that for all natural numbers $m \geq 2$ and for all $y_1, y_2, \dots, y_m \in Y$,

$$\bigcap_{i=1}^m R\{y_i\} \neq \emptyset. \quad (219)$$

- CM 15(A:B): Any class which contains a $\subseteq$ -maximal subclass with the property A contains a $\subseteq$ -maximal subclass with the property B, where $A \neq B$ and $B \neq \bar{D}$. (The possible values for A and B are D, $\bar{D}$, J, $\bar{J}$, K and $\bar{K}$). (230)
- CMR 15(A:B): For any relation R and class X, if there exists a $\subseteq$ -maximal subclass $Y \subseteq X$ such that $A_R[Y]$ then there exists a $\subseteq$ -maximal subclass $Z \subseteq X$ such that $B_R[Z]$, where $A \neq B$ and it is not the case that $A = JK$ and $B = \bar{D}$, or $A = JK$ and $B = \bar{U}$. (The possible values for A and B are D, $\bar{D}$, J, $\bar{J}$, K, $\bar{K}$, U, $\bar{U}$, DJ, DK, DU, JK, JU, and KU.) (230)
- CM 16(A:B): Every class can be linearly ordered and CM 15(A:B), where A and B have the same values as in CM 15(A:B). (230)
- CMR 16(A:B): Every class can be linearly ordered and CMR 15(A:B), where A and B have the same values as in CMR 15(A:B). (230)
- CM 17: Every class X contains a $\subseteq$ -maximal subclass Y such that for all $s, t \in Y$, $s \neq t$, either $s \in t$ or $t \in s$. (205)
- CA: Every partially ordered class contains a $\subseteq$ -maximal anti-chain. (205,212)

LIST OF FORMS RELATED TO THE AXIOM OF CHOICE

In the text we mention a number of statements related to the axiom choice, some weaker, some stronger and some whose relative strength is unknown. We shall list them in the six categories described below.

I. This list of statements has two parts, (a) those statements S such that $AC \rightarrow S$ in NBG and $S \rightarrow AC$ in NBG° ; and (b) those statements S such that $AC \rightarrow S$ in NBG° and $S \rightarrow AC$ in NBG. Each of these statements have been discussed in the text, so here we shall give its name followed in parentheses by the page number where it occurs in the text.

- (a)
 - 1. INJ (73)
 - 2. PROJ (74)
 - 3. M 19 (87)
 - 4. AC 20 (88)
 - 5. AC 21 (88)
 - 6. $(\forall \alpha)DCC(\alpha)$ (89)
- (b)
 - 1. PW (73)
 - 2. LW (73)
 - 3. MC 1 (74)
 - 4. MC 2 (74)
 - 5. MC 3 (74)
 - 6. M 8($\bar{K}$) (74)
 - 7. M 15($A:\bar{K}$), for $A = D, \bar{D}, J, \bar{J}, K$. (77)
 - 8. AC_Q^P , for $P = \emptyset, Q = LO, DLO$; $P = LO, Q = \emptyset, LO, DLO$. (78)

9. MC_Q^P , for $P = \emptyset, Q = LO, DLO; P = LO, Q = \emptyset, LO, DLO.$
(78)
10. ORD (78)
11. W-ORD (78)
12. TCL (78)
13. INF 2 (84)
14. U_Q^P , for $P = LO, Q = LO, WO; P = WO, Q = LO.$ (85)
15. $M\ 5(U, VC)$, for $U = AS \& TR, AS, TR.$ (86)
16. $M\ 5(C, UN)$ (86)
17. $M\ 5(U, VC)$ for $U = RA, D.$ (86)
18. T 11 (90)
19. T 12 (90)
20. T 13 (90)
21. B (119)
22. AL 20(F) (119)
23. AL 21(F) (119)
24. B' (121)
25. AL 25 (128)
26. TOP 6(P,Q), where $P = D\ 2, L\ 1, L\ 2, A\ 1$ or $A\ 2$
and $Q = D\ 1, B\ 2$ or $B\ 3$; and $P = D\ 2$ and $Q = A\ 2.$
(169,172)
27. TOP 5(Q), where $Q = D\ 1, B\ 2, B\ 3,$ or $A\ 2.$ (169,172)
28. TOP 6(P,Q), where $P = W\ 2, B\ 1,$ or S and $Q = W\ 1$;
and $P = B\ 1$ and $Q = W\ 2, D\ 1$ or $D\ 2.$ (169,173)
29. TOP 6(S,W 2) (169,173)
30. TOP 6(P,Q), where $P = B\ 2, B\ 3, S,$ or C and $Q = D\ 1$;
and $P = C$ and $Q = W\ 1$ or $W\ 2.$ (169,174)

31. TOP 5(D 2) (169,174)

32. AN 4 (181)

II. In this set we shall list statements S such that AC implies S in NBG° , but it is not known if S implies AC in NBG . In parentheses following each statement we shall give the page number where it is mentioned in the text.

1. (The Nielson-Schreier Theorem). Every subgroup of a free group is free. (xiii)
2. (Basis Theorems for Vector Spaces).
 - (a) There exists a field F such that every vector space over F has a basis. (125)
 - (b) There exists a field F such that for every vector space V over F , if B is a basis of V and X any linearly independent subset of V , then there is a subset $B' \subseteq B$ such that $X \cap B' = \emptyset$ and $X \cup B'$ is a basis of V . (125)
3. For all sets x and y either $\mathcal{P}(x) \prec \mathcal{P}(y)$ or $\mathcal{P}(y) \prec \mathcal{P}(x)$. (xiii)
4. For all infinite cardinals κ , $\kappa^2 = \kappa^3$. (xiii)
5. (The Partition Principle). For each set x and partition P of x , $P \prec x$. (xiii,7)
6. ($T_3(\kappa)$, for κ an infinite cardinal). If κ is an infinite cardinal, then for every set of cardinals x such that $|x| = \kappa$ there exist $\mu_1, \mu_2 \in x$ such that $\mu_1 < \mu_2$ or $\mu_2 < \mu_1$. (22)
7. ($M_5(A, TR + \text{Reflexive})$). For any set x and relation R there exists a $\subseteq$ -maximal subset $y \subseteq x$ such that R is transitive and reflexive on y . (41,44)
8. If B is a Boolean algebra and $S \subseteq B$ such that S is closed with respect to $\wedge$ (meet) then there is a $\subseteq$ -maximal proper ideal I such that $I \cap S \subseteq \{0\}$. (101)

9. For each cardinal $\kappa \geq \aleph_0$ there is a cardinal μ such that $\kappa < \mu$ and $(\forall v)(v < \mu \rightarrow v \leq \kappa)$ (Ie. $\kappa <_3 \mu$). (150)
10. For each pair of cardinals κ and μ , the greatest lower bound of κ and μ , $\kappa \cap \mu$, exists. (153)
11. (TOP 6(W 2,Q)). Each T_2 space which has the property W_2 also has the property Q , where $Q = B_1, B_2, L_1, L_2, A_1$, or A_2 . (169,176)

III. The statements in this list are either equivalent or implied by the Boolean Prime Ideal Theorem (BPI). None are provable in NBG. In parentheses following each statement we shall write E, if it is equivalent, and I, if it is implied by BPI, followed by the page number where the statement appears in the text. The statement 1. below is BPI.

1. (BPI) Every Boolean algebra contains a proper prime ($\subseteq$ -maximal) ideal. (E; xii,99)
2. (BPI') If I is a proper ideal in a Boolean algebra then I can be extended to a proper prime ($\subseteq$ -maximal) ideal. (E; 101)
3. If B is a Boolean algebra and $S \subseteq B$ such that $0 \notin S$ and S is closed with respect to $\wedge$ (meet) then there is a proper prime ($\subseteq$ -maximal) ideal disjoint from S . (E;101)
4. Every distributive lattice with an identity element and at least one other element contains a prime ideal. (E; 101, 102)
5. (The Tychonoff Compactness Theorem for Hausdorff (T_2) spaces.) The product of compact T_2 spaces is compact in the product topology. (E;167,168)
6. (The Hahn-Banach Theorem, HB) Let p be a real-valued function defined on the vector space X over the reals $\mathbb{R}$ such that p is a sublinear functional. Suppose f is a linear functional defined on a subspace $S \subseteq X$ such that

$f(s) \leq p(s)$ for all $s \in S$. Then there is a linear functional F defined on X such that $F(x) \leq p(x)$ for all $x \in X$ and $F(s) = f(s)$ for all $s \in S$. (I;177)

7. (Alaoglu's Theorem)

(a) (AL) The unit sphere in the dual of a normed vector space over the reals is compact in the weak* topology.

(E; 178)

(b) (ALC) The unit sphere in the dual of a normed vector space over the reals is convex-compact (quasi-compact) in the weak* topology. (I (ALC $\leftrightarrow$ HB); 178)

8. If $\langle L, \leq \rangle$ is a Boolean algebra with identity 1, then there exists a two-valued measure μ on L such that $\mu(1) = 1$. (E; 184)

IV. Each of the statements in the list below is implied by the $2\kappa = \kappa$ Principle which is the first statement in the list. As far as we know none are provable in NBG. In parentheses following each statement we give the page number where it occurs in the text.

1. (The $2\kappa = \kappa$ Principle) For all infinite cardinals κ , $2\kappa = \kappa$. (xii)

2. $(\forall \kappa \geq \aleph_0)(\forall \mu)(\kappa + \kappa = \kappa + \mu \rightarrow \mu \leq \kappa)$. (140)

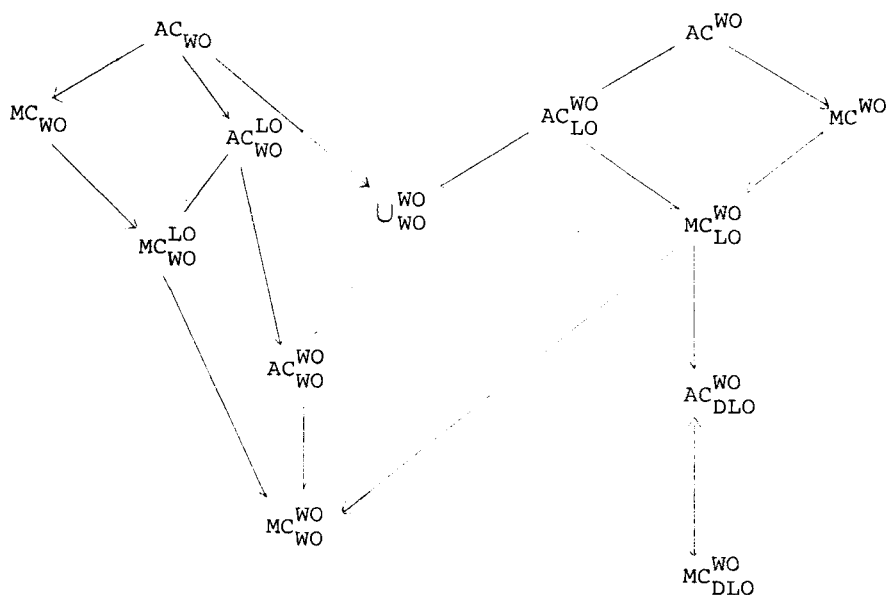
3. $(\forall \kappa, \mu)(\kappa + \mu \leq \kappa + \kappa \rightarrow \mu \leq \kappa)$. (141)

4. $(\forall \kappa_1, \kappa_2)(\forall \mu_1, \mu_2 \geq \aleph_0)[(\kappa_1 < \mu_1 \wedge \kappa_2 < \mu_2 \wedge \kappa_1 + \mu_1 = \kappa_2 + \mu_2) \rightarrow \mu_1 = \mu_2]$. (145)

V. Each of the statements listed below is implied by the axiom of choice in NBG°, but does not imply the axiom of choice in NBG. As far as we know, none are provable in NBG. In parentheses following each statement we give the page number where it appears in the text.

1. (The Principle of Dependent Choices, DC) If $R \neq \emptyset$ is a relation such that $\mathcal{R}(R) \subseteq \mathcal{D}(R)$ then there is a function f with domain ω such that for all $n \in \omega$, $\langle f(n), f(n+1) \rangle \in R$. (19)
2. (The Kinna-Wagner Principle, $KW 1 \leftrightarrow KW 2$)
 - (a) (KW 1) If x is a set of sets such that for each $u \in x$, $|u| \geq 2$ then there is a function f such that for each $u \in x$, $\emptyset \neq f(u) \subset u$. (83)
 - (b) (KW 2) For each set x there is an ordinal α such that $x \prec \mathcal{P}(\alpha)$. (83. Also see statement (1) on p 25).
3. Every set can be linearly ordered. (54,65)
4. (The Axiom of Choice for Well Ordered Sets, AC_Q^P .) For each set x , if x is a P -ordered set of non-empty sets such that each $u \in x$ can be Q -ordered, then there is a function f such that for each $u \in x$, $f(u) \in u$, where $P = WO$ and $Q = \emptyset, LO, DLO$, or WO ; and $P = \emptyset$ or LO and $Q = WO$. (78,85)
5. (The Multiple Choice Axiom for Well Ordered Sets, MC_Q^P). For each set x , if x is a P -ordered set of non-empty sets such that each $u \in x$ can be Q -ordered then there is a function f such that for each $u \in x$, $f(u)$ is a non-empty finite subset of u , where $P = WO$ and $Q = \emptyset, LO, DLO$ or WO ; and $P = \emptyset$ or LO and $Q = WO$. (78)
6. (U_{WO}^{WO}) The union of a well ordered set of well ordered sets can be well ordered. (85)

The following implications hold in NBG° between the statements in 4., 5., and 6. above.



7. If x is a set such that for each $u \in x$, u can be linearly ordered, then there is a function f such that for each $u \in x$, $f(u)$ linearly orders u . (82)
8. (Yos' Theorem, LT) Let A and X be non-empty sets, R a binary relation on A , $M = \langle A, R \rangle$, and $\mathcal{F}$ an ultrafilter on X , then $M^X/\mathcal{F}$ and M are elementarily equivalent. (166)
9. (The Krein-Milman Theorem)
 - (a) (KM) Let K be a compact convex set in a locally convex topological vector space. Then K has an extreme point. (177)
 - (b) (KMC) Let K be a convex compact (quasicompact) convex set in a locally convex topological vector space. Then K has an extreme point. (178)

10. For each ordinal α , $\aleph_{\alpha+1} \leq 2^{\aleph_\alpha}$. (26)
11. (INF 1) For every set x , if every infinite subset of x is D-infinite, then x can be well ordered. (84)

VI. This list contains the remaining statements which are related to the axiom of choice and are mentioned in the text. In parentheses following each statement we shall give the page number where it appears in the text, followed by a short statement describing its relative strength.

1. (The Generalized Continuum Hypotheses, GCH). For each cardinal $\kappa \geq \aleph_0$, there is no cardinal μ such that $\kappa < \mu < 2^\kappa$. (xii; $GCH \rightarrow AC$ in NBG° , but AC does not imply GCH in NBG .)
2. (The Continuum Hypotheses, CH). The cardinality of the set of real numbers is $\aleph_1$. (xii; $GCH \rightarrow CH$ in NBG° , but AC and CH are independent in NBG .)
3. (UP) All ultrafilters are principal. (167; $UP \rightarrow LT$ and $AC \rightarrow \neg UP$).
4. (PR) For every relation R there is a relation $S \subseteq R$ such that $\mathcal{I}(S) = \mathcal{I}(R)$ and for each $x \in \mathcal{I}(S)$, $S''\{x\}$ is a set. (194; $NBG \vdash PR$ but PR is not provable in NBG° .)
5. (LV) The universe can be linearly ordered. (216; A class form of the axiom choice implies LV in NBG° , but LV does not imply any set form of the axiom of choice in NBG and LV is not provable in NBG .)
6. For each pair of proper classes X and Y , either $X \prec Y$ or $Y \prec X$. (188; A class form of AC implies 6. and 6. implies a set form of AC in NBG° , but it is not known if 6. implies a class form of AC in NBG .)
7. If S is a class of sets of the same cardinality then there is a function which associates with each pair of elements of S a bijection mapping one onto the other. (192; A class form of AC implies 7. and 7. implies a set form of AC in NBG° , but it is not known if 7. implies a class form of AC in NBG .)

8. On can be injected into each proper class. (188; A class form of AC implies 8. in NBG° , but 8. does not imply any set form of AC in NBG and 8. is not provable in NBG.)

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